NANOMETRE: A unit of length, which is equal to 1/1,000,000,000th of a metre (one-billionth of a metre). The symbol for nanometre is nm, 1 nm = 10^{-9} metre. A carbon atom, for example, is 0.34 nm wide, a piece of paper is about 100,000 nanometres thick and human hand is 100 million nanometres long. Most current digital devices use processors, sensors and memory chips based on 45 and 60 nm, but Intel, makes transistors now based on a 14 nm manufacturing process. Also, the next generation of microprocessor technology is released by Intel, with transistors using a 10 nanometre manufacturing process. Over 10 billion transistors can now be packed onto a single chip.

NANOPORE: 1. A very small hole with dimensions of only a few nanometres, for example, soil pores having dimensions in nanometers. 2. A protein nanopore is set in an electrically resistant polymer membrane to measure current in order to identify a molecule; used to study the physical properties of biomolecules.

NANOSCOPIC FILM: A film which has lateral dimensions in the range of 0.1 nm to 100 nm.

NANOTECHNOLOGY: The branch of technology that involves materials on the atomic or molecular level with dimensions and tolerances less than 100 nanometres to provide a fundamental understanding of phenomena and materials at the nano scale level and to create and use structures, devices and systems such as computer chips that have novel properties and functions and are thousands of times smaller than current technologies permit. Quantum mechanical effects play a large role in the study of nanotechnology. The development of devices that are small, light, self-contained, use little energy and that will replace larger microelectronic equipment is one of the first goals of nanotechnology.

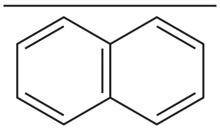
N

NANOTUBE: A cylinder made out of a length of extremely thin metallic or semiconducting cylinder, capped at one end consisting of a rolled-up layer of fullerene-structured carbon atoms. It is used to strengthen plastics and carbon fibres and as storage cells in Nantero's non-volatile memory chips. Nanotubes have the potential for making ultra-strong fabrics as well as reinforcing structural materials in buildings, cars and airplanes; and may replace silicon in electronic circuit. They are expected to be used in the construction of sensors and display screens. In 1998, IBM and NEC created nanotube transistors and three years later, IBM created a NOT gate using two nanotube transistors. Nanotube has a tensile strength 10 times greater than steel at about one quarter the weight and is considered the strongest material for its weight.

NAPALM (NAPHTHENIC AND PALMITIC ACIDS): A substance used in incendiary bomb flame throwers, made by forming a gel of petrol with aluminium soaps (aluminium salts as palmitic acid).

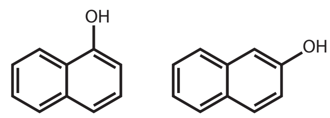
NAPHTHA: A mixture of hydrocarbons forming a flammable liquid obtained by the fractional distillation of petroleum or coal tar. It is a colourless to reddish brown volatile aromatic liquid, very similar to petrol and obtained by distilling coal tar (yielding aromatic products) or shale oil or from the refining and cracking of petroleum. The mixture depends on the hydrocarbons used and the temperature at which it is boiled. Full range naphtha consists of 5- to 12-carbon hydrocarbons boiled between 30 °C and 200 °C. Light naphtha consists of 5- to 6-carbon hydrocarbons boiled between 30 °C and 60 °C. Heavy naphtha consists of 6- to 12-carbon hydrocarbons boiled between 90 °C and 200 °C. It is used as a solvent and as a starting material for cracking into more volatile products such as petrol.

NAPHTHALENE (BICYCLO[4.4.0]DECA-1,3,5,7,9-PENTENE; ANTIMITE): A white, volatile, crystalline solid, aromatic hydrocarbon with the chemical formula $C_{10}H_8$, density 1.14 g/cm³, molecular weight 128.2, melting point 80.26 °C (353 K) and boiling point 218 °C (491 K). It is more reactive than benzene and produced from coal tar. It is the primary ingredient of mothballs and obtained from crude oil. It is a raw material for making certain synthetic resins, mothballs, indigo and certain dyes and also used as a mild disinfectant and as an insecticide.



Chemical formula of Naphthalene

NAPHTHOLS (NAPHTHYL GROUP): Any of two phenols, α-naphthol, composed of 1-hydroxynaphthalene and β-naphthol, which is 2-hydroxynaphthalene derived from naphthalene. The chemical formula is $C_{10}H_7OH$, relative density 1.28, melting point 123-124°C (396-397 K) and boiling point of 295°C (568 K). It is prepared by reacting naphthalene with sulfuric acid and hydrolysing the resultant sulfate ester by heating it with sodium hydroxide solution. They differ in the position of the -OH group. The most important one is β-naphthol with the -OH in the 2-position. It is a white solid used in rubber as an antioxidant. Naphthalene -2-ol or β- naphthol will couple with diazonium salts at the 1-position to form red azo compounds, a reaction used in testing for the presence of primary amines (by making the diazonium salt and adding napthalen-2-ol).



1-Naphthol ***2-Naphthol***

NAPHTHYLACETIC ACID (1-NAPHTHALENEACETIC ACID; α-NAPHTHALENEACETIC ACID, NAA): A white, odourless, synthetic auxin or plant hormone, with the molecular formula $C_{12}H_{10}O_2$, molecular weight 186.2066 g/mol and melts at 135 °C. It is soluble in organic solvents and slightly soluble in water. It is used as a plant growth regulator such as to stimulate the rooting of cuttings from stems and leaves and to promote flowering in plants. It is slightly toxic and in higher concentrations can be toxic to plants and have adverse effects such as inhibiting growth and development. Some of its functions on plants include the following: the stimulation of cell elongation, cell division in the cambium, differentiation of phloem and xylem, root initiation on stem cuttings and lateral root development in tissue culture; mediating the tropistic response of bending in response to gravity and light; driving apical meristem dominance; delaying leaf senescence, inducing fruit setting and growth in some plants; delaying ripening of fruits; etc.

NAPIER, JOHN (1550-1617): Scottish theologian, mathematician, physicist and astronomer, best known for the invention of Naperian logarithms and the notation of the decimal point in writing numbers. He also contributed to the theorems in spherical trigonometry, by introducing Napier's rules and Napier's analogies used in solving spherical triangles problems. He also invented the **Napier's bones**, a sort of slide rule used for multiplication and division using rods.

NAPIER'S BONES: An early mechanical calculating device consisting of a set of graduated rods that was used to do multiplication and division. Using the devise, multiplication was reduced to additions and division to subtractions. It was devised by Scottish mathematician, John Napier (1550-1617).

NAPIERIAN LOGARITHM (NAPERIAN LOGARITHM; NATURAL LOGARITHMS): A term often used to mean the natural logarithm, as first defined by John Napier. In natural or Napierian logarithms it is the exponent of base e, where e is the irrational number with the value 2.71828. It is generally written as $\ln(x)$, where $\ln(x)$ is the exponent to which e must be raised to get x. It is also written as $\log_e(x)$ or sometimes, if the base of e is implicit, written as $\log(x)$.

NAPIFORM (NAPACEOUS): Shaped like a turnip.

NAPPE: 1. Either of the two parts into which a cone is divided by the vertex. 2. A large-scale fold in rocks, thrown up by mountain-building processes and transported over large distances. The Alps are a system of several such nappes.

NARCISSISM: In psychology, it is a mental disorder in which a person has an exaggerated sense of his/her attributes and a deep need for admiration and a lack of empathy for others. This may affect relationships with other people.

NARCISSISTIC NUMBER (ARMSTRONG NUMBER; PERFECT NUMBER): An n-digit number, which is equal to the sum of its digits raised to the nth power. For instance, 371 is narcissistic because $3^3 + 7^3 + 1^3 = 371$ and 9474 is narcissistic because

$9^4 + 4^4 + 7^4 + 4^4 = 9474$. The smallest narcissistic number of more than one digit is $153 = 1^3 + 5^3 + 3^3$. The largest narcissistic number (in base 10) is 115132219018763992565095597973971522401, which is the sum of the 39th powers of its digits.

NARCOLEPSY: A disorder marked by excessive daytime sleepiness, uncontrollable compulsion to sleep and cataplexy, a sudden loss of muscle tone, usually lasting up to half an hour. It is the second-leading cause of excessive daytime sleepiness after obstructive sleep apnoea. It occurs even after getting enough sleep at night. The unusual sleep pattern can affect lifestyle.

NARCOTIC: Any drug that induces stupor, sleep and relieves pain, especially morphine and other opiates. Such narcotics are addictive and cause dependence. Their medical use is strictly controlled. Encephalins and endorphins are forms of narcotics produced biologically by the human body; they induce a sense of euphoria and also have pain-relieving effects very similar to those of morphine.

NARES (NOSTRILS; NARIS): The paired openings of the nasal cavity in vertebrates. All vertebrates have nares, which open to the exterior, from the point where they bifurcate to the external opening. In birds and mammals, they contain branched bones or cartilages called **turbinates**, whose function is to warm air on inhalation and remove moisture on exhalation. Internal nares are present only in air breathing vertebrates including lung fish and open into the mouth cavity. In mammals, they open posteriorly, beyond the secondary palate.

NARICORNS: The raised, tough, horny nostrils found on top of the bill of a bird.

NARROWBAND: In communications, it is a term, used to refer to a data transfer that has a slow or small transfer rate of 50 bps and 64 kbps.

NASA (NATIONAL AERONAUTICS AND SPACE ADMINISTRATION): An US government agency, founded in 1958 and responsible for civilian programs in aeronautics research and space exploration. Its headquarters is in Washington, DC and its main installations include the Kennedy Space Centre on Merritt Island in Florida; the Johnson Space Centre in Houston, Texas; the Jet Propulsion Laboratory in Pasadena, California; the Goddard Space Flight Centre in Beltsville, Maryland; and the Marshall Space Flight Centre in Huntsville, Alabama. NASA's Mercury, Gemini and Apollo programmes resulted in landing the first astronauts on the Moon in the space vehicle Apollo 11 in 1969.

NASAL CAVITY, NASAL FOSSA, NASAL CHAMBER: The large fluid filled cavity in the head of a vertebrate that is lined by a membrane rich in sensitive olfactory receptors. It conditions the air to the other areas of the respiratory tract by warming or cooling it to within 1 degree of body temperature. It is connected to the exterior by external nostrils and in air-breathing vertebrates to the respiratory system by internal nares. **Nasal bone** is any of a pair of narrow, oblong bones that form the bridge and root of the nose. The cavity is the uppermost part of the respiratory system and provides the passage for inhaled air from the nostrils to the nasopharynx and the rest of the respiratory tract. The nasal cavity is lined with a mucous membrane which produces mucous to assist in keeping the nose moist and preventing bleeding from dryness. There are tiny hairs in each nostril, which help filter the air, blocking dirt and dust from entering the lungs.

NASCENT: Starting to develop, but not fully formed.

NASCENT HYDROGEN (ATOMIC HYDROGEN): A reactive form of hydrogen made of individual hydrogen atoms which are not bound together like ordinary hydrogen into molecules. It is generated in the reaction mixture for example, by the action of acid on zinc. Nascent hydrogen can reduce elements and compounds that do not readily react with normal hydrogen. The commonest sources of nascent hydrogen are zinc and dilute hydrochloric acid, tin and concentrated hydrochloric acid, sodium and alkanol, sodium amalgam and water.

NANSEN'S FORMULA: The identity, da = { $\mathbf{F}^{-T}$ det$\mathbf{F}$ dA}, where da and dA are elements of area in the current and reference configuration respectively and $\mathbf{F}$ is the deformation gradient.

NASTIC MOVEMENTS: A non-growth movement response by plants to a stimulus that is independent of the direction of the stimulus or the movement of plant parts in response either to certain external stimuli such as light, temperature or to internal growth stimuli. Examples are the opening of tulip flowers to a rise in temperature (thermonasty), the opening of silk cotton tree (*Ceiba pentandra*) flowers at night (photonasty) and the folding up and drooping of leaves of the sensitive plant (*Mimosa pudica*) when lightly touched (haptonasty).

NATALITY (BIRTHRATE): The rate of reproduction in mammals such as humans in a specified community or area over a specified period of time. The birthrate is often expressed as the number of live births per 1,000 of the population per year.

NATANT: Swimming or floating in water.

NATIONAL AERONAUTICS AND SPACE ADMINISTRATION (NASA): An US government agency, founded in 1958 and responsible for civilian programs in aeronautics research and space exploration. Its headquarters is in Washington, DC and its main installations include the Kennedy Space Centre on Merritt Island in Florida; the Johnson Space Centre in Houston, Texas; the Jet Propulsion Laboratory in Pasadena, California; the Goddard Space Flight Centre in Beltsville, Maryland; and the Marshall Space Flight Centre in Huntsville, Alabama. NASA's Mercury, Gemini and Apollo programs resulted in landing the first astronauts on the Moon in the space vehicle Apollo 11 in 1969.

NATIONAL GRID: The network of cables carried overhead on pylons or buried under the ground and interconnects electrical power stations nation-wide.

NATIVE: In biology, it is a plant or animal that is in its own natural locality.

NATIVE CODE: In programming, it refers to a code that is written to run on a specific processor. In order to run the code on a different processor, an emulator must be used to trick the program into thinking it has a different processor. If an emulator is used, the code almost always runs more slowly than it would in its native environment.

NATIVE COMPILER: An application that converts one language, commonly a high-level language, into the binary or the computer's native language.

NATIVE ELEMENT: An element that occurs in nature uncombined with other elements and not as a compound in its ore or any non gaseous element that occurs naturally, uncombined with any other element(s). They are divided into three groups: metals, semimetals and nonmetals. There are about 20 elements that are found as minerals in the native state. They include bismuth, cadmium, chromium, indium, iron, nickel, tellurium, tin, titanium and zinc, as well as two groups of metals: the gold group and the platinum group. The gold group consists of gold, copper, lead, mercury and silver. The platinum group consists of platinum, iridium, osmium, palladium, rhodium and ruthenium. The semi metals include the arsenic group arsenic, antimony and bismuth; and the tellurium group include tellurium and selenium. The native nonmetals are sulfur and carbon in the forms of graphite and diamond.

NATIVE FILE: A file in its original file format that has not been converted to a digital image or other file format. The native file format is the default file format used by a specific software application. The native file format of an application is proprietary and these types of files are not meant to be transferred to other applications. An example is by saving a Word document, the file format is native to the Microsoft Word application and may not be recognised by other programs. But when it is saved as "Save As rich text (.rtf) file", it is not native to Microsoft Word, but can still be opened by the Microsoft Word program.

NATIVE LANGUAGE (HOST LANGUAGE): In computer science, it refers to the language understood by a computer. For example, binary is the language understood by the computer.

NATIVE METAL (FREE METAL): Any of the pure or alloy metallic elements found in nature. They include bismuth, cadmium, chromium, indium, iron, nickel, tellurium, tin, titanium and zinc, as well as two groups of metals: the gold group and the platinum group. The gold group consists of gold, copper, lead, mercury and silver. The platinum group consists of platinum, iridium, osmium, palladium, rhodium and ruthenium. Only gold, silver, copper and the platinum metals occur in nature in larger amounts. Alloys found in native state are brass, bronze and silver-mercury and gold-mercury amalgam.

NATIVE STATE: The occurrence of an element in an uncombined or free state in nature, for example, gold.

NATRIURETIC PEPTIDE: Any of several peptide hormones that causes natriuresis, the excretion of an excessively large amount of sodium in the urine. The natriuretic peptides are produced by the heart and vasculature. A-type natriuretic peptide is secreted largely by the atrial myocardium in response to dilatation. B-type natriuretic peptide is manufactured mainly by the ventricular myocardium. C-type natriuretic peptide is produced by endothelial cells that line the blood vessels. B-type natriuretic peptide is useful in the diagnosis of heart failure. The finding of a low level of B-type natriuretic peptide tends to exclude heart failure.

NATRON: Naturally-occurring hydrated sodium carbonate, found on the bed of dried-out soda-lakes with the chemical formula $Na_2CO_3.10H_2O$. It is a mixture of sodium carbonate decahydrate and about 17% sodium bicarbonate (nahcolite or baking soda, $NaHCO_3$) along with small quantities of household salt (halite, sodium chloride) and sodium sulfate.

NATURAL ABUNDANCE (NA): Relative proportion of one isotope to the total of the various isotopes in a naturally occurring sample of an element. For example, uranium has three naturally occurring isotopes: ^{238}U, ^{235}U and ^{234}U. Their respective NA is 99.2745%, 0.72% and 0.0055%.

N

NATURAL CLASSIFICATION: A classification of organisms according to relationships based on descent from a common ancestor. Most taxonomists strive towards the production of natural classifications, although their approaches and interpretations of the data may vary greatly. Natural classifications can best be achieved by exploring the fields of cytogenetics, microanatomy and phytochemistry in addition to morphology, anatomy and plant geography, thus broadening the base on which classifications are founded.

NATURAL DEDUCTION: A system of formal logic that permits the assumption of premises of an argument and does not have any axioms. It uses sequents to record which assumptions are operative at any stage. This contrasts with the axiomatic systems that use axioms to express the logical laws of deductive reasoning.

NATURAL DENSITY (ASYMPTOTIC DENSITY): A method for measuring the size of natural numbers. A subset A of positive integers has asymptotic density α, where $0 \leq \alpha \leq 1$, if the proportion of elements of A among all natural numbers from 1 to n is asymptotic to α as n tends to infinity.

NATURAL DISASTER: A naturally occurring event such as hurricanes, floods, fires, earthquakes, and tornadoes that leads to great damage or loss of life.

NATURAL EPIMORPHISM (NATURAL HOMOMORPHISM): An epimorphism from a group G to its factor group, G/ N, where N is a normal subgroup of G, given by mapping an element to its left (or right) coset xN or Nx. Analogous epimorphisms exist for rings and modules with respect to ideals and submodules respectively.

NATURAL FIBRES: Fibres such as cotton, jute, silk, wool and asbestos obtained from plant, animal or mineral sources. Asbestos is the fibrous form of several minerals and hydrous silicates of magnesium and also the fibrous forms of calcium and iron. Fibres such as wool and jute obtained from animals are classified as protein fibres and those obtained from plants are cellulosic fibres. The use of asbestos in consumer textiles has been banned, because of its carcinogenic effects.

NATURAL FREQUENCY: 1. The frequency at which a mechanical system will vibrate freely or the frequency at which a system that has been moved from its resting position will oscillate about a position if there are no resistive forces. 2. The frequency at which resonance occurs in an electrical circuit.

NATURAL GAS: Naturally occurring mixture of gaseous hydrocarbons found in porous sedimentary rocks in the earth's crust, usually in association with petroleum deposits. It consists chiefly of methane [CH_4] (about 85%), ethane [C_2H_6] (up to about 10%), propane [C_3H_8] (about 3%) and butane [C_4H_{10}]. Carbon dioxide, nitrogen, oxygen, hydrogen sulfide and sometimes helium may also be present. Natural gas like petroleum originates in the decomposition of organic matter. It is widely used as a fuel and also to produce carbon black and some organic chemicals.

NATURAL GRAPHITE: A mineral found in nature. It consists of graphitic carbon regardless of its crystalline perfection. Some natural graphites, often in the form of large flakes, show very high crystalline perfection. Occasionally, they occur as single crystals of graphite. Commercial natural graphite is often contaminated with other minerals, such as silicates and may contain rhombohedral graphite due to intensive milling.

NATURAL GROUP: A group of organisms of any taxonomic rank that are believed to be descended from a common ancestor. For example, man and the apes are often regarded as a natural group, descended from fossil ancestors, the dryopithecines.

NATURAL HAZARD: Naturally occurring phenomenon capable of causing destruction, injury, disease or death to humans and property. Examples include avalanche, drought and famine, earthquakes, floods, hurricanes, tornadoes, tropical storms and wild fire. Natural hazards occur globally and can play an important role in shaping the landscape. Natural hazards become natural disasters when people's lives and livelihoods are destroyed. Human activities such as blocking the natural flow of a river or stream by erecting structures can trigger natural hazards such as floods.

NATURAL HISTORY: The scientific study of objects and living organisms such as plants or animals in their natural habitat or the study of all natural phenomena.

NATURAL IMMUNITY (GENETIC IMMUNITY; INHERENT IMMUNITY; NATURAL IMMUNITY; NONSPECIFIC IMMUNITY): Immunity that does not arise from a previous infection or vaccination, but occurs naturally as a result of a person's genetic constitution or physiology.

NATURAL INSECTICIDES: Plant extracts such as nicotine, pyrethrum and neem, which are toxic to insects and are made by plants as defences.

NATURAL KEYBOARD (ERGONOMIC KEYBOARD): A Microsoft keyboard, introduced in 1994 for Mac OS X and Windows computers that lays out the keys in a more natural position by conforming more closely to the shape of human hands, making the keys closer to the finger tips and easier to reach than a traditional keyboard. This makes typing more comfortable and reduces muscle strain and related problems and keeps the wrists and fingers comfortable while typing for hours on end. An example is the Microsoft Natural Ergonomic Keyboard 4000 introduced in September 2005, with a Leatherette cushioning wrist rest built onto the keyboard.

NATURAL KILLER CELL (NK CELL): A type of cytotoxic lymphocyte that constitutes a major component of the innate immune system. It consists of a lymphoid cell that recognises and destroys tissue cells infected with pathogenic organisms. Before the more specific antigen specific mechanisms of the adaptive immune response are mobilised, they play a significant role in plays in the rejection of tumours and cells infected by viruses. They become activated in response to interferons or the release of cytokines by macrophages, bind to target cells and release cytotoxic granules onto the surface of their targets.

NATURAL LOGARITHM (NAPIERIAN LOGARITHM; (NAPERIAN LOGARITHM): A term often used to mean the natural logarithm, as first defined by John Napier. In natural or Napierian logarithms it is the exponent of base e, where e is the irrational number with the value 2.71828. It is generally written as ln(x), where ln(x) is the exponent to which e must be raised to get x. It is also written as $\log_e(x)$ or sometimes, if the base of e is implicit, written as log(x).

NATURAL NUMBER (WHOLE NUMBER): Any whole number greater than zero. Examples are 1, 2, 3, 4, etc. Natural numbers are often denoted $\mathbb{N}$.

NATURAL ORDER: A historical category more or less equivalent to the modern **family**. Fifty eight natural orders were listed by Linnaeus in 1764, many corresponding closely to present-day families, for example, Umbelliferae and Compositae. However, the International Rules of Nomenclature state that natural orders cannot be used in place of family.

NATURAL RADIOACTIVITY: The spontaneous disintegration of naturally occurring radioisotopes to produce more stable isotope with emissions of radiation and energy. The rate of disintegration is not influenced by chemical reactions or any normal changes such as temperature and pressure in their environment.

NATURAL SELECTION: An evolutionary process whereby gene frequencies in a population change through certain individuals producing more descendants than others because they are better able to survive and reproduce in their environment.

NATURAL TRANSFORMATION: A mapping between two functors that preserves the internal structure.

NATURAL TRANSMUTATION: The conversion of one chemical element into another. Examples of natural transmutation are radioactive decay, nuclear fission, and nuclear fusion.

NATURAL UNITS: A system of units of measurements, used principally in particle physics, in which all quantities that have dimensions involving length, mass and time are given dimensions of a power of energy, which is usually expressed in electrovolts. This is equivalent to setting the rationalised Planck constant and the speed of light both equal to unity.

NATURALISED: In biology, it refers to organisms that are introduced from elsewhere but now established.

NATURE: 1. The living world, all organisms and naturally formed features of the landscape such as mountains and rivers or the whole universe. 2. Typical qualities and characteristics of a person or an animal.

NATURE RESERVE (NATURAL RESERVE; NATURE PRESERVE; NATURAL PRESERVE): Area or land, of importance for wildlife, flora, fauna or features of geological or other special interest set aside to protect a habitat and the wildlife that lives within it, with only restricted admission for the public.

NATURE TRAIL: A path through the countryside labelled with signs to draw attention to important and interesting features about plants, animals and the environment.

NATURE VERSUS NURTURE: The combined effects of inherited factors (nature) and environmental factors (nurture) on the development of an organism. Geneticists are considering the question of what percentage of the trait is controlled by genes and what percentage is by the environment.

NATURE-NURTURE CONTROVERSY (ENVIRONMENT-HEREDITY CONTROVERSY): A long-standing dispute among philosophers and psychologists over whether nature or nurture is more important in the development of living organisms, especially human beings. Nurture is defined as the relative importance of environment including upbringing, experience and learning; and nature is defined as the physical and personality traits determined by the genes of an organism.

NAUCUM: The soft and fleshy part of a drupe.

NAUPLIUS: 1. The free-swimming larvae of many marine and freshwater crustaceans. It has an unsegmented body with a single eye at the front mandible, antennae and three pairs of limbs. 2. A genus of flowering plants in the daisy family.

NAUTICAL MILE (SEA MILE): An internationally agreed non-SI standard unit of distance used in sea and air navigation, equalling the average length of one minute of arc on a great circle of the Earth. The general unit varies from 1842 metres at the Equator to 1862 metres near the Poles. The internationally adopted nautical mile is equal to 1,852 metres with the symbol M, NM, Nm or nmi. The knot is a unit of speed of 1 nautical mile per hour.

NAUTILOID: Having the shape of a spiral, as seen in the nautilus shell.

NAUTILUS (CHAMBERED NAUTILUS; PEARLY NAUTILUS): A cephalopod mollusc of the genus *Nautilus*, especially *N. pompilius*, found in the Indian and Pacific oceans and having a spiral, pearly-lined shell with a series of air-filled chambers.

NAVEL (UMBILICUS; BELLY BUTTON): A small depression or a protrusion in the middle of the stomach of humans, where the umbilical cord was attached at birth. All placental mammals have a navel. It may be useful in distinguishing between identical twins in the absence of other identifiable marks.

NAVEL ORANGE: A large, sweet and flavourful orange, usually seedless with a navel-like formation at the stylal end. It has a thick bright-orange skin making it easy to peel.

NAVEL-ILL (JOINT-ILL): A disease of young livestock, especially newborn calves, kids and lambs. It causes abscesses at the navel and swelling in some joint.

NAVICULAR: 1. Having the shape of a boat. 2. The bone in front of the talus on the inner side of the foot.

NAVIER-STOKES EQUATION: An equation describing the flow of a Newtonian Fluid. The Navier-Stokes equation can be written in the form dv/dt + (v grad)v =1/p grad p + $h2v$, where v is the velocity, p is the density and p is the pressure, respectively, of the fluid. The equation can be derived using fluid mechanics or in certain cases from kinetic theory. It requires approximation techniques for a solution in all but the simplest problems.

NAVIGATION: 1. The science and technology of directing a ship, plane or other craft from one place to another by finding the position, course and distance travelled. It involves locating the navigator's position compared to known locations or patterns. 2. The ability of animals or insects to navigate. Although many animals navigate by following established routes or known landmarks, many animals can navigate without such aids.

NAVIGATION BAR: A set of graphic images in a row or column located at the top of a webpage used to link the users to major topic sections on a Web site.

NAVIGATION MAP (SITE MAP): In computing, it is a specialised tool to help users find their way around a website; it is an overview of the pages within a website. Site maps of smaller sites may include every page of the website, while site maps of larger sites often only include pages for major categories and subcategories of the website. Site maps help visitors or search engines get to any popular section of a web site or see what the web site has to offer.

NAVIGATION PANE: It is found on the left portion of the "Open" or "Save as" windows and lists all of the drives that are mapped to the user's computer, as well as items such as History, Desktop and Downloads that used to be on the Places bar. It was introduced by Microsoft Windows Vista to replace Places bar. Places bar is a left-hand section of a Microsoft Windows "Open" or "Save" window that contains folders such as History, My documents, Desktop, Favorites, Network and other custom shortcuts that is used to access common sections of a computer.

NAVIGATIONAL SATELLITES: Spacecrafts designed to assist in air, land and oceanic navigation. They include satellites such as Transit, NAVSTAR and GLONASS. Navigation satellites were developed in the 1950s and rely on the Doppler Effect to calculate the position of vessels emitting a radio signal.

NE'EMAN, YUVAL (1925 –2006): A renowned Israeli theoretical physicist, military scientist and a politician. He was educated at the Israel Institute of Technology at Haifa, where he graduated in engineering in 1945. He went to the Ecole de Guerre in Paris and, while serving as a military attaché at the Israeli embassy in London, gained his PhD in physics from the University of London in 1962. One of his greatest achievements in physics was the discovery of the classification of hadrons through the SU(3) flavour symmetry, now named the Eightfold Way, which was also proposed independently by Murray Gell-Mann. This SU(3) symmetry laid the foundation of the quark model, proposed by Gell-Mann and George Zweig in 1964 (independently of each other). The SU(3) theory successfully predicted the mass of the omega-minus particle observed for the first time in 1964. From 1961 to 1963 Ne'eman was scientific director of the Saraq Research Establishment of the Israeli Atomic Energy Commission and from 1963 was head of the physics department and an associate professor at Tel Aviv University. In 1964 he became a full professor and was also appointed vice-rector of the university. He was a minister in the Israeli government in the 1980s and early 1990s.

NEANDERTHAL MAN (NEANDERTHAL; NEANDERTAL): A form of fossil man that appeared in Europe and parts of western and central Asia about 100,000 years ago. It was thought to be a different species (*Homo neanderthalensis*) from modern man but is now generally regarded as a subspecies of *Homo sapiens*.

N

NEAP TIDES: Moderate tides with high tides a little lower and low tides a little higher than average that occur during the first and third quarter Moon phases. They occur twice a month when the Sun, Earth, and Moon form a 90-degree angle.

NEAR: A binary operator used in searches of computer text that returns true if its operands (usually two words) occur within a specific proximity to each other and false if otherwise.

NEAR FIELD COMMUNICATION (NFC): A set of standards developed primarily for smartphones and other similar electronic devices, allowing two-way communication between the devices by touching them together or bringing them into proximity. The range of NFC is very short, usually only a few centimetres.

NEAR INFRARED SPECTROPHOTOMETRY: A method of establishing tissue composition, used in agriculture to assess the quality of meat and of grain crop.

NEAR POINT (PRESBYOPIA): The nearest point at which the human eye can focus on an object or the closest position to the eye to which an object may be brought and still be seen clearly. As the lens loses its elasticity of the crystalline lens and becomes harder, with advancing age, the extent to which accommodation can bring a near object into focus decreases. The first signs of presbyopia occur between the ages of 40 and 50. It is characterised by eyestrain, difficulty seeing in dim light and problems focusing on small objects or fine print.

NEAR-EARTH ASTEROID: An asteroid whose orbit occasionally brings it close to Earth. Most, if not all of such asteroids appear to have come from the asteroid belt and have been given new orbits through collisions with other asteroids or the gravitational influence of Jupiter.

NEAR-EARTH OBJECT (NEO): An asteroid or extinct short-period comet in an orbit that may bring it close to the Earth. NEOs have perihelion distances of less than 1.3 astronomical units and include near-Earth asteroids belonging to the Apollo, Amor and Aten groups. Three NEOs – Ganymed, Eros and Eric have diameters greater than 10 km, while estimates suggest there may be at least 100,000 NEOs measuring 100 metres or more across. Many of these will collide with the inner planets over the next 10 million years or so.

NEAR-FIELD SCANNING OPTICAL MICROSCOPY (NSOM/SNOM): A form of scanning probe microscopy in which a probe with a very small aperture is used giving high resolving power.

NEAREST POINT: A point not in a given subset of a metric space that has minimal distance from the points of the subset; a rest approximation. Such a point exists whenever the subset is compact, but it is usually not unique.

NEBULA: A cloud of gas and dust in space or a fixed, extended and somewhat fuzzy white haze observed in the sky with a telescope. Many of these can now be resolved into clouds of individual stars and have been identified as galaxies.

NEBULA FILTER: A special type of filter made so as to enhance the contrast of many nebulae and thereby enable the observer to see nebulae that are otherwise not very visible in light-polluted locations. They can also enhance the visibility of some nebulae even at very dark observing sites. Nebula filters are manufactured with vacuum-deposition technology to create coatings that transmit very specific wavelength regions and block the rest of the visible spectrum to increase contrast for certain nebulae.

NEBULAR HYPOTHESIS (SOLAR NEBULAR HYPOTHESIS): A hypothesis that the Solar System evolved from a nebula cloud made from a collection of dust and gas. It was first developed in the 18th century by Emanuel Swedenborg, Immanuel Kant and Pierre-Simon Laplace.

NEBULOSE: Having cloud-like markings; indistinct, as in a fine, diffuse inflorescence.

NECESSARY: 1. In logic, of a statement, formula, etc., true under all interpretations or in all possible circumstances. 2. In logic, of an inference, having a conclusion that is true whenever the premises are true; valid. 3. In logic, of a property, essential, so that without it, the entity cannot continue to be what it is. 4. In logic, of a proposition, determined to be true by its meaning, so that its denial would be self-contradictory. 5. In logic, of a condition, entailed by the truth of some statement or the obtaining of some state of affairs, that is required to be true as a precondition for the latter to be true, so that if the necessary condition is false, then what it was a condition for must also be false.

NECESSITY: 1. Logically necessary, true or being essential. 2. The operator in modal logic that indicates that the expression it takes as argument is true in all possible worlds. It is usually denoted by $\Box$ or L.

NECK: 1. The highly flexible part of the body that connects the head to the shoulders and allows the head to turn and flex in all directions. The neck supports the weight of the head and protects the nerves that carry sensory and motor information from the brain down to the rest of the body. 2. The narrow part of an object such as a bottle similar to the neck in shape or position. 3. The slender portion of the archegonium through which the male gamete has to travel to reach the female gamete prior to fertilisation.

NECK COLLAR: A leather band put round the neck of a horse or cow to hold the animal in a stall.

NECK ROT: A disease affecting bulb onion during storage. The onion becomes soft and begins to rot from the green leaves downwards.

NECROSIS: The premature death of cells which can be caused by external factors such as infection, toxins or trauma. It often involves denaturation of proteins and may be preceded by a change in the appearance of cells and their content.

NECROTROPH: A parasitic organism that obtains its nutrition from its host's dead tissues and cells. Necrotrophic organisms produce lytic enzymes which kill the host.

NECTAR: A sugary or sweet liquid produced by some flowering plants from a specialised gland called nectaries, usually situated near the base of the flower. It attracts pollinating insects and other animals such as bees, butterflies, moths, hummingbirds and bats.

NECTAR GUIDE: Patterns seen on petals in some species of flowers that guide pollinators towards the nectaries. In some plants, they are visible only under ultra-violet. When in ultra-violet light, the flowers have a darker centre at where the nectaries are located. This is believed to make flowers more attractive to pollinators. An example is the sunflower. These patterns are sometimes also visible to humans. An example is the Dalmatian toadflax (*Linoria genistifolia*), which has yellow flowers with orange nectar guides.

NECTARINE: Name for a tree (*Prunus persica* var. *nectarina*) of the family Rosaceae (rose family) and for its fruit, a smooth-skinned variety of the peach. They are classified in the division Magnoliophyta, class Magnoliopsida order Rosales, family Rosaceae. The nectarine is a rounded fruit with a single central groove. Its smooth skin is blushed with hues of ruby, pink and gold throughout. The flesh is perfumed with aromatics, overtly juicy when ripe and golden coloured with red bleeds at the skin and surrounding the central rough pit. A ripe nectarine's texture is soft with a melting quality, its flavours balanced with layers both bright and sweet. There are two types of nectarines, yellow-fleshed and white fleshed. White-fleshed nectarines are typically low or sub-acid, while yellow-fleshed nectarines are both sweet and tart.

NECTARY GLAND: A gland of some flowers, producing sugary liquid called nectar located at the base of the petal. The nectar attracts insects that pollinate the flower. Flowers with nectaries have guides which lead to the nectar gland. Nectar is an economically important item, the sugar source for honey. It is also useful in agriculture and horticulture because the adult stages of some predatory insects feed on nectar.

NECTARIVORE (NECTIVOROUS): Animals, birds or insects that rely on nectar as a source of food. Some nectarivores, in the process of feeding on the nectar of plants do a valuable service by pollinating the flowers. Examples of nectarivores are hummingbirds, sunbirds and hawkmoths, carpenter bees and honeybees.

NEEDLE: 1. A small thin piece of polished steel pointed at one end and with a hole for thread at the other, used for sewing. 2. A piece of pointed hollow end of a syringe used for injections. 3. An object similar in shape to a needle, appearance or use. Examples are the thin pointed leaf of a pine tree, a pointed rock or peak.

NEEL TEMPERATURE (MAGNETIC ORDERING TEMPERATURE, TN): The temperature at which the susceptibility of an antiferromagnetic material has a maximum value or the temperature above which an antiferromagnetic material becomes paramagnetic. At that temperature, the thermal energy becomes large enough to destroy the macroscopic magnetic ordering within the material. The susceptibility increases with temperature, reaching a maximum at the Neel temperature, after which it abruptly declines.

NEEM (NIM): A fast-growing tree (*Azadirachta indica*) in the mahogany family, Meliaceae. It occurs all over the tropical and semi-tropical regions. It is drought resistance and normally thrives in areas with sub-arid to sub-humid conditions. It has long compound leaves crowded near the ends of the branches. Each leaf has up to 29 or 31 leaflets which are deeply serrated, sharply pointed and curved like a scythe. The white and fragrant flowers are arranged in drooping axillary panicles. The fruit is a smooth (glabrous) olive-like drupe which varies in shape from elongate oval to nearly roundish. Margosa oil, which is an extract of the seed of the Neem tree is widely used for the treatment of leprosy and skin. Every part of the tree has a medicinal value. Dried leaves put in drawers and cupboards keep out moths and cockroaches. Neem timber resembles mahogany, in that it is beautifully mottled, hard and heavy.

NEGATION: 1. A statement of denial or contradiction in logic, especially an assertion that a particular proposition is false. 2. The conversion of a binary digit or bit, into its opposite. For example, a 1 converted into a 0 is considered a negation. A complete negation of an 8-bit binary number may look similar to: Binary Number: 01100101; Negation: 10011010. 3. The act of a computer or computer hardware device denying access or a process from occurring.

NEGATIVE: 1. In mathematics, a quantity or value less than zero, defined as the result of subtracting natural numbers from zero and usually denoted by the minus (-) symbol, for example, negative 3 is written as -3. 2. A direction opposite to that of a positive one, having the same magnitude but an opposite sense to an equivalent positive quantity. 3. Of an angle, measured in a clockwise direction, especially from the positive direction of the x-axis of a coordinate system. 4. In logic, of a categorial proposition, denying the satisfaction by the subject of the predicate, as in 'some men are irrational' or 'no dogs have wings'. Also, the term is used for an expression that contains a privative term or a negation sign. 5. In physics, containing or producing the type of electric charge carried by electrons. 6. In photography, a reverse image which when printed is again reversed, restoring the original scene.

NEGATIVE ACKNOWLEDGMENT (NAK): In computer science, it is a signal sent among computers or network devices connected to a network indicating that the signal was improperly received or corrupted during the transmission of data.

NEGATIVE ADSORPTION: The depletion of one or more components in an interfacial layer.

NEGATIVE BINOMIAL DISTRIBUTION: The probability distribution of the number of failures, X, before the kth success in a sequence of Bernoulli trials where the probability of success at each trial is p and probability of failure is $q = 1 - p$. The distribution is given by: $P(X = x) = \binom{k + x - 1}{x} p^k q^x$, $0 \le x \le \infty$. The mean and variance are given by: Mean $= \frac{kq}{p}$, variance $= \frac{kq}{p^2}$.

NEGATIVE CHARGE: Having a surplus of electrons; having a lower electric potential. For example, if a sphere has a negative charge of 1×10^{-10} coulomb, it has 6.24×10^8 electrons more than needed to neutralise its atoms. Negatively charged substances are repelled from negative and attracted to positively charged substances. A positive charge has a deficiency of electrons and has a higher electric potential. Positively charged substances are repelled from other positively charged substances, but are attracted to negatively charged substances. A charge possessed by electrons in ordinary matter, may be produced by rubbing dissimilar nonmetallic substances such as resin and wool briskly together. The S.I. unit of quantity of electricity or charge is the coulomb, with the symbol q or Q. The coulomb is defined as the quantity of charge passing through an electrical conductor carrying one ampere in one second. The charge of a proton is equivalent to about 6.242×10^{18} and that of an electron is approximately -1.602×10^{-19} C.

NEGATIVE EXPONENTIAL DISTRIBUTION: A continuous probability distribution, whose density function is given by $f(x) = ae^{ax}$, where $a > 0$ for $x < 0$ and $f(x) = 0$ for $x > 0$. The mean and the standard deviation are $\frac{1}{a}$ and $\frac{1}{a^2}$ respectively.

NEGATIVE EXPONENT PROPERTY: A method of calculating negative exponents by determining the value as if the exponent was positive and then taking the reciprocal of that result as:

$(ab)^{-1} = \frac{1}{ab}$ and $\left(\frac{1}{ab}\right)^{-1} = ab$.

NEGATIVE FEEDBACK: 1. A feedback that reduces the output of a system. In electronic amplifiers, stability is achieved and distortion reduced by using a system in which the input is decremented in proportion as the output increases. Examples of negative feedback are the action of cold on a thermostat in a refrigeration system to limit the output or the accumulation of toxic waste products by a growing population of bacteria. 2. In biology, it is a primary mechanism of homeostasis, whereby a change in a physiological variable that is being monitored triggers a response that counteracts the initial fluctuation. There are several negative feedback in biological system to regulate and maintain homeostasis. Examples are the regulation of hormone synthesis, blood glucose levels, body temperature and baroflex in blood.

NEGATIVE HYPERGEOMETRIC DISTRIBUTION: Sampling without replacement from a population consisting of r elements of the same kind and $N - r$ of another, if two elements corresponding to that selected are replaced each time, then the probability of finding x elements of the first kind in a random sample of n elements is given by $P(x) = \frac{\binom{r+x-1}{x}\binom{N-r+n-x-1}{n-x}}{\binom{N-n-1}{n}}$. The mean of the distribution is $\frac{Nr}{N}$ and the variance is $\left(\frac{nr}{N}\right)\left(\frac{1-\frac{r}{N}}{N+1}\right)$.

NEGATIVE ION: An atom, radical, molecule or molecular moiety in the vapour phase which has gained one or more electrons thereby acquiring an electrically negative charge. The use of the term anion as an alternative is not recommended because of its connotations in solution chemistry.

NEGATIVE NUMBER: A number less than zero. On a number line, any number to the left of zero is negative. Negative numbers are always written with a minus sign before the number, for example, -4. Scales often display negative numbers. Subtracting a negative number from negative infinity (-∞) gives negative infinity (-∞).

NEGATIVE RATIONAL NUMBER: A rational number is negative if its numerator and denominator are of opposite signs such that, one of them is a positive integer and another one is a negative integer. A negative rational number is less than 0. Examples are -1, -2, -3, -4 and -2/3. It can be expressed as the quotient of x/y, where x or y is a negative integer. In a negative rational number, the negative sign is either put out in front of the fraction or with the numerator like -2/3 or 2/-3.

N

NEGATIVE REACTION: In botany, it is a tropism in which the organism moves, or the member grows, from a region where the stimulus is stronger to one where it is weaker.

NEGATIVE SEMIDEFINITE MATRIX: A Hermitian matrix in which all eigenvalues are non-positive.

NEGATIVE STAINING: A staining technique that relies on the property of certain components of the specimen not to take up the dye or negative stain. These, then show up light against a dark background. For example, in light microscopy, bacteria can be made visible by mixing them with a dark dye such as nigrosin or Indian ink. The method is used widely in electron microscopy to study viruses and certain large molecules.

NEGATIVE TERMINAL: The pole of a battery through which current enters from an external circuit.

NEIGH: A sound made by a horse.

NEIGHBOURHOOD: 1. In mathematics, also called **∈-neighbourhood** in a Euclidean or metric space, it is the open set of all points with distance from a given point strictly less than a specified value; the set of points $\{x: d(x, a) < \epsilon\}$, written N(∈, a) . An open ∈-neighbourhood is also called an **open ball** or **open sphere**. In this case, a function is said to have a limit as x tends to a if there is a p such that for every $\epsilon > 0$, there exists a $\delta > 0$ such that $f(x) \in N(\epsilon, p)$ for all x such that $x \in N(\delta, a)$. 2. Any set in a topological space that contains an open set that contains the given point. In a Euclidean or metric space, it is any set that contains an ε-neighbourhood, so that every ball is an open neighbourhood, but an open neighbourhood is not necessarily a ball. A closed ∈-neighbourhood of a point a in metric space is defined by $\{x: d(x, a) \leq \epsilon\}$. 3. **Punctured neighbourhood**, also known as **deleted neighbourhood** is a neighbourhood of a point, without the point itself; that is the point has been deleted. It is usually written with a prime to indicate deletion, as $N'(\delta, a)$ in a metric space or $U'(a)$ more generally in a topological space. A punctured ∈-neighbourhood of a, written as $N'(\epsilon, a)$, is $N(\epsilon,a)\setminus\{a\}$. 4. **Neighbourhood of infinity** is a neighbourhood of an ideal point added in a compactification. Thus]-r, ∞] is an open neighbourhood of +∞ on the real line.

NEIGHBOURBOOD BASE: A collection of neighbourhoods that form a base for a topology, so that every open set of the topology can be expressed as a union of some of these neighbourhoods.

NEIGHBOURING GROUP PARTICIPATION: The direct interaction of the reaction centre (usually, but not necessarily, an incipient carbenium centre) with a lone pair of electrons of an atom or with the electrons of a σ- or π-bond contained within the parent molecule but not conjugated with the reaction centre. A distinction is sometimes made between n-, σ- and π-participation. A rate increase due to neighbouring group participation is known as **anchimeric assistance**. **Synartetic acceleration** is the special case of anchimeric assistance ascribed to participation by electrons binding a substituent to a carbon atom in a β-position relative to the leaving group attached to the α-carbon atom. The term synartetic acceleration is not widely used.

NEKTON: Aquatic organisms such as squid, marlin, jellyfish, turtles and whales that swim actively through the water without being affected by water currents.

NEMATIC CRYSTAL: Any of various liquids in which the atoms or molecules are regularly arrayed in one dimension or two dimensions, the order giving rise to optical properties, such as anisotropic scattering, associated with the crystals. Liquid crystals can be divided into thermotropic, lyotropic and metallotropic phases. Thermotropic and lyotropic LCs consist of organic molecules. Thermotropic LCs exhibit a phase transition into the LC phase as temperature is changed. Lyotropic LCs exhibit phase transitions as a function of both temperature and concentration of the LC molecules in a solvent (typically water). Metallotropic LCs are composed of both organic and inorganic molecules and their LC transition depends not only on temperature and concentration, but also on the inorganic-organic composition ratio. Liquid crystals can be found in electronic displays. Lyotropic liquid-crystalline phases are found in abundance in many proteins and cell membranes, in solutions of soap and various related detergents, as well as tobacco mosaic virus.

NEMATOCIDE: A substance that destroys nematodes or roundworms. Although some species of nematodes help to control pests in gardens by feeding on them, some other species, may attack plants. Examples of nematicides are DiTera, Enzone and Furadan. Margosa oil, which is an extract of the seed of the Neem tree (*Azadirachta indica*) and marigold extracts can be used as natural nematicides.

NEMATOCYST (CNIDOCYST; CNIDA): A capsule-like structure found within the nemotocyte (cnidocyte) of cnidarians containing a thread-like, coiled, hollow tube with a poisonous or paralysing substance, used for capturing prey and as defence mechanism. It is characteristic of cnidarians such as jellyfishes, sea anemones, corals and hydrae.

NEMATODA: A phylum of pseudocoelomate invertebrates comprising the roundworms or threadworms which live in the soil as well as in freshwater and marine environments. They are characterised by a smooth, elongated, narrow cylindrical unsegmented body which tapers at the anterior and posterior ends. They shed their tough outer cuticle four times in their life to allow growth. Microscopic free-living forms play an important role in destruction and recycling of organic matter. In humans and some other mammals, the parasitic nematodes cause diseases such as lymphatic filariasis (caused by Fillaria, *Wuscheria bancrofti*), which can develop into elephantiasis, if left untreated; guinea worm disease (caused by *Dracunculus medinensis*), causing ulcerative lesions on the legs and feet; and trichinosis (caused by *Trichinella*), whose larvae invade and encyst in muscle tissue through eating undercooked meat, usually pork.

NEMATODE: Any type of worm such as hookworms, pinworms and eelworms belonging to the phylum Nematoda that has a smooth unsegmented body. They include plant and animal parasites and free-living forms found in soil, freshwater, saltwater and even vinegar and beer malts. Nematodes are pointed at both ends with a tough, smooth, outer skin. Microscopic nematodes help break down and recycle dead plants and animals. Some are parasites that cause diseases.

NEMATODE DISEASE: A disease of alimentary tract and lungs, caused by nematodes. Infection is transmitted from one group of animals to another by means of infective larvae in herbage.

NEMERTEA (NEMERTINEA; NEMERTINE): A phylum of soft-bodied worm-like animals known as ribbon worms or proboscis worms. About 1,400 species are known; most are marine, but some are freshwater and a few live on land. Most species grow to a length of 20 mm but one, the Bootlace worm, *Lineus longissima*, attains a length of 55 metres, making it one of the longest (if not the longest) animals known.

NEO-DARWINISM (MODERN EVOLUTIONARY SYNTHESIS; NEW SYNTHESIS; THE MODERN SYNTHESIS; NEO-DARWINIAN SYNTHESIS): The current theory of the process of evolution, formulated between 1920 and 1950 that combines evidence from classical genetics with the Darwinian theory of evolution by natural selection. It makes use of modern knowledge of genes and chromosomes to explain the source of genetic variation upon which selection works.

NEO-LAMARCKISM: Any of the comparatively modern theories of evolution, which are based on Larmarck's theory and responds to environmental influence can be inherited and transmitted through the action of natural selection.

NEO-PYTHAGOREAN MEANS: One of ten means defined by the followers of Pathagoras in terms of proportions, of which the first three are the Pythagorean means, which corresponds to the arithmetlc, geometric and subcontrary or harmonic means. The fourth mean is the counter-harmonic mean, expressed as $\frac{a^2+b^2}{a+b}$.

NEOCORTEX (NEOPALLIUM; ISOCORTEX): An area of the cerebral cortex of the vertebrate brain, forming the outer layer of the cerebral hemispheres. It is made up of six layers, labelled I to VI. It is most heavily developed in mammals in which it forms a surface layer covering most of the forebrain. It is also the major centre for the coordination of sensory and motor information.

NEODYMIUM: The yellowish metallic element of the lanthanide series with the symbol Nd, atomic number 60, relative atomic mass 144.24, melting point 1024 °C (1297 K) and boiling point 3074 °C (3347 K). It is a component of misch metal and some other alloys. It was discovered in 1885 by Austrian chemist Carl von Welsbach. It is used as a component in the alloys used to make high-strength neodymium magnets that are widely used in such products as microphones, professional loudspeakers, in-ear headphones and computer hard disks, where low magnet mass (or volume) or strong magnetic fields are required. It is used in lasers and its rose-coloured compounds are used in colouring glass and ceramics.

NEOLITHIC (NEOLITHIC AGE/ERA/PERIOD; NEW STONE AGE): A period in the development of human technology, beginning in the Middle East approximately 9000 BC and lasting until 6000 BC, during which man first developed agriculture, using wild and domesticated crops and herding of livestock, made pottery and polished stone tools and developed metal tools.

NEON: A rare, inert, colourless, odourless, gaseous element belonging to Group 18 (VIIIA) of the Periodic Table, the group of the noble gases. Its symbol is Ne, atomic number 10, atomic weight 20.180, melting point -248.67 °C (24.48 K) and boiling point -245.95 °C (27.2 K). Although it is colourless, it glows reddish orange in an electric discharge and used in electric advertising signs, television tubes and in lasers.

NEON LAMP: A cold-cathode small lamp consisting of a pair of electrodes, treated to emit electrons freely, sealed in a glass bulb containing neon gas at low pressure.

NEON TUBE: An electron tube in which neon gas is ionised by the flow of electric current through long lengths of gas tubing, to produce a luminous red glow discharge. It is used chiefly in outdoor advertising signs.

NEONATE: A newborn baby, who is under 28 days of age. During this time, the child is at highest risk of dying. It is thus crucial that appropriate feeding and care are provided during this period, both to improve the child's chances of survival and to lay the foundations for a healthy life.

NEOPALLIUM (NEOCORTEX; ISOCORTEX): An area of the cerebral cortex of the vertebrate brain, forming the outer layer of the cerebral hemispheres. It is made up of six layers, labelled I to VI. It is most heavily developed in mammals in which it forms a surface layer covering most of the forebrain. It is also the major centre for the coordination of sensory and motor information.

NEOPLASM: Any new abnormal growth of cells forming either a harmless tumour or a malignant one.

NEOPRENE: A synthetic rubber of high molecular weight made from the polymerisation of chloroprene. It is much more resistant to heat, light, oxidation and petroleum than an ordinary rubber. It is used in a number of applications including shoe soles, hoses, adhesives, gaskets, seals, foamed articles and rocket fuels.

NEOTENY: The retention of some juvenile characteristics in an animal that seems otherwise mature. An example is the retention of gills in some salamanders. Certain environmental conditions can inhibit the completion of metamorphosis. Low temperature or lack of available iodine retards the action of the thyroid gland. If environmental conditions improve, neoteny is reversible and the larvae can complete metamorphosis and attain normal maturity. It has been suggested that new species could arise in this way. Neoteny is thought to have been an important mechanism in the evolution of certain groups of animals such as man, who is believed to have developed from the juvenile form of apes on the grounds that facially, he resembles a young ape.

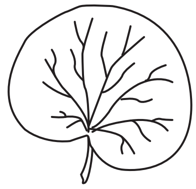
NEOTYPE: A plant specimen chosen to act as a standard taxonomic reference point following the irretrievable loss of all type material. The neotype should match the original description as closely as possible and ideally should come from the same locality as the original material.

NEPER (NAPIER): A unit used in telecommunication to express a ratio of powers, voltages and currents. It is used to denote the attenuation of amplitude A_1 to amplitude A_2 as N nepers, where $N = \log_e(A_2/A_1)$. One (1) neper = 8.686 decibels.

NEPHELOMETRY: The measurement of the turbidity or cloudiness of a liquid by measuring the reduction in the intensity of the incident light after it passes through the sample being measured by the use of Nephelometers. They generally have a light source, often a laser and a detector to measure the amount of light scattered by suspended particles. They are used in a number of fields including water quality control and blood analysis for protein content.

NEPHRIDIUM: 1. A simple tube-shaped organ in many invertebrate organisms and earthworms for excreting waste matter into the gut or out of the body. 2. An embryonic organ in a vertebrate animal which develops into kidney. The most primitive type is known as a **protonephridium** consisting of a system of flame cells occuring in platyhelminthes and rotifiers. The metanephridium, which occurs in the earthworm and some other annelids, opens into the coelem by combining with a coloemduct. The internal opening is known as **nephrostome** and the external opening is the **nephridiopore**.

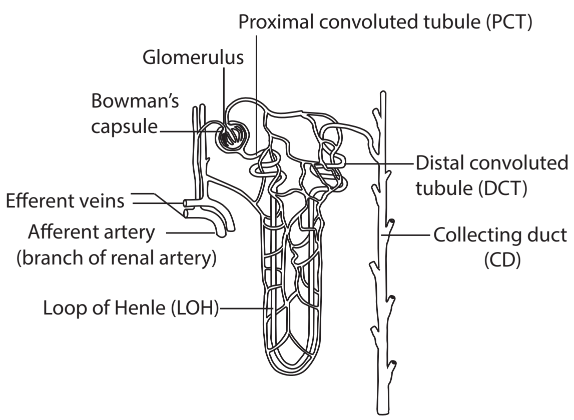
NEPHRITIS: Acute or chronic inflammation of the kidney caused by infection, degenerative process or vascular disease. The most prevalent form of acute nephritis is glomerulonephritis. This condition affects mostly children and teenagers. It is inflammation of the glomeruli or small round filters located in the kidney. Acute pyelonephritis is a bacterial infection of the kidney and renal pelvis, following sepsis or lower urinary tract infection. It affects mostly adults. A third type of nephritis is hereditary nephritis, a rare inherited condition.



Nephroid leaf

NEPHROID: 1. In mathematics, it refers to an epicycloid for which the diameter of the fixed circle is two times the diameter of the rolling circle. 2. In biology, reniform; having the shape of a kidney, as occurs in some leaves.

NEPHRON: Any of the numerous filtering units in vertebrate kidney consisting of Bowman's capsule, the first (proximal) and second (distal) convoluted tubules with the connecting loop of Henle, which opens into a collecting tubule that extracts waste materials from the blood to form urine. Many contents of the blood are filtered from the glomerulus into the Bowman's capsule. Water is reabsorbed in the collecting duct and the resulting concentrated solution of nitrogen-containing waste matter (urea in most mammals) plus inorganic salts drain from the collecting duct. The **malpighian body** or **malpighian corpuscle** is the filtration unit of the vertebrate nephron at the expanded end of the nephron. It consists of the glomerulus surrounded by the Bowman's capsule that opens into a tubule.



Nephron

NEPHROTOXIN: Any toxic substance such as chemicals and medication that affects the kidney. Common examples are mercury salt and certain herbicides such as paraquat and some mycotoxins such as aflatoxin.

NEPTUNE: The eighth planet from the Sun, with an average distance of 2.8 billion (4.5 billion). It was discovered in 1846 and named after the Roman god of the sea. It has no solid surface and orbits the Sun every 164.8 years. The mass of Neptune is 1.02 × 10^{26} kg, which is 17 times the mass of Earth, with a volume of 6.3 × 10^{13} km^3 (57 times that of Earth's) and 12% stronger gravity at the top of its atmosphere. It has an equatorial diameter of 49,528 km. Neptune consists largely of hydrogen and helium. It has no solid surface; its interior is believed to consist of a fluid mixture of rock, ices and gas. Its atmosphere contains substantial amounts of methane gas, whose absorption of red light causes Neptune's deep blue-green colour. Neptune has a system of rings, made up largely of dust-size particles and at least 13 moons, the largest is Triton, almost as big as Earth's Moon.

NEPTUNIUM: A silvery, ductile radioactive element in Group 3 (IIIB) of the Periodic Table with the chemical symbol Np, atomic number 93 and melting point 637 °C (910 K). It was synthesised as the first transuranium element in 1940 by bombardment of uranium with neutrons to produce neptunium-239. It is reactive and forms many binary compounds, for example, with hydrogen, carbon, nitrogen, phosphorus, oxygen, sulfur and the halogens. It has several isotopes; the lighter isotope neptunium-237 (^{237}Np) is a long-lived alpha emitter with half-life of 2.14 × 10^6 years. Neptunium is used mainly for neutron detection.

NEPTUNIUM SERIES: One of the four radioactive series, with nuclei that undergoes a series of alpha decay and beta decay until a stable nucleus is achieved. The neptunium series is artificially produced by nuclear reactions and have short half-lives. It has a half-life of 2 million years headed by neptunium-237 and ends with stable bismuth-209.

NERITIC ZONE (COASTAL OCEAN; SUBLITTORAL ZONE): The region of the sea extending from the low tide mark to the edge of the continental shelf, which is less than 200 metres deep. The neritic zone has generally well-oxygenated water, low water pressure and relatively stable temperature and salinity levels and coupled with light, phytoplankton and floating sargassum making the neritic zone rich in sea life. It is approximately the maximum depth for organisms carrying out photosynthesis.

NERIUM: A genus of flowering evergreen shrub or a small tree belonging to the family Apocynaceae, consisting of only one known species, *Nerium oleander*, a plant that thrives in warm subtropical regions and can tolerate drought for a long time. It can grow to between 2 and 6 metres tall. It has two cotyledons in the seed that usually appear at germination. The stem is erect and splay outwards as it matures. The dark green leaves are opposite or whorled, thick and leathery, narrow lanceolate and grow in pairs opposite each other or in whorls of three. The flowers are fragrant with white, red, pink or purple colours that grow in clusters at the end of each branch. The fruits are long, narrow capsules. The plant is used as an ornamental plant in landscapes, in parks and along roadsides. All parts of the plant are poisonous, but its extracts are used in controlled doses to treat some conditions such as muscle cramps, asthma, corns, menstrual pain, epilepsy, paralysis, skin diseases, heart problems and cancer. It is also used as an insecticide and rat poison.

NERNST EFFECT (NERNST-ETTINGSHAUSEN EFFECT): A thermoelectric or thermomagnetic phenomenon in which a temperature gradient along an electric conductor or semiconductor placed in a perpendicular magnetic field causes a potential difference to develop in the third perpendicular direction between opposite edges of the conductor.

NERNST EQUATION: An equation that relates the redox potential to the standard redox potential and the concentration of the oxidised and reduced form of two half-cells. It is expressed mathematically as $E_{cell} = E^{\circ}_{cell} - \frac{RT}{ZF}\ln Qr$, where E_{cell} = cell potential at absolute temperature (T in K), E_{cell} = standard cell potential, R = ideal gas constant (8.314 J/K/mol), T = absolute temperature, F = Faraday's constant, Z = number of electrons transferred and Qr = reaction quotient.

NERNST HEAT THEOREM: A statement of the third law of thermodynamics in a restricted form; which states that if a chemical change takes place between pure crystalline solids at absolute zero there is no change of entropy between the final and initial substances.

NERNST-EINSTEIN EQUATION: An equation relating the limiting molar conductivity Λm_0 to the ionic diffusion coefficients. The Nernst-Einstein equation is mathematically described as $\wedge m_0 = (F2/Rt)(v+z+2D+v--22D-)$, where F is the Faraday constant, R is the gas constant temperature, v+ and v- are the number of cations and anions per formula unit if electrolyte, z + and z- are the valences of the ions and D+ and D- are the diffusion coefficients of the ions. It is applied to calculate the ionic diffusion coefficients from experimental determination of conductivity.

NERNST, (HERMANN) WALTHER (1864-1941): German physical chemist noted for his work on electrochemistry and chemical thermodynamics and regarded as one of the founders of modern physical chemistry. He is remembered as the discoverer of the third law of thermodynamics, for which he was awarded the 1920 Nobel Prize for Chemistry. He also invented an improved electric light and an electronically amplified piano.

NERVATION: The arrangement of veins or nerves in an organ.

NERVE (PERIPHERAL NERVE): A strand of neurons enclosed in a sheet of connective tissues joining the central and the automatic nervous system, with receptor and effector organs. Each nerve is a cordlike structure that contains many axons referred to as "fibres". A nerve may carry only motor nerve fibre (motor nerve) or only sensory fibre (sensory nerve) or it may be mixed and carry both types. Although nerve fibres are in close proximity within the nerve, their physiological responses are independent of each other.

NERVE AGENTS (NERVE GASES): A class of highly toxic organic chemicals or organophosphates which contain phosphorus. They affect the nervous system by interfering with the action of acetylcholine, a neurotransmitter regulated by the enzyme acetycholinesterase (AChE). Acetylcholine is a common neurotransmitter found in the central and peripheral nervous system. The nerve agents, which are organophosphates, bind to acetycholinesterase (AChE) and block its action, stopping the buildup of acetylcholine. Examples of nerve agents are tabun (GA) with the chemical formula $C_5H_{11}N_2O_2P$, sarin (GB) with the chemical formula $C_4H_{10}FO_2P$, soman (GD) with the chemical formula $C_7H_{16}FO_2P$ and cyclosarin (GF) with the chemical formula $C_7H_{14}FO_2P$. An exposure to small amounts of nerve gas can kill in minutes. Poisoning by a nerve agent leads to contraction of pupils, profuse salivation, convulsions, involuntary urination and defecation and eventual death by asphyxiation as control is lost over respiratory muscles. Although the chemical was originally developed in 1854 and was to be used to control insects against crops, it was first made for military use by Germany in 1936. There are two series of agents: the G-series and the V-series. The G-series (G stands for German) was produced by German scientists. The main ones are tabun (GA) and cyclosarin (GF). The V-series, of which the most common is VX, was discovered at Porton Down in the UK in 1952. Nerve agents are classified as chemical weapons of mass destruction by the UN. The VX is the most dangerous. Although tabun is classified as the least deadly, it is still very dangerous.

NERVE CELL (NEURON): A specialised cell which is the basic functional unit of the mammalian nervous system and capable of rapidly transferring impulse or information between different parts of the body. A neuron or nerve cell usually consists of a cell body, axon and dendrite that transmit nerve impulses. The main part is the nerve fibre which is made up of a long threadlike axon covered in an insulating coat and conducts impulses from the cell body to other cells. A nerve fibre carries impulses in one direction only. Sensory neurons carry information from sense organs to the central nervous system. Motor neurons carry impulses from the central nervous system to muscles and glands. The dendrite transmits impulses received from other cells at synapses to the cell body.

NERVE CORD: A large bundle of nerve fibres, which run down the longitudinal axis of the body. It forms an important part of the central nervous system. Most invertebrates have a pair of solid nerve cords, situated ventrally and bearing segmentally arranged ganglia. All animals of the phylum Chordata have a dorsal hollow nerve cord, which forms the spinal cord,

NERVE FIBRE: The axon of a neurone together with the tissue associated with it such as myelin sheath. The length and diameter of nerve fibres are very variable even within the same organism.

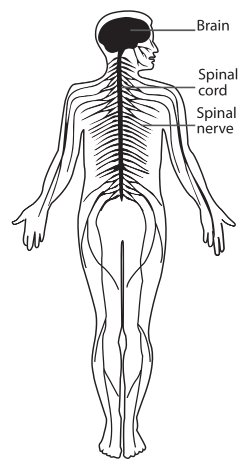
NERVE GROWTH FACTOR: A polypeptide produced by neurons and their supporting tissues that stimulates the growth of neurons or any of various polypeptides that promote the growth and maintenance of neurons, including neurotrophins.

NERVE IMPULSE: A wave of physical and chemical excitation along a nerve fibre in response to a stimulus, accompanied by a transient change in electric potential in the membrane of the fibre.

NERVE NET: A network of nerve cells connected with each other by synapses or fusion found in members of the cnidaria and echinodermata phyla. It helps them to respond to physical contact and to detect food and other chemicals. Although the nerve net allows the animal to respond to its environment, it has trouble alerting the animal from where the stimulus is coming.

NERVIFORM: Resembling a nerve. **Nervose** means having prominent nerves.

NERVINES: Agents or substances containing properties which tone down the nerves, as well as relieve pains and regulate the body's nervous system. They include herbs such as lavender, hops, lemon balm, chamomile, passionflower, wild lettuce, kola, coffee and green/black tea.



Human nervous system

NERVOUS SYSTEM: The system of interconnected nerve cells of most invertebrates and all vertebrates or the system of cells and tissues in multicellular animals by which information is conveyed between sensory cells and organs and effectors such as muscles and glands. The central nervous system of vertebrates such as humans contains the brain, spinal cord and retina. Its function is to receive, transmit and interpret information and then to formulate appropriate responses for the effector organ. The peripheral nervous system consists of sensory neurons, clusters of neurons called ganglia and nerves connecting them to each other and to the central nervous system.

NESQUEHONITE: A mineral form of magnesium carbonate trihydrate having the chemical formula $MgCO_3\,3H_2O$. It is a colourless orthorhombic mineral consisting of hydrated magnesium carbonate.

NESSLER'S REAGENT: A solution of mercury(II) iodide (HgI_2) in potassium hydroxide, named after Julius Nessler (1827 - 1905). It is generally prepared from potassium iodide and mercury(II) chloride. It is used in testing for ammonia with yellow colouration indicating the presence of ammonia. At higher concentrations, a brown precipitate may form.

NEST: A structure prepared by many animals for the protection of their eggs and young or for sleeping purposes. In social insects, the nest provides the home of the whole colony and may have special structures for temperature control and ventilation. The sleeping nests of, for example, the great apes, are commonly no more than crudely woven hammocks of twigs and branches. The sleeping nests of other mammals, which may be used for hibernation, are as complex and woven as any breeding nests. Both these and the nests used by birds and mammals for breeding, must protect the animals within from both weather and predators. Nests can be built of mud, leaves, twigs, down, paper and various kinds of human garbage.

NEST BOX: An open-fronted box in which a hen lays egg. The box may be a single unit or part of a series of boxes.

NESTED: Of a sequence of sets or intervals, having the property that each set is contained in the preceding set. In Euclidean space, if nested intervals are non-empty, bounded and closed, there must be at least one point common to all. This is called the nested interval theorem.

NET: 1. A flattened three-dimensional figure which can be turned into the solid by folding it. 2. In mathematics, also called **Moore-Smith sequence**, it is a generalisation of the concept of a sequence to permit talk of convergence in non-metrizable topological spaces. A net of a set S is a mapping from a directed set D into S. 3. A two-dimensional lattice (in a particular range of surface coverage and temperature) into which the adsorbate is ordered, especially in some cases of localised adsorption.

NET ASSIMILATION RATE (NAR): The increase of implant dry mass per unit leaf area per unit time typically measured in kilograms per square centimetre per day. It enables comparisons of the efficiency of production between plants of different sizes.

NET BIOME PRODUCTION (NBP): Net gain or loss of carbon from a region. NBP is equal to the net ecosystem production minus the carbon lost due to a disturbance, for example, a forest fire or a forest harvest.

NET BLOTCH: A fungal disease of barley, with dark brown patches on the leaves.

NET CARBON DIOXIDE EMISSIONS: Difference between sources and sinks of carbon dioxide in a given period and specific area or region.

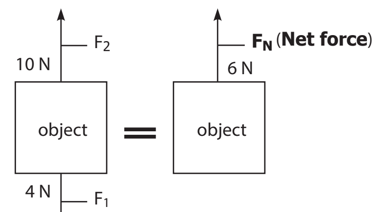
NET CONVERGENCE (MOORE-SMITH CONVERGENCE): In topology, the property of a net of a sets with respect to a directed set D, that a sequence $\{x_d\}$, where x_d is the element of S associated with d in D, is eventually in every neighbourhood V of some x, in the sense that there exists d such that x_e lies in V for all e in D with $e \geq d$; the net is then said to converge to x.

NET CURRENT: The sum of cathodic and anodic partial currents.

NET ECOSYSTEM PRODUCTION (NEP): Net gain or loss of carbon from an ecosystem, defined as the difference between gross primary production (organic carbon fixed in an ecosystem by photosynthesis) and total carbon lost through heterotrophic respiration. NEP represents the total amount of organic carbon in an ecosystem available for storage, export as organic carbon, or nonbiological oxidation to carbon dioxide through fire or ultraviolet oxidation. Whether NEP is greater or less than zero determines if an ecosystem is autotrophic or heterotrophic.

NET ELECTRIC CHARGE: In electrophoresis, it is the algebraic sum of the charges present at the surface of the particle divided by the elementary charge of the proton.

NET FORCE: The single force that has the same effect both in magnitude and direction as all the original forces acting together. The net force is calculated by adding all the forces vectorially, taking into consideration the forces in the positive and negative x and y directions. This is done by the resolution of the various forces in the x and y directions. The direction of the net force is that of the combined forces. If the forces are parallel but opposite, the net (resultant) force is obtained by subtration (algebraic addition). If there are several forces acting on the object, the net force for each direction must be determined.



Net force of forces acting on an object

NET IONIC EQUATION: An equation that indicates only the ionic species that actually take part in a chemical reaction.

NET PRIMARY PRODUCTIVITY (NPP): The rate at which producer (usually plants) biomass is created in a community. Net primary production (NPP) is the gross production minus losses due to autotrophic or plant respiration per unit time and it represents the actual new biomass that is available for consumption by heterotrophic organisms.

NET SECONDARY PRODUCTIVITY (NSP): The rate of production of biomass by heterotrophs, which feed on plant products or other heterotrophs.

NET SHAPING: Production of an object in or as close as possible to, its final shape prior to ceramisation.

NET WEIGHT: The weight of a quantity or product deducted from its package.

NET-VEINED (NETTED): Possessing veins that are branched and form a network. This is seen in leaves of most dicotyledonous plants.

NETIQUETTE (CYBER ETHICS, NET ETIQUETTE): A term used to describe the unwritten rules of internet courtesy. Netiquette includes rules to maintain civility in discussions and also special guidelines unique to the electronic nature of forum messages. For example, netiquette advises users to use simple formats because complex formatting may not appear correctly for all readers.

NETMASK: A 32-bit mask used to divide an IP address into subnets so that only the host computer part of the address remains. There are three standard classes of IP addresses: Class A with 255.0.0.0 as netmasks; Class B with 255.255.0.0 as netmasks; and Class C with 255.255.255.0 as netmasks. Two bits are always automatically assigned. For example, in 255.255.225.0, "0" is the assigned network address; and in 255.255.255.255, "255" is the assigned broadcast address and cannot be used.

NETRIN: Any of several related proteins that act as long-range chemical cues to guide the growth of axons in vertebrates. Netrins are secreted and diffuse through the extracellular matrix, creating a concentration gradient that influences how the growth cone of the developing axon navigates through the tissues. Specific netrin receptors on the growth cone bind the netrin molecules.

NETTLE: A flowering plant, classified as *Urtica dioica*, which possesses stinging hairs. The family comprises over 1,000 species placed in 45 genera, which includes the mulberry family and the elm family.

NETWORK: 1. In mathematics, also known as a **graph**, it is a set of nodes or vertices that are connected together. The connections between the nodes are called **edges** or **links**. A **directed network** or a **directed graph**, (**digraph**) has edges that are pointing in only one direction (directed), with the edges typically drawn as arrows indicating the direction. If all edges are bidirectional or undirected, the network is called **undirected network** or **undirected graph**. 2. A method of connecting computers so that they can share data and peripheral devices, such as printers or a system of interconnected terminals on-line to one or more computers. 3. Highly ramified structure in which essentially, each constitutional unit is connected to each other and to the macroscopic phase boundary by many paths through the structure, the number of such paths increasing with the average number of intervening constitutional units. The paths must, on average be co-extensive with the structure.

NETWORK ADAPTER (NETWORK CARD, NIC): A hardware unit that connects a device to a communication line. For wide area networks (WAN), adapters connect routers to the specific type of connection (T1, BRI) that is installed. For local area networks (LAN), adapters connect workstations to the LAN (Ethernet or TokenRing) cabling. These include Ethernet cards, internal Wi-Fi chips and external wireless transmitters.

NETWORK ADDRESS TRANSLATION (NAT): An Internet standard that enables a local-area network (LAN) to use one or more IP addresses for internal traffic and a second for external. A router or firewall rewrites the source and/or destination Internet addresses in a packet as it passes through, typically to allow multiple hosts to connect to the Internet via a single external IP address. It translates the IP addresses of computers in a local network to a single IP address. This address is often used by the router that connects the computers to the Internet. The router can be connected to a DSL modem, cable modem, T1 line or even a dial-up modem. When other computers on the Internet attempt to access computers within the local network, they only see the IP address of the router. This adds an extra level of security, since the router can be configured as a firewall, only allowing authorised systems to access the computers within the network.

NETWORK ADMINISTRATOR: An individual or group of individuals responsible for the maintenance and operation of a network or server.

NETWORK APPLICATION PROGRAM: In networking, a program used to connect and communicate with adapters on a network, enabling users to perform application-oriented activities and to run other application programs.

NETWORK COMPUTING: The use of a scalable distributed computing infrastructure that encompasses the key elements of today's networking technologies, such as systems and network management; the Internet and intranets; clients and servers; application programs; databases; transaction processing; and various operating systems and communication protocols.

NETWORK FLOW: In mathematics, it is a directed graph in which each arc (edge) has a capacity such that the amount of flow it receives is less than or cannot exceed its capacity so that the total amount of flow entering and exiting each intermediate node (vertice) balances. This is applicable, for example, in routing telephone calls; or to direct traffic such as aeroplanes at very busy airports in order to maximise the value of the network. A source in the network flow has only outgoing flow and a sink, has only incoming flow.

NETWORK INTERFACE CARD (NIC): The card that physically makes the connection between the computer and the network cable. These cards typically use an Ethernet connection and are available in 10, 100 and 1000 Base-T configurations. A 100 Base-T card can transfer data at 100 Mbps. The cards come in ISA and PCI versions and are made by companies like 3Com and LinkSys.

NETWORK OPERATING SYSTEM (NOS): An operating system that runs on a network server to facilitate communication and sharing of network resources and to manage security and other network functions among multiple computers. A network operating system can also be installed on a hardware operating the functions of the network layer (layer 3) of the ISO, OSI model, such as Cisco IOS (Internetwork Operating System).

NETWORK PLUS (N+): A certification programme, developed by the Computing Technology Industry Association (CompTIA) that certifies individuals for knowledge in TCP/IP, managing a network, network operating systems and installing and upgrading networks. The examination is in two parts, the first covering computer hardware and software in general and the second covering a specific operating system.

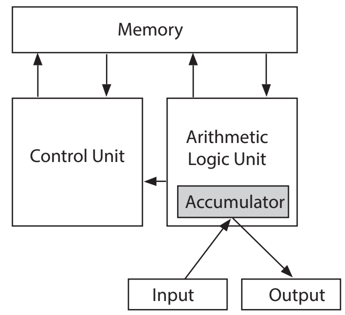
NETWORK TIME PROTOCOL (NTP): Internet standards for synchronising clocks between computer systems as it is important for computer systems to have the correct date and time, which is especially important for online transaction processing systems such as banks. NTP servers base their own clocks on the times transmitted from super-accurate atomic clocks. Other computers, which represent clients, periodically check their own time with that of the NTP server, to remain in accordance.

NETWORK TOPOLOGY: In communication networks, a topology is usually a schematic description of the arrangement of a network, including its nodes and connecting lines. There are two ways of defining network geometry: the physical topology, which is the interconnected structure of a local area network (LAN); and the logical or signal topology, which defines the way the data passes through the network from one device to the other. The main network topologies are star topology, ring topology and linear bus topology. In addition some configurations of the three main topologies also exist. Examples are decentralised networks, the full mesh network and partial mesh networks.

NETWORK TRAFFIC (DATA TRAFFIC; TRAFFIC): In telecommunications, it is the amount of data moving across a network at a given point of time. It is the total volume of cells, blocks, frames, packets, calls, messages, or other units of data carried over a circuit or network, or processed through a switch, router, or other system.

NETWORK-ATTACHED STORAGE (NAS): A file-level computer data storage connected to a computer network providing data access to other systems on the network. By using NAS, computers can store and access data using a centralised storage location. Instead of each computer sharing its own files, the shared data is stored on a single NAS server. This provides a simpler and more reliable way of sharing files on a network. NAS is often manufactured as a specialised computer built from the ground up for storing and serving files rather than simply a general purpose computer being used for the role.

NEUMANN ARCHITECTURE: A computer architecture conceived by the mathematician John von Neumann in 1945, comprised of an arithmetic logic unit (ALU), memory, input/output and a control unit. It forms the core of nearly every computer system in use today.



von Neumann's Computer Architecture

NEUMANN CONDITION: A boundary condition that is imposed on an ordinary or a partial differential equation to specify the values that the derivative of a solution is to take on the boundary of the domain: the normal derivative, $\partial u/\partial n$, of u, defined as $\nabla u.n$, is given at each point.

NEUMANN FUNCTION: In partial differential equations, the function given by $Y_v(x) = \dfrac{\cos v\pi\, J_v(x) - J_{-v(x)}}{\sin v\pi}$, where $J_v(x)$ is the Bessel function and v is the order of the Neumann function.

NEUMANN LINES: Found in some iron meteorites, they are groups of very fine parallel lines that cross each other at various angles. Irons containing Neumann lines can easily be cleaved in three mutually perpendicular directions. They are named after their discoverer, the German crystallographer and mathematical physicist Franz Ernst Neumann (1798–1895) who first studied them in 1848.

NEUMANN, VON JOHN (1903-1957): Hungarian-born American mathematician, who is best known for the founding of game theory and also made major contributions to a number of fields including mathematical economics, computer science, quantum theory, set theory, group theory, operational calculus, probability, mathematical logic and the foundations of mathematics. He proposed and embodied the "Stored Program Concept". He is reputed as the father of the structure of the digital computer system in which a single relatively large memory is used to hold or store both instructions and data. A diagram of his computer architecture is shown in the diagram.

NEUMANN'S LAW (FARADAY-NEUMANN LAW): It states that the magnitude of an electromagnetically induced emf (E) is given by E = dΦ/dt, where Φ is the magnetic flux. This is a quantitative statement of Faraday's second law of electromagnetic induction.

NEURAL: 1. Of or relating to functions and parts of a body related to the nervous system. 2. Of, relating to or located on the same side of the body as the spinal cord.

NEURAL CREST: A group of embryonic cells along the border where the neural tube pinches off from the ectoderm. The cells migrate to various parts of the embryo and form the pigment cells in the skin, bones of the skull, the teeth, the adrenal glands and parts of the peripheral nervous system. The most conspicuous of the neural crest derivatives are the melanocytes, cells in the deep layers of the epidermis that contain pigment and are responsible for skin colouration.

NEURAL GROOVE: Dorsal, longitudinal groove that forms in a vertebrate embryo. It is bordered by two neural folds and is preceded by the neural-plate stage and followed by the neural-tube stage.

NEURAL NETWORK: An emerging technology in the area of machine information processing and decision making. It is an information-processing device that consists of a large number of simple nonlinear

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processing modules, connected by elements that have information storage and programming functions that attempts to mimic the structure of nerve cells (neurons) in the human brain.

NEURAL PLATE: A strip of ectoderm, above the notochord that lies along the central axis of the early embryos of chordate and develops into the central nervous system. The neural plate folds and forms the neural tube, a process called **neurulation**. During this stage when the neural tube is forming, the embryo is called **neurula**.

NEURAL SPINE: A projection that points backward from the neural arch of a vertebra.

NEURAL TUBE: A hollow tube of tissue in the early embryo of vertebrate that subsequently develops into the brain and spinal cord.

NEURAL TUBE DEFECT (NTD): Any of a group of birth defects that affect the brain, spine or spinal cord caused by incomplete closure of the neural tube. Types of NTD include spinal bifida, where part of the spinal cord and its membranes are exposed through a gap in the backbone; and anencephaly in which the major portion of the brain, skull and scalp does not develop during embryonic development. Babies with anencephaly are usually either stillborn or die shortly after birth. Factors that increase the risk of having NTD babies include obesity, poorly controlled diabetes and NTD history in the family.

NEURALGIA: Pain originating in a nerve and characterised by sudden sharp, often electric shock-like pain or exacerbation of pain. Nerves commonly affected include the digital nerves of the toes and intercostal nerves. Neuralgia may be due to inflammation or trauma.

NEURITE: Any projection from the cell body of a neuron, especially an immature neuron. As the neuron develops, neurites can become either dendrites or axons.

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NEURITIS (PERIPHERAL NEUROPATHY): Any disorder of the peripheral nervous system which interferes with sensation, the nerve control of muscle or both. Causes of neuritis include mechanical (injury, pressure), vascular (occlusion of a vessel or haemorrhage into nerve tissue), infectious (invasion by microorganisms), toxic (metallic or chemical poisoning, alcoholism) or metabolic (vitamin deficiencies, pernicious anaemia).

NEUROBLASTOMA: Malignant tumour that develops from immature nerve cells. It develops most often in the adrenal gland; as well as in several areas of the body such as the abdomen and in the chest, neck and near the spine.

NEUROENDOCRINE SYSTEM: Any of the systems of dual control of certain activities in the body of some higher animals by nervous and hormonal stimulation. For example, the posterior pituitary gland and the medulla of the adrenal gland receive direct nervous stimulation to secrete their hormones, whereas the anterior pituitary gland is stimulated by factors released from the hypothalamus.

NEUROFIBRIL: Any of the very fine fibres that occur in the cell bodies, axons and dendrites of neurons. It includes neurofilament and neurotubules, microtubules that play role in the transport of proteins and other substances within the cytoplasm.

NEUROFILAMENT: A type of intermediate filament found in the axons of nerve cells. They serve as elements of the cytoskeleton supporting the axon of cytoplasm.

NEUROHAEMAL ORGAN: A discrete body formed by a cluster of neurosecretory-cell terminals, where neurohormones are released into an adjacent blood space or capillary bed. They are found widely in both vertebrates and invertebrates.

NEUROHORMONE: A chemical that is released by neurons that act at a specific distant site or sites. Examples of neurohormones are TRH (thyrotrophin releasing hormone), dopamine or epinephrine. Dopamine, epinephrine and norepinephrine can be both neurotransmitters and neurohormones depending on the action. For example, epinephrine acting on the adrenal glands is a neurohormone. Epinephrine acting in the brain is a neurotransmitter. The neurohormones in most mammals include oxytocin and vasopressin, both of which are produced in the hypothalamic region of the brain and secreted into the blood by the neurohypophysis (part of the pituitary gland).

NEUROLEPTIC DRUG (ANTIPSYCHOTIC DRUG): Any of the class of drugs that acts on the nervous system by simultaneously modifying agitation and neuronal activity. These drugs are usually prescribed for mental problems such as schizophrenia or bipolar disease.

NEUROLOGY: A branch of medicine concerned with diseases of the brain, spinal cord and peripheral nervous system. These include multiple sclerosis, epilepsy, migraine, stroke, Parkinson's disease, neuritis, encephalitis, meningitis, brain tumours (gliomas), muscular dystrophy and myasthenia gravis.

NEUROMODULATION: The treatment that delivers either electricity or drugs to nerves in order to change their activity. It can be used for a variety of diseases and symptoms. For example, neuromodulation can be used to treat spinal cord damage, headaches, Parkinson's disease, chronic back pain and even deafness.

NEUROMODULATOR: A chemical agent that is released by a neuron and diffuses through a local region of the central nervous system, acting on neurons within that region. It generally has the effect of modulating the response to neurotransmitters.

NEUROMUSCULAR JUNCTION (NMJ): The point, where a muscle fibre comes into contact with a motor neuron carrying nerve impulse from the central nervous system. The development of the neuromuscular junction depends on signal molecules, such as agrin, which recruits components required for the synaptic connections.

NEURON (NERVE CELL): A specialised cell which is the basic functional unit of the mammalian nervous system and capable of rapidly transferring impulses or information between different parts of the body. A neuron or nerve cell usually consists of a cell body, axon and dendrite that transmit nerve impulses. The main part is the nerve fibre which is made up of a long threadlike axon covered in an insulating coat and conducts impulses from the cell body to other cells. A nerve fibre carries impulses in one direction only. Sensory neurons carry information from sense organs to the central nervous system. Motor neurons carry impulses from the central nervous system to muscles and glands. The dendrite transmits impulses received from other cells at synapses to the cell body.

NEURON THEORY (NEURON DOCTRINE): The hypothesis, now accepted, proposed in 1891 by H. Waldeyer-Hartz that the nervous system consists of nerve cells (neurons), which are functionally linked at synapses but are physically separate.

NEURONAL NETWORK: A circuit interconnecting neurones. All types of behaviour depend on information in the form of nerve impulses, being transmitted via synapse between individual neurons through neuronal network. Such networks range from the relatively simple connection of a reflex arc to the highly complex circuits involved in processing information in the brain. However, their activity depends primarily on the number of constituent neurons and the configuration of their connecting synapses.

NEUROPEPTIDE: Small protein-like signalling neuronal molecules used by neurons to communicate with each other to influence the activity of the brain in specific ways and are thus involved in particular brain functions such as analgesia, reward, food intake, learning and memory. Examples of neuropeptides are hypothalamic releasing hormones, antiduiretic hormone and the gastric peptides released from the cells in the duodenal walls.

NEUROPEPTIDE Y (NPY): A 36-amino acid neuropeptide that is widespread throughout the animal kingdom and in various body systems. It plays important roles in energy metabolism and in regulating the activity of the heart and blood vessels. It is released from cells in the hypothalamus to stimulate appetite and feeding behaviour. Its release is triggered by the gut hormone ghrelin and inhibited by the hormone leptin. It modulates the effects of adrenaline and noradrenaline.

NEUROPHYSIN: Either of two cysteine-rich proteins that bind to oxytocin and antidiuretic hormones in the neurohypophysis of the pituitary gland and are secreted with them. They have no hormonal activity and are cleaved from their respective hormone following their secretion.

NEUROPIL: A dense felt-like mass of microscopic threadlike axons, dendrites and extensions of glial cells within the grey matter of the central nervous system or within ganglia outside the cells. The cell bodies of the neurons that give rise to the axons and dendrites may be dispersed within the neuropil or segregated into distinct zones around it.

NEUROPSYCHOLOGY: The science that is concerned with the study of the physical brain and behaviour, specifically those disorders of perception, memory, language and behaviour that result from brain injury or disease. Neuropsychologists may work in clinical settings closely with doctors, including neurologists to evaluate and treat people with various types of nervous system disorders such as stroke, Parkinson's disease, Alzheimer's disease, learning disabilities and traumatic brain injuries.

NEUROSECRETORY CELLS: A type of neuron or nerve cell, present in most multicellular animals. They are found in the hypothalamus, where they receive impulses from other parts of the brain, but transmit these signals to the pituitary gland by neurohormones that are released into the blood. Neurosecretory cells are distinguished from other neurons by the unusually large size of the cell nucleus, axon endings and the cell itself. **Neurosecretion** is the secretion of neurohormones by neurosecretory cells which possess characteristics of both nerve cells and endocrine cells.

NEUROSIS (PSYCHONEUROSIS): Any psychological emotional disorders, such as anxiety, depression and phobias with no loss of reality.

NEUROTICISM: Personality dimension characterised by anxiety, fear, moodiness, worry, envy, frustration, jealousy and loneliness.

NEUROTOXIN: 1. A chemical that physically damages, reduces or alters the functions of the nerve. An example is 16-hydroxydopamine, which damages the nerve terminal of the sympathetic neurons. 2. Venom produced by some snakes, for example, cobras which affects the functioning of the nervous system.

NEUROTRANSMITTER (TRANSMITTER): A chemical such as acetylcholine or dopamine released by neurons to transmit information across the junction (synapse) that separates one nerve cell (neuron) from another nerve cell or a muscle that diffuses across a synapse and thus transmits impulses between nerve cells or between nerve cells and effector organs such as muscles.

NEUROTROPHIN (NT): Any of several growth factors that promote the development, maintenance and repair of neurons. The nerve growth factor (NGF), is essential for the normal growth and survival of neurons of the sympathetic nervous system and also sensory neurons.

NEUROTROPHIN RECEPTOR: A protein on the surface of a cell that interacts with a neurotrophin molecule outside the cell. There are two types: low affinity neurotrophin receptors, which are found on both neurons and other cell types; and high affinity neurotrophins which are restricted to neurons. Low affinity receptors bind neurotrophins to provide a local store of these molecules or they interact with high affinity receptors in mediating the effects of neurotrophins. In some cases, binding of neurotrophin to low affinity receptors can trigger cell death (apoptosis).

NEUSTON: A collective term for organisms that colonise the surface of an aquatic habitat, particlarly in freshwater habitat and include pond-skater, certain water-beetles and floating plants.

NEUTER: An organism that does not possess either male or female reproductive organ. In plants, having no stamens or pistils; or in animals, possessing undeveloped sexual organs.

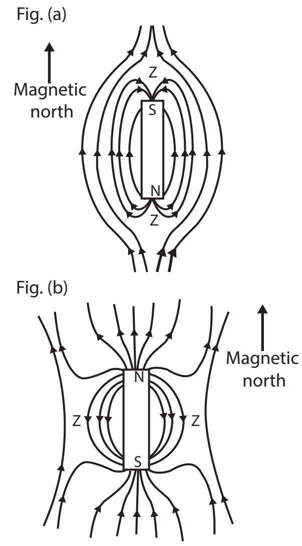
NEUTRAL: 1. A term describing a liquid or solution that does not affect litmus indicator and is neither an acid nor an alkali. 2. Of or relating to a compound that does not ionise in solution. 3. A term describing an object that is not charged. An atom is neutral when it has the same number of electrons and protons. 4. Achromatic or indicating a colour, such as grey, black or white, that lacks hue.

NEUTRAL ELEMENT (IDENTITY; IDENTITY ELEMENT): For a set endowed with a binary operation, a member of the set such that the result of applying the operation to that element and any member of the set is equal to the latter member. A multiplicative identity is a unity; an additive identity is often called a zero. For example, the identity for multiplication in ordinary arithmetic is 1, since $x \times 1 = 1 \times x = x$, and identity for addition in ordinary arithmetic is 0 since $0 + x = x + 0 = x$.

NEUTRAL EQUILIBRIUM (INDIFFERENT EQUILIBRIUM): A property of the steady state of a system which exhibits neither instability nor stability according to the particular criterion under consideration. A disturbance introduced into such equilibrium will thus be neither amplified nor damped.

NEUTRAL GENE HYPOTHESIS: The hypothesis that most genetic variation in natural populations is not maintained by selection because most alleles have equal fitnes.

NEUTRAL MUTATION: 1. A mutation that has no effect on the Darwinian fitness of its carriers. 2. A mutation that has no phenotypic effect.



Neutral points

Z= nuetral points, S= south pole, N= North pole

NEUTRAL OXIDE: Oxide that has neither acidic nor basic properties, for example, carbon monoxide (CO) or nitrous oxide (N_2O).

NEUTRAL POINT: A point in space at which a particle experiences no net gravitational force. In theory, a unique neutral point would exist between any two static bodies. In practice, when two objects, such as Earth and the Moon are orbiting each other, there are five points, known as Lagrangian points, at which gravitational and centrifugal forces are exactly in balance.

NEUTRAL SOIL: Soil in which the surface layer, to plough depth is in the range from slightly acid to slightly alkaline, usually considered to be in the range of pH values from 6.6 to 7.3.

NEUTRAL SOLUTION: A compound or solution that is neither acidic nor basic or a solution of pH 7 or any aqueous solution in which the activities of the hydrogen and hydroxide ions are equal. Such a solution does not have any effect on litmus indicator. Examples are water, sugar or sodium chloride solutions, and human blood.

NEUTRAL THEORY OF MOLECULAR EVOLUTION: A theory, introduced by Motoo Kimura in the late 1960s and early 1970s that suggests that most evolutionary changes at the molecular level are due to the random process of genetic drift acting on mutations, rather than natural selection.

NEUTRAL VARIATION: Genetic diversity that confers no apparent selective advantage.

NEUTRAL WIRE: The wire, coloured blue or black, in a mains electricity lead that has zero electric charge or potential. The potential difference between the neutral and each of the other conductors are approximately equal in magnitude and equally spaced in phase.

N

NEUTRAL-DENSITY FILTER (ATTENUANCE FILTER; ATTENUATOR): An optical filter, which reduces the radiant power of a light beam by a constant factor over all wavelengths within its operating range.

NEUTRALISATION (NEUTRALIZATION): A reaction in which equimolar amounts of a base and an acid react to form salt and water. With strong acids and bases the essential reaction is the combination of hydrogen ions with hydroxyl ions to form water. It is an exothermic reaction. To **neutralise** is to react an acid with a base to form a solution that is neutral.

NEUTRALISM: In biology, it describes a situation in which two species populations co-exist with neither population being affected by its association with the other.

NEUTRINO: A particle that has no charge and that travels at the speed of light. Neutrinos are thought to have zero mass when stationary.

NEUTRINO ASTRONOMY: A branch of astronomy that studies neutrinos emitted by objects, using detectors based at Earth observatories. Nuclear reactions in stars and supernova explosions produce copious amounts of neutrinos, of which a very few may be detected by a neutrino telescope.

NEUTRON: A subatomic particle contained in the nucleus of all atoms except hydrogen, discovered by James Chadwick in 1932. It has no charge and has approximately the same mass as a proton. A nucleus with an excess of neutrons is radioactive. The extra neutrons convert to protons by beta decay. Neutrons are part of the nucleus of all atoms, except hydrogen. In the nucleus of an atom it is stable, but when free it decays into a proton, an electron and an antineutrino and have a mean lifetime of approximately 1.0×10^3 seconds as free particles. It has a mass of 1.674×10^{-24} grams (1,838 times that of the electron and slightly greater than that of the proton). Its rest mass is $1.674\ 92716 \times 10^{-27}$ kilogram.

N

NEUTRON BEAM MACHINE: Nuclear reactor or accelerator producing a system of neutrons, which can "see" through metals. It is used in industry to check molecular changes in metal under stress.

NEUTRON BOMB (ENHANCED RADIATION WEAPON, ERW): A small tactical nuclear weapon consisting of hydrogen bomb that kills by radiation without destroying buildings and other structures. Its explosion is typically about one-tenth as powerful as that of a comparable fission-type atomic bomb because standard thermonuclear weapons create increased explosive yield by capturing their neutron radiation.

NEUTRON DENSITY: The number of free neutrons divided by the containing volume. Partial densities may be defined for neutrons characterised by such parameters as energy and directions.

NEUTRON DIFFRACTION: A type of elastic scattering, in which the neutrons exiting the experiment have more or less the same energy as the incident neutrons. It provides similar structural information as electron diffraction. Atomic structure of solids, liquids or gases are scattered by passing high flux beams of thermal neutrons and measuring the intensity of scattered neutrons in varied directions. This process has given rise to a technique, analogous to x-ray diffraction techniques using a flux of thermal neutrons from a nuclear reactor to steady solid state phenomena. Of the three types of diffraction methods, neutrons are unique in that they have a magnetic moment and are therefore sensitive to magnetic ordering in a solid. Thermal neutrons have average kinetic energies of about 0.025 eV (4×10^{-21}J) giving them an equivalent wavelength of about 0.1 nanometre, which is suitable for the study of interatomic interference. There are two types of scatterings: (i) **Nuclear scattering** between the neutrons and the atomic nucleus. Interaction with the nucleus gives diffraction patterns that complement those from x rays. X rays, which interact with the extra nuclear electrons, are not suitable for investigating light elements such as hydrogen, whereas neutrons do give diffraction patterns from such atoms because they interact with nuclei. (ii) **Magnetic scattering** between the magnetic moments of the neutrons and the spin and orbital magnetic moments of the atoms. Although neutrons are uncharged, they carry a spin and therefore interact with magnetic moments, including those arising from the electron cloud around an atom. Neutron diffraction can therefore reveal the microscopic magnetic structure of a material. This has provided valuable information on antiferromagnetic and ferromagnetic materials. Neutron beams interact more strongly with nuclei than do X-rays and neutron diffraction is more useful than X-ray diffraction for determining proton positions. Neutrons interact with a solid to a much lesser degree than X-rays and therefore have advantages in studying materials that are damaged by X-rays and in cases where a large penetration depth is desired. The two main uses of neutron scattering are in refining molecular structures that have been mostly determined by X-ray diffraction and in polymer characterisation using small-angle neutron scattering (SANS).

NEUTRON DRIP: The rapid increase in the abundance of free neutrons that occur, when matter becomes sufficiently dense that electrons are absorbed into nuclei.

NEUTRON INTERFEROMETER: A type of interferometer that uses the interference caused by the wave nature of neutrons to investigate many phenomena described by quantum mechanics. They have been used to demonstrate the spinor nature of wave functions for fermions.

NEUTRON MULTIPLICATION: The process in which a neutron produces on the average more than one neutron in a medium containing fissile material.

NEUTRON NUMBER: The number of neutrons in the nucleus of an atom that is the nucleon number minus proton number. Its symbol is N. Only 58 stable nuclides have odd neutron number, compared to 200 with even neutron number. Neutron number is usually written as a subscript to the right of the element symbol, for example, ${}^{14}_{6}C_{8}$.

NEUTRON OPTICS: The study of aspects of neutron behaviour that have wavelike characteristics. Like all elementary particles, neutrons can be made to display wavelike, as well as particle-like behaviour. They can be reflected and refracted and they can scatter, diffract and interfere, like light or any other type of wave.

NEUTRON REST MASS: Atomic fundamental physical constant m_n, which is equal to $1.6749286 \times 10^{-27}$ kg.

NEUTRON SCATTERING: The change in direction of neutrons caused by collision with nuclei in a material used as a scientific probe. Neutrons readily interact with atomic nuclei and magnetic fields from unpaired electrons, making a useful probe of both structure and magnetic order. Neutron scattering falls into two basic categories, elastic and inelastic. **Elastic scattering** is when a neutron interacts with a nucleus or electronic magnetic field but does not leave it in an excited state (the emitted neutron has the same energy as the injected neutron). Scattering processes that involve an energetic excitation or relaxation by the neutron are **inelastic** (the injected neutron's energy is used or increased to create an excitation or by absorbing the excess energy from a relaxation) and consequently the emitted neutron's energy is reduced or increased respectively.

NEUTRON STAR: A very small super-dense star composed mostly of neutrons. They are thought to form when massive stars explode as supernovae, during which the protons and electrons of the star's atoms merge owing to intense gravitational collapse to make neutrons.

NEUTRON TEMPERATURE (NEUTRON ENERGY): A concept used to express the energies of neutrons that are in thermal equilibrium with their surroundings, assuming that they behave like a monatomic gas. The neutron temperature T, on the Kelvin scale is given by $T = \frac{2E}{3K}$, where E is average neutron energy and K the Boltzmann constant.

NEUTROPHIL (GRANULOCYTES; POLYMORPHONUCLEAR NEUTROPHILS, PMNs): A type of white blood cell (leucocyte) that has a lobed nucleus and granular cytoplasm. It engulfs bacteria and releases various substances such as lysozyme and oxidising agents. Neutrophils are the most abundant type of white blood cells in mammals and form an essential part of the innate immune system.

NEUTROPHILIC ORGANISMS: Organisms that prefer a neutral medium for growth.

NÉVÉ: Compacted snow that lasts from year to year, representing one of the earliest stages in the development of a glacier. Further compaction, until there is no air left in the snow, results in the formation of firn.

NEW BLOOD: Genetic variation brought into a breed. An example is by introducing a new male to a flock or herd.

NEW MATHEMATICS: The approach to mathematical education in which the principles of set theory are introduced at an elementary level as the foundations of arithmetic.

NEW MOON: One of the four phases of the Moon, during which it is directly between the Earth and the Sun and invisible as its dark side is toward the Earth, or seen only as a narrow crescent.

NEW SYNTHESIS (EVOLUTIONARY SYNTHESIS; MODERN SYNTHESIS; NEO-DARWINIAN SYNTHESIS): A consolidation of the results of various lines of investigation from the 1920s through the 1950s that supported and reconciled the Darwinian theory of evolution and the Mendelian laws of inheritance in terms of natural selection acting on genetic variation. It is a union of ideas from several biological specialties which provides a widely accepted account of evolution.

NEWCASTLE DISEASE: A highly infectious disease of the nervous and respiratory system found in birds. It mainly affects chickens and it is characterised by paralysis, muscle and thigh bone tumours and yellowish white diarrhoea.

NEWMAN PROJECTION: A type of projection in which a molecule is viewed along a bond. It is useful in alkane stereochemistry to visualise carbon-carbon chemical bond. It is named after Melvin Spencer Newman, an American chemist of Ohio State University, who introduced it in 1952.

NEWSGROUP: In computing, it is a worldwide discussion about a particular subject consisting of notes written to a central Internet site and redistributed through Usenet.

NEWTON: The SI unit of force, with the symbol N, named after Sir Isaac Newton (1642-1727), British physicist and mathematician. One newton is the force needed to accelerate an object with a mass of one kilogram by one metre per second squared.

NEWTON DIAGRAM: A vector diagram used to show the relationship between initial and final velocities or momenta in a two-particle scattering process. It is commonly used to relate laboratory and centre-of-mass coordinates.

NEWTON-COTES FORMULAE (NEWTON-COTES QUADRATURE RULES): A class of numerical quadrature methods for numerical integration that extend the trapezoidal rule and Simpson's rule. They may be closed if the interval $[x_1, x_n]$ is included in the fit, open if the points $[x_2, x_{n-1}]$ are used or a variation of these two. If the formula uses n points (closed or open), the coefficients of terms sum to n - 1.

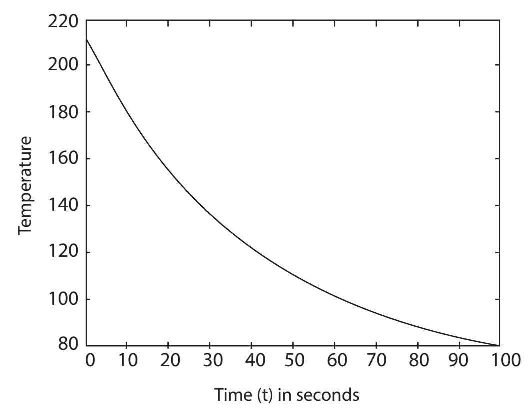
NEWTON, SIR ISAAC (1642-1727): English scientist and mathematician, who made major contributions in mathematics and theoretical and experimental physics. He discovered the law of universal gravitation and invented calculus, an honour he shares with Leibniz. He studied at Trinity College, Cambridge University, where he became the Lucasian Professor of Mathematics from 1669 to 1701. During that time, he discovered the law of universal gravitation, developed calculus in 1665-1666 and discovered that white light is composed of all the colours of the spectrum. In the late 1660's he constructed a reflecting telescope. He formulated the three laws of motion, which describe classical mechanics. He showed that his principle of universal gravitation provided an explanation both of falling bodies on the Earth and of the motions of planets, comets and other bodies in the heavens. He represented his university in Parliament (1689-90 and 1701-02) and was president of the Royal Society from 1703 until his death. In 1696, Newton left Cambridge and took charge of the British Mint in London. The SI unit of force, newton, with the symbol N is named after him.

NEWTON'S FORMULA: For a lens, the distance p and q between two conjugate points and their respective foci is given by $pq = f^2$, where f is the focal length for the lens. It is named after Sir Isaac Newton (1642-1727), British physicist and mathematician.

NEWTON'S CONSTANT (GRAVITATIONAL CONSTANT; UNIVERSAL GRAVITATIONAL CONSTANT): The numerical factor relating force, mass and distance in Newton's theory of gravitation that yields the force one body exerts on another when multiplied by the product of the masses of the two bodies and divided by the square of the distance between them. Its symbol is G. The attractive force, F, between two bodies of mass m_1 and m_2 and the distance between them, r is expressed as $F = \frac{Gm_1m_2}{r^2}$. The numerical value of G is expressed as $G = \frac{Fr^2}{m_1m_2} = 6.67259(85) \times 10^{-11} m^3 kg^{-1} s^{-2}$.

NEWTON'S IDENTITIES: The formulae expressing the powers of the sums of all the roots of a polynomial in terms of the coefficients of the polynomial, without actually finding these roots.

NEWTON'S LAW OF COOLING: Formulated by Sir Isaac Newton (1642-1727), British physicist and mathematician, it states that the rate at which a body loses heat is proportional to the difference in temperature between the body and the surrounding. Mathematically, this is given by $\frac{dQ}{dt} \propto (T_2 - T_1)$ or $\frac{dQ}{dt} = -k(T_2 - T_1)$, where $\frac{dQ}{dt}$ is change in heat with respect to time, $T_2 - T_1$ change in temperature and k is a constant.



Newton's Law of Cooling

NEWTON'S LAW OF GRAVITY: Formulated by Sir Isaac Newton (1642-1727), British physicist and mathematician, it states that there is a force of attraction between any two massive particles in the universe. For any two point masses m_1 and m_2, separated by a distance d, the force of attraction F is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. Mathematically, it is expressed as

$F = \frac{Gm_1m_2}{d^2}$, where G is the gravitational constant.

NEWTON'S LAWS OF MOTION: Three fundamental principles which form the bases of classical mechanics. They are: (i) A body continues in a state of rest or uniform motion in a straight line unless it is acted upon by external force or an object at rest stays at rest and a moving object continues moving at the same speed in the same straight line unless acted upon by an external resultant force. (ii) The rate of change of momentum of a moving body is proportional to and in the same direction as the forces acting on it, expressed as $F = \frac{d(mv)}{dt}$, where F is the applied force, v is the velocity of the body and m it's mass. If the mass remains constant, $F = m\frac{dv}{dt}$ or $F = ma$, where a is the acceleration. (iii) If one body exerts a force on another, there is an equal and opposite force, called a reaction, exerted on the first body by the second or to every action, there is an equal and opposite reaction.

NEWTON'S METHOD: A method of obtaining rough approximations to the values of the roots of an equation $f(x) = a$ by repeatedly computing in one dimension $x_{NEW} = x_{OLD} - \frac{f(x_{OLD})}{f'(x_{OLD})}$. This corresponds to approximating the function by its tangent and exhibits a quadratic rate of convergence if the initial estimate is sufficiently good. It was named after Sir Isaac Newton, British physicist and mathematician and Joseph Raphson, an English mathematician.

NEWTON'S RINGS: Interference fringes formed by placing a slightly convex lens on a flat glass plate or the irregular patterns produced by thin film interference between a projected transparency and its cover glass. It was named after Sir Isaac Newton (1642-1727), British physicist and mathematician.

NEWTONIAN FLUID: A fluid in which the components of the stress tensor are linear functions of the first spatial derivatives of the velocity components. These functions involve two material parameters (taken as constants throughout the fluid, although depending on ambient temperature and pressure). It was named after Sir Isaac Newton (1642-1727), British physicist and mathematician, who first deduced the concept.

N

NEWTONIAN MECHANICS: The system of mechanics that relies on Newton's laws of motion. Newtonian mechanics is applicable to bodies moving at speeds relative to the observer that are small compared to the speed of light. It was named after Sir Isaac Newton (1642-1727), British physicist and mathematician.

NEWTONIAN PHYSICS: Physics based on Newton's laws of motion as defined by English physicist and mathematician Isaac Newton (1642-1727), the formulation of quantum theory or relativity theory.

NEWTONIAN TELESCOPE: A type of reflecting telescope invented by the British physicist and mathematician, Sir Isaac Newton (1643-1727), in which the light reflected from a concave mirror is reflected again by a plane mirror making an angle of 45° with the telescope axis, so that it passes through a hole in the side of the telescope containing the eyepiece. Newton's first reflecting telescope was completed in 1668 and is the earliest known functional reflecting telescope. The Newtonian telescopes' simple design makes them very popular with amateur telescope makers.

NEXINE: The inner layer of the exine in a pollen grain.

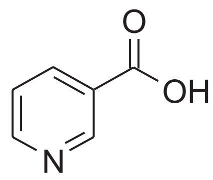
NEXT-TO-HIGHEST OCCUPIED MOLECULAR ORBITAL (NHOMO): Together with second-lowest unoccupied molecular orbital (SLUMO), they are subjacent orbitals, which in certain cases are significant in frontier-orbital theory.

NEYMAN-PEARSON LEMMA: In statistics, the theorem that of all tests of a given hypothesis of the same significance level, the likelihood ratio test is the most powerful. It is named after Jerzy Neymann (1894-1981) and Egon Sharpe Pearson (1895-1980).

NFC (NEAR FIELD COMMUNICATION): A wireless communication technology that enables communication between two electronic devices such as smartphones and other enabled devices when they are touched or brought within a close proximity (about 4 cm) of each other. This is achieved by magnetic field induction using special chips embedded in them. It can be used for contactless payments at supermarkets and such outlets, transfer photos and other data, etc.

NIBBLE: In computer science, four binary digits or half of an eight-bit byte; a byte of four bits.

NIACIN (NICOTINIC ACID; VITAMIN B3, VITAMIN PP): B complex vitamin with the molecular formula $C_6H_5NO_2$, density 1.473 g cm^{-3} and melting point 237°C (510 K). It is found in meat and dairy products. It is a colourless, water-soluble solid, a derivative of pyridine, with a carboxylic group (COOH) at the 3-position. Its deficiency causes pellagra.



Chemical structure of Niacin

NICCOLITE (KUPFERNICKEL; NICKELINE): A coppery-red mineral consisting of nickel arsenide (NiAs) containing 43.9% nickel and 56.1% arsenic. Its specific gravity ranges from 7.3 to 7.7. It is a minor nickel ore often found with pyrrhotite. It crystallises in the hexagonal system.

NICHE: The relational position of a particular species or population in an ecological community. It includes all the chemical, physical and biological components, such as what it eats, the time of the day it feeds, temperature, moisture, the parts of the habitat that it uses, for example, trees or open grassland and the way it reproduces and how it behaves.

NICHE OVERLAP: The extent to which two species require similar resources which specifies the strength of the competition between the two species.

NICHROME: Name for a group of non-magnetic alloy of nickel and chromium. They possess good resistance to oxidation, have high melting point and high resistivity. They are used as resistance wire in heating elements such as in hair dryers, electric ovens, toasters, in the explosives and fireworks industry, as a bridgewire in electric ignition system, such as electric matches and model rocket igniters. An example is Nichrome 80 which is composed of 80% Nickel and 20% Chromium.

NICKEL: A lustrous, silvery-white, hard, malleable and ductile transition magnetic metallic element. Its symbol is Ni, atomic number 28, atomic mass 58.70, melting point about 1,453°C (1726 K), boiling point about 2,732°C (3005 K) and specific gravity 8.902 at 25°C. It occurs in igneous rocks and as a free metal. Its chief ores are pentlandite and pyrrhotite. It also occurs in garnierite, millerite and niccolite. It dissolves slowly in dilute acids but, like iron, becomes passive when treated with nitric acid. It does not tarnish and therefore is much used for alloys, electroplating, in the Edison and nickel-cadmium (NiCad) storage batteries, for coinage, in paints, varnishes and ceramics.

NICKEL CARBONYL (TETRACARBONYLNICKEL): A colourless, flammable, poisonous co-ordination compound in which the oxidation states of nickel is zero. Its chemical formula is $Ni(CO)_4$ with a boiling point of 43°C (316 K). It is soluble in alcohol and concentrated nitric acid and insoluble in water. It is used as a starting material in the synthesis of a wide variety of nickel compound and in gas plating.

NICKEL ORE: Any mineral ore such as pentlandite and pyrrhotite and also garnierite, millerite and niccolite from which nickel is obtained.

NICKEL SILVER (GERMAN SILVER; ALPACCA): An alloy of nickel and copper and often zinc, used for coinage and for making cutlery and hollow-ware. It consists of 60% copper, 20% nickel and 20% zinc or 52% to 80% copper, 10% to 35% zinc and 5% to 35% nickel. It may also contain a small percent of lead and tin.

NICKEL-CADMIUM CELL (NICD; NICAD): A type of secondary or rechargeable battery with a nickel (nickel(III) oxide-hydroxide) anode, a cadmium cathode and an alkaline electrolyte (potassium hydroxide). It is widely used in cordless appliances.

NICKEL-IRON ACCUMULATOR (EDISON ACCUMULATOR): A type of secondary or rechargeable battery having a positive plate of nickel oxide and a negative plate of iron both immersed in an

electrolyte of potassium hydroxide. The reaction on discharge is: $2NiO(OH).H_2O + Fe \rightarrow 2Ni(OH)_2 + Fe(OH)_2$. The reverse occurs during charging. Each cell gives an e.m.f of about 1.2 volts and produces about 100 kJ per kilogram during each discharge. The nickel-cadmium cell is a similar device with a negative cadmium electrode, $2NiO(OH).H_2O + Cd \rightarrow 2Ni(OH)_2 + Cd(OH)_2$.
It is often used as a dry cell.

NICKEL-PLATING: A process in which a thin coating of nickel is electroplated on another metal for decorative purposes, to provide corrosion resistance, etc.

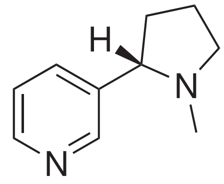
NICKELIC COMPOUNDS: Compounds of nickel in its +3 oxidation state, for example, nickelic oxide is nickel(III) oxide (Ni_2O_3).

NICKELOUS COMPOUNDS: Compounds of nickel in its +2 oxidation state, for example, nickelous oxide is nickel(II) oxide (NiO).

NICOL PRISM: An optical device for producing plane polarised light, constructed of two prisms of calcite cemented together with Canada balsam. An ordinary beam of light entering the crystal undergoes double refraction (split into two parts, each of which is affected in a different way). One of the split rays, called ordinary ray, undergoes total reflection at the Canada-balsam joint and is turned off from its course to pass out at one side of the crystal. The other ray, the extraordinary ray, passes on through the crystal. By means of this device a beam of light can be polarised or a beam of polarised light can be subjected to analysis. The principle involved has been applied to the microscope in the illumination of the field. It was devised in 1828 by William Nicol (1768 – 1851). It has now been superseded by Polaroid filters, etc.

NICOTIANA: A genus of about 67 species of annual but in some varieties perennial herbaceous plants and shrubs that belongs to the Solanaceae (nightshade) family. It is native to the Americas, Australia, South West Africa and the South Pacific, but some species are now cultivated in different parts of the world. It is characterised by large fleshy leaves and numerous sticky hairs and active biochemicals of nicotine, nornicotine and anabasine, the contents varying with the species and varieties. The two commercially important species are *Nicotiana tabacum*, commonly known as tobacco, cultivated in a temperature range of between 20-30 °C for the cured leaves, which are manufactured into cigarettes, cigars and other products; and *Nicotiana rustica*, cultivated mainly for the high concentration of nicotine in its leaves, used for the manufacture of insecticides. Some species cultivated in gardens as ornamental plants include *Nicotiana alata* with the common names winged tobacco and jasmine tobacco, with a lovely fragrance in the evenings and nights; *Nicotiana Sylvestris,* with strongly fragrant pendant clusters of long-tubed, trumpet-shaped, white flowers; and *Nicotiana mutabilis* with white flowers that turn pink and finally magenta. *Nicotiana* species have also been used in plant breeding and genetic research. Medicinally, the leaves of some species such as *Nicotiana tabacum* are used externally in the treatment of rheumatic swelling, skin diseases and scorpion stings. The juice of the leaves can be rubbed on the body as an insect repellent. The fine-ground tobacco leaves called snuff is inhaled into the nose by some people; and the leaves are also chewed as a stimulant.

NICOTINAMIDE ADENINE DINUCLEOTIDE (NAD+): The oxidised form of nicotinamide adenine dinucleotide (NAD), a coenzyme found in all living cells, derived from the B vitamin nicotinic acid that participates in many biological dehydrogenation reactions. The compound is a dinucleotide, since it consists of two nucleotides joined through their phosphate groups, with one nucleotide containing an adenine base and the other containing nicotinamide. NAD is characteristically loosely bound to the enzymes concerned. It normally carries a positive charge and can accept one hydrogen atom and two electrons to become the reduced form, NADH. NADH is generated during the oxidation of food, especially by the reactions of the Krebs cycle. It then gives up its two electrons (and a single proton) to the electron transport chain, thereby reverting to NAD^+ and generating three molecules of adenosine triphosphate (ATP) per molecule of NADH. Nicotinamide adenine dinucleotide phosphate (NADP) differs from NAD only in possessing an additional phosphate group. It functions in the same way as NAD although anabolic reactions generally use NADPH (reduced NADP) as a hydrogen donor rather than NADH. Enzymes tend to be specific for either NAD or NADP as coenzyme.



Chemical Structure of Nicotine

NICOTINE [3(1-METHYLPYRROLIDIN-YL)-PYRIDINE]: A colourless, poisonous, oily liquid alkaloid with a pungent odour and an acrid taste found in tobacco with the molecular formula $C_{10}H_{14}N_2$, density 1.01 g/cm³, melting point -79 °C (194 K) and boiling point 247 °C (520 K). In low concentration (about 1mg) nicotine is a stimulant and acts primarily on the autonomic nervous system in mammals. It is the chief addictive ingredient in cigarettes and cigars and in snuff. A stick of cigarette, when smoked inhales approximately 3 mg of nicotine, which increases the heart rate, constricts the blood vessels and acts on the central nervous system, imparting a feeling of alertness and well-being. In larger doses nicotine is a highly toxic and used as an insecticide, fumigant and vermifuge.

NICOTINE ADENINE DINUCLEOTIDE PHOSPHATE (NADP+): A substance to which electrons are transferred from photosystem I during photosynthesis. The addition of electrons reduces NADP, which acquires a hydrogen ion to form NADPH, which is a storage form of energy that can be transferred to the Calvin Cycle for the production of carbohydrate.

NICOTINIC ACETYLCHOLINE RECEPTORS (nAChRs): The two main classes of acetylcholine receptors that have their effect mimicked by nicotine. Nicotinic receptors are found at neuromuscular junctions in skeletal muscle and at post ganglionic cells in ganglia of both divisions of the vertebrate autonomic nervous system. They are fast acting receptors containing their own intrinsic ion channel, which is directly activated by the receptor agonists.

NICOTINIC ACID (NIACIN; VITAMIN B3): Water-soluble vitamin of the B-complex, with the chemical formula $C_6H_5NO_2$, found in meat, fish and cereals. Its deficiency in the diet leads to the disease pellagra. It can be manufactured by plants and animals from the amino acid tryptophan.

NICTITATING MEMBRANE: A clear or translucent membrane, which forms a third eyelid in amphibians, reptiles, birds and some animals such as camels, polar bears and seals. It can be drawn across independently of the other eyelids, thus clearing the eye surface and giving added protection without obscuring the continuity of vision.

NICTOTINAMIDE ADENINE DINUCLEOTIDE (NAD^+): An important oxidised coenzyme found in all living cells. It consists of a hydrogen and electron carrier in redox (reduction-oxidation) reactions. In metabolic reactions, nicotinamide adenine dinucleotide may be oxidised (donate electrons to other molecules) to NAD^+, or reduced to NADH (accepts electrons from other molecules). Reduction-oxidation (redox) reactions are important in cellular respiration and other biochemical pathways as an efficient way to transfer energy from one molecule to another.

NICOTINAMIDE ADENINE DINUCLEOTIDE PHOSPHATE ($NADP^+$): An important oxidised coenzyme that acts as a hydrogen and electron carrier in various redox reactions. It provides energy to the many reactions in a cell, such as lipid and nucleic acid synthesis, which requires NADPH (the reduced form of $NADP^+$) as a reducing agent.

NIDATION: The embedding of a fertilised mammalian egg into the wall of the uterus (womb), where it continues its development.

NIDICOLOUS: 1. Of hatchlings, remaining in the nest for some time after hatching as they depend on the parents for feeding and protection; this also applies to some animals. 2. Sharing the nest of another species of animal.

NIDIFUGOUS: It refers to birds or animals that leave the nest or birth place within a few days after being hatched or after birth.

NIDO-: An affix used in names to denote a nest-like structure, especially a boron skeleton that is almost closed.

NIDULENT: Lying loose in a pulp; lying within a cavity.

NIFE CELL (NICKEL-IRON BATTERY; NiFe BATTERY; EDISON CELL): A storage battery having nickel(III) oxide-hydroxide cathode and an iron anode in potassium hydroxide electrolyte. It is very robust, has long life battery, which withstands overcharging, overdischarging and short-circuiting.

NIGHT BLINDNESS (NYCTALOPIA): A disease caused mainly by deficiency of vitamin A (retinol), in which one is incapable of seeing properly in the dark or dim light. It can exist from birth or caused by injury.

NIGHT SOIL: Human excreta, collected and used for fertiliser in some parts of the world.

NIGHT TEMPERATURE: The temperature prevailing during the period of darkness, which can greatly affect the growth of many plants. For example, tomatoes have been shown to grow better if the night temperature is significantly lower than the day temperature. It is thought that low night temperatures inhibit respiration.

NIGHT-BREAK EFFECT: A phenomenon exhibited by many plants whereby a period of light, administered artificially during the night, even if only of short duration or low intensity, will interfere with the normal flowering rhythm of the plant. The precise effects vary with the time that has elapsed since the light phase but generally, under short-day conditions, a light treatment inhibits flowering in short-day plants and promotes it in long-day plants.

N

NIGHT-SKY LIGHT: The faint, diffuse glow of the night sky. It comes from four main sources: airglow, diffuse galactic light, zodiacal light and the light from these sources scattered by the troposphere.

NIGHTSHADE: A family of plants making up the genus *Solanum* and the family Solanaceae, in the order Solanales. It includes crop and garden plants such as potato, tomato, petunia, tobacco, chilli pepper and eggplant and also many poisonous plants. They generally have flowers that have five petals and small berries. The poisonous nightshades contain alkaloids of three major types: tropane, found in belladonna, jimsonweed and henbane; pyridine, in tobacco; and steroid, in some members of the nightshade genus. Nightshades have been used in medicine as stimulants, narcotics, pain relievers, poisons and antidotes for such agents as opium and snake venom.

NIGRESCENT: Becoming black or dark.

NIKODYM SET (IMPOSSIBLE SET; LINEARLY ACCESSIBLE SET): A subset of the unit square with unit area, such that for every point in the set there is a line that intersects the set only in this point.

NILPOTENT: Of a matrix, function or ring element, having the property that some integral power of the given quantity is zero. For example, the matrix $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ is nilpotent since $\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}\begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.

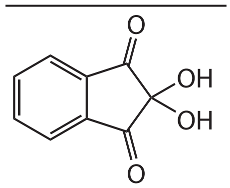
NILPOTENT GROUP: An "almost abelian" group that has a central series of finite length, whose lower central series terminates in the trivial subgroup after finitely many steps and whose upper central series terminates in the whole group after finitely many steps.

NIMBOSTRATUS CLOUD: An amorphous, dark grey, rainy cloud layer reaching almost to the ground; 300 to 600 m with vertical velocities of 0.05 to 0.2 m s^{-1}.

NINE POINT CENTRE: The circumcentre of the medial triangle in a triangle or the centre of a circle containing nine significant points of the circle.

NINE POINT CIRCLE (FEUERBACH'S CIRCLE; EULER'S CIRCLE; NINE-POINT CIRCLE; TERQUEM'S CIRCLE): The circle that passes through nine significant midpoints in a triangle. The points are: the midpoint of each side of the triangle, the foot of each altitude and the midpoint of the line segment from each vertex of the triangle to the orthocentre, where the three altitudes meet.

NINHYDRIN (2,2-DIHYDROXYINDANE-1,3-DIONE): A compound with the chemical formula $C_9H_6O_4$ that reacts with amino-acid to give a blue colour. It is used commonly in chromatography to analyse the amino-acid content of protein. It is also used to detect latent finger and palm prints. When ninhydrin solution is applied to the finger, the terminal amines or lysine residues in peptides and proteins sloughed off as fingerprints react with ninhydrin.



Chemical Structure of Ninhydrin

NINTENDO: A successful and well known Japanese video game hardware manufacturer and software publisher which introduced worldwide popular video games, for example, Game Boy (sold over 150 million), Super Nintendo Entertainment System (sold more than 49 million systems worldwide), Game Boy Pocket, Nintendo 64, etc.

NIOBIUM: A soft, grey white, ductile and malleable metallic element in Group 5 (VB) of the Periodic Table with the symbol Nb, atomic number 41, relative atomic mass 92.906, melting point about 2,468 °C (2741 K), boiling point 4,742 °C (5015 K) and specific gravity 8.57 at 20 °C (293 K). It reacts readily at high temperatures with oxygen, carbon, the halogens, nitrogen and sulfur. Niobium is widely distributed in nature. It occurs in the minerals columbite and tantalite, together with tantalum. Niobium is one of the five major refractory metals such as tungsten, molybdenum, tantalum and rhenium. Refractory metals have very high resistance to heat and wear. Niobium is used with iron and other elements in stainless steel alloys and also in alloys with a variety of nonferrous metals, such as zirconium. Niobium alloys are often used in applications such as pipeline construction and for jet engines and rockets, in cutting tools and for making superconductor magnets.

NISSL GRANULES (NISSL BODIES; TIGROID BODY): Particles seen within the cell bodies of neurons, named after Franz Nissl, German neurologist (1860-1919). They are sites for protein synthesis. They are rich in RNA and stain strongly with the basic dyes.

NIT: A unit of luminance equal to one candela per square metre. Ten thousand nits are equal to one stilb.

NITID: Bright; lustrous.

NITRAMINES: Amines substituted at N with a nitro group (a contracted form of *N*-nitroamines). They are thus amides of nitric acid and the class is composed of nitramide, $O_2N_2H_2$ and its derivatives formed by substitution.

NITRATE (TRIOXONITRATE(V) ION): Any conjugate base of nitric acid with the chemical formula NO_3^- consisting of one central nitrogen atom surrounded by three identical oxygen atoms in a trigonal planar arrangement with a molecular mass of 62.0049 g/mol. They are the most water soluble salts known. Nitrates of various kinds are used in explosives, in the chemical industry, in curing meat and as inorganic fertilisers. It is a form of nitrogen used by plants.

NITRATE POLLUTION: The contamination of water by nitrates. It results from increased use of artificial fertilisers containing nitrates.

NITRATE REDUCTASES: Enzymes that catalyse the oxidation of nitrate (NO^{-3}) to nitrite (NO^{-2}). This reaction is critical for the production of protein in most crop plants, as nitrate is the predominant source of nitrogen in fertilised soils. Nitrate reduction is called **assimilatory** when incorporated into the biomass, or **dissimilatory** when it is excreted from the cell.

NITRATING MIXTURE: A mixture of concentrated sulfuric and nitric acids. It is used mainly to introduce nitro groups into the molecules of aromatic compounds. The nitro group can subsequently be converted into or replaced by others to make commercial nitro compounds such as the explosive cellulose trinitrate (nitroglycerine), trinitrotoluene (TNT) and picric acid (trinitrophenol).

NITRATION: A chemical reaction in which a nitrogen group is incorporated into a chemical structure, to make a nitro compound. A hydrogen or halogen atom is often replaced by the nitro group. The three general reactions are: C nitration, in which the nitro group attaches itself to a carbon atom (reaction 1); O nitration, an esterification reaction, in which an ON bond is formed to produce a nitrate (reaction 2); and N nitration, in which a NN bond is formed (reaction 3).

NITRE (SALT PETRE): Potassium nitrate mineral, brittle, colourless to white, with vitreous lustre. Its chemical formula is KNO_3 with a specific gravity of 2.109 and hardness on Mohs scale of 2. It occurs in massive, granular or earthy forms. It is found on and just under the ground in desert regions. It is used in explosives and as a preservative and also widely used for curing meat.

NITRENE (AZACARBENE; AZENE; IMENE; IMINE RADICAL): A species of the type HN: or RN:, analogous to a carbene. The nitrogen atom has four nonbonding electrons, two of these electrons are paired while the other two have spins that may be either antiparallel (singlet state) or parallel (triplet state).

NITRIC ACID (TRIOXONITRATE(V) ACID): A fuming corrosive poisonous liquid with the chemical formula HNO_3, density 1.5129 g cm^{-3}, melting point -42 °C (231 K) and boiling point 83 °C (356 K). It is a highly corrosive acid, dissolving most metals and a strong oxidising agent. It is obtained by the oxidation of ammonia or the action by sulfuric acid on potassium nitrate. It may be prepared in the laboratory by the distillation of a mixture of an alkali metal nitrate and concentrated sulfuric acid. The industrial production is by the oxidation of ammonia to nitrogen monoxide, followed by oxidation of nitrogen monoxide to nitrogen dioxide and the reaction of nitrogen dioxide with water to form nitric acid. The nitrogen monoxide is recycled. Nitric acid is a strong acid (highly dissociated in aqueous solution) and dilute solutions behave much like other mineral acids. Concentrated nitric acid is a strong oxidising agent. Concentrated nitric acid also reacts with several minerals to give the oxo acid or oxide. Nitric acid is generally stored in dark brown bottles because of the photolytic decomposition to dinitrogen tetroxide.

NITRIC OXIDE (NITROGEN MONOXIDE; NITROGEN (II) OXIDE): A colourless gas formed by the combustion of nitrogen and oxygen in the presence of energy: energy + N_2 + $O_2 \rightarrow$ 2NO. Its density is 1.3402 g dm^{-3}, melting point -164 °C (109 K) and boiling point -152 °C (121 K). It combines readily with oxygen or air to form nitrogen dioxide (NO_2), which can again be separated by ultraviolet light to produce nitric oxide and highly reactive oxygen atoms. These oxygen atoms combine with hydrocarbons producing noxious compounds that irritate the membranes of living organisms and destroy vegetation. Large amounts of nitric oxide are created by internal-combustion engines and manufacturing processes. It is also produced in tissues from muscular oxygen and the amino-acid arginine by the enzyme nitric oxide synthase and diffuses to neighbouring cells, where it stimulates formation of the intracellular messenger cyclic GMP. Its effects also include relaxation of smooth muscle and dilation of blood vessels. It also inhibits platelet aggregation and adhesion, acts as a neurotransmitter in the central nervous system and is necessary for penile erection and may influence neuronal development. Despite its usefulness, nitric oxide can have a toxic effect on body cells and has been implicated in Huntington's disease and Alzheimer's disease.

NITRIDES: Compounds of nitrogen with a more electropositive element. Certain electropositive elements such as lithium, magnesium and calcium react directly with nitrogen to form ionic nitrides containing the N^{3-} ion. Transition elements form a range of interstitial nitrides such as manganese nitride (Mn_4N) and tungsten nitride (W_2N), which can be produced by heating the metal in ammonia. Boron nitride is a covalent compound having macromolecular crystals.

NITRIDING: A metallurgical process through which the surface of a steel, aluminium, titanium, molybdenum and some other metals are modified, usually to harden the surface by introducing nitrogen unto the surface. The three main techniques are: **gas nitriding** by heating the metal in ammonia gas; **salt bath nitriding** by dipping the hot metal in a bath of molten sodium cyanide; and **plasma nitriding**, in which intense electric fields are used to generate ionised molecules of the gas around the surface to be nitrided. Applications include gears, crankshafts, camshafts, cam followers, valve parts, extruder screws, die-casting tools, forging dies, extrusion dies, injectors and plastic-moulding tools.

NITRIFICATION: A biological chemical process in which ammonium in plant and animal wastes and dead remains is oxidised at first to nitrites (NO^-_2) and then to nitrates (NO^-_3) which can be absorbed by roots. These reactions are effected mainly by the nitrifying bacteria Nitrosomonas and Nitrobacter respectively. Unlike ammonia, nitrates are readily taken up by plant roots. Nitrification is therefore a crucial part of the nitrogen – containing compounds and often applied as fertiliser to soils deficient in this element. It also plays an important role in the removal of nitrogen from municipal wastewater. **Nitrify** is to convert nitrogen or nitrogen compound into nitrites and nitrates.

NITRIFICATION INHIBITORS: Chemical compounds that are added to nitrogen fertiliser to slow down the release of nitrate in ammonia, ammonium-containing, or urea-containing fertilisers to nitrate form by nitrifying bacteria in soil. Nitrate, unlike ammonium, can cause loss from soil by leaching and denitrification. Examples of chemical compounds used as nitrification inhibitors are thiourea (CH_4N_2S), dicyandiamide ($C_2H_4N_4$) and 3, 4-dimethylpyrazole phosphate ($C_5H_{11}N_2O_4P$).

NITRIFYING BACTERIUM: A soil bacterium that converts ammonia to nitrites and then nitrates, making nitrogen available to plants.

NITRILE RUBBER (BUNA-N; PERBUNAN; NBR): A copolymer of 1,2-butadiene or 1,3-butadiene and propenonitrile. Nitrile rubbers are commercially important synthetic rubbers because of their resistance to oil and many solvents and also ability to withstand a range of temperatures from -40°C (23 K) to 108°C (381 K). It is used in the automotive and aeronautical industry to make fuel and oil handling hoses, seals and grommets. Nitrile butadiene is also used to create moulded goods, footwear, adhesives, sealants, sponges, expanded foams and floor mats.

NITRILES (CYANIDES): Organic compounds containing the cyano group, -CN bound to an organic group. Nitriles are made by reaction between potassium cyanide and haloalkanes in alcoholic solution, such as $KCN + CH_3 \rightarrow CH_3CN + KCl$. An alternative method is dehydration of amides: $CH_3CONH_2 - H_2O \rightarrow CH_3CN$. They can be hydrolysed to amides and carboxylic acids and can be reduced to amine. They are used as insecticides, in making pigments, in electroplating and case hardening and in refining gold and silver by the cyanide process.

NITRIMINES (N-NITROIMINES): Compounds having the structure $O_2NN{=}CR_2$.

N

NITRITE: Salt or ester of nitrous acid, containing the nitrite ion $(NO_2)^-$. Nitrites are used as preservatives and as colouring agents in cured meat such as bacon and sausages. It is an oxo acid.

NITRO COMPOUND: An organic compound that contains one or more nitro functional groups ($-NO_2$) in the basic molecular structure. Nitro compounds are usually made by nitration; and used as commercial explosives, chemical intermediates and solvents.

NITROALKANE (NITROPARAFFIN): A type of organic compound of the general formula $C_nH_{2n+1}NO_2$. They are colourless, pleasant-smelling liquid made by treating a haloalkane with silver nitrate. They can be reduced to amines by the action of tintroalkanes such as nitromethane (CH_3NO_2,). They are used as high-performance fuels for internal combustion engines and rockets, as chemical intermediates and as polar solvents.

NITROBACTER: Rod-shaped, pear-shaped or pleomorphic gram-negative and chemoautotrophic bacteria, which plays an important role in nitrogen cycle by oxidising nitrite into nitrate in soil. It uses energy from the oxidation of nitrite ions, NO_2^-, into nitrate ions, NO_3^-.

NITROBENZENE: A poisonous, flammable, pale yellow, liquid aromatic compound with the chemical formula $C_6H_5NO_2$, melting point 5.85 °C (279 K) and boiling point 210.9 °C (484 K). It is slightly soluble in water, very soluble in ethanol, ether and benzene. It is prepared by treating benzene with a mixture of nitric and sulfuric acids, in the resulting nitration reaction, one hydrogen in the benzene molecule is replaced with a nitro group, NO_2. It is used to make aniline and dye, in the production of analgesic acetaminophen or paracetamol, in shoe and floor polishes, leather dressings and paint solvents to mask unpleasant odours.

N

NITROCELLULOSE (CELLULOSE NITRATE; FLASH PAPER): A highly flammable material made by treating cellulose (wood pulp) with concentrated nitric acid. Despite the alternative name *nitrocellulose*, the compound is in fact an ester containing $CONO_2$ groups, not a nitro compound, which would contain $C–NO_2$. It is used in making explosives, plastics and varnishes.

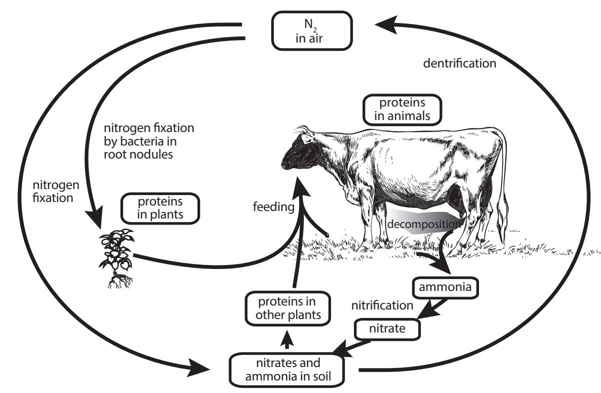
NITROGLYCERINE(1,2,3-TRINITROXYPROPANE; GLYCERYL TRINITRATE; TRINITROGLYCERINE): A heavy, colourless, oily explosive liquid obtained by nitrating glycerol. Its chemical formula is $C_3H_5N_3O_9$, density 1.6 g cm^{-3} (at 15 °C, 288 K), melting point 14° C (287 K) and boiling point 50-60 °C (323-333 K). It is the active ingredient in the manufacture of explosives (dynamite). It has been used by the military as an active ingredient and a gellatiniser for nitrocellulose in some solid propellant, such as cordite and ballistite. In medicine, it is used as vasodilator to treat heart conditions such as angina and chronic heart failure.

NITROGEN: A colourless gaseous element belonging to Group 15 of the Periodic Table with the symbol N, atomic number 7, atomic weight 14.0067, melting point -209.86 °C (63.29 K), boiling point -195.8 °C (77.35 K) and density 1.25 grams per litre at standard temperature and pressure (STP). It forms about 80% of the Earth's atmosphere by volume and an essential constituent of proteins and nucleic acids in living organism. Nitrogen is obtained for industrial purposes by fractional distillation of liquid air. Pure nitrogen can be obtained in the laboratory by heating a metal azide. There are two natural isotopes: nitrogen -14 and nitrogen -15 (about 3%). The gas is diatomic and relatively inert, reacting with hydrogen at high temperatures and with oxygen in electric discharges. The element is used in the Haber process for making ammonia and is also used to provide an inert atmosphere in welding and metallurgy. It also forms nitrides with certain metals.

NITROGEN ASSIMILATION: The incorporation of nitrogen into organic cell substances by living organisms. Organisms like plants, fungi and certain bacteria that cannot fix nitrogen gas (N_2) assimilate nitrate or ammonia for their needs.

NITROGEN COMPOUND: A substance such as a fertiliser containing mostly nitrogen with other elements.

NITROGEN CYCLE: The series of natural processes by which nitrogen from air and soil in suitable form enters plants and animals and used by them and eventually returned to the air and soil. Nitrogen is essential to plant and animal proteins. The nitrogen cycle consists of four natural processes: (i) **Nitrogen fixation process** in which nitrogen gas from the air which cannot be used by organisms is made into ammonia by nitrogen-fixing bacteria and cyanobacteria in the soil and by lightening and used by plants. (ii) **Denitrification process** in which certain soil bacteria act on nitrogen compounds to produce gas that enters the atmosphere. (iii) **Decay process** in which bacteria and fungi convert organic nitrogen compounds, chiefly proteins and urea of dead plants and animals to ammonia and ammonium compounds in the soil. (iv) **Nitrification process** in which ammonia in the soil is oxidised by nitrifying bacteria to form nitrate compound such as are absorbed by plants and used to make proteins.



The nitrogen cycle

NITROGEN DEFICIENCY: A lack of nitrogen in the soil, especially in areas where organic matter is low, resulting in thin weak growth of plants. It can occur when woody material such as sawdust is added to the soil. Nitrogen deficiency can be prevented in the short term by using mowed grass as mulch or foliar feeding with manure and in the longer term by building up levels of organic matter in the soil.

NITROGEN DIOXIDE (NITROGEN(IV) OXIDE): A highly toxic, brown gas with the chemical formula, NO_2, density 2.62 g dm^{-3} and boiling point 21 °C (294 K). It is made by the action of concentrated nitric acid on copper or oxygen on nitrogen monoxide. It exists in equilibrium with its dimer, dinitrogen tetroxide, N_2O_4. It is used as an oxidising agent and in the manufacture of nitric acid. It is often present in smog and exhaust gas from vehicles.

NITROGEN FERTILISER/FERTILIZER: Fertiliser containing mainly nitrogen. There are at least eleven forms of nitrogen fertiliser that are commercially available around the world. Examples are ammonium nitrate (NH_4NO_3), ammonium sulfate ($(NH_4)_2SO_4$); urea $CO(NH_2)_2$; calcium nitrate $5Ca(NO_3)_2NH_4NO_310H_2O$; and anhydrous ammonia (NH_3). Nitrogen fertilisers enable farmers achieve the high yields that drive modern agriculture. Once in the soil it is either dissolved into soil solution and/or adsorbed onto soil particles (clay surfaces or humus material). It must be placed, where plant roots will contact and absorb it. Once in the plant roots it can be translocated within the plant and used to build plant components such as protein molecules. Each form of nitrogen fertiliser has features that can make it more or less subject to losses from the soil-plant system or positionally available to crops.

NITROGEN FIXATION: 1. The process by which free nitrogen (N_2) in the atmosphere is combined with other elements to form inorganic compounds that is converted into ammonia by the action of micro-organisms (prokaryotes-bacteria and cyanobacteria). Some bacteria

(*Rhizobium, Bradyrhizobium*) are able to fix nitrogen in mutualistic association with cells in the roots of leguminous plants such as peas, groundnuts and beans. Also, in the filamentous plants, the bacterium *Frankia*, fix nitrogen in mutualistic association with root nodules of non-legume trees. Some non-symbiotic free-living soil bacteria such as *Azotobacter* and *Clostridium* also fix nitrogen. 2. Any of the processes, such as the Haber process that is used to combine atmospheric nitrogen with other elements for use mainly in the preparation of fertilisers or other products.

NITROGEN FIXING PLANTS: Plants which form symbiotic association with bacteria that convert nitrogen from the air into ammonia. Examples include the legume family (Fabaceae) with members such as clover, soybeans, alfalfa, lupines and peanuts.

NITROGEN MONOXIDE (NITRIC OXIDE; NITROGEN(II) OXIDE): A colourless gas with the chemical formula NO, melting point -163.6 °C (109.55 K) and boiling point -151.8 °C (121.35). Nitric oxide readily combines with oxygen or air to form nitrogen dioxide (NO_2), which can again be separated by ultraviolet light to produce nitric oxide and highly reactive oxygen atoms. It is soluble in water, ethanol and ether. It is made commercially by the catalytic oxidation of ammonia and used for making nitric acid. It is a neutral oxide.

NITROGEN MUSTARDS: A group of nitrogen compounds similar to sulfur mustard. They were used as chemotherapy agents in cancer treatment. Like sulfur mustard, they are powerful blistering agents. Large quantity was made during World War II, although none were used in combat.

NITROGEN OXIDES: Generally written as NOx, it is a generic term that refers to binary compounds of oxygen and nitrogen. These include NO, NO_2, NO^-_3, N_2O, N_2O_3, N_2O_4 and N_2O_5. These oxides are produced during combustion, especially at high temperature, for example, from car exhausts, aircraft and factories. Nitrogen dioxide (NO_2) is known to contribute to acid rain. When NOx and volatile organic compound (VOC) react in the presence of sunlight, they produce photo chemical smog.

NITROGEN-FREE EXTRACT COMPOUND (NFE): The chemical analysis of animals feeding stuffs, the nitrogen-free extract consists mainly of soluble carbohydrate and starch. It is calculated by adding crude protein, fat, water, ash and the fibre and the sum subtracted from 100.

NITROGEN-GAS LASER (NITROGEN LASER): A gas laser which operates in the ultraviolet range (typically 337mm) using molecular nitrogen as the gain medium, pumped by an electrical discharge. One of the uses of nitrogen laser is in measurement of air pollution, for example, LIDAR technology (Light Detection and Ranging).

NITROGEN(I) OXIDE (DINITROGEN MONOXIDE; NITROUS OXIDE; LAUGHING GAS; N_2O): A colourless, non-flammable gas at room temperature, with a slightly sweet odour and taste with the chemical formula N_2O, density 1.977 g/L (gas), melting point -90.86 °C (182.29 K) and boiling point -88.48 °C (184.67 K). It is known as "laughing gas" due to the euphoric effects of inhaling it. It is used in surgery and dentistry for its anaesthetic and analgesic effects, as an oxidiser in rocketry and in motor racing to increase the power output of engines. Nitrous oxide is a major greenhouse gas and air pollutant, contributing about a tenth as much as carbon dioxide to global warming.

NITROGEN(II) OXIDE (NITRIC OXIDE; NITROGEN MONOXIDE): Nitrogen(II) oxide is the International Union of Pure and Applied Chemistry's (IUPAC) name for **nitric oxide** with the chemical formula NO. It is a colourless gas formed by the combustion of nitrogen and oxygen in the presence of energy: energy + N_2 + $O_2 \rightarrow$ 2NO. Its density is 1.3402 g dm^{-3}, melting point -164 °C (109 K) and boiling point -152 °C (121 K). It combines readily with oxygen or air to form nitrogen dioxide (NO_2), which can again be separated by ultraviolet light to produce nitric oxide and highly reactive oxygen atoms. These oxygen atoms combine with hydrocarbons producing noxious compounds that irritate the membranes of living organisms and destroy vegetation. Large amounts of nitric oxide are created by internal-combustion engines and manufacturing processes. It is also produced in tissues from muscular oxygen and the amino-acid arginine by the enzyme nitric oxide synthase and diffuses to neighbouring cells, where it stimulates formation of the intracellular messenger cyclic GMP. Its effects also include relaxation of smooth muscle and dilation of blood vessels. It also inhibits platelet aggregation and adhesion, acts as a neurotransmitter in the central nervous system and is necessary for penile erection and may influence neuronal development. Despite its usefulness, nitric oxide can have a toxic effect on body cells and has been implicated in Huntington's disease and Alzheimer's disease.

NITROGEN(IV) OXIDE (NITROGEN DIOXIDE): A highly toxic, brown gas with the chemical formula, NO_2, density 2.62 g dm^{-3} and boiling point 21 °C (294 K). It is made by the action of concentrated nitric acid on copper or oxygen on nitrogen monoxide. It exists in equilibrium with its dimer, dinitrogen tetroxide, N_2O_4. It is used as an oxidising agent and in the manufacture of nitric acid. It is often present in smog and exhaust gas from vehicles.

NITROGEN(V) OXIDE (NITROGEN PENTOXIDE; DINITROGEN PENTOXIDE): Nitrogen(v) oxide is the International Union of Pure and Applied Chemistry's (IUPAC) name for **nitrogen pentoxide** with the chemical formula N_2O_5, density1.642 g/cm^3 (18 °C), melting point 41 °C (314 K) and boiling point 47°C, 320 K (sublimates).It is one of the binary nitrogen oxides, a family of compounds that only contain nitrogen and oxygen. It is an unstable and potentially dangerous oxidiser that once was used as a reagent, when dissolved in chloroform for nitrations but has largely been superseded by nitronium tetrafluoroborate (NO_2BF_4). It is a rare example of a compound that adopts two structures depending on the conditions: most commonly it is a salt, but under some conditions it is a polar molecule.

NITROGENASE: An important enzyme complex, unique to certain prokaryotes that is present in those microorganisms that are capable of fixing atmospheric nitrogen. Nitrogenase catalyses the conversion of atmospheric nitrogen into ammonia, which can be used to synthesise nitrites, nitrate or amino acid. Dinitrogenase reductase and dinitrogenase are the two main enzymes within the nitrogenase.

NITROGENOUS BASE: A basic compound containing nitrogen. There are two major types: purines and pyrimidines. The term is used especially for organic ring compounds such as adenine, guanine, cytosine and thiamine which are constituents of nucleic acids.

NITROGENOUS WASTE: Any metabolic waste product that contains nitrogen. Urea and uric acid are the most common nitrogenous waste products in terrestrial animals. Fresh water fish excrete ammonia and marine fish excrete both urea and trymethylamine oxide.

NITROGLYCERINE (1,2,3-TRINITROXYPROPANE; TRINITROGLYCERIN; TRINITROGLYCERINE; GLYCERYL TRINITRATE): A heavy, colourless, oily explosive liquid obtained by nitrating glycerol. Its chemical formula is $C_3H_5N_3O_9$, density 1.6 g cm^{-3} (at 15 °C, 288 K), melting point 14 °C (287 K) and boiling point 50-60 °C (323-333 K). It is the active ingredient in the manufacture of explosives (dynamite). It has been used by the military as an active ingredient and a gellatiniser for nitrocellulose in some solid propellant, such as cordite and ballistite. In medicine, it is used as vasodilator to treat heart conditions such as angina and chronic heart failure.

NITROSAMINES: A group of carcinogenic compounds with the general formula RR'NNO, where R and R' are side groups with a variety of possible structures. They are found in trace amounts in mushrooms, fermented fishmeal and smoked fish and in pickled foods, where they are formed by reaction between nitrite and amines. Nitrosamines are components of cigarette smoke causing cancer in a number of organs, particularly in the liver, kidneys and lungs. An example of a nitrosamine is dimethynitrosamine, which has two methyl side groups (CH_3^-)

NITROSOMONAS: A genus comprising rod shaped chemoautotrophic bacteria, found in soil, sewage, freshwater and on building surfaces, especially in polluted areas that contain high levels of nitrogen compounds. It oxidises ammonium to nitrite.

NITROSONIUM ION (NITRYL ION): The positive ion NO^+ present in certain salts such as the chlorates (NO^+ClO_4) and the borofluoride (NO^+BF_4). The nitrogen atom is bonded to an oxygen atom with a bond order of 3, $[N{\equiv}O]^+$ and the overall diatomic species bears a positive charge usually obtained as the following salts; $NOClO_4$ and $NOSO_4H$.

NITROUS ACID (DIOXONITRIC(III) ACID): A weak acid known only in solution and in the gas phase with the chemical formula HNO_2. It is prepared by the action of acids upon nitrites, preferably using a combination that removes the salt as an insoluble precipitate, for example, barium nitrite ($Ba(NO_2)_2$) and sulfuric acid (H_2SO_4). The solutions are unstable and decompose on heating to give nitric acid and nitrogen monoxide. Nitrous acid can function both as an oxidising agent (forms NO) with I^- and Fe^{2+} or as a reducing agent (forms NO_3^-) with, for example, Cu^{2+}, which is the most common of the two functions. Nitrous acid is widely used for the preparation of diazonium compounds in organic chemistry.

NITROUS OXIDE (NITROGEN(I) OXIDE; DINITROGEN MONOXIDE; LAUGHING GAS; N2O): A colourless, non-flammable gas at room temperature, with a slightly sweet odour and taste and the chemical formula N_2O, density 1.977 g/L (gas), melting point -90.86 °C (182.29 K) and boiling point -88.48 °C (184.67 K). It is known as "laughing gas" due to the euphoric effects of inhaling it. It is used in surgery and dentistry for its anaesthetic and analgesic effects, as an oxidiser in rocketry and in motor racing to increase the power output of engines.Nitrous oxide is a major greenhouse gas and air pollutant, contributing about a tenth as much as carbon dioxide to global warming.

NITRYL ION (NITRONIUM ION): The ion NO_2^+, found in mixtures of nitric acid and sulfuric acid and solutions of nitrogen oxides in nitric acid. It is a reactive cation (NO_2^+) created by the removal of an electron from the paramagnetic nitrogen dioxide molecule or the protonation of nitric acid. It is stable enough to exist in normal conditions but is extensively an electrophile in the nitration of other substances. A few stable nitronium salts that can be isolated are NO_2^+, ClO_4^-, IO_4^- and $NO_2^+BF_4^-$. Nitryl ions generated in situ are used for nitration in organic chemistry.

NIVEN NUMBER (HARSHAD NUMBERS): Any whole number that is divisible by the sum of its digits. For example, 126 is a Niven number because, the sum of its digits 1 + 2 + 6, is 9 and 9 goes into 126 exactly 14 times.

NIVEOUS: White; resembling snow.

NO BOUNDARY CONDITION: The idea that the universe is finite but has no boundary (in imaginary time). It was proposed by Stephen Hawking, former Lucasian Professor of Mathematics at the University of Cambridge and predicts that the universe would start at a single point, like the North Pole of the Earth. But this point would not be a singularity, like the Big Bang. Instead, it would be an ordinary point of spacetime, like the North Pole is an ordinary point on the Earth. According to the no boundary proposal, the universe would have expanded in a smooth way from a single point. As it expanded, it would have borrowed energy from the gravitational field, to create matter.

NO CARRIER: In telecommunications, it is an error message transmitted from an answering modem to a calling modem when a telephone line drops unexpectedly because of line noise or other interruptions.

NO CARRIER ADDED: A preparation of a radioactive isotope which is essentially free from stable isotopes of the element in question.

NO-CLONING THEOREM: A theorem stating that it is not possible to copy quantum information about a quantum state because examining a quantum state alters that state.

NO-HAIR THEOREM: A popular name for general relativistic theorems, which postulates that all black hole solutions of the Einstein-Maxwell equations of gravitation and electromagnetism in general relativity can be completely characterised by only three externally observable classical parameters: mass, electric charge and angular momentum. All other information about the matter which forms a black hole or is falling into it, "disappears" behind the black-hole event horizon and is therefore permanently inaccessible to external observers.

NO-OBSERVED-ADVERSE-EFFECT-LEVEL (NOAEL): A method used in a non-clinical risk assessment of organisms, such as people or animals based on experiment or observation that represents the highest tested dose of a substance to which an organism is exposed that has no harmful or adverse health effects on it.

NO-OBSERVED-EFFECT-LEVEL (NOEL): The greatest concentration or amount of a substance, found by experiment or observation, that causes no alterations of morphology, functional capacity, growth, development or life span of target organisms distinguishable from those observed in normal (control) organisms of the same species and strain under the same defined conditions of exposure.

NO-REGRETS POLICY: Policies that would generate net social benefits whether or not there is climate change to reduce the risk of global warming, which might pose a serious threat. No-regrets opportunities for greenhouse gas emissions reduction are defined as those options whose benefits such as reduced energy costs and reduced emissions of local/regional pollutants equal or exceed their costs to society, excluding the benefits of avoided climate change. **No-regrets potential** is defined as the gap between the market potential and the socio-economic potential.

NO-TILL: A process in which seed is planted directly into the residue of the previous crop, without ploughing. Herbicides are used to control weeds, resulting in reduced soil erosion and preservation of soil nutrients.

NOBEL PRIZE: An annual international prize awarded to persons who have made the greatest contributions in the fields of physics, chemistry, medicine, literature, world peace and economics. It is named after Alfred Benhard Nobel (1833-96), Swedish chemist, inventor and philanthropist. He left the major portion of his $9 million estate to be set up as a fund for annual prizes. It was first awarded in 1969. Each prize consists of a gold medal, a sum of money and a diploma with the citation of award. The amount of money available for each prize varies from year to year. In 2011, the Nobel Prize in Chemistry was awarded to the Israeli chemist Dan Shechtman "for the discovery of quasi crystals". Dan Shechtman (1941-) is the Philip Tobias Professor of Materials Science at the Technion (Israel Institute of Technology), an Associate of the US Department of Energy's Ames Laboratory and Professor of Materials Science at Iowa State University.

NOBELIUM: A synthesised, radioactive, metallic element of the actinide series in Group 3 (IIIB) of the Periodic Table with the symbol No, atomic number 102, relative atomic mass 259 and melting point 827 °C (1100 K). It is synthesised by bombarding curium (Cm) with carbon nuclei. Thirteen radioactive isotopes are known, the most stable, nobelium-259 has a 58-min half-life.

NOBLE GASES (INERT GASES): A group of six gaseous, colourless, odourless and nonflammable chemical elements constituting the Group 8 of the Periodic Table. In their order of increasing atomic weight, they are helium (He), neon (Ne), argon (Ar), krypton (Kr), xenon (Xe) and the radioactive radon (Rn). They occur in tiny amounts in the atmosphere. Their uses include in lighting, welding and space exploration.

NOBLE METAL: Highly unreactive metals, which are resistant to corrosion and oxidation in moist air. They tend to be precious, often due to their rarity in the Earth's crust. In order of increasing atomic number, they include ruthenium, rhodium, palladium, silver, osmium, iridium, platinum and gold.

NOCICEPTION (NOCIOCEPTION; NOCIPERCEPTION): The process by which the body generally detects and perceives injury or noxious stimuli as pain. The skin is richly supplied with nerve endings that respond to excessive heat or cold, tissue damage or harmful chemicals and thus signal the body to take defensive action. Similar nerve endings serve the muscles, joints, bones and visceral organs.

NOCTILUCENT CLOUD: A luminous cloud composed of ice that forms in the upper atmosphere at about 83 km. They are visible on summer nights, particularly when sunspot activity is low.

NOCTURNAL: 1. Occurring in the night. 2. In botany, it refers to flowers that open during the night. 3. In zoology, it describes animals that perform their main activities at night.

NOCTURNAL LIGHT: One of the 6 categories of unidentified flying object (UFO) in the classification scheme devised by J. Allen Hynek in the 1950s. It is defined as any light or lights in the night sky that cannot be explained in terms of aircraft, astronomical objects or any other familiar luminous source.

NODAL LINE: The line connecting the ascending node and descending node of an orbit. Rotation of an orbit's nodal line in the same direction as the object is moving is called **nodal progression**. Rotation in the opposite sense is called **nodal recession**.

NODAL PLANE: A plane of a system in which the value of the orbital wave-function equals zero. This plane defines a region of zero electron density for the orbital. For example, in Py orbital xz constitutes its nodal plane.

NODAL POINTS: Two points on the axis of a system of lenses. If the incidental ray passes through one, the emergent ray will pass through the other.

NODDING: 1. Having branches or flower heads that bend to one side and downward, as occurs in Nodding daffodils. 2. To lower and raise the head quickly, as in agreement.

NODE: 1. In botany, a point on the stem of a plant from which one or more leaves arise or a small knot-like protuberance, as those found on the root of many leguminous plants. The node is often swollen or slightly enlarged. 2. In anatomy, natural thickening or bulge in an organ or part of the body, for example, the sinoatrial nodes that control the heartbeat and the lymph nodes. 3. In mathematics, a point at which two or more branches of a graph meet. The term is also use for a vertex of a tree, network or digraph. 4. In phylogenetics, it is a branch point in a phylogram. 5. A position in a standing wave pattern at which there is no vibration. Points at which there are maximum vibrations are called **antinodes**. 6. In computer science, it is a terminal or a client on a computer network.

NODE OF RANVIER: A gap, approximately 1 micrometre in length, between two of the Schwann cells that make up an axon's myelin sheath, serving as a point for generating a nerve impulse.

NODULE: 1. A growth that occurs on the roots of certain plants such as beans, groundnuts, peas, soyabeans and other legumes which contains the mutualistic nitrogen-fixing bacteria, Rhizobium. 2. A rounded mineral concretion such as chert and flint, clay ironstone and phosphorites that is distinct from and may be separated from, the formation in which it occurs. Nodules usually are elongated and have a knobby irregular surface. They are generally oriented parallel to the bedding. Manganese-rich nodules are found on the ocean floor.

NODULIN: Proteins produced in root hairs or nodules in response to rhizobial infection.

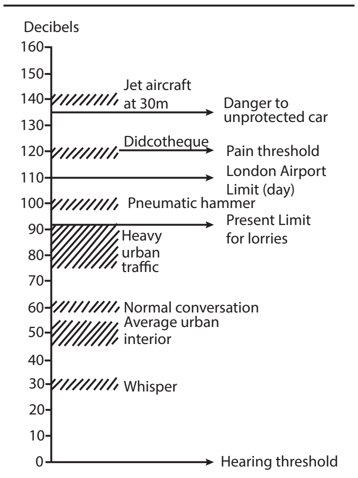
NODULOSE (NODIFORM): Having minute knobs or nodules.

NOETHER'S THEOREM: A theorem proved by German mathematician, Amalie Emmy Noether in 1915 and published in 1918. It states that every continuous symmetry under which a Lagrangian or Hamiltonian is invariant in form is associated with a conservation law. For example, invariance of a Langrangian under time displacement implies the conservation of energy. Not all conservation laws are associated with topology, particularly for solitons. It does not automatically follow that if there is a conserved quantity in a classical field theory associated with Neother's theorem, there is a conserved quantity in the corresponding quantum field theory; this raises the possibility of an anomaly.

NOETHERIAN MODULE: A module that satisfies the ascending chain condition so that every strictly increasing chain of submodules is finite. This is equivalent to satisfying the maximum condition that every non-empty set of submodules has a maximal member and, even if the module is not unitary, to every submodule, including the module itself, being fintitely generated. For example, the set $\mathbb{Z}$ of integers form a Noetherian $\mathbb{Z}$-module, but the rationals do not.

NOETHERIAN RING: In mathematics, a ring in which every strictly ascending chain of left or right ideals is finite. A ring is left-Noetherian if it satisfies the ascending chain condition on left ideals; it is right-Noetherian if it satisfies the ascending chain condition on right ideals and is Noetherian if it is both left- and right-Noetherian. When considered as a (left or right) R-module, it is a Noetherian module.

NOISE: Any signal or sound within a useful frequency band in a communication channel that does not convey useful information. It is measured on a decibel scale ranging from the threshold of hearing (0 dB) to the threshold of pain, which is 130 dB. A whisper registers about 30dB and a heavy hammer on a steel plate registers about 100db.



Decibel scale of sounds

NOISE POLLUTION: Irritating, distracting or physically dangerous human-created noise such as those created by aircraft, railways, trucks, buses, motorcycles, construction equipment, etc. to which people are exposed in their environment and over which they usually have no control.

NOMAD: 1. A cell that migrates or wanders from its site of formation. Certain types of phagocytes are nomads. 2. A plant that grows in pastures. 3. A member of a community of people called **nomads** or **nomadic people** that moves from one place to another, rather than settling down in one location, usually returning to the various locations at various times of the year. Nomadism is not the same as migration, in which people move permanently to another location.

NOMADIC AGRICULTURE (NOMADIC FARMING): A type of keeping livestock in which a small community of herders called nomads move from one place to another in search of new grazing grounds and water for the animals without a fixed range, usually returning to the various locations at various times of the year. Nomadic herders are usually composed of a small number of people, usually family members with no permanent home. An example of such people is the Fulani of West Africa. Nomads often live in arid and semiarid parts of Africa, Asia and Europe and in the tundra regions of Asia and Europe.

NOMARSKI MICROSCOPE (DIFFERENTIAL-INTERFERENCE CONSTRAST MICROSCOPE): A type of light microscope that is useful for viewing live transparent unstained specimen such as cells or microscopic organisms. The shadow-cast images give the illusion of depth to the outlines and surface features of organelles or other structures. An incident beam of plane-polarised light is split into parallel beams by a prism so that different parts of the beam pass through closely adjacent areas of the specimen.

NOMENCLATURE: A systematic method of naming plants or animals or chemical compounds; or a system of names used in an art or science.

NOMINAL DATA: A set of data is said to be nominal if the values/observations belonging to it can be assigned a code in the form of a number where the numbers are simply labels. Nominal data can be counted but cannot be ordered or measured. For example, in a data set, males could be coded as 0, females as 1.

NOMINAL SCALE: In statistics, a discrete classification of data or different types of variables labelling variables, without any quantitative value. Data are neither measured nor ordered but subjects are merely allocated to distinct categories. For example, a record of students' course choices constitutes nominal data that could be correlated with, for example, school results.

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NOMINAL VARIABLE (CATEGORICAL VARIABLE): A variable whose values are not numerical but names or labels that have two or more categories. Examples are gender (e.g., male or female) and colour (e.g., red, green, blue).

NOMINATED SERVICE: Artificial insemination with semen from a named and tested male animal.

NOMOGRAM (NOMOGRAPH): A graph that allows a third variable to be measured when the values of two related variables are known. It consists of three lines each with its own scale, each line representing the values of a variable over a specified range. A ruler laid between two points on two of the lines enables the value of the third to be read off the third line. It used in engineering, industry and the natural and physical sciences.

NOMOPHILOUS: Growing in woodland or pasture.

NON SEQUITUR: 1. Latin for it does not follow. 2. Used in logic, an invalid argument of which the conclusion does not follow from its premises. Also, the conclusion of such an argument, a statement which does not follow from what was previously said.

NON-: A prefix used for "not' or the negation of some property; for example, non-biodegradable refers to materials that are not broken down or decomposed by biological processes.

NON-AQUEOUS PHASE LIQUID (NAPL): A hazardous organic liquid solution which may be lighter or denser than water. NAPLs are hydrophobic or do not dissolve in water. Examples are dry cleaning fluids, petrol and other petroleum products. If the NAPL is denser than water, it is called DNAPL (dense non-aqueous phase liquid) and will tend to sink once it reaches the water table. An example of DNAPL is chlorinated hydrocarbons (e.g., chloroform, dichloromethane, dichloroethene, etc.). If the NAPL is lighter than water, it is called LNAPL (light non-aqueous phase liquid) and will tend to float on water. Examples of LNPLs are lubricants, petrol, toluene and other petroleum products.

NON-ADIABATIC COUPLING: This is a momentum coupling between two adiabatic potential-energy surfaces.

NON-ATOMIC: In mathematics, of a measure or measure ring, having the property such that no member of the measure space is an atom (a measurable set which has positive measure and contains no set of smaller but positive measure).

NON-BARYONIC MATTER: Matter that, unlike all the kinds of matter, is not made of baryons (including the neutrons and protons found in all atomic nuclei). Proposed as a possible constituent of dark matter, it could come in two forms, classified as cold non-baryonic matter or hot non-baryonic matter.

NON-BENZENOID AROMATIC: An aromatic compound in which the number of carbon atoms in the aromatic ring is not equal to six. Examples are the cyclopentadienyl anion ($C_5H_5^-$) and the tropylium cation ($C_7H_7^+$).

NON-BIODEGRADABLE: Materials that are not broken down or decomposed by biological processes such as action of microbes or enzymes.

NON-BONDING ORBITAL (NON-BONDING MOLECULAR ORBITAL, NBMO): A molecular orbital derived only from an atomic orbital of one atom. It lends neither stability nor instability to a molecule or ion when electrons are present.

NON-BREAKING SPACE, NBSP (FIXED SPACE; HARD SPACE): A special space character that acts like a letter or digit used in HTML and word processing to create a space in a line that cannot be broken by word wrap or in the case of HTML create multiple spaces.

NON-CALORIMETRIC THERMOPHYSICAL MEASUREMENTS: These include (p, V, T) measurements, saturated density measurements and any other measurements which give information on the (p, V, T) surface of a fluid.

NON-CENTRIFUGAL SUGAR: A dark semi-solid sugar made by boiling the juices obtained from crushed sugar cane.

NON-CODING DNA (ANTISENSE STRAND): The strand of DNA that does not carry the information necessary to make a protein. The non-coding strand is the mirror image of the coding strand.

NON-COMMUTATIVE GEOMETRY: A type of geometry in which the basic elements are non-cumulative objects, such as matrices, rather than points. There are many applications of the subject in theoretical physics.

NON-COMPETITIVE INHIBITOR: A reversible inhibitor of enzyme activity in which the inhibitor does not compete with natural substrates for the active site but rather binds to a different site on the enzyme. This distorts the conformation at the enzyme active site and prevents the natural substrate from binding to it, eliciting the particular physiological condition associated with its binding.

NON-CONDUCTOR: A substance such as ceramic that does not conduct electricity because of lack of free electrons or ions.

NON-CONSERVATIVE FORCE: It is the type of force for which work done depends on the path taken. Examples are the kinetic frictional force.

NON-CONSTRUCTIVE: In mathematics, of a proof or definition, not enabling a way to construct an example or the conclusion to be proved or the object to be constructed in a finite number of steps, since, for example, it involves quantifying over an infinite domain, as does the axiom of choice.

NON-CONTAGIOUS: Diseases of an animal or human that cannot be spread to other individuals. Examples of such diseases are hypertension and diabetes mellitus. Contagious diseases or diseases easily spread from one person to another, either directly or indirectly include common cold and human immunodeficiency virus (HIV).

NON-CYCLIC ELECTRON FLOW (NON CYCLIC ELECTRON FLOW; NONCYCLIC ELECTRON FLOW): A route of electron flow during the light reactions of photosynthesis that involves photosystem II and photosystem I to produce adenosine triphosphate (ATP), NADPH and oxygen. The combined action of photosystem II and photosystem I in which electrons flow in a linear manner to produce NADPH.

NON-CYCLIC PHOTOPHOSPHORYLATION (NONCYCLIC PHOTOPHOSPHORYLATION): The light reactions of photosynthesis in plants, in which an electron donor is required and oxygen is produced as a waste product. It consists of two photoreactions, resulting in the synthesis of adenosine triphosphate (ATP) and NADPH. In this process, an excited electron from Photosystem II flows down an electron transport chain from excited P680 to the oxidised form of P700, creating a proton gradient between the stroma and thylakoids that generates ATP.

NON-DEFINING ATTRIBUTES: Characteristics such as colour, orientation, or size that do not define the geometric aspects of a shape.

NON-DENUMERABLE: In mathematics, infinite and not able to be put in one-to-one correspondence with the positive integers; uncountable as its cardinal number is larger than that of the set of all natural numbers.

NON-DESTRUCTIVE ACTIVATION ANALYSIS: An activation analysis procedure in which, after the irradiation, no chemical and physical operations are applied which cause a change of any properties of the sample.

NON-DETERMINISTIC POLYNOMIAL TIME ALGORITHM (NP-DECISION PROBLEM): In mathematics, a computational problem, to which there is a yes or no answer, such as the satisfiability problem, for which a polynomial time algorithm exists to verify any guessed solution.

NON-DIRECTIONAL HYPOTHESIS (TWO-TAILED TEST): In statistics, it is the standard test of significance to determine if there is a relationship between variables in either direction (whether the given sample is greater or less than a particular range of values). For a significance level of 0.05, half (0.025) is used to test the statistical significance in one direction and another half for testing statistical significance in the other direction. If the testing sample lies in the null hypothesis, then the null hypothesis is rejected and the alternative hypothesis is accepted. If, for example, the null hypothesis states that the mean is equal to 10, the alternative hypothesis is that the mean is less than or greater than 10. The region of rejection then consists of a range of numbers that are less than or greater than 10.

NON-DOMINATED: In mathematics, of a point in a partial ordering, minimal or maximal.

NON-DRAINING: An adjective describing a chain macromolecule that behaves in a hydrodynamic field as though the solvent within the domain of the macromolecule were virtually immobilised with respect to the macromolecule.

NON-ELECTROLYTES: Substances such as sugar and alcohol that do not ionise in solution and therefore cannot conduct electricity.

NON-ELEMENTARY: In mathematics, of a proof, having no formula in terms of elementary functions and using methods of complex analysis.

NON-EMPTY: In mathematics, of a set, having members or elements or at least one member.

NON-EQUILIBRIUM REACTION: A reaction in which the concentration of the substrates and products is smaller than the value of the equilibrium constant, so that the value of ΔG is large and negative. These reactions are important in providing directionality in pathways in that the flux through the reaction is usually controlled by allosteric factors.

NON-EQUILIBRIUM STATISTICAL MECHANICS: The statistical mechanics of systems that are not in thermal equilibrium. One of the main purposes of non-equilibrum statistical mechanics is to calculate transport coefficients and inverse transport coefficient, such as conductivity and viscosity, from first principles and to provide a basis for transport theory. The non-equilibrium systems easiest to understand are those near thermal equilibrium. For systems far from equilibrium such as chaos and self-organisation it can arise due to nonlinearity.

NON-EQUILIBRIUM THERMODYNAMICS: The thermodynamics of systems not in thermal equilibrium. Non-equilibrium thermodynamics is concerned with transport processes and with the rates of chemical reactions.

NON-EUCLIDEAN GEOMETRY: A type of geometry that does not comply with the basic postulates of Euclidean geometry, particularly a form of geometry that does not accept Euclid's postulate that only one straight line can be drawn through a point in space parallel to a given straight line. Examples of non-Euclidean geometries include the hyperbolic and elliptic geometry, which are contrasted with a Euclidean geometry.

NON-EXPANSIVE MAPPING: A Lipschitz function on a metric space, of which the Lipschitz constant does not exceed 1.

NON-FERMI LIQUID: A many-fermion system that, unlike the electrons in ordinary metals, cannot readily be described in terms of quasi particles. Non-Fermi liquid (NFL) is a metallic state which deviates from Fermi liquid theory. An example of a non-Fermi liquid is the system of interacting fermions in one-dimension, called **Luttinger liquid**.

NON-GRAPHITIC CARBON: All varieties of solids consisting mainly of the element carbon with two-dimensional long-range order of the carbon atoms in planar hexagonal networks, but without any measurable crystallographic order in the third direction (c-direction) apart from more or less parallel stacking. Some varieties of non-graphitic carbon convert on heat treatment to graphitic carbon (graphitisable carbon) but some others do not (non-graphitisable carbon).

NON-GRAPHITISABLE CARBON: A non-graphitic carbon which cannot be transformed into graphitic carbon solely by high-temperature treatment up to under atmospheric pressure or lower pressure. The term non-graphitisable is limited to the result of heat treatment without additional influence of foreign matter or neutron radiation. Non-graphitisable carbon can be transformed into graphitic carbon by a high-temperature process via intermediate dissolution in foreign matter and precipitation under high pressure or by radiation damage.

NON-IMPACT PRINTER (NIP): A printer that operates without physically striking a ribbon onto paper like the earlier printers such as daisywheel printers, and therefore operate without much noise. Examples of non-impact printers are Laser, LED and inkjet.

NON-ISOTOPIC LABELLING: Labelling in which the resulting product has a different chemical composition from the initial one.

NON-LINEAR MODEL: A model that is non-linear in the parameters, $y = \alpha d^{\varphi x_1} + \beta e^{\beta_1 x}$. Non-linear model can be converted into linear model by linearisation.

NON-LINEARITY: A nonlinear system is one in which there is no simple proportional relation between cause and effect, therefore, the change of the output is not proportional to the change of the input. The climate system contains many such non-linear processes, resulting in a system with a potentially very complex behaviour. Such complexity may lead to rapid climate change.

NON-LINEAR OPTICAL EFFECT: An effect brought about by electromagnetic radiation the magnitude of which is not proportional to the irradiance. Non-linear optical effects of importance to photochemistry are harmonic frequency generation, lasers, Raman shifting, upconversion and others. **Non-linear optics** is a branch of optics concerned with the optical properties of matter subjected to intense electromagnetic fields. For nonlinearity to manifest itself, the external field should not be negative compared to the internal fields of the atoms and molecules of which the matter consists. Lasers are capable of achieving this and non-linear optics has been largely developed due to the invention of the laser.

NON-LINEARITY ERROR: An error caused by any deviation from linearity of the response of the detector to the measured radiant power.

NON-LOGICAL AXIOMS: Axioms valid only in a given structure for a theory, for example, $x + y = y + x$. Unlike logical axioms, non-logical axioms are not tautologies.

NON-MARKET IMPACTS: Impacts that affect ecosystems or human welfare, but that are not directly linked to market transactions, for example, an increased risk of premature death.

NON-MEASURABLE: Of a set, not belonging to a given measure algebra; in particular, not Lebesgue measurable.

NON-METAL: A substance whose typical physical properties in the solid state include brittleness and an inability to conduct electricity and whose physical and chemical properties are opposite to those of metals. Nonmetal tends to accept electrons and form negative ions and its oxide is acidic. At low temperatures, non-metals are poor conductors of both electricity and heat, as few electrons move through the material. If the conduction band is near the valence band it is possible for non-metals to conduct electricity at high temperatures, but in contrast to metals, the conductivity increases with increasing temperature. Examples of non-metals are gases such as argon, chlorine, fluorine and helium; bromine (liquid); and solids such as boron, carbon, iodine, phosphorus and silicon.

NON-METALLIC CHARACTER: The tendency of elements to accept electrons during chemical reactions and form a negative ion and to have a high attraction for electrons within a compound. The most reactive non-metals are located in the upper right portion of the periodic table. Fluorine is the most reactive non-metal.

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NON-NEWTONIAN FLUID: Any fluid that does not conform to Newton's law of viscosity or any fluid with viscosity, which is not independent of flow rate. In a non-Newtonian fluid, the relation between the shear stress and the shear rate is different and can even be time-dependent. Therefore, a constant coefficient of viscosity cannot be defined. Examples of non-Newtonian fluids are some salt solutions, molten polymers, ketchup, custard, toothpaste, starch suspensions, paint, blood and shampoo.

NON-OHMIC DEVICE: Any device whose resistance does not remain constant over a wide range of potential differences. Examples are filament lamp, light emitting diode (LED), thermistor and a diode.

NON-ORGANIC CROPS: Crops that are not produced according to guide-lines restricting the use of fertiliser and other practices.

NON-PARAMETRIC STATISTICS: Statistical procedures that allow the process data from small samples, on variables about which nothing is known concerning their distribution. Specifically, non-parametric statistical methods are developed to be used in cases where the researcher knows nothing about the parameters of the variable of interest in the population (hence the name nonparametric). Non-parametric methods do not rely on the estimation of parameters (such as the mean or the standard deviation) describing the distribution of the variable of interest in the population. Non-parametric models differ from parametric models in that the model structure is not specified, but is instead determined from data. Therefore, these methods are also sometimes called parametric free methods or distribution free methods.

NON-PERSISTENT PESTICIDE: Compounds which readily break down quickly in the environment and do not enter the food chain. Examples are cholinesterase-inhibiting pesticides, which include the organophosphates and carbamates, chlorinated phenols, herbicides, pyrethroids and fungicides. Non-persistent pesticides are less harmful to the environment because they do not build up but they have to be applied more often to be effective.

NON-POINT SOURCE POLLUTION: Pollution that enters waterways. Sources of such pollutions include farmlands and residential areas (excess fertilisers, herbicides and insecticides, salt from irrigation, etc.); oil, grease and toxic chemicals from urban runoff and energy production; surface mining (acid drainage from mines); waste disposal sites; construction sites; etc. There are four main forms of non-point source pollution, which are sediments, nutrients, toxic substances and pathogens.

NON-POLAR COMPOUND: A compound that has covalent molecules with no permanent dipole moment. Most of these compounds are insoluble in water. Examples are methane, ethane, ethene and benzene.

NON-POLAR SOLVENT (NON POLAR SOLVENT): Any solvent whose molecules have roughly the same electrical charge on all sides. Such solvents are typically hydrocarbons, such as petrol, pentane and hexane, which can only dissolve other hydrocarbons, such as oils, grease and waxes.

NON-POLARISED INTERPHASES: Interphases for which the exchange of common charged components between the phases proceeds unhindered.

NON-RADIATIVE DECAY: The disappearance of an excited species due to a radiationless transition.

NON-REDUCING SUGAR: Disaccharide sugar such as sucrose that cannot donate electrons to other molecules and therefore cannot act as a reducing agent. When tested with Benedict's reagent it does not produce a positive result or a change in colour. The linkage between the glucose and fructose units in sucrose, which involves aldehyde and ketone groups, is responsible for the inability of sucrose to act as a reducing sugar.

NON-REFLEXIVE: 1. In logic, of a relation, neither reflexive nor irreflexive; holding between some members of its domain and themselves and failing to hold between others. 2. Of a Banach space, not reflexive; not canonically identifiable with its second dual Banach space; possessing a unit ball that is not compact in the weak topology.

NON-REM SLEEP (NREM SLEEP): A stage in sleeping characterised by decreased metabolic activity such as slow and regular breathing; low heart rate; and regular and low blood pressure. The sleeper is relatively still and has no dreams. NREM sleep is divided into 4 stages: stage 1 is characterised by drowsiness, stage 2 by light sleep and stages 3 and 4 by deep sleep, followed by rem sleep. In adults, about 75-80% of sleep consists of NREM sleep.

NON-RENEWABLE RESOURCE: A natural resource that exists in limited supply and cannot be replaced after it has been used up; or any natural resource that cannot be replenished by natural means at the same rate that it is consumed. Examples are fossil fuels, such as oil, natural gas and coal.

NON-RENORMALISATION THEOREMS (NON-RENORMALIZATION THEOREMS): Theorems in quantum field theory providing that the correct result of a calculation is given by the first term in perturbation theory, with Feynman diagrams used to perform calculations involving renormalisation of the theory, with no corrections due to higher-order terms in the perturbation series. Non-renormalisation theorems occur when perturbation theory is capable of giving a general non perturbative result for the theory. Examples of non-renormalisation occur in calculations involving chiral anomalies and in certain theories involving super symmetry.

NON-RESIDUAL HERBICIDES (KNOCKDOWN HERBICIDES): Herbicides such as the non-selective paraquat and glyphosate, that do not leave an active residue in the soil and are quickly deactivated in the soil as they will damage most plants. They are either broken down or bound to soil particles, becoming less available to growing plants.

NON-SELECTIVE HERBICIDE: A chemical compound such as glyphosphate, which destroys a wide range of vegetation that it comes into contact with. It is used to clear vegetation in industrial sites, railway embankments, etc. Some selective herbicides may become nonselective if used at very high rates.

NON-SINGULAR: Of a linear transformation or matrix, possessing an inverse. A square matrix is non-singular if and only if its determinant is non-zero.

NON-SPECIFIC ADSORPTION: Ions approach an interface differently depending on the forces in play. Ions are non-specifically adsorbed (positively or negatively) when they are subjected in the interphase only to long-range coulombic interactions (attraction or repulsion). They are believed to retain their solvation shell and in the position of closest approach to the interface they are separated from it by one or more molecular layers.

NON-STANDARD: 1. Of a mathematical system, elementarily equivalent, but not isomorphic, to the usual model for a given set of axioms; for example, the non-standard real numbers have infinitesimal elements. 2. Of an element of the non-standard real numbers, not corresponding to an element of the ordinary real numbers; for example, every infinitesimal and every infinite number is non-standard.

NON-STANDARD ANALYSIS: A branch of mathematics that formulates analysis using a rigorous notion of an infinitesimal number. It introduces hyperreal numbers to allow for the existence of "genuine infinitesimals," which are numbers that are less than 1/2, 1/3, 1/4, 1/5, ..., but greater than 0.

NON-STANDARD REAL NUMBER: An element of a non-standard model of the axioms of the real numbers. Finite non-standard real numbers correspond to the sum of an ordinary real number (its standard part) and an infinitesimal (its non-standard part).

NON-STOICHIOMETRIC COMPOUND (BERTHOLLIDE COMPOUND): A chemical compound in which the elements do not combine in simple ratios. For example, rutile (titanium (IV) oxide) is often deficient in oxygen, typically having a formula $TiO_{1.8}$.

NON-SYMMETRIC: In logic, of a relation, not symmetric, asymmetric, nor antisymmetric; holding between some pairs and failing to hold for some other pairs of members, x and y, of its domain when it holds between y and x and thus having both symmetric and asymmetric restrictions. For example, "... is a brother of ..." is symmetric on its restriction to males.

NON-THERMAL RADIATION: Electromagnetic radiation given off by particles for reasons other than their thermal energy. The commonest example in astrophysics is synchrotron radiation, produced by the acceleration of electrons or other charged particles by a magnetic field. The spectrum of non-thermal radiation is different from that predicted by Planck's law for a blackbody.

NON-TRANSITIVE: In logic, of a relation, neither transitive nor intransitive; holding between some pairs and failing to hold between other pairs of members, x and z, of its domain when it is given to hold both between x and y and between y and z. For example, "... is a half-brother" is non-transitive, since the half-brother of my half-brother may be myself, my full brother, another half-brother or no relation of mine.

NON-TRIVIAL: 1. In mathematics, of a substructure of a given algebraic structure, neither empty nor a trivial pathological example; sometimes it is also required that the substructure be proper. For example, a non-trivial subgroup of a group does not consist only of the identity of the group, nor a non-trivial ring of its zero. 2. Of a solution of an equation, not zero; for example, $\sqrt{3}$ is the non-trivial solution to $x^3 = 3x$.

NON-UNIFORM CORROSION: Corrosion is non-uniform if the time average of the corrosion carried through a unit area depends on its position on the surface. Non-uniform corrosion can be due to inhomogeneities of structure or of composition of the corroding material or to inhomogeneities of the environment. Special cases of non-uniform corrosion such as pitting or intergranular corrision are sometimes called **localised corrosion**.

NON-VOLATILE RANDOM ACCESS MEMORY (NON-VOLATILE RAM; NVRAM): A computer storage medium such as dynamic RAM (DRAM) and static RAM (SRAM) chips that are backed up by a battery or to non-volatile chips such as flash memory. Flash memory devices such as EEPROM stores data using electrical charges that maintain their state without electrical power and does not need a battery or other power source to retain data.

NON-ZERO RING: In mathematics, a ring that is not the zero ring.

NONA- (ENNEA-): A prefix that denotes nine. For example, a nonagon or enneagon is a two-dimensional nine-sided polygon that has nine angles and sides.

NONAGONAL NUMBER (ENNEAGONAL NUMBER): A polygonal number that represents a nonagon. The nonagonal number for n is given by the formula $n(7n\text{-}5)/2$. Nonagonal numbers include 1, 9, 24, 46, 75, 111, 154, 204, 261, 325, 396, 474, 559, 651, 750, etc.

NONAHYDRATE: A crystalline compound that has nine moles of water per mole of compound.

NONANOIC ACID (PERARGONIC ACID): A clear oily liquid carboxylic acid with the chemical formula $CH_3(CH_2)_7COOH$, density 0.900 g/cm^3, melting point 12.5 °C (285.65 K) and boiling point 254 °C (527 K). It is found as esters in oil of pelargonium and certain esters. They are used as flavourings and also in the preparation of plasticisers and lacquers and also as a herbicide.

NONARY: A base nine (9) numeral system, typically using the digits 0-8, but not the digit 9. They include 1, 2, 3, 4, 5, 6, 7, 8, 10, 11, 12, 13, 14, etc.

NONBONDED INTERACTIONS: Intramolecular attractions or repulsions between atoms that are not directly linked to each other, affecting the thermodynamic stability of the chemical species concerned.

NONCONVEX UNIFORM POLYHEDRON: A uniform polyhedron of a type obtained by relaxing the conditions used to produce the Archimedean solids (which have regular convex faces and identical convex vertices) to allow both nonconvex faces and vertex types, as in the case of the Kepler-Poinset solids. The condition that every vertex must be identical, but the faces need not be, gives rise to 53 nonconvex uniform polyhedra. An example is the great truncated dodecahedron, obtained by truncating the corners of the great dodecahedron at a depth which gives regular decagons.

NONDISJUNCTION: A phenomenon that occurs when a pair of homologous chromosomes does not separate in meiosis but migrate to the same pole of the cell. This results in an uneven number of chromosomes being present in the daughter cells. The daughter cell with an uneven number of chromosomes is said to be **aneuploid**. Klinefelter's syndrome is a result of nondisjunction of the sex chromosomes.

NONFERROUS METAL: Any metal other than iron or any alloy that does not contain iron. It includes aluminium, copper, lead nickel, tin, zinc or their alloys.

NONLINEAR SYSTEM: Systems which are not characterised by linear or first-order equations, but are governed by any variety of complex, reciprocal relationships or feedback loops. Thus, sensitive dependence on initial conditions in chaotic systems illustrates the extreme nonlinearity of these systems. The components are interactive, interdependent and exhibit feedback effects.

NONLINEAR VIDEO EDITING: In computing, it is a digital video editing method that processes compressed data stored on a hard disk; providing high-quality editing on a desktop computer.

NONPOLAR COVALENT BOND: A type of covalent bond in which electrons are shared equally between two atoms of similar

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electronegativity of less than 0.5. It usually occurs between two non-metals.

NONPOLAR MOLECULES: A molecule in which a bonding pair is shared equally. In most cases the atoms of the molecule are the same. These molecules would be composed either of elements having nonpolar covalent bonds or polar covalent bonds that cancel each other out. They are hydrophobic and are poorly soluble in water.

NONRELATIVISTIC QUANTUM THEORY: A system of mechanics that was developed from quantum theory and is used to explain the properties of atoms and molecules. It brings quantum mechanics and special relativity together to account for subatomic phenomena. Using the energy quantum as a starting point it incorporates Heisenberg's uncertainty principle and the de Broglie wavelength to establish the wave-particle duality on which Schrödinger's equation is based.

NONSENSE CODON (STOP CODON; TERMINATION CODON; TERMINATOR; NONSENSE TRIPLETS): 1. An amino acid specifying codon that has been converted to a stop codon by mutation. 2. Any of the nucleotide triplets UAA, UAG and UGA that signal termination of translation and release of the polypeptide chain from the site of synthesis or ribosomal site. These three nonsense codons are called amber (UAG), ochre (UAA) and opal (UGA). 3. A codon for which no normal tRNA molecule exists. 4. Triplets of nitrogenous base sequences or codons that do not code for any protein.

NONSENSE CORRELATION: In statistics, a correlation supported by data but having no basis in reality, such as between incidence of the common cold and ownership of televisions.

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NONSENSE MUTATION: A type of point mutation that changes an amino acid codon to one of the three stop codons, resulting in a truncated and usually nonfunctional protein. The three nonsense codons are called amber (UAG), ochre (UAA) and opal (UGA).This can cause genetic diseases such as Duchenne muscular dystrophy, in this case, caused by a nonsense mutation in gene responsible for dystrophin.

NONSENSE TRIPLET: A codon that causes the premature termination of a growing polypeptide chain, thus producing incomplete protein fragments. They define the end of a polypeptide chain. There are three such triplets, UAG, UGA and UGG.

NONSTANDARD REALS (HYPERREAL NUMBER): Any of the set of numbers formed by the addition of infinite numbers and infinitesimal numbers to the set of real numbers. Because hyperreals represent an extension of the real numbers, R, they are usually denoted by *R.

NONSYNONYMOUS MUTATION (MISSENSE MUTATION): A type of point mutation that converts one codon to another by specifying a different amino acid. This results in incorporation of the wrong amino acid in the polypeptide chain during translation and consequently a potentially faulty protein.

NONVASCULAR PLANTS: Plants lacking lignified vascular tissue (xylem and phloem), vascularised leaves and having a free-living, photosynthetic gametophyte stage that dominates the life cycle. Examples are the mosses and liverworts. These plants have no distinct roots, stems or leaves.

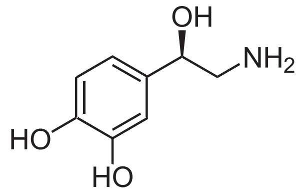
NONVOLATILE: 1. That which does not have a measurable vapour pressure. 2. Not prone to evaporation, especially at low temperature.

NOON: The time of day, when the Sun crosses the observer's meridian and is at its highest point above the horizon. At this point, the Sun lies due south of an observer in the Northern Hemisphere and due north of an observer in the Southern Hemisphere. The observed crossing of the meridian is known as **apparent noon**. Between one observed meridian crossing and the next is an apparent solar day. However, because Earth's orbit is elliptical and Earth's spin axis is tilted with respect to the ecliptic, the length of the apparent solar day varies throughout the year. To overcome this problem, the mean solar day is defined as the yearly average of the apparent solar day and used to define a fictitious mean sun, which moves along the celestial equator at a constant rate. The passage of this mean sun across the meridian defines mean noon. At Greenwich, the mean noon defines 12.00 hours Greenwich Mean Time, also known as Universal Time.

NOOTROPICS (COGNITIVE ENHANCERS; SMART DRUG): Any drug or combination of nutrients such as vitamins, amino acids, minerals and sometimes and herbs that is said to enhance the functioning of the brain such as improvement of memory, learning, attention, concentration, problem solving, reasoning, social skills, decision making and planning, etc. The drugs have been used to treat people with neurological or mental disorders, but there is no scientific evidence to suggest that these drugs have significant effect on healthy people.

NOR-: Affix used to denote the elimination of one methylene group from a side chain of a parent structure (including a methyl group).

NORADRENALINE (NOREPINEPHERINE): A hormone produced by the adrenal glands and also secreted from nerve endings in the sympathetic nervous system as a chemical transmitter of nerve impulses. Many of its general action are similar to those of adrenaline but it is more concerned with maintaining normal body activity than with preparing the body for emergencies.



Chemical Structure of Noradrenaline

NORFOLK CROP ROTATION SYSTEM (NORFOLK FOUR-COURSE SYSTEM): A system of farming, practiced in Norfolk County, England in the 18^{th} century, in which a field was sown annually with a different crop. For example, in a four-year rotation system, wheat was grown in the first year, turnips in the second, followed by barley, with clover and ryegrass undersown, in the third. The clover and ryegrass were grazed or cut for feed in the fourth year. The turnips were used for feeding cattle and sheep in the winter. It provided a well-balanced system for building up and maintaining soil fertility, controlling weeds and pests, providing continuous employment and profitability.

NORFOLK ISLAND PINE (CHRISTMAS TREE; *Araucaria heterophylla*): An evergreen coniferous tree belonging to the Araucariaceae family with about 19 species in genus. It is native to Norfolk Island. It is a large dioecious trees growing up to 50-60 metres in height with huge erect stems, having horizontal branches. It has a symmetrical pyramid shape of whorled, horizontal branches, widely spaced, in formal ascending order to its crown. The dark green leaves (needles) are awl-shaped, covering all sides of the branches, changing their shape slightly in the upper branches as the tree ages. The bark is grey and rough. The male and female cones produced are found on separate trees. The female cones, which have edible seeds, are found high on the top of the tree. The male cones are smaller and narrow. The tree is cultivated as an ornamental plant in certain parts of the world such as Australia, Brazil, Chile, New Zealand, Peru, Portugal, South Africa and Spain. Young trees are often grown as houseplants in winters when they cannot survive the cold outside temperatures. The wood of large trees is used in construction, furniture and shipbuilding.

NORITE: A basic, coarsely crystalline plutonic igneous rock containing the plagioclase mineral labradorite as the main component and differing from gabbro by having orthopyroxene as the main ferromagnesian mineral present. Minor components can include biotite and hornblende. Norite is generally found in layered igneous intrusions formed by progressive crystal fractionation. Norite is sometimes associated with anorthosite and troctolite. It is found extensively in the highland regions of the Moon.

NORM: 1. A generalisation of the concept of magnitude of any vector space. The norm of a vector x is denoted by $\|x\|$. It is a real number associated with the vector and positive or zero. If k is a real number, then $\|kx\| = k\|x\|$ and $\|x + y\| \leq \|x\| + \|y\|$. 2. Of an algebraic number, the product of all the conjugates to the given number; for example, the norm of an algebraic number of the form $a + b\sqrt{d}$ is $a^2 - db^2$ for any integers a, b and d. 3. Norm- is used as a prefix with respect to a given norm or in the topology induced by a given norm. For example, a norm-convergent or norm-bounded function is convergent or bounded with respect to some norm; a norm-compact function is compact with respect to the norm topology.

NORMABLE: Of a topological linear space, compatible with a norm; such that there exists a norm for which the topology imposed by the norm is the given topology.

NORMAL: 1. In mathematics, perpendicular, especially a perpendicular to a line tangent to a plane curve or to a plane tangent to a space curve. 2. In chemistry, a solution having a concentration of one gram equivalent per dm^3. 3. In chemistry, aliphatic hydrocarbon with unbranched chains of carbon atom.

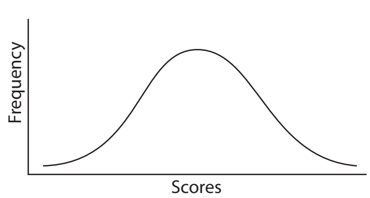
NORMAL CLOSURE: In mathematics, the smallest normal subgroup of a group, G, that contains a given non-empty subset, X. It is usually denoted X^G, (X^G) or $(X)^G$.

NORMAL CURVATURE: Of a surface at a point, the curvature of a normal section of the surface at the given point. A positive sign is taken if the principal normal to the section points in the same direction as the normal to the surface and a negative sign is taken otherwise.

NORMAL DERIVATIVE: In mathematics, the normal derivative of a function defined on a curve or surface is the directional derivative taken in the direction normal or orthogonal to a given curve or surface in space. If the normal direction is denoted by $\mathbf{n}$, then the directional derivative of a function f is sometimes denoted as: $\frac{\partial f(x)}{\partial n} = \nabla f(x) \cdot \mathbf{n}$.

NORMAL DEVIATE: The ratio of the deviation, d, to the standard deviation, σ. It is used to assess experiments, where the results do not fall into a limited number of classes but show a continuous range, for example, the recording of crop yield in response to various fertiliser levels. To find whether any deviation is significant, its probability is looked up in the normal distribution table. Its symbol is c.

NORMAL DISTRIBUTION (GAUSSIAN DISTRIBUTION): A probability distribution, $f(x)$, of a random variable, x that is assumed by many statistical methods. This is given by $f(x) = \frac{1}{\sigma\sqrt{2\pi}} \exp\left[-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}\right]$, where μ is the mean and σ^2 is the variance. All normal distributions are symmetric and have bell-shaped density curves with a single peak when plotted on a graph, and also known as **normal curve**, **normal distribution curve** or **Gausian curve**. The area of a vertical section of the curve represents the probability that the random variable lies between the values that delimit the section and these probabilities can be found from statistical tables. The mean, median and mode all occupy the same high middle point of the curve, which is perfectly symmetrical around this point.



Normal Curve

NORMAL EQUATIONS: The simultaneous equations involving experimental unknowns derived from a larger number of observation equations that describe the best least square approximation for the distance of a vector from a set of n vectors.

NORMAL EXTENSION FIELD: In mathematics, an algebraic extension field such that the group of automorphisms leaving the base field fixed (a relative automorphism) is such that no other element is left fixed by the entire set of automorphisms. Alternatively, if any irreducible polynomial has one root in the extension, it has all its roots in the extension.

NORMAL FACTOR: In mathematics, any of the factor groups of members of a normal series for a group by the next member of the series.

NORMAL FAMILY: 1. A set of continuous functions that are precompact, especially in the topology of uniform convergence on compact subsets of a given set in Euclidean space. 2. A family of complex functions that are analytic on a common domain and of which any infinite sequence of members contains a subsequence converging uniformly on compact sub-regions either to an analytic limit or identically to infinity.

NORMAL FAULT: A fault that occurs where two blocks of rock are pulled apart, as by tension causing the hanging wall to move downward.

NORMAL FORM: In mathematics, a canonical form to which a given structure or object may be reduced, especially of a matrix or in logic.

NORMAL MATRIX: A matrix that commutes with its adjoint or conjugate; the matrix expression of a normal operator on the Hilbert space C^n. In other words, a complex square matrix A is normal such that $AA^* = A^*A$, where A^* is the conjugate transpose of A. The Jordan normal form of a normal matrix is a diagonal matrix.

NORMAL MODE OF VIBRATION: A basic vibration of a polyatomic molecule. All vibrational motion of a polyatomic molecule can be treated as a superposition of the normal modes of vibration of the molecule. If N is the number of atoms in a molecule, the number of modes of vibration is 3N - 5 for a linear molecule and 3N - 6 for a nonlinear molecule. Each of these vibrational modes has a characteristic frequency, although it is possible for some of them to be degenerate. For example, in a linear triatomic molecule there are four normal modes of vibrations since 3N – 5 = 4 for N = 3. These vibrational modes are: (a) symmetric stretching (breathing) vibrations; (b) antisymmetric stretching vibrations; and (c) two bending vibrations, which are degenerate

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NORMAL MODES: A convenient set of simple vibrational motions of an oscillatory system.

NORMAL NUMBER: In mathematics, a real number whose expansion to a given base is such that each block of digits of fixed length tends to occur equally often as the number of digits in the expansion tends to infinity. In base 10, one out of every 100 two-digit blocks should on average be 37 and one in ten single-digit blocks should be 7. Almost all numbers are normal in every base (are absolutely normal) but it cannot be proved that any naturally occurring irrational numbers are so.

NORMAL OPERATOR: In mathematics, an operator that commutes with its adjoint such that $AA^* = A^*A$. Examples of normal operators are unitary operators, $N^* = N^{-1}$; Hermitian operators or self-adjoint operators, $N^* = N$; Skew-Hermitian operators, $N^* = -N$; and positive operators, $N = MM^*$ for some M, therefore N is self-adjoint.

NORMAL SALT: A salt in which all of the acid hydrogen atoms have been replaced by a metal or the hydroxide radicals of a base are replaced by an acid radical, for example, sodium carbonate (Na_2CO_3) or sodium chloride (NaCl).

NORMAL SECTION: In mathematics, a section of a surface by a plane containing both a normal to the surface and a prescribed tangent at a given point.

NORMAL SERIES: In mathematics, a finite sequence of subgroups $\{G_k\}$ of a given group, G, beginning with the group itself and terminating in the trivial subgroup, such that each G_{k+1} is a normal subgroup of G_k. If no two members of the sequence are identical, the normal series is said to be without repetitions and the factor groups G_k / G_{k+1} are the **normal factors** of the series.

NORMAL SPAN (SPAN): In estimating lengths using body parts, it is the distance from the end of the thumb to the end of the index (first) finger of an outstretched hand. Using an adult hand, it is approximately 15 centimetres.

NORMAL STRESS: In mathematics, that part of a stress vector, $\mathbf{t}(\mathbf{n})$, that acts along the unit outward normal, $\mathbf{n}$, to the surface of a fluid and equal to $[\mathbf{t}(\mathbf{n}) .\mathbf{n})]\ \mathbf{n}$.

NORMAL SUBGROUP (INVARIANT SUBGROUP): In mathematics, a subgroup that is left invariant by all inner automorphisms of the group or equivalently, bas identical left- and right-cosets in the group. This occurs as the kernel of some homomorphism of the given group into another group (this may also define monoids). For example, the centre of a group is a normal subgroup.

NORMAL SUBRING: In mathematics, it is a two-sided ideal.

NORMAL TEMPERATURE AND PRESSURE (NTP): Former name for standard temperature and pressure (STP). Standard temperature is the freezing point of pure water, 0 °C or 273.15 K. The standard pressure is the pressure exerted by a column of mercury (symbol Hg) 760 mm high, often designated 760 mm Hg. This pressure is also called one atmosphere and is equal to 1.01325×10^6 dynes per sq cm.

NORMAL TOPOLOGICAL SPACE: A topological space in which for every pair of disjoint closed sets there is a pair of disjoint open sets in which they are respectively contained. If this follows when the sets are merely disjoint from each other's closures, the space is said to be **completely normal**.

NORMALISATION (NORMALIZATION): 1. The process of taking data from a problem and reducing it to a set of relations while ensuring data integrity and eliminating data redundancy. 2. In mathematics, the process of putting in some normal form, especially to divide a non-zero vector by its norm, to divide a polynomial by its leading coefficient or to change the variable in a probability distribution so that its mean becomes zero and its variance unity. 3. In computer science, floating-point representation, to express (a number) in the standard form as regards the position of the radix point, usually immediately following the first non-zero digit. 4. An annealing process in which steel or ferrous alloy is heated to a suitable temperature above the transformation temperature range followed by cooling in still air. This promotes the formation of uniform internal structure and the elimination of internal stress.

NORMALISE (NORMALIZE): In metallurgy, to heat (steel) above a critical temperature, usually between 800 - 900 °C and allowing it to cool in still air to relieve internal stresses. This refines the grain size and improves the mechanical properties.

NORMALISER (NORMALIZER): In mathematics, of a given subset of a group, the set of elements, g, of the group, such that $g^{-1}Hg = H$, where H is the given subset. Where G is the underlying group, the normaliser of the subset H is denoted $N_G(H)$.

NORMALISING/NORMALIZING SELECTION (STABILISING/ STABILIZING SELECTION): A type of natural selection in which genetic diversity decreases as the population stabilises on a particular trait value. This is probably the most common mechanism of action for natural selection. Stabilising selection commonly uses negative selection or purifying selection to select against extreme values of the character.

NORMALLY: A term used to describe a chemical solution containing an equivalent weight of solute in grams per litre of solution or aliphatic hydrocarbons with unbranched chains of carbon atoms.

NORMED SPACE: In mathematics, a vector space having a norm.

NORTH: One of the four cardinal directions or main points of the compass, lying to the left of a person facing the sunrise or the opposite of south perpendicular to east and west. By convention, the top side of a map is north.

NORTH ATLANTIC DRIFT (NORTH ATLANTIC CURRENT; ATLANTIC SEA MOVEMENT): Warm ocean current in the North Atlantic Ocean. It merges with the Gulf Stream at latitude 40°N and longitude 60°W. It warms the coast and affects the climate of northwest Europe.

NORTH ATLANTIC OSCILLATION (NAO): A natural weather phenomenon in the North Atlantic Ocean consisting of opposing variations of barometric pressure near Iceland and the Azores. It is an irregular fluctuation of atmospheric pressure over the North Atlantic Ocean that has a strong effect on winter weather in Europe, Greenland, northeastern North America, North Africa, and northern Asia. It is irregular in that it can occur on a yearly basis, or decades apart. On average, a westerly current, between the Icelandic low pressure area and the Azores high pressure area, carries cyclones with their associated frontal systems towards Europe. However, the pressure difference between Iceland and the Azores fluctuates on time scales of days to decades, and can be reversed at times. It is the dominant mode of winter climate variability in the North Atlantic region, ranging from central North America to Europe.

NORTH POLE: 1. The northernmost point of the Earth's geographic axis, located at latitude 90° N in the Arctic Ocean and covered with drifting pack ice, where all meridians or longitudes converge. 2. The north-seeking magnetic pole of a straight or rectangular shaped magnet.

NORTHBRIDGE (NB): A chipset or an integrated circuit such as Intel or VIA that is responsible for communications between the CPU interface, AGP, PCI and the memory. It is one of the two chips that control sthe functions of the chipset. The other is the Southbridge.

NORTHERN BLOT: A technique used to identify and locate mRNA sequences that are complementary to a piece of DNA called a probe.

NORTHERN CROSS: A group of the five brightest stars that form a large cross in the constellation Cygnus. The stars are: Deneb (Alpha Cyg), Albireo (Beta Cyg), Sadr (Gamma Cyg), Delta Cyg and Gienah (Epsilon Cyg).

NORTHERN LIGHTS (AURORA BOREALIS): Aurora that appears in the Northern Hemisphere skies characterised by streamers of reddish or greenish light. It is caused by collisions between gaseous particles in the Earth's atmosphere with charged particles released from the Sun's atmosphere.

NOSE: The protuberance on the face of some vertebrates that contains the nostrils and part of the nasal cavity. It, therefore, forms part of the olfactory system and the external opening of the respiratory system.

NOSEBAND: A broad leather band worn around a horse's nose and above the bit, used to prevent a horse from opening its mouth too wide.

NOSEBLEED (EPISTAXIS): Bleeding or haemorrhage from the nose, usually due to rupture of small blood vessels in the nose that may be brought about by injury to the nose or face, infection, tumours, irritation from foreign bodies, hypertension, alcohol abuse, inherited bleeding problems or excessive nose blowing or picking during a cold.

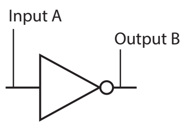
NOSTRIL (NARIS): The opening of the nasal cavity of vertebrates, in which cells sensitive to smell are located in vertebrates. In birds and mammals, they contain branched bones or cartilages called turbinates, whose function is to warm air on inhalation and remove moisture on exhalation.

NOT: "NOT" is used in logic to indicate the negation of a statement. The statement "NOT A" is false if A is true and it is true if A is false. The symbol $\sim A$ or $-A$ or $\bar{A}$ are used to represent NOT. The operation of NOT can be described by the truth table in the diagram.

A	NOT A
T	F
F	T

A truth table describing the operation of NOT

NOT GATE (INVERTER; NOT CIRCUIT): The most basic of all the logical gates that has only one input and one output. It changes the input A so that a 1 input becomes a 0 output and a 0 input becomes a 1 output. In other words, it has a high-voltage output signal if the input signal is low and if the input signal is high it has a low-voltage output signal.



INPUT	OUTPUT
A	NOT A
0	1
1	0

Not gate symbol

NOTATION: The use of different signs and symbols to represent mathematical statements.

NOTE: A musical sound of specified pitch or a representation of such a sound in music.

NOTEBOOK COMPUTER (LAPTOP): A small, lightweight portable computer weighing less than 3 kilograms with a flat-panel display which is back-hinged and folds open for viewing. It is powered by rechargeable batteries. Most notebook computers also have connections for external monitors, keyboards and pointing devices and some come with docks that facilitate attachment to these peripherals.

NOTOCHORD: The stiff but flexible rod that lies beneath the nerve cord and above the alimentary canal of all embryonic and carval chordates, including vertebrates. In adult vertebrates it is largely replaced by the vertebral column.

NOVA: Any of a class of stars whose luminosity temporarily increases by several thousands up to a million times normal.

NOWHERE DENSE SET: In mathematics, a set whose closure has empty interior; for example, any finite set of reals is nowhere dense.

NOWHERE DIFFERENTIABLE FUNCTION: In mathematics, a continuous function on the real line that is not differentiable at any point in its domain. An example is the Weierstrass function.

NOXIOUS: Harmful to health or life, especially by being poisonous.

NOZZLE: A narrow or tapering part at the end of a tube or pipe, used to direct or control the flow of a liquid or gas.

NOZZLE AREA RATIO: The ratio of a nozzle's throat area to the exit area. The ideal nozzle area ratio allows the gases of combustion to exit the nozzle at the same pressure as the ambient pressure.

NOZZLE EFFICIENCY: The efficiency with which the nozzle converts the potential energy of a burned fuel into kinetic energy for thrust.

NOZZLE EXIT ANGLE: The angle of divergence of a nozzle.

NP: In mathematics, the class of NP-decision problems for which an answer can be verified in polynomial time.

NP-COMPLETE PROBLEM: In mathematics, a decision problem, such as the satisfiability problem for which a polynomial time algorithm can exist if and only if $P = NP$. An answer to the $P = NP$ question would determine whether problems that can be verified in polynomial time, like the subset-sum problem, can also be solved in polynomial time. If $P \neq NP$, it would mean that there are problems in NP (such as NP-complete problems) that are harder to compute than to verify: they could not be solved in polynomial time, but the answer could be verified in polynomial time.

NP-DECISION PROBLEM (NON-DETERMINISTIC POLYNOMIAL TIME ALGORITHM): In mathematics, a computational problem, to which there is a yes or no answer, such as the satisfiability problem, for which a polynomial time algorithm exists to verify any guessed solution.

NP-HARD PROBLEM: In mathematics, a decision problem that whether in NP, a polynomial time reduction to an NP-complete problem exists or not. Examples of such problems include the Hamiltonian cycle and traveling salesman problems.

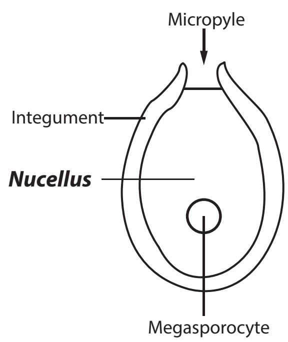
NPK: Initials for the three major elements, nitrogen, phosphorous, potassium, with which fertilisers are made. Different amounts of these three elements are used to suit particular soils and crops.

NREM SLEEP (NON-REM SLEEP): A stage in sleeping characterised by decreased metabolic activity such as slow and regular breathing; low heart rate; and regular and low blood pressure. The sleeper is relatively still and has no dreams. NREM sleep is divided into 4 stages: stage 1 is characterised by drowsiness, stage 2 by light sleep and stages 3 and 4 by deep sleep, followed by rem sleep. In adults, about 75-80% of sleep consists of NREM sleep.

NTLDR (NT LOADER): A small piece of software loaded from the hard drive boot sector that displays the Microsoft Windows OS startup menu and helps Microsoft Windows NT/2000/XP load. This portion of the boot sequence was replaced by Bootmgr and winload.exe in Windows Vista and later versions of Windows.

NUCAMENTUM: Looking like a catkin; indehiscent.

NUCELLUS: The tissue that makes up the greater part of the ovule of seed plants or the central portion of an ovule called the megasporangium in which the embryo sac develops. It is a nutritive tissue surrounding the embryo sac and in turn enclosed by the integuments. In certain flowering plants, it may persist after fertilisation and provide nutrients to the embryo.



NUCIFEROUS: Producing nuts.

NUCLEAR APPARATUS (CHROMATIN BODY; NUCLEAR REGION): A region containing the cell's genetic information in prokaryotic cells. Unlike the nucleus in eukaryotic cells, it is not surrounded by a membrane.

NUCLEAR BARRIER: A region of high potential energy that a charged particle must pass through in order to enter or leave an atomic nucleus.

NUCLEAR BATTERY: A single cell or battery of cells in which the energy of particles emitted from the atomic nucleus is converted internally into electrical energy.

NUCLEAR BINDING ENERGY: The energy required to break up a nucleus to its protons and neutrons.

NUCLEAR DETECTION SATELLITES: Spacecraft designed to detect nuclear explosions on the ground or in the atmosphere. Although primarily intended to monitor nuclear treaty compliance, these satellites have also been used to detect and observe galactic events, such as supernovae.

NUCLEAR DISINTEGRATION: Nuclear decay involving a splitting into more nuclei or the emission of particles.

NUCLEAR DIVISION: The first in the process of cell division by which a nucleus divides into two, mitosis and meiosis, which is necessary for growth, asexual production or sexual reproduction. Mitosis is responsible for growth and asexual production, and meiosis is responsible for sexual reproduction and the resultant cell is a gamete.

NUCLEAR ENERGY: Energy from the inner core or nucleus of an atom or energy obtained from nuclear fission or nuclear fusion. The energy is released as a result of nuclear reaction.

NUCLEAR ENVELOPE (NUCLEAR MEMBRANE): The double membrane that separates the nucleoplasm of an eukaryotic cell from the cytoplasm, but contains nuclear pores by which the necleoplasm and the cytoplasm communicate. The membranes consist of lipid bilayers that are separated by a perinuclear space. The outer layer is continuous with the rough endoplasmic recticulum and is structurally and functionally distinct from the inner membrane.

NUCLEAR FACTOR kappaB (NF-kappaB; NF-κB): A protein complex that controls transcription of DNA involved in rapid cellular responses to stresses, such as injury or invading pathogens and in many aspects of cell proliferation and survival. It is found in the cells of organisms ranging from flies to mammals and has a key role in such disorders as systemic shock, cancer, arthritis, chronic inflammation, asthma, neurodegenerative diseases and heart disease.

NUCLEAR FISSION: The split up of heavy atomic nucleus into two lighter nuclei emitting several small particles that is accompanied by nuclear radiation and the release of a large amount of energy. The splitting into two parts called **fission products**, subsequently emit either two or three neutrons, releasing a quantity of energy equivalent to the difference between the rest mass of the neutrons and the fission products and that of the original nucleus. Controlled nuclear fission is employed in generating electricity, scanning machines and to undertake research work. An example of nuclear fission is: ${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{139}_{56}Ba + {}^{94}_{36}Kr + 3{}^{1}_{0}n.$

NUCLEAR FORCE: A strong attractive force between nucleons in the atomic nucleus that holds the nucleus together.

NUCLEAR FUEL: A substance that will sustain a fission chain reaction so that it can be used as a source of nuclear energy. The fissile isotopes are uranium -235, uranium -233, plutonium -241 and plutonium -239.

NUCLEAR FUSION: The process in which two lighter atomic nuclei are fused together to form a heavier one, with the release of large amounts of energy. This is in contrast to nuclear fission reactions, where a neutron is used to break up a large nucleus, but in nuclear fusion the two reacting nuclei themselves have to be brought into collision. An example of nuclear fusion is: $4{}^{1}_{1}H \rightarrow {}^{4}_{2}He + 2{}^{0}_{1}e + energy.$

NUCLEAR GRAPHITE: A polygranular graphite material for use in nuclear reactor cores consisting of graphitic carbon of very high chemical purity. High purity is needed to avoid absorption of low-energy neutrons and the production of undesirable radioactive species.

NUCLEAR HEART SCAN: A noninvasive test that uses radioactive tracers to delineate the hearts chambers and major vessels. It may be used to detect a heart attack, heart muscle function and coronary artery disease. During a nuclear heart scan, a safe, radioactive material called a **tracer** is injected through a vein into the bloodstream. The tracer then travels to the heart and releases energy, which special cameras outside of the body detect. The cameras use the energy to create pictures of different parts of the heart.

NUCLEAR ISOMERISM: A condition in which atomic nuclei with the same number of neutrons and protons have different lifetimes. This occurs when nuclei exist in different unstable quantum states, from which they decay to lower excited states or to the ground state, with the emission of gamma-ray photons.

NUCLEAR MAGNETIC RESONANCE (NMR): The absorption of electromagnetic radiation at a suitable precise frequency by a nucleus with nonzero magnetic moments in an external magnetic field. The phenomenon occurs if the nucleus has nonzero spin, in which case it behaves as a small magnet. Thus, for hydrogen (spin of ½), there are two possible states in the presence of a field, each with a slightly different energy. NMR can be used for the accurate determination of nuclear moments. It can also be used in a sensitive form of magnetometer to measure magnetic fields. In medicine, **nuclear magnetic resonance imaging** (MRI) has been developed, in which images of tissue are produced by magnetic resonance techniques. The main application of NMR is as a technique for chemical analysis and structure determination, known as NMR spectroscopy.

NUCLEAR MEMBRANE (NUCLEAR ENVELOPE): An intracellular structure, consisting of two concentric membranes, enclosing the nucleoplasm and separating it from the cytoplasm. It is perforated with holes, called nuclear pores, to facilitate and regulate the exchange of materials (for example, proteins and RNA) between the nucleus and the cytoplasm. In between these two membranes is a space called perinuclear space. The outer membrane is continuous with the endoplasmic reticulum. The inner membrane is associated with a network of intermediate filaments called nuclear lamina, which is made of lamin. The lamina acts as a site of attachment for chromosomes. It also acts like a shield for the nucleus.

NUCLEAR MODELS: Any model that attempts to describe the atomic nucleus. Because of the complexity of the many-body problem in the atomic nucleus, various models have been suggested to explain various aspects of the entity. In the shell model, the nucleons are regarded as almost independent particles. In the liquid drop model all the nucleons in the nucleus are regarded as acting collectively in much the same way as the molecules in a liquid. As these models appear to be physically very different, it is necessary to distinguish the circumstances under which they are applicable. Considerable insight is provided by considering the interactions between the quasiparticles and collective excitations in the many nucleon problems. This enables more complicated models, which incorporate both the shell model and the liquid-drop model, to be constructed.

NUCLEAR MOMENT: A property of atomic nuclei in electric moments resulting from deviations of the nuclear charge distribution from spherical symmetry. Magnetic moments are a consequence of the intrinsic spin and the rotational motion of nucleons within the nucleus.

NUCLEAR OVERHAUSER EFFECT, NOE (OVERHAUSER EFFECT): An effect through which the resonance intensities can be magnified in nuclear magnetic resonance spectroscopy by transferring spin polarisation from one spinning nuclei to another through spin relaxation techniques. NOE is observed through space and not through bonds, hence the signal to be intensified should be within certain proximity for the spin polarisation transfer to be possible.

NUCLEAR PARTICLE: A nucleus or any of its constituents in any of their energy states.

NUCLEAR PHYSICS: The study of the properties of the nucleus of the atom including the structure of nuclei, nuclear forces, the interactions between particles and nuclei and the study of radioactive decay.

NUCLEAR PORE: Opening in the nuclear envelope of a cell that allows the exchange of materials between the nucleus and the cytoplasm. There are about on average, 2000 nuclear pore complexes in the nuclear envelope of a vertebrate cell, but it varies depending on cell type and the stage in the life cycle.

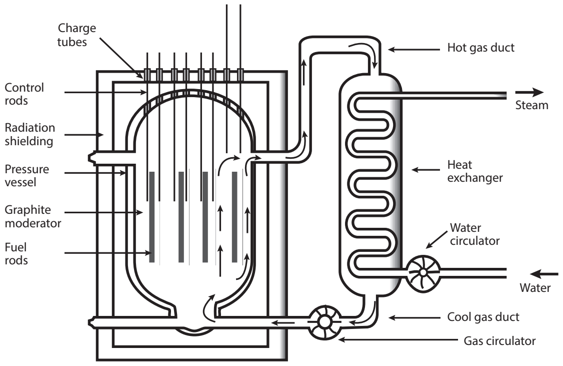
NUCLEAR POWER: The supply of useful energy from nuclear reactions or the utilisation of the fission reactions in a nuclear power reactor to produce energy in the form of steam for electric power production, for ship propulsion or for process heat. Fission reactions provide intensive sources of energy. For example, the fission of an atom of uranium yields about 200 MeV (million electron volts), whereas the oxidation of an atom of carbon releases only 4 eV (electron volts).

NUCLEAR PROPULSION: The use of energy released by a nuclear reaction to provide thrust directly, as distinct from nuclear-electric propulsion. A nuclear propulsion system derives its thrust from the products of nuclear fission or fusion.

NUCLEAR RADIATION: Radiations given off during the decay of a radioisotope. Examples are alpha, beta or gamma radiation.

NUCLEAR REACTION: A process in which two nuclei or nuclear particles collide to produce products different from the initial particles. Sometimes a nuclear reaction is accompanied by energy release.

NUCLEAR REACTOR: A device that uses fission to produce energy or radioisotopes in a sustained and controlled way. Neutrons released in one fission reaction may strike other heavy nuclei, causing them to fission. The rate of this chain reaction is controlled by introducing materials, usually in the form of rods that readily absorb neutrons. Typically, control rods made of cadmium or boron are gradually inserted into the core if the series of fissions begins to proceed at too great a rate, which could lead to meltdown of the core. The heat released by fission is removed from the reactor core by a coolant circulated through the core. Some of the thermal energy in the coolant is used to heat water and convert it to high-pressure steam used for electric power production, for ship propulsion or for process heat.



Nuclear reactor

NUCLEAR SAFETY: Measures to avoid nuclear and radiation accidents and absorption of radiant energy by humans in the operation of nuclear reactors and in the production and disposal of nuclear weapons and nuclear waste.

NUCLEAR STABILITY: The ability of an isotope to resist decay or fission. The stability of an atomic nucleus depends on its binding energy per nucleon, E. The energy represents the average energy that needs to be supplied to liberate a nucleon from the nucleus. Nuclei with deeper binding energy wells are more stable and occur at intermediate mass numbers.

NUCLEAR TESTING: The process of carrying out a test on a nuclear weapon to determine its effectiveness, yield and explosive capability. This is usually done secretly underground or in very remote areas, but is easily detected by the seismic shock waves it produces. The first nuclear weapon test was done by the United States at the Trinity site on July 16, 1945, with a yield equivalent to about 20 kilotons of trinitrotoluene (TNT). These tests produce large quantities of radioactive fallout, which is toxic and hazardous to the environment. Because of this, a test ban treaty was signed by three nations including UK, US and the Soviet Union in 1963 prohibiting all tests carried out in the atmosphere, underwater or in outer space except those carried out underground.

NUCLEAR TIMESCALE: The time a star takes to convert its usable hydrogen into helium. This may be measured in a few millions of years (large, hot stars) or many billions of years (small, cool stars).

NUCLEAR TRANSFER: A technique used in cloning animals in which a nucleus from a donor cell is inserted in an egg cell which is then stimulated to develop into an embryo. This technique was successfully used to produce the famous Dolly, the sheep in 1997. A single embryo cell is injected into an unfertilised egg cell, from which the chromosomes have been removed by micropipette. Fused recipient and donor cells are then stimulated by electrical pulses to begin dividing and form an embryo, like a normal fertilised egg cell. The embryo is then implanted into the uterus of a surrogate mother to continue its development. Cows are commonly cloned to select those that have the best milk production.

NUCLEAR TRANSFORMATION: The change of one nuclide into another with a different proton number or nucleon number.

NUCLEAR TRANSITION: For a nucleus, a change from one quantised energy state into another or a nuclear transformation.

NUCLEAR TRANSMUTATION: The conversion of an element into another by nuclear reaction. Natural transmutation occurs when radioactive elements spontaneously decay over a long period of time and transform into other more stable elements. Artificial transmutation occurs in machinery that has enough energy to cause changes in the nuclear structure of the elements. Nuclear transmutation is considered as a possible mechanism for reducing the volume and hazard of radioactive waste.

NUCLEAR WARFARE (ATOMIC WARFARE): War involving the use of nuclear weapons. It is vastly more destructive in range and extent of damage than conventional warfare. It can have severe long-term effects, primarily from radiation release but also from possible atmospheric pollution leading to nuclear winter that could last for decades, centuries or even millennia after the initial attack. An example is the atomic bombs dropped on Hiroshima and Nagasaki in Japan by the United States in 1945, during World War II, the only time it had been used.

NUCLEAR WASTE: The radioactive and toxic by-product of nuclear energy and nuclear weapons industries.

NUCLEAR WEAPONS: Weapons in which an explosion is caused by nuclear fission, nuclear fusion or a combination of both. Example of a fission weapon is the atomic bomb or A-bomb. Thermonuclear weapon and hydrogen bomb or H-bomb are examples of weapons produced by nuclear fusion. Neutron bomb is an example of a weapon produced by fission-fusion.

NUCLEAR-CYTOPLASMIC RATIO (NUCLEUS-CYTOPLASM RATIO; NUCLEUS CYTOPLASM RATIO; NC RATIO): A measure of the size of a cell nucleus in relation to the cytoplasm. The nuclear-cytoplasmic ratio is often used as an index in the comparison of cells from normal and abnormal tissue. Cultured cancer cells show an increase in the nuclear-cytoplasmic ratio.

NUCLEAR-ELECTRIC PROPULSION: A form of electric propulsion in which the electrical energy used to accelerate the propellant comes from a nuclear power source, such as a space-based nuclear reactor.

NUCLEARITY: The number of central atoms joined in a single coordination entity by bridging ligands or metal-metal bonds is indicated by dinuclear, trinuclear, tetranuclear, polynuclear, etc.

NUCLEASE: An enzyme that breaks down nucleic acids to nucleotides.

NUCLEATING AGENT: A material either added to or present in a system, which induces either homogeneous or heterogeneous nucleation.

NUCLEATION: 1. A process in a physical system or a mathematical model such as a cellular automaton or a statistical model, whereby a bubble or other structure appears spontaneously at a random or unpredictable spot. 2. The process of forming a nucleus. It is the initial process in crystallisation, in which ions, atoms or molecules arrange themselves in a pattern characteristic of a crystalline solid, forming a site in which additional particles deposit as the crystal grows.

NUCLEATION AND GROWTH: A process in a phase transition in which nuclei of a new phase are first formed, followed by the propagation of the new phase at a faster rate.

NUCLEIC ACID: A type of complex organic molecule found in the nucleus of a living cell. It is made up of a long chain of nucleotides. There are two types known as deoxyribonucleic acid (DNA), which forms the basis of heredity; the other, ribonucleic acid (RNA) plays several important roles in the process of transcribing genetic information from DNA into proteins.

NUCLEIC ACID HYBRIDISATION: The association in vitro of two complementary nucleic-acid strands to form a hybrid double strand.

NUCLEIC-ACID PROBE (NUCLEIC ACID PROBE): In DNA technology, it is a nucleic-acid molecule that is complementary to another nucleic-acid sequence. When labelled in some manner, as with a radioisotope, molecules of the probe hydrogen bond to the complementary sequence wherever it occurs and can be used to identify complementary segments present in the nucleic-acid sequences of various microorganisms.

NUCLEOCYTOPLASMIC LARGE DNA VIRUS (NCLDV, GIANT VIRUS): Any of the five families of large eukaryotic DNA viruses characterised by large capsids and containing large genomes.

NUCLEOFUGE: A leaving group that carries away the bonding electron pair. For example, in the hydrolysis of an alkyl chloride, Cl^- is the nucleofuge. The tendency of atoms or groups to depart with the bonding electron pair is called **nucleofugality**.

NUCLEOHISTONE: A complex formed between the polynucleotides of DNA and basic proteins called histones, which only occur in the nuclei of eukaryotic organisms. Nucleohistone complexes are visible as nucleosomes.

NUCLEOID (NUCLEAR REGION): The part of a cell of the bacterium or a blue-green alga (prokaryotic cell) that contains the genetic material DNA and therefore controls the activity of the cell. It corresponds to the nucleus of the more advanced eukaryotic cells but is not bound by a membrane.

NUCLEOLAR ORGANISER: A segment of a chromosome of an eukaryotic cell containing genes that encode ribosomal RNA. In the non-dividing cell it is associated with the nucleolus in the assembly of ribosomes. The nucleolar consists of numerous tandem repeats of a single gene. Each repeat is transcribed simultaneously to form multiple copies of a precursor RNA molecule containing all the larger ribosomal RNA subunits.

N

NUCLEOLUS (NUCLEOLE): A small round body inside the nucleus, composed of RNA and protein associated with the formation of ribosomal RNA and ribosomes. It forms around the nucleolar organiser, which encodes most of the segments of ribosomal RNA. Ribosomal proteins migrate to the nucleolus from their assembly sites in the cytoplasm and are packaged into ribonucleotide proteins, which then return to the cytoplasm where they become mature ribosomes.

NUCLEON: A particle found in the nucleus of an atom. Protons and neutrons are nucleons.

NUCLEON EMISSION: A decay mechanism in which a particularly unstable nuclide regains some stability by the emission of a nucleon or a proton or neutron.

NUCLEON NUMBER (MASS NUMBER; ATOMIC MASS NUMBER): The total number of nucleons or protons and neutrons in the nucleus of an atom. Its symbol is A.

NUCLEONICS: The technological aspects of nuclear physics, including the designs of nuclear reactors, devices to produce and detect radiation and nuclear transport systems.

NUCLEOPHILE: An electron-rich reactive species which donates an electron pair in organic reactions. Nucleophiles are often Lewis bases. They are either negative ions, for example, Cl^- or molecules that have electron pairs such as NH_3. Nucleophiles always seek a positive centre.

NUCLEOPHILIC SUBSTITUTION REACTION: A type of reaction in which the first step in the reaction mechanism involves the displacement of group by another nucleophile. For example, in $CR_3Cl + OH^- \rightarrow CR_3OH + Cl^-$, the nucleophile is the OH^- ion.

NUCLEOPLASM (KARYOPLASMS; NUCLEAR SAP): The material contained within the nucleus of a cell or that part of the nuclear contents other than the nucleolus. It is a highly viscous liquid that surrounds the chromosomes and nucleoli.

NUCLEOPROTEIN: Any compound present in cells of living organisms that consists of a nucleic acid (DNA or RNA) combined with a protein. Chromosomes consist of nucleoprotein (DNA and proteins, mostly histones), as do ribosomes (RNA and protein).

NUCLEOSIDE: Any of a class of organic compounds, including structural subunits of nucleic acids. Each consists of a molecule of a five-carbon sugar (ribose in RNA, deoxyribose in DNA) and a nitrogen-containing base, either a purine or a pyrimidine. The base uracil occurs in RNA, thymine in DNA and adenine, guanine and cytosine in both, as part of the nucleosides uridine, deoxythymidine, adenosine or deoxyadenosine, guanosine or deoxyguanosine and cytidine or deoxycytidine. Nucleosides usually have a phosphate group attached, forming nucleotides. Usually obtained by decomposition of nucleic acids, nucleosides are important in physiological and medical research. Those that are not part of nucleic acids include puromycin and certain other antibiotics produced by fungi.

NUCLEOSIDE DIPHOSPHATE SUGARS (NDP-SUGARS): Compounds formed by the reaction of a phosphorylated hexose sugar with a nucleoside triphosphate (NTP) in the general reaction: NTP + sugar 1 -phosphate ↔ NDP-sugar + pyrophosphate. NDP sugars are important as high-energy glycosyl donors in the synthesis of most polysaccharides, for example, the enzyme starch synthase utilises uridine diphosphate glucose (UDP-glucose) and adenosine diphosphate glucose (ADP-glucose) as substrates. UDP-glucose is the most important of the NDP-sugars. It can be synthesised both from glucose 1-phosphate using the reaction shown above or from sucrose using the enzyme sucrose synthase. Besides their importance as glycosyl donors, NDP-sugars are important intermediates in the interconversion of monosaccharides and their derivatives. For many enzymes catalysing sugar interconversions, notably the epimerases, NDP-sugars are substrates rather than the sugars themselves or their phosphate esters.

NUCLEOSOME (NU BODY): The fundamental unit of chromatin material of which eukaryotic chromosomes are made. It consists of a core histone protein around which are coiled about 160 base pairs of DNA. Consecutive nucleosomes are joined by uncoiled linker DNA, like beads on a string. This string is coiled into a 30 nm diameter solenoid, which undergoes further coiling in the fully condensed chromosome.

NUCLEOSYNTHESIS: The synthesis of chemical elements by nuclear processes.

NUCLEOTIDASE: A hydrolytic enzyme that catalyses the breakdown of nucleotide. It is present in the epithelial cells of small intestines and plays an important role in the digestion of protein.

NUCLEOTIDE: An organic compound consisting of nitrogen containing purine or pyrimidine base linked to a sugar and a phosphate group.It is the basic building block of nucleic acids, such as DNA and RNA and are made up of long chains of nucleotide. DNA molecule consists of nucleotides in which the sugar component is deoxyribose. RNA molecule has nucleotides in which the sugar is a ribose. The most common nucleotides are divided into purines and pyrimidines based on the structure of the nitrogenous base. In DNA, the purine bases include adenine and guanine while the pyrimidine bases are thymine and cytosine. RNA includes adenine, guanine, cytosine and uracil instead of thymine (thymine is produced by adding a methyl to uracil). Aside from serving as precursors of nucleic acids, nucleotides also serve as important cofactors in cellular signaling and metabolism.

NUCLEOTIDE BASES: The heterocyclic pyrimidine and purine compounds which are constituents of all nucleic acids. Adenine (A), guanine (G) and cytosine (C) are found in both DNA and RNA, thymine (T) is found (primarily) in DNA and uracil (U) only in RNA.

NUCLEUS: 1. The nucleus of a cell is a body embedded in the cytoplasm of all eukaryote cells that contains the genetic material DNA organised into chromosomes. The contents of the nucleus constitute the nucleoplasm. In certain protozoans there are two nuclei per cell, a macronucleus or meganucleus concerned with vegetative functions and a micronucleus involved in sexual reproduction. 2. **Atomic nucleus** is the central region of an atom that contains all the positive charge and essentially all the mass. It consists mainly of protons and neutrons. It was discovered in 1911 by Ernest Rutherford (1871 – 1937), a New Zealand-born British physicist. Nuclear physics is the branch of physics concerned with the study and understanding of the atomic nucleus. 3. **Cometary nucleus** is a central and relatively small part of a comet. It is made up of about 75 percent ices (including frozen water, ammonia and carbon dioxide) and about 25 percent dust and rock. 4. **Galactic nucleus** is the small, bright, central part of a galaxy. If the nucleus gives off considerably more energy than can be accounted for by its stellar contents, it is said to be an **active galactic nucleus** (**AGN**).

NUCLIDE: A type of atom as characterised by its atomic number and its neutron number or one or more atomic nuclei identifiable as being of the same element by having the same number of protons and neutrons and the same energy content. A nuclide refers only to a particular nuclear species, for example, the nuclides uranium-235 and plutonium-239 are fissile. The term is also used for a type of nucleus.

NUCLIDE SYMBOL: The symbol for an atom, in which E is the symbol of an element; Z is its atomic number and A is its mass number, expressed as ${}^{A}_{Z}E$.

NUCLIDIC MASS: The rest mass of a nuclide expressed in atomic mass units.

NUDICAUL: In botany, it means having a stem that is leafless.

NUGGET: A small lump of gold found ready-formed in the earth. Nuggets occur in alluvial deposits where river-borne particles of the metal have adhered to one another.

NUISANCE THRESHOLD: That concentration of an air pollutant that is considered objectionable. In the case of a substance with an objectionable odour, it is the smallest concentration of the substance which can be detected by a human being (nose).

NULL: In mathematics, empty or equal to zero or having measure zero. For example, in a normed vector space the null vector is the zero vector.

NULL CHARACTER: A character with all its bits set to zero. Therefore, it has a numeric value of zero and can be used to represent the end of a string of characters, such as a word or phrase. This helps programmers determine the length of strings. In practical applications, such as database and spreadsheet programs, null characters are used as fillers for spaces.

NULL GRAPH: A graph having no vertices, its order is zero. The term is also used for an empty graph.

NULL HYPOTHESIS: The hypothesis that is being tested in a hypothesis testing situation. Often the null hypothesis is of the form "There is no variation between two quantities".

NULL MEASURE (ZERO MEASURE): In mathematics, a measure of a subset of n-dimensional Euclidean space for which, for every $\varepsilon > 0$, there is a covering of the subset by rectangles whose volume is less than ε. Any set of 0 measure is called a null set.

NULL METHODS: A method of making a measurement in which the quantity to be measured is balanced by another similar reading by adjusting the instrument to read zero.

NULL SEQUENCE: In mathematics, a Cauchy sequence having a zero limit.

NULL SET: 1. In mathematics, a set having zero measure. Alternatively, the set with no elements or members, denoted by ø. 2. The set of arguments of a function for which the value of the function is zero. For example, the Cantor set and all countable sets on the line are null sets for Lebesgue measure.

NULL SPACE: In mathematics, the null set of a linear operator.

NULL STRING: In computing, it is a character string containing that no data. It is used in some programming languages to denote the last of a series of values.

NULL VECTOR: 1. A vector with zero components. 2. A matrix A is a nonzero vector x with the property that Ax = 0.

NULLITY: In mathematics, the dimension of the kernel or null set of a matrix or operator.

NUM LOCK (NUMBER LOCK): A key that is typically located near the keypad on the right side of a keyboard. It is a toggle key, which toggles the input of certain keys, depending on whether Num Lock is on or off.

NUMBER: Basic element of mathematics used for counting, measuring, solving equations and comparing quantities. They fall into several categories: **counting numbers** also known as **natural numbers** are the familiar 1, 2, 3, etc.; **whole numbers** are the counting numbers and zero; **integers** are the whole numbers and the negative counting numbers; and the **rational numbers** are all possible quotients formed by integers, including fractions. These numbers can be symbolically represented by terminating or repeating decimals. Irrational numbers cannot be represented by fractions of integers or repeating decimals and must be represented by special symbols such as $\sqrt{2}$, e and pi. Together, the rational and irrational numbers constitute the **real numbers**, which form an algebraic field, as the complex numbers. While the counting numbers and rational numbers come about as the means of counting, calculating and measuring, the others arose as means of solving equations.

N

NUMBER AND WORD NOTATION: In mathematics, writing the significant digits of a large number and using words for the place value. The number 45,000,000, in number and word notation is 45 million.

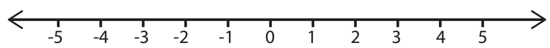
NUMBER BOND: A pair of numbers with a particular total, for example, 1 + 9, 2 + 8, 3 + 7, 4 + 6, 5 + 5.

NUMBER CONCENTRATION: Number of entities of a constituent in a mixture divided by the volume of the mixture. Its symbol is C or n.

NUMBER FLOW RATE: Number of defined particles or elementary entities of a defined component, crossing a cross-section divided by the time. Its symbol is q_N.

NUMBER GRID: A table in which consecutive numbers are arranged in rows, usually 10 columns per row. A move from one number to the next within a row is a change of 1; a move from one number to the next within a column is a change of 10. **Number grid puzzle** is a piece of a number grid in which some, but not all, of the numbers are missing, used to practice place-value concepts.

NUMBER LINE (REAL LINE): An infinite line on which each point represents a real number by its distance from a fixed origin, the axis of a one-dimensional coordinate system.



Number line

NUMBER MODEL: In mathematics, it refers to a format of an equation which includes addition, subtraction, multiplication and division. In other words, it is a number sentence, expression, or other representation that models a number story or situation. Every equation in mathematics is a notation of number model.

NUMBER PATTERN: A pattern that can be described in terms of numerical relationships.

NUMBER SEQUENCE PATTERN: The pattern in the number system that determines the number names for numbers greater than 20, for example, numbers from 21 to 29 follow the pattern of naming the decade number, twenty, and then the digit name, one, two, three, four, five, six, seven, eight, or nine.

NUMBER SENTENCE: A statement used in mathematics, which has numbers instead of words, for example, 2 + 3 = 5.

NUMBER SQUARE: A square grid with numbers stated in order.

NUMBER STORY: A short story that illustrates a mathematical equation, making it easier for young students to understand the equation involved. For example, the equation 5 + 2 = 7 can be told as a story about five birds sitting on a tree that were joined by two more birds.

NUMBER SYSTEM: A way of counting using a particular base. The familiar decimal number system is a base 10 system, because it uses ten different digits, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9 and combinations of these. Other number systems, used for special purposes or by certain cultures, are the binary (base 2), trinary (base 3), duodecimal (base 12), hexadecimal (base 16), vigesimal (base 20) and sexagesimal (base 60).

NUMBER THEORY (HIGHER ARITHMETIC): A branch of mathematics concerned with the abstract study of the structure of number systems and the properties of positive integers or whole numbers. It is more advanced than basic arithmetic, algebra, geometry and trigonometry. An important area in number theory is the analysis of prime numbers.

NUMBER-DISTRIBUTION FUNCTION: A distribution function in which the relative amount of a portion of a substance with a specific value or a range of values, of the random variable(s) is expressed in terms of mole fraction.

N

NUMERAL: A symbol or a combination of symbols, that describes a number.

NUMERATOR: The number above the line in a fraction, which indicates the number of parts of a whole. For example, if an orange is cut into eight equal pieces and three of them are taken off, then three of the eight parts that make up a whole pie is taken off. That is, three-eighths is taken off and this can be written as ⅜. The number 3 is the numerator.

NUMERICAL: 1. Containing or using constants, coefficients, terms or elements represented by numbers; for example, $4x^2 + 4y = 2$ is a numerical equation. 2. Also known as **absolute**, of a value, it is a term used in mathematics to refer to the distance of a value from zero or the magnitude or value of numbers, without regard to their sign. The absolute value of a number is denoted by ||, for example, For example, the numbers -3 and 3 have the same absolute value, written as |3| = |-3| = 3.

NUMERICAL ANALYSIS: A branch of applied mathematics concerned with methods for solving complicated equations using arithmetic operations, often so complex that they require a computer, to approximate the processes of analysis (calculus). The arithmetic model for such an approximation is called an **algorithm**, the set of procedures the computer executes is called a **programme** and the commands that carry out the procedures are called **code**. An example is an algorithm for deriving by calculating the perimeter of a regular polygon as its number of sides becomes very large.

NUMERICAL ECCENTRICITY (ECCENTRICITY): Denoted by ε, it is the ratio of the linear eccentricity to half the length of the major axis of a conic, that is constant for a family of similar curves. In other words, it is a measure of how closely a conic section resembles a circle. This ratio is constant for any particular conic section. For an ellipse the eccentricity is less than 1, for a hyperbola it is more than 1 and for a parabola it is unity (1).

NUMERICAL QUANTIFIER: 1. In logic, any of the sequence of quantifiers, $(\exists_n x)$, read as "there are at least n Fs". The first member, $(\exists_1 x)$, is defined by $(\exists_1 x)\ Fx \equiv (\exists x)\ (Fx)$ to be equivalent to the existential quantifier, $(\exists x)$ and the sequence is defined recursively by $(\exists_{n+1} x) Fx \equiv (\exists_n x)(Fx) \& (\exists y)(Fy \& x \neq y))$, that is, "there are at least $n + 1$ Fs" is equivalent to "there are at least n Fs different from some F". 2. **Exact numerical quantifier** is a restriction on the above, definable in terms of it as $(\mathbf{n}\ x)\ Fx \equiv (\exists_n x)\ Fx\ \&\ \text{-}(\exists_{n+1} x)\ Fx$. Alternatively, $(\mathbf{1}x)Fx$ can be defined as the unique quantifier, $(\exists! x)\ Fx \equiv (\exists x)\ (Fx\ \&\ (\forall y)\ (Fy \rightarrow x = y))$, whence by recursion $(\mathbf{n+1}x)\ Fx \equiv (\mathbf{n}x)\ (Fx\ \&\ (\mathbf{1}y)\ (Fy\ \&\ x \neq y))$.

NUMERICAL RANGE: For a matrix or for a continuous linear operator, T on a Hilbert space, it is the convex set of values in the complex plane traced out by $\langle Tx, x\rangle$ for $\| x \| \leq 1$.

NUMERICAL TAXONOMY (TAXOMETRICS): A classification method based on the numerical analysis of the variation of a large number of characters in a group of organisms. It is assumed that a classification will be more predictive the more characters on which it is based. It is also assumed that, to begin with, each character is of equal weight although some characters may later be weighted. Initially, a matrix of data is compiled of operational taxonomic units (OTUs) against characters so that for every OTU the state of each of perhaps 50 or more characters is recorded. This matrix can be subjected to a variety of mathematical analyses, which provide a measure of the similarity or dissimilarity between all the OTUs. The end product is usually one or more dendrograms.

NUMERICAL VALUE (ABSOLUTE VALUE): A term used in mathematics to refer to the distance of a value from zero or the magnitude or value of numbers, without regard to their sign. The absolute value of a number is denoted by ||, for example, For example, the numbers -3 and 3 have the same absolute value, written as |3| = |-3| = 3.

NURSE: 1. To manage, guide or supervise somebody or something with care and devotion. 2. A person educated and licensed in the practice of nursing or combining many different disciplines, including aspects of biology and psychology to promote the restoration and maintenance of health. Types of nurses include **hospital nurse**, who works under the direct supervision of a doctor on patients; the **public nurse**, who works outside the hospital giving expert nursing care in the community; the **occupational health** or **industrial nurse** works in business and industry to improve health and prevent accidents among employees; and the **nurse educator** combines nursing and teaching in hospital schools and colleges. 3. In zoology, it is an insect that looks after the young or the larvae in a colony of social insects such as ants or bees.

NURSE CELL: Any of the cells present in the ovary that supplies nourishment and support to the developing egg cell. An example is the Sertoli cell.

NURSE CROP: A crop used to protect another crop until it can get established.

NURSE TISSUE: Metabolically active tissue that is used in tissue culture work to stimulate the growth of single cells that cannot be grown using defined media. The nurse tissue, which is often callus tissue, may be separated by a piece of filter paper through which growth substances and nutrients can diffuse.

NURSERY BED: A specially prepared portion of land used for raising seedlings before transplanting. It acts as a temporary home for young plants until they are eventually planted in a permanent garden.

NUT: A dry single-seeded fruit that develops from more than one carpel and is indehiscent or does not split open when the fruit is matured. Most so-called nuts are not nuts but types of fruits, for example, coconut is a drupe; groundnuts are legumes, the shells are pods and the content is a seed. The true nuts are the acorn, chestnut and hazelnut.

NUTATION: 1. In botany, it is also known as **circumnutation**, which is a spiral movement exhibited by the tips of certain stems during growth. It enables a climbing plant to find a suitable support. It sometimes occurs in tendrils and flower stalks. 2. In astronomy, it is an irregular periodic oscillation of the Earth's poles caused by the varying gravitational pulls of the Sun and Moon.

NUTLET: 1. A small nut. 2. A unit of the lobes or fragment of the mature ovary of some members of the Boraginaceae, Verbenaceae and Lamiaceae.

NUTMEG TREE (*Myristica fragrans*): An aromatic, evergreen, dioecious tropical tree with a greyish-brown smooth bark, having a lot of yellow juice. It can grow 5 - 13 metres high. The seed of the tree is also known as nutmeg. Native to the Moluccas and Banda Islands in the South Pacific, it is now cultivated in tropical regions, especially Indonesia, Grenada in the West Indies and Sri Lanka. The branches spread in whorls and the dark green leaves are pointed and arranged alternately along the branches and are borne on leaf stems about 1 centimetres long. The pale yellow, waxy, fleshy and bell-shaped flowers are dioecious, small in axillary racemes. The male flowers are 5 to 7 mm long; and the female flowers are up to 1 centimetres long. The fruits are fleshy, drooping, yellow, smooth, 6 to 9 centimetres long with a longitudinal ridge. When ripe, the succulent yellow fruit coat splits into two valves revealing a purplish-brown, shiny seed (nutmeg) surrounded by a red aril (mace). The tree is propagated asexually by grafting or marcotting. It takes 7-9 years to flower and reaches full production after twenty years. The plant produces seeds which are egg-shaped with a reddish covering or aril. The seed is usually used in a powdered form to flavour many dishes. Essential oils are extracted by steam distillation from the powdered nutmeg. The essential oils obtained are used in the pharmaceutical and perfumery industries.

NUTRACEUTICAL (FUNCTIONAL FOOD; MEDICINAL FOOD;): A food designed to be medically beneficial or to have a health-promoting or disease-preventing property beyond the basic function of supplying nutrients. It helps to protect against serious conditions such as diabetes, cancer or heart disease.

NUTRICULTURE (HYDROPONICS; WATER CULTURE): Techniques of growing plants by supplying mineral nutrient solutions in water directly to the roots of the plants, without soil or other media.

NUTRIENT: Any substance that is required for the nourishment of an organism, providing a source of energy or structural components. In animals, nutrients form part of the diet and include the major nutrients including carbohydrates, proteins and lipids as well as vitamins and certain minerals. Plant nutrients are derived from carbon dioxide in the atmosphere and water absorbed from the soil by the roots.

NUTRIENT CYCLE: Transfer of nutrients from one part of an ecosystem to another through one or more organisms. Plants, for example, take up nutrients such as calcium and potassium from the soil through their root systems and store them in leaves. When the leaves fall, they are decomposed by bacteria and the nutrients are released back into the soil where they become available again for root uptake.

NUTRIENT MEDIUM (CULTURE MEDIUM): A mixture of nutrients used to support the growth of microorganisms or of the cells, tissues or organs of animals or plants. Such media may be solid, if mixed with a gelling agent such as agar or liquid.

NUTRIGENOMICS: The study of how genetic and environmental influences act together on an animal. It is concerned with how DNA is transcribed into mRNA and then to proteins and provides a basis for understanding the biological activity of food components.

NUTRITION: 1. The process of taking food components to obtain energy, which is necessary for the growth of body tissues and organs and to remain healthy. 2. Nourishment or foods which support growth of organisms. Substances essential for nutrition processes include carbohydrates, proteins and fats and also some vitamins and minerals. The two main types of nutrition are autotrophic nutrition for plants and heterotrophic nutrition for animals. 3. The branch of science concerned with food and nourishment. It includes the study of the nutrients that each organism must obtain from its environment for the growth of body tissues and organs, to remain healthy and to reproduce. An expert in nutrition is called a **nutritionist**.

NUTRITIVE VALUE: The degree to which food is valuable in promoting health. It is the ratio of digestible protein to other digestible components in the feed. **Nutritive** is containing nutrient(s) or concerned with or promoting nutrition.

NUX: A nut; or a fruit with a hard shell or rind.

NVRAM (NON-VOLATILE RAM; NON-VOLATILE RANDOM ACCESS MEMORY): A computer storage medium such as dynamic RAM (DRAM) and static RAM (SRAM) chips that are backed up by a battery or to non-volatile chips such as flash memory. Flash memory devices such as EEPROM stores data using electrical charges that maintain their state without electrical power and does not need a battery or other power source to retain data.

N

NYCTAGINACEAE: Also called the **four-o'clock family**, it is a family of flowering plants, in the Caryophyllales order (pink, or carnation order) containing about 30 genera with about 400 species of herbs, shrubs, and trees. They are widely distributed in tropical and sub-tropical regions with some in warm temperate areas. Examples are the genus *Bougainvillea*, *Boerhavia*, and *Mirabilis* (four o'clocks). One species, *Boerhavia diffusa* (common names, red spiderling, spreading hogweed, or tarvine), for example, has some medicinal properties used to treat health problems like anaemia, arthritis, indigestion, cough and asthma, abdominal pain, impotence, liver disorders, renal diseases, skin diseases, etc. It also rejuvenates the body, enhances the body's immunity and increases appetite.

NYCTALOPIA (NIGHT BLINDNESS): A disease caused mainly by deficiency of vitamin A (retinol), in which one is incapable of seeing properly in the dark or dim light. It can exist from birth or caused by injury.

NYCTANTHOUS: A plant that flowers only at night.

NYCTIGAMOUS: A term used for flowers that open at night and close during the day.

NYCTINASTY: Nastic movements of plant organs in response to the changes in light and temperature that occur between day and night and vice versa. Examples are the opening and closing of many flowers and the folding together of leaflets of plants at night. Nyctinastic movements are controlled by the circadian clock and the light receptor phytochrome.

NYCTITROPIC: Movement or positioning of plant organs at or just before nightfall, unlike those maintained during the day. Such a movement in plant often involves the folding or drooping of the leaves. Its advantage is that it reduces the radiation of heat.

NYLON: Any synthetic plastic material composed of polyamides of high molecular weight and usually, but not always, manufactured as a fibre. It is used in the manufacture of moulded articles, textiles, medical sutures, carpets and ropes.

NYMPH: The sexually immature or juvenile stage or form of insects such as grasshoppers, dragonflies and cockroaches that do not have a pupa stage in their life cycle. The nymph develops directly into an adult but the wings and reproductive organs are not developed.

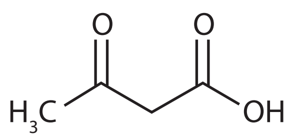
NYMPHAEA (WATER LILIES): A genus of aquatic flowering plants of 46 species in the family Nymphaeaceae that makes up the water lilies, many of which are cultivated as ornamental plants. The leaves grow from the rhizome on long petioles. Most of them float on the surface of the water. The blades have smooth or spine-toothed edges and can be rounded or pointed. The flowers rise out of the water or float on the surface, opening during the day or at night. *Nymphaea lotus* with the common names tiger lotus, green tiger lotus or red tiger lotus is an aquatic flowering plant with a white flower that opens at night. It is native to Central and West Africa, Madagascar as well as Egypt, thriving in a plain sand or fine gravel substrate. It has a pubescent peduncle and petiole. The plant grows from tubers that remain dormant during the dry season. The leaves are broad and wide, dentate with a notch at the petiole. The plant is viviparous, producing new plantlets from tubers that emerge from the flowers. *Nymphaea thermarum*, is the smallest water lily in the world and the only *Nymphaea* to grow in damp mud rather than water. Water lilies ecological importance is that they provide shades to reduce the growth of algae in ponds and lakes. They are also ornamental plants. The young leaves and unopened flower buds are eaten as vegetable. The seeds are also eaten for their high starch, protein and oil contents.

NYQUIST THEOREM (SAMPLING THEOREM): A theorem, which states that a sound wave should be sampled no less than twice per period in order to properly represent that sound wave. It is used in the conversion of analogue signals to digital ones. This is due to the fact that if a sound wave at a certain frequency is sampled less than twice in its period, it will most likely be interpreted as a much lower frequency, referred to as aliasing. It is an undesirable condition, which, for example, in images, produces a jagged edge or stair-step effect; and for sound, produces a buzz.

NYSTAGMUS: A condition that causes involuntary, rapid movement of one or both eyes. The eye(s) may move from side to side (horizontal nystagmus), up and down (vertical nystagmus) or in a circular motion or rotary (rotary or torsional nystagmus). It is often accompanied by reduced visual acuity. They may also have problems with depth perception that can affect their balance and coordination. Nystagmus can be aggravated by fatigue and stress. Also, looking out of a moving vehicle at passing objects induces nystagmus. It is caused by abnormal functioning of the part of the brain or inner ear that regulates eye movement and positioning. The condition can be either congenital or acquired. Congenital nystagmus is called infantile nystagmus syndrome (INS) and may be caused by an inherited genetic condition. Acquired (or acute) nystagmus can develop at any stage of life. It is often the result of injury or disease. Acquired nystagmus is typically caused by events that affect the labyrinth. Causes of acquired nystagmus include the following: stroke, certain medications or drugs such as sedatives and antiseizure medications like phenytoin (Dilantin), excessive alcohol consumption, head injury or trauma, diseases of the eye, diseases of the inner ear, vitamin deficiency (specifically B12 or thiamine), diseases of the brain including multiple sclerosis or brain tumours and diseases of the central nervous system.

O

3-OXOBUTANOIC ACID (ACETOACETIC ACID; DIACETIC ACID): A colourless organic compound having the chemical formula $C_4H_6O_3$. It is the simplest of the beta-keto acid group and an unstable compound, decomposing into propanone and carbon dioxide. The acid can be prepared from its ester, ethyl 3-oxobutanoate. In the body, it occurs as **acetoacetate** and is one of the ketone bodies formed in metabolism of certain substances, particularly in the liver in the oxidation of fats. It is excreted in the urine in uncontrolled diabetes mellitus and starvation and in bovine acetonaemia and pregnancy toxemia of cows and ewes.



Chemical Structure of 3-oxobutanoic Acid

O STAR (O-TYPE STAR): A spectral-type classification in the Draper catalogue of stars. It is a massive, luminous, blue star of spectral type O with prominent spectral lines of ionised helium and surface temperatures of between 30,000 °C and 50,000 °C (30,273.15K - 50,273.15K) and masses of 20-100 solar masses.

***O*-ETHYL S-[2-(DIISOPROPYLAMINO)ETHYL] METHYLPHOSPHONOTHIOATE (VX):** A highly toxic colourless oily liquid with the chemical formula $C_{11}H_{26}NO_2PS$, density 1.0083 g/cm^{-3}, melting point -50 °C (223 K) and boiling point 298 °C (571 K). It is a deadly nerve gas several times more toxic than sarin but less volatile. It is an organophosphorous compound and is probably one of the most toxic of all nerve agents. Its production and stockpiling was banned by the Chemical Weapons Convention of 1993.

O-NOTATION (BIG O-NOTATION): In computer science, the term is used to describe the performance or complexity of an algorithm.

OAK: A deciduous or evergreen hardwood temperate tree or shrub, comprising of about 450 species belonging to the genus *Quercus* in the beech family. Oak trees typically grow to a height of 30-40 metres. They are found in North America, Europe and in tropical regions of Asia. Each species have unique features. Some of the species include the *Quercus Robur* (common oak), *Quercus Rubra* (red oak) and *Quercus Alba* (White Oak). The leaves are broad, thin and flat and are spirally arranged, with lobed margins in many species, often leathery on top and pale underneath. Some have serrated or toothed leaves or entire leaves with smooth margins. Oaks produce both male and female flowers. The male flowers are generally tiny and yellow that grow along a slender stalk called a **catkin**. The fruit is a nut called an **acorn**, borne in a cup-like structure known as a **cupule**. Each acorn contains one seed (rarely two or three) The wood or timber of oak has great strength and hardness and is very resistant to insect and fungal attack because of its high tannin content. It is generally used for flooring, furniture making, construction, millwork (doors, banisters, etc), barrel making (mainly for aging or maturation of wine), boat construction, etc. The seed is used for animal feed and for making flour for baking. It is also roasted, ground and used as a coffee substitute. However, the leaves and acorns are poisonous to cattle, horses, sheep and goats if eaten in large amounts due to the toxin tannic acid, which causes kidney damage and gastroenteritis.

OASIS: 1. An isolated area of land with vegetation in an otherwise arid region made fertile by the presence of water. Oasis provides habitat for animals and even humans if the area is big enough. Oasis receives water from underground supplies or aquifers such as an artesian aquifer, where water can reach the surface naturally by pressure or by man-made wells. 2. Abbreviation for **Organisation for the Advancement of Structured Information Standards**. It is a non-profit, global consortium that supports the development and adoption of e-business standards. Standards regulated by the OASIS include protocols, file formats and markup languages. Hardware and software companies often work with OASIS to develop and institute standards that are efficient and effective.

OAT: A hardy cereal crop (*Avena sativa*), grown in the temperate regions in most types of soil. Oats have high carbohydrates. They also provide protein, fat, calcium, iron and B vitamins. They are used as food for humans and animals and the straw is used as animal bedding.

OB ASSOCIATION: An area in space in which a group of very young and hot massive stars of spectral types O and B, which have not yet dispersed predominate and emit strong ultraviolet radiation, which ionises the surrounding hydrogen.

OB-: A prefix meaning inverted, for example, **obcordate** describes a leaf shaped like an inverted heart.

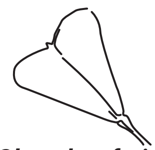
OBCLAVATE: 1. Inversely club shaped. 2. Elongated and narrowing toward the apex and gradually thickened toward base, with the attachment at the broad end.

OBCOMPRESSED: Compressed opposite the normal way, usually, flattened from back to front rather than from side to side. Examples are fruits of pennycress or peppergrass.

OBCONIC (OBCONICAL): Inverted cone shaped with the attachment at the narrow end. This describes some fruits, ovaries or other parts.

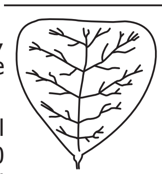
OBCORDATE (OBCORDIFORM): Heart shaped, with the attachment at the narrower end. This is usually used to describe leaves and fruits.

Obcordate leaf



Obcordate fruit

OBDELTOID: In botany, of leaves, triangular and attached to the stem via a tip.



Obdeltoid leaf

OBDIPLOSTEMONOUS: A term used to describe stamens that are inserted in two whorls with the outer opposite the petals and the inner opposite the sepals. Obdiplostemonous stamens are seen in some members of the Caryophyllaceae.

OBELIA: An ocean hydrozoan consisting of mainly marine and some freshwater animal species that have both the polyp and medusa stages in their life cycle. It forms colonies on ships' hulls, piles and rocks. It belongs to the phylum Coelenterata (Cnidaria). They are generally found up to 200 metres from the water's surface.

OBELLIPTIC (OBELLIPTICAL): Almost elliptic, but having the distal end somewhat larger than the proximal end, as occurs in some leaves.

OBERON: The second largest moon or natural satellite of Uranus with a diameter of about 1400 kilometres. It was discovered by William Herschel in 1787.

OBESITY: The condition of having excess body fat resulting in having a body mass, which is 20% greater than recommended for a relevant height. Generally, it is determined by body mass relative to height and expressed as mass (kg)/height (kg/m^2). It is considered to have serious consequences for physical and mental health and risk factor for health disorders such as diabetes mellitus, hypertension, coronary heart disease, stroke, arthritis and some forms of cancer. Obesity is difficult to treat and it is considered a major public health risk.

OBFUSCATION: The process of concealing software code or instructions of what the code performs. It may be done to help prevent the program from being modified or stolen by other users or to prevent another program from reading the code and understanding its true malicious intentions.

OBJECT: 1. The source or radiation affected by a system to form an image, the rays come from the object but are seen to diverge from the image. 2. In computer science, it is a block of information such as an onscreen graphic that can be selected and manipulated. In object-oriented programming, objects include data and the procedures necessary to operate on that data.

OBJECT CODE (MACHINE CODE; MACHINE LANGUAGE): A collection of binary digits or bits that the computer reads and interprets. Object code is the only language a computer is capable of understanding.

OBJECT COMPUTER: In data communications, it refers to a computer capable of receiving a fax, modem or other data transmission.

OBJECT EMBEDDING: Placing an object such as graphic, video clip and spreadsheet from a source file into a destination file. When an object from a source file is embedded into a document, it becomes part of the destination file.

OBJECT LANGUAGE: The language described by or being investigated by another language, called the metalanguage.

OBJECT LINKING AND EMBEDDING (OLE): A compound document standard developed by Microsoft Corporation that allows an object to be embedded into a document. It is a feature of application programs that create objects such as graphic, video clip and spreadsheet with one application and then link or embed them in a second application.

OBJECT PROGRAMME (OBJECT ROUTINE; TARGET PROGRAM): The machine code translation of a program prepared by an assembler or a compiler after acting on a programmer-written source programme.

OBJECT-ORIENTED PROGRAMMING (OOP): A computer programme system based on objects in which data are closely linked to the procedures that operate on them. The programs are composed of self-sufficient modules (objects) containing all the information needed to manipulate a data structure.

OBJECTIVE: The aim or purpose for undertaking a programme, experiment or scientific research.

OBJECTIVE FUNCTION: In mathematics, it is an equation that is to be optimised in an optimisation problem, especially one with certain constraints and with variables that need to be minimised or maximised.

OBJECTIVE LENS (OBJECT GLASS): The first lens or mirror in an optical system such as a microscope, which is nearest to the object being examined and through which light passes or is reflected.

OBLANCELATE: In botany, it is a term used especially of leaves that are lanceolate with the more pointed end at the base.

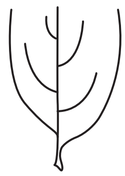
Oblancelate leaf

OBLATE SPHEROID (OBLATE ELLIPSOID): An ellipsoid produced by rotating an ellipse through 360° about its minor axes so that the diameter of its equatorial circle exceeds the length of the axis of revolution.

OBLATENESS: Flattened at the poles, as a spheroid generated by the revolution of an ellipse about its shorter axis with two of its axes of symmetry having an equal length greater than that of the third. For example, the Earth and the other planets of the Solar System are oblate spheres, with the polar diameter of the Earth being about 45 km longer than its equatorial diameter.

OBLIGATE: A term used to describe a specific requirement for a particular environmental factor and being unable to survive if it is not available. For example, an **obligate parasite** cannot grow in the absence of a host; it can grow or reproduce only when in a living host, for example, rickets and chlamydia grow in eukaryotic cells; *Bdellovibro* grows in bacterial cells. All viruses are obligate parasites. **Obligate anaerobes** are organisms that do not utilise oxygen for respiration and therefore do not grow or even die in the presence of oxygen.

OBLIQUE: 1. Not horizontal or vertical, but sloping, slanting or not going directly to the point. 2. Of, or relating to geometric lines or planes that is neither parallel nor perpendicular. 3. Of, or relating to geometric figures, especially a triangle, not having a right angle. 4. Having sides of unequal length or form, for example, an oblique leaf; or having unequal leaf bases, slanting, one side larger, wider or rounder than the other. Example is found in American Elm and Common Hackberry which have an oblique leaf base. 4. Situated in a slanting position, not transverse or longitudinal for example, oblique muscles or ligaments.



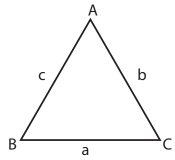
Oblique leaf base

OBLIQUE ANGLE: An angle that is not a right angle.

OBLIQUE COORDINATE: A coordinate system in which the axes are not perpendicular.

OBLIQUE FACTORS: A term used in factor analysis for common factors that are allowed to be correlated.

OBLIQUE TRIANGLE: A triangle that is not a right-angled triangle. Examples are an acute triangle and obtuse triangle. Problems related to oblique triangles can be solved using two theorems of geometry that give useful laws of trigonometry. These are called the "law of sines" and the "law of cosines." The law of sines is stated as $\frac{\sin\hat{A}}{a}=\frac{\sin\hat{B}}{b}=\frac{\sin\hat{C}}{c}$ or $\frac{a}{\sin\hat{A}}=\frac{b}{\sin\hat{B}}=\frac{c}{\sin\hat{C}}$.



Oblique Triangle

The law of cosines is stated as:

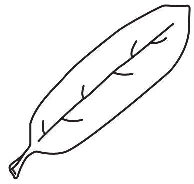
$c^2=a^2+b^2-2ab\cos\hat{C}$,

$a^2=b^2+c^2-2bc\cos\hat{A}$ and

$b^2=a^2+c^2-2ac\cos\hat{B}$.

OBLIQUITY: 1. A deviation from the perpendicular or horizontal. 2. Also known as **obliquity of the ecliptic**, it is a term used in astronomy to describe the angle between the plane of the Earth's orbit and that of the celestial equator, equal to approximately 23.5°. Obliquity also refers to the angle that a planet's rotational axis makes with its orbital plane. It gives a good indication of how extreme the seasons would be on a given world. Earth's obliquity of 23.5° means that at the summer solstice, the North Pole tilts toward the Sun by 23.5° and at the winter solstice, it tilts away by the same amount. This leads to fairly extreme temperature changes.

OBLONG: 1. Also known as **rectangle**, it is a shape having two pairs of straight, unequal sides and four right angles. 2. In botany, it refers to a leaf that is rounded at each end with parallel sides.



Oblong Leaf

OBLONG NUMBER (PRONIC NUMBERS): Any number that is the product of two consecutive integers. For example, 0, 1, 2, 3, 4, 5, 6, 7, ... give the products 0, 2, 6, 12, 20, 30, 42, ...

OBOVATE: In botany, it refers to a leaf that is egg-shaped and flat, with the narrow end attached to the stalk.

Obovate leaf

OBOVOID: Inversely ovoid and attached to the stem by the narrower end.

OBPYRAMIDAL: Inverted pyramid with the attachment at the tapering end.

OBSERVATION: An act or instance of gathering information actively by viewing or noting a fact or occurrence for some scientific or other special purpose. Observation is made using the senses. Observation can also involve the recording of data via the use of instruments such as microscopes, scanners or transmitters to extend the vision or hearing. It is important in science as scientists use it to collect and record data, in order to formulate and test hypotheses and theories. It includes the following steps: (i) Identify and observe a natural phenomenon and ask questions about it; (ii) make a hypothesis; (iii) make predictions about logical consequences of the hypothesis; (iv) test the predictions by controlled experiment, natural experiment, an observational study or a field experiment; (v) create conclusions on the basis of data or information gathered in the experiment; (vi) establish a theory based on repeated verification of the results.

OBSERVATIONAL STUDY: A type of study in which individuals are observed or certain outcomes are measured without influencing or interfering with the outcome. Since the investigator does not control the assignment of treatments there is no way to ensure that similar subjects receive different treatments. An example is the study of the relationship between smoking and lung cancer.

OBSERVATORY: A location used for observing and studying terrestrial and celestial events such as the stars and the weather by scientists.

OBSIDIAN: A jet black or dark coloured natural glass of volcanic glass, with high silica composition, which induces a high viscosity and polymerisation degree of the lava content formed by the rapid cooling of viscous lava. It is commonly found within the margins of rhyolitic lava flows known as **obsidian flows**. Obsidian was formerly used for weapons, implements, tools ornaments and mirrors.

OBSOLESCENT (OBSOLETE): In biology, it is an organ or structure, which is imperfectly developed and likely non-functional, especially in comparison with other individuals or related species.

OBSTETRICS: The branch of medicine and surgery concerned with the care of women during pregnancy and childbirth. It is usually linked with gynaecology.

OBTURATOR: 1. A disk or plate that closes an opening. 2. A small glandular structure which is attached to the pollinia of members of the Asclepiadaceae and Orchidaceae and closes the opening to the anther. 3. Any of the two muscles that covers the outer front part of the pelvis on each side and involved in movements of the thigh and hip.

OBTUSE: 1. In botany, the term is used for a blunt or rounded leaf apex. 2. Of or relating to an angle that is greater than a right angle (90°) but less than 180°; an obtuse angle. An **obtuse triangle** is a triangle that contains one interior angle that is greater than a right angle (90°).

Obtuse leaf

OBVERSE: In logic, a categorial statement obtained from another by changing its predicate from positive to negative or vice versa and negating the whole statement. For example, the obverse of "all cattle are herbivorous" is "no cattle is non-herbivorous".

OBVOLUTE: Relating to a type of venation in which two leaves are folded together in the bud so that the margins overlap, with one half of each being external and the other half internal.

OCCAM'S RAZOR: An axiom proposed by William of Occam (William Ockham - 1280-1349), a British philosopher that, when alternative hypotheses exist, the one requiring the fewest assumptions should be preferred. In other words, it is a scientific principle which states that the simplest explanation for a phenomenon is usually the most desirable explanation.

OCCIPITAL BONE: One of the four main bones that form the cranium, the others being the frontal bone and the two parietal bones. The occiptal bone lies at the back of the skull and meets the parietals at the lamboid suture.

OCCIPITAL CONDYLE: A bony knob that protrudes from the occipital bone of the skull and articulates with the atlas (first cervical vertebra). Humans have a pair of occipital condyles, one on each side of the foramen magnum opening at the base of the skull. Most fishes that cannot move their heads do not have occipital condyles. In insects, it is a projection on the posterior border of their heads that articulates with the lateral neck plates.

OCCIPITAL LOBE: The posterior lobe of each cerebral hemisphere of the mammalian brain responsible for receiving and interpreting visual information.

OCCLUSION: 1. The trapping of some pockets of liquids in a crystal during crystallisation. 2. The absorption or adsorption of a liquid or gas by a solid such that atoms or molecules of the gas occupy interstitial positions in the solid lattice. 3. In dentistry, it is the alignment of the teeth of the upper and lower jaws when the jaw is closed. 4. In medicine, occlusion is an obstruction or a closure of a passageway or a vessel. 5. Also known as occluded front, it is weather formed when a cold air mass (cold fronts) overtakes a warm air mass (warm front). It may occur in a mid-latitude depression, lifting the wedge of warm air off the ground and meeting the cold air ahead of the warm front. If this overtaking air is colder than the cold air which is ahead of it, it will undercut it, forming a cold occlusion and if it is warmer than the cold air which is ahead of it, rides over it forming a warm occlusion. It brings cloud and rain as the warm air rises along the front, cooling and condensing as it does so. The point where the front and the occluded front meet is called the triple point. On a weather map occluded fronts are indicated by a purple line with alternating semicircles and triangles pointing in direction of travel.

OCCULTATION: The temporary apparent disappearance from view of a celestial body such as a star or planet by a body passing across the line of sight in the solar system. It usually refers to stars occulted by the Moon, but it also applies to stars occulted by major or minor planets occulting each other. An example is a solar eclipse.

OCCULTING DISK: A small metal disk used in a telescope to block the view of a bright object in order to allow observation of a fainter one.

OCCUPATIONAL PSYCHOLOGY: The study of human behaviour at work. Occupational psychologists apply psychological knowledge, theory and practice to deal with problems in organisations, advising on management difficulties, and investigating the relationship between humans and machines. They aim to help an organisation get the best performance from their employees and also to improve employees' job satisfaction.

OCEAN: The continuous body mass of salt water that covers 70% of the Earth's surface. The five main oceans are Atlantic, Pacific, Indian, Arctic and Antarctic oceans.

OCEAN BASIN: One of two major provinces of the deep ocean floor, lying over 2 km in depth. The mid-ocean ridges form the other province. Together they constitute 50% of the Earth's surface. The deep ocean basin is underlain by a basaltic crust, about 7 km thick and is covered in sediment and dotted by low abyssal hills.

OCEAN CONVEYOR BELT: A constantly moving system of deep-ocean circulation by which water circulates around the entire global ocean. It is driven by wind, temperature and salinity.

OCEAN CURRENT: The horizontal and vertical movements of ocean waters, produced by gravity, wind force and water density variation. The direction of ocean current is determined by Coriolis forces. Examples of major ocean currents are the Kurushio current in the North Pacific and the Gulf Stream in the North Atlantic.

OCEAN PLANET: A hypothetical planet that is completely covered by ocean.

OCEAN RIDGE: A series of mountain ranges on the seabed with a length of more than 80,000 kilometres indicating the pressure of a constructive plate margin (where tectonic plates are moving apart and magma rises to the surface).

OCEANIC TRENCH (DEEP OCEAN TRENCH; DEEP-SEA TRENCH): A long narrow steep-sided depression in the ocean floor with depths of about 7,300 to over 11,000 metres, that typically forms in locations where one tectonic plate subducts another. These trenches are the deepest parts of the ocean floor. Examples are the Cayman Trench in the Caribbean Sea with a depth of 7,686 metres and the Puerto Rico Trench in the Atlantic Ocean with a depth of 8,605 metres. The deepest trench on Earth is the Challenger Deep of the Mariana Trench in the Pacific near Guam, with a depth of 11,034 metres below sea level.

OCEANARIUM: A large exhibition tank filled with saltwater in which aquatic animals and plants are kept as if they are in their natural environments.

OCEANIC CRUST: A layer, mostly composed of different types of basalts (type of volcanic rock) extending 5 - 10 kilometres beneath the ocean floor. Geologists' term for basalts, the rocks of the oceanic crust is **sima**, which stands for silicate and magnesium, the most abundant minerals in the oceanic crust. Oceanic crust is dense, almost 3 g/cm^3. It is denser than the continental crust which is 2.7 g/cm^3.

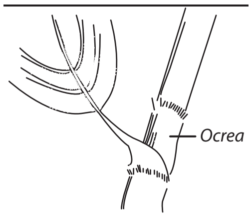
OCEANIC ZONE: The region of the open sea beyond the edge of the continental shelf, where the depth is greater than 200 metres. It includes 65% of the ocean's open water. It has a wide range of terrain such as very deep crevices, ocean basins and volcanoes.

OCEANOGRAPHY: The study of all aspects of the world's oceans and seas including their origin, composition, structure, history and wildlife and other organisms. Oceanography is useful in forecasting weather and climate, in exploitation of the Earth's resources and in the studies of the effects of pollutants.

OCELLUS: 1. A simple eye that occurs in insects and other invertebrates. It consists of light-sensitive cells and a single cuticular lens but unable to focus clearly. 2. An eyespot or a marking that resembles an eye, for example, the markings on the tail feathers of a male peacock.

OCHRE (OCHER): 1. A mineral of clay and iron(III) oxide (Fe_2O_3) used as a light yellow to brown pigment. They were one of the earliest pigments used by mankind. 2. A yellowish-orange colour.

OCHREA (OCREA): A cup-shaped structure formed by sheathing stipules at the leaf base. Ochreas are characteristic of the Polygonaceae giving the familiar swollen joints on the stems of such plants. **Ocreate** means possessing an ocrea; sheathed.



Ocreate structure

OCHROLEUCOUS: Yellowish white; cream-coloured.

OCR (OPTICAL CHARACTER RECOGNITION; OPTICAL CHARACTER READER): A process of putting a text into a computer by means of a dominant reader in which a software attempts to identify characters by comparing shapes to those stored in the software library. It avoids the need to retype already printed material for data entry.

OCREOLA: A tiny stipular sheath around the secondary divisions of the inflorescence in some members of the buckweed family (Polygonaceae). **Ocreolate** means having minute sheathing stipules.

OCTA-: A prefix denoting the number eight. For example, an octagon is a two-dimensional eight-sided, eight angled geometric figure.

OCTAD: 1. Also known as **ogdoad**, it is a set or group of eight elements. 2. In chemistry, it is an element, atom or group having a valence of eight.

OCTADECENOIC ACID (CIS-OCTADEC-9-ENOIC ACID; (9*Z*)-OCTADEC-9-ENOIC ACID; (9*Z*)-OCTADECENOIC ACID; (*Z*)-OCTADEC-9-ENOIC ACID; OLEIC ACID): A straight chain unsaturated fatty acid with the chemical formula $C_{17}H_{33}COOH$, density 0.895 g/ml, melting point 13-14 °C (286 K) and boiling point 360 °C (633 K). The glycerides of these acids are found in many natural fats and oils such as butter fat, groundnut oil and soya bean oil. It is used in making soap.

OCTAGON: A two-dimensional eight-sided, eight angled geometric figure with sides of the same length and internal angles of the same size. The internal angle at each vertex of a regular octagon is 135° and the sum of all the internal angles is 1080°.

OCTAHEDRITE (ANATASE): A rare blue or black mineral, with the chemical formula TiO_2 that consists of titanium oxide in tetragonal crystalline form. It is the commonest type of iron meteorite, which occurs in veins in igneous rocks.

OCTAHEDRO-: An affix used in names to denote six atoms bound into an octahedron.

OCTAHEDRON: A regular solid geometrical figure comprising eight plane triangular faces, 12 edges and six tetrameric vertices. In a regular octahedron the faces are congruent equilateral triangles just like two equal pyramids opposing each other on the same square base.

OCTAHYDRATE: A crystalline hydrate containing eight molecules of water per formula unit of compound.

OCTAL NUMBER SYSTEM (OCTONARY NUMBER SYSTEM): A number system to the base eight used in computing. For example, 27_8 is $(2 \times 8^1) + (7 \times 8^0) = 16 + 7 = 23$. The highest digit that can appear in the octal system is seven; it uses the digits 0 to 7.

OCTANDROUS (OCTOSTEMONOUS): Having eight stamens.

OCTANE: A highly flammable liquid hydrocarbon, which is a component of petrol with the chemical formula C_8H_{18}, density 703 mg mL^{-1}, melting point -56.79 °C (216.36 K) and boils at 260 °C (533 K). It exists in 18 structurally different forms. It is used as a fuel and solvent.

OCTANE NUMBER: A rating that indicates the tendency to knock when a fuel is used in a standard internal combustion engine under standard conditions. It is defined as the percentage of isooctane required in a blend with normal heptane to match the knocking behaviour of the gasoline being tested. Fuels with high numbers are less likely to cause knocking. For example, a highly knock-resistant is octane or 2,2,4-trimethylpentane (C_8H_{18}) is assigned a rating of 100 on the octane scale and normal heptane (C_7H_{16}), with very poor knock resistance, represents zero on the scale.

OCTANOIC ACID (CAPRYLIC ACID): A colourless oily carboxylic acid, used in the manufacture of dyes and perfumes with the chemical formula $C_7H_{15}COOH$, density 0.910 g/cm^3, melting point 16.7°C (290 K) and boiling point 239.7 °C (513 K). It occurs naturally in the milk of some mammals and also as a minor constituent of coconut oil and palm kernel oil. Its uses include in the production of esters used in perfumery, in the manufacture of dyes, in the treatment of some bacterial infections and as a sanitiser in commercial food processing and also as algaecide, bactericide and fungicide.

OCTANT: Any of the eight divisions of space determined by the three coordinate planes defined by the signs of the coordinates. For example, to determine a wind direction accurately, it is given as one of the eight octants (N, NE, E, SE, S, SW, W, NW).

OCTAVE: The perfect interval between two musical notes that have fundamental frequencies in the ratio 2:1 in terms of the eight notes of the diatomic scale**.** For example, the range of the male voice is roughly an octave below that of the female voice.

OCTAVES (NEWLANDS LAW): A law which classifies elements by grouping them in increasing atomic weights (starting with lithium) in horizontal rows of eight elements with each new row directly beneath the previous one. Any one element has properties similar to those of the elements eight places in front of it and eight places behind it in the list. This law predated Mendeleev's periodic table.

OCTET: 1. A stable group of eight electrons in the outer shell of an atom, for example, as in an atom of an inert gas. 2. In computing, it is a series of exactly eight bits often used in protocol and other technical specifications for computing machinery or networks. 3. The group of eight spin-half baryons.

OCTET RULE: A chemical rule stating that in the formation of compounds, atoms of low atomic numbers (less than 20) tend to combine such that they have 8 valence electrons, giving them the same electronic configuration as noble gases.

OCTOGYNOUS: Having eight pistils or styles.

OCTOLOCULAR: Having eight locules.

OCTOPETALOUS: Having eight petals.

OCTOPUS: Any of numerous eight-limbed carnivorous cephalopod molluscs of the genus *Octopus* or related genera, found in seas and oceans worldwide. It has a rounded soft body, eight arms with each bearing two rows of suckers, a large distinct head and a strong beaklike mouth. The largest species is the Giant North Pacific octopus, with adults having a 4.3 m arm span and a weight of about 15 kg. Octopuses are highly intelligent and are considered the most intelligent of the invertebrate.

OCTORADIATE: Having eight ray flowers.

OCTOSEPALOUS: Having eight sepals.

OCTOSTICHOUS: In eight vertical ranks or rows, as seen in some leaves.

OCTUPOLE: 1. A set of eight point charges that has zero net charge and which does not have neither a dipole moment nor a quadrupole moment. An example of an octupole is methane molecule (CH_4). Octupole interactions are much smaller than quadrupole interactions and very much smaller than dipole interactions. 2. Two electric or magnetic quadrupoles having charge distributions of opposite signs and separated from each other by a small distance. 3. Any device for controlling beams of electrons or other charged particles, consisting of eight electrodes or magnetic poles arranged in a circular pattern, with alternating polarities. It is commonly used to correct aberrations of quadrupole systems.

OCULAR: 1. Of or concerned with the eye or vision. 2. In optics, it is the lens or system of lenses in an optical instrument that is positioned nearest to the eye, especially a compound microscope.

OCULOMOTOR NERVE: Either of the third pair of cranial nerves, which originate in the midbrain and controls all the muscles that move the eye, except for two: the **superior oblique muscle**, which rotates the eyeball downward and outward and is controlled by the trochlear nerve; and the **lateral rectus muscle**, which moves the eye outward and is controlled by the abducens nerve. The oculomotor nerve also carries parasympathetic fibres to the muscles that constrict the pupil (sphincter pupilae), the ciliary muscle (which focuses the eye) and the muscle that raises the upper eyelid.

ODD FUNCTION: A function $f(x)$ is odd if $f(-x) = -f(x)$ for all x. For example, if $f(x) = x^3$, $f(-x) = -x^3$, then $f(x) = x^3$ is an odd function. Some other examples of an odd function are
$f(x) = \sin x$, $f(x) = \sin(-x) = -\sin x$,
$f(x) = \tan x$, $f(-x) = \tan(-x) = -\tan x$.

ODD NUMBER: An integer which, when divided by two, gives a remainder of one. In other words, it is a number that is not exactly divisible by two and therefore, leaves a remainder of one.

ODD PART: The largest odd factor of a given positive integer, n, often written odd(n) or Od(n) and defined as $\mathrm{Odd}(n)=\frac{n}{2^{b(n)}}$, where $b(n)$ is the exponent of the exact power of 2 dividing n. For example, the odd part of 52 is 13 and odd(16) = 1.

ODD PERMUTATION: In mathematics, a permutation derived from the natural order by an odd number of transpositions. For example, the permutation (1, 3, 2) is odd but (3, 1, 2) is even, since it can be achieved by exchanging 1 and 3 and then exchanging 1 and 2.

ODD-EVEN NUCLEUS: An atomic nucleus with an odd number of protons and an even number of neutrons.

ODD-ODD NUCLEUS: An atomic nucleus containing an odd number of protons and an odd number of neutrons.

ODD-PINNATE: A compound leaf with leaflets arranged in opposite sides of an elongated axis called the rachis and a single terminal leaflet at the tip and therefore have an odd number of leaflets. Examples are the leaves of the rose plant, some acacia, mockernut hickory, pecans, etc.

ODDS: The ratio of the probabilities of two possible states of binary variable.

ODDS RATIO: The ratio of the odds for a binary variable in two groups of subjects, for example, boys and girls. If the two possible states of the variable are labelled "success" and "failure" then the odds ratio is a measure of the odds of a success in one group relative to that in the other. When the odds of a success in each group are identical then the odds ratio is equal to one. It is expressed as: Odds Ratio = $\frac{DC}{BE}$, where D, C, B and E are appropriate frequencies in the two-by-two contingency table formed from the data.

ODONATA: An order of the class *Insecta* containing the dragonflies and damselflies, most of which occur in tropical regions. They are characterised by a head with large compound eyes and wings with clear or transparent membranes traversed by networks of veins.

ODONTOBLAST: A biological cell of neural crest origin and part of the outer surface of the dental pulp, responsible for producing the dentine (the substance under the tooth enamel) of a vertebrate's teeth.

ODONTOCETES: Any whale of the suborder Odontoceti, characterised by an asymmetrical skull, a single blowhole, and rows of teeth. Odontocetes include dolphins, killer whales, and sperm whales, which feed mainly on fish, squid, and crustaceans.

ODONTOID PROCESS: A tooth-like projection that rises perpendicularly from the second cervical vertebra (axis) to engage with the first (atlas), acting as a pivot for the head to turn in a horizontal plane.

ODORANT: A substance responsible for the perception of a smell and detected by olfactory organs, such as the nose.

ODORIFEROUS: Giving out a distinctive smell or odour.

ODOUR: The characteristic smell of something. It is activated when airborne molecules stimulate receptors of the first cranial nerve that mammals detect by the sense of olfaction.

ODOUR THRESHOLD: The concentration of a compound which produces an odour which is detectable by the nose of a human being. For certain compounds, this threshold is very low, for example, 1 part in 10^9 for certain sulfides.

OEDEMA (DROPSY): An abnormal-build up of excess serous fluid between cells, tissue spaces or cavities of the body causing swelling of the affected part. **Localised oedema** occurs in conditions such as inflammation or from skin rashes caused by allergies. **Generalised oedema** can result from a variety of pathological conditions, including kwashiorkor and diseases of the kidneys, heart, veins or lymphatic system, which affect water balance in the cells, tissues and blood. It can also occur as side effect of certain drugs and as reaction to toxic chemicals. A generalised subcutaneous oedema is called **anasarca**. An excess of fluid in the pleural spaces is referred to as **hydrothorax**; that in the pericardial sac is **hydropericardium**; if it occurs in the peritoneal cavity it is called **ascites**.

OEDIPUS COMPLEX: A term coined by Austrian physician and psychologist, Sigmund Freud (1856-1939) that defines the unconscious antagonism of a son to his father, whom he sees as a rival for his mother's affection.

OERSTED: The centimetre–gram–second (cgs) unit of magnetic field strength, which is equivalent to 103/4 A m^{-1}, named after Hans Christian Oersted (1777-1851), Danish physicist and chemist. Its symbol is Oe. A field has a strength of one oersted if it exerts a force of one dyne on a unit magnetic pole placed in it. The International System of Units (SI) is the currently used system of units, and the ampere/metre (A m^{-1}) has replaced the oersted (Oe).

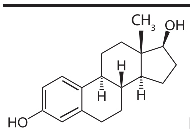
O

OERSTED, HANS CHRISTIAN (1777-1851): Danish physicist and chemist, who in 1820 observed that electric current in a wire, deflected a magnetised compass needle, a phenomenon that inspired the development of electromagnetic theory. In the same year, he discovered piperine, a pungent component of pepper, an important contribution to chemistry. He also discovered the preparation of metallic aluminium in 1825. In 1824, he founded a society devoted to the spread of scientific knowledge among the general public. In 1932, to honour him, the **oersted** was adopted as the physical unit of magnetic field strength.

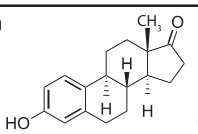
OESOPHAGUS (GULLET): The muscular canal that extends from the pharynx to the stomach through which food passes from the pharynx to the stomach. It is about 23 cm long. Passage of food moved by means of wavelike contractions called **peristalsis** along its length to the stomach.

OESTRADIOL: The hormone that mainly controls the follicular phase (proliferative phase), the phase of the oestrous cycle during which the Graafian follicles mature and the lining of the uterus thickens under the influence of oestrogen. In humans and great apes it is the menstrual cycle.

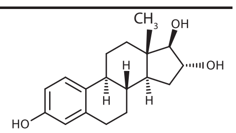
OESTROGEN: Any of a class of hormones that control the female reproductive cycle and the development of secondary sexual characteristics in female. Oestrogen also refers to various synthetic hormones that mimic their effects. There are three major types: oestradiol, oestrone and oestriol. They are produced mainly by the ovaries and the placenta. The adrenal glands and the testes also secrete smaller amounts. Oestrogens affect the ovaries, vagina, fallopian tubes, uterus and mammary glands and play crucial roles in puberty, menstruation, pregnancy and parturition (labour). They also influence the structural differences between female and male bodies. Loss of oestrogens diminishes mating desires and other behavioural patterns.



Oestradiol



Oestrone



Oestriol

OESTROUS CYCLE: The cycle of recurring physiologic reproductive activities shown by most sexually mature female mammals (except most primates), which are not pregnant. Some animals such as dogs have only one ovulation during a breeding season while others such as the ground squirrels experience ovulation repeatedly during the breeding season until impregnated. During this period the female secretes pheromones that signal her receptivity to the males. Her genital area may become swollen and she may also give behavioural signals. There are four phases: (i) **Pro-oestrus** when follicles of the ovary start to develop, lasting from one day to 3 weeks, depending on the species. Some animals may experience vaginal secretions that could be bloody. The female is not yet sexually receptive. (ii) **Oestrus** is the phase when the female is sexually receptive (in heat) or in ovulation and becomes sexually attractive to the male and is ready to mate. (iii) **Metoestrus** (luteal phase) in which the corpus luteum develops from ruptured follicle. The uterine lining begins to secrete small amounts of progesterone. This phase may last from 1 to 5 days. (iv) **Dioestrus** in which progesterone secreted by corpus luteum prepares the uterus for implantation. The length of the cycle depends on the species: larger mammals have single annual cycle accompanied by well-defined breeding season known as **monoestrous**. The males have a corresponding cycle of sexual activity. Other species may have many cycles per year known as **polyoetrous**. Males belonging to the polyoestrus species may be sexually active all the time.

OFF DIAGONAL (SECONDARY DIAGONAL): A set of entries in a square matrix that runs from the upper right to lower left elements. For an entry in a square matrix, A = (a_{ij}), an element a_{ij} is an off-diagonal entry if a_{ij} is not on the diagonal, i.e., if $i \neq j$. Alternatively, any other related sequence of elements of a matrix, such as the elements above the main diagonal.

OFFAL: Internal organs of a butchered animal including the brain, tongue, blood, viscera, loose fat, etc. Offal may be used as human or animal food and can also be processed for use as fertiliser or fuel.

OFFICE (MICROSOFT OFFICE): In computer science, it is a collection or package of software programs developed for Windows and Macintosh systems. All Office editions include the three standard programs, Word, Excel and PowerPoint and may also include the Microsoft Office suites: Microsoft OneNote, Microsoft Outlook, Microsoft Publisher and Microsoft Access.

OFFICIAL STATISTICS: Numerical data collected and published by government establishment or private agencies working for the government. Official statistics provide an indispensable element in the information system of a democratic society, serving the government, the economy and the public with data about the economic, demographic, social and environmental situation.

OFFLINE: Disrupted or disconnected internet connection to a computer. When a device is offline it is unable to send or receive information through that device.

OFFLINE BROWSING: In computing, downloading and copying Web pages onto a computer so that they can be viewed without being connected to the Internet. During off-peak hours, telephone charges are lower and the network responds faster, making offline browsing an economical way of using the Internet.

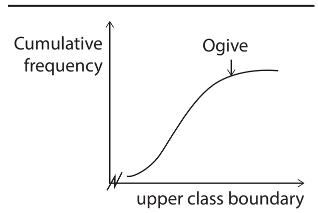
OFFSET: 1. A short, lateral shoot, either a stolon or a sucker that develops from an axillary bud near the base of the stem and gives rise to a daughter plant at its apex. It is a type of short runner used for vegetative reproduction. The term is also applied to bulbils and cormlets that form at the side of the parent bulb or corm. 2. Also known as **offset address**, it is a value added to a base address to produce a second address in low level programming. The resulting address, therefore, is relative to some other addresses. 3. Also known as **gutter**, it is the amount of space along the edge of a paper in desktop publishing that allows for binding.

OFFSET ADDRESS: In low level programming, it is the number of address locations that can be added to a base address in order to produce a second address.

OFFSET PRINTING: The most common commercial method of printing in which the inked image on a printing plate is imprinted on a rubber cylinder and then transferred (offset) to paper or other material. Because of the flexibility of the rubber cylinder, the offset printing method permits printing on wood, cloth, metal, leather and rough paper.

OFFSHOOT: 1. Something that derives its existence from a particular source. 2. A branch or a lateral shoot that arises from a main stem.

OFFSPRING (PROGENY): A group of sexually- or asexually-reproduced individuals from the same parents or ancestors; or the child of a particular parentage.



Ogive (cumulative frequency curve)

OGIVE: 1. A curve or graph that represents the cumulative frequencies of a set of values. 2. The roundly tapered end of a two-dimensional or three-dimensional object.

OHM: The SI unit of electrical resistance. It is defined as the resistance in a circuit between two points when a potential difference of one volt between them produces a current of one ampere. One ohm equals 1 volt/ampere. Its symbol is (Ω).

OHM, GEORGE SIMON (1787 – 1854): A German physicist, best known for formulating Ohm's law in 1827. He discovered that the flow of electric current through a conductor is directly proportional to the potential difference or voltage and inversely proportional to the resistance. The unit of electrical resistance, the ohm with the symbol Ω was named after him.

OHM'S LAW: A law, which states that the current (I) flowing in a metallic conductor is directly proportional to the potential difference (V) across its ends provided temperature and other physical conditions remain constant. The other physical conditions include the length of wire, cross sectional area and the type of material (for example, copper or aluminium). It is expressed as $V = IR$. Ohm's law holds true for only resistive circuits without the effects of inductances, capacitances to the flow of current.

OHMIC DEVICE: A device that follows Ohm's Law: V = IR with a resistance that remains constant over a wide range of potential differences. Examples include a wire, a heating element or a resistor.

OHMIC HEATING (JOULE HEATING; RESISTIVE HEATING): The process by which the passage of an electric current through a conductor releases heat. It was first studied by James Prescott Joule, English physicist in 1841. Joule heating is caused by interactions between the moving particles that form the current, mainly electrons and the atomic ions that make up the body of the conductor. Charged particles in an electric circuit are accelerated by an electric field but give up some of their kinetic energy each time they collide with an ion. The increase in the kinetic or vibrational energy of the ions manifests itself as heat and a rise in the temperature of the conductor. This method finds practical applications in the food processing industry, in which an electric current is passed through foodstuffs to sterilise them before packing. It is also applied in incandescent light bulb when the filament is heated by joule heating that it glows white with thermal radiation, in electric stoves and other electric heaters. Power stations reduce the transmission loss associated with Joule heating of the power lines, by transmitting power at very low current and very high voltage instead of high current and low voltage.

OHMMETER: A portable instrument used for measuring the value of a resistance in ohms. An example is a multimeter which is also used for measuring current and voltages. Ohmmeters are generally used in electrical troubleshooting and rectifying problems such as measuring the resistance of circuits and components, continuity of circuits, earth leakage problems, etc. One type used in measuring the high resistance of electrical materials in the order of a few megohms to infinity at 1000 volts is called **megohmmeter** or **megger** (trade name). It has a built-in, hand-operated generator for providing the voltage needed to measure the electrical continuity, the insulation resistance and the resistance between a component such as a motor and an electrical ground.

OIDIUM (ARTHROSPORE): A spore that forms by the organised fragmentation of a hypha as seen, for example, in Endomyces.

OIL: Any of various viscous substances that is liquid at room temperature and insoluble in water. Oils may be solids such as fats and waxes. Natural plant and animal oils are either volatile mixtures of terpenes and simple esters, for example, essential oils (lemon oil, eucalyptus oil, coconut oil, etc.) or glycerides of fatty acids. Mineral oils such as petroleum are mixtures of hydrocarbons. Some oils are used in industries and food processing. Soybean, sunflower and safflower oils which are also used in processing foods are also used in paints and varnishes. Some specialty oils and oil derivatives are also used in leather dressing and textile manufacture.

OIL CROP: A crop such as coconut, palm or sunflower grown for the oil it produces and exploited commercially.

OIL GLAND: A gland such as a sebaceous gland that secretes an essential oil.

OIL IMMERSION: A technique used in light microscopy to increase the resolution of a microscope. The objective lens and the specimen are immersed in a transparent oil of high refractive index, thereby increasing the numerical aperture of the objective lens.

OIL MEAL: A by-product of vegetables such as linseed, cottonseed or soybean from which the oil has been pressed, forming oil cake which is ground into small particles for feeding livestock.

OIL OF WINTERGREEN (METHYL SALICYLATE; METHYL 2-HYDROXYBENZOATE; BETULA OIL; GAULTHERIA OIL; WINTERGREEN OIL): A colourless liquid that is soluble in water, with an odour of wintergreen and the molecular formula $C_8H_8O_3$, density 1.174 g/cm³, melting point -9 °C (264 K) and boiling point 223 °C (496 K). It occurs in the essential oils of some plants such as wintergreen or birch and is manufactured from salicylic acid. It is easily absorbed through the skin and used in medicine for treating muscular and sciatic pain. Because of its attractive smell it is also used in perfumes and food flavourings.

OIL PALM: An economically valuable palm tree (*Elaeis guineensis*), native to West Africa. The American oil palm *Elaeis oleifera* is native to tropical Central America and South America. It is a monocotyledonous plant and can grow to 20 m tall. The leaves are pinnate and reach between 3-5 m long. It is widely cultivated for its pulpy fruits and the seeds called palm kernel. Oil is extracted from both the pulp of the fruit called palm oil and the kernel called palm kernel oil. Both oils are used in making foods such as margarine as well as soap and candles.

OIL REFINERY (PETROLEUM REFINERY): A process plant, where crude oil is processed and refined into more useful petroleum products, such as petrol, diesel fuel, asphalt base, heating oil, kerosene and liquefied petroleum gas. Petroleum products are usually grouped into three categories: light distillates (LPG, gasoline, naphtha), middle distillates (kerosene, diesel), heavy distillates and residuum (heavy fuel oil, lubricating oils, wax, asphalt).

OIL SAND (TAR SAND; BITUMINOUS SAND): A sandstone or porous carbonate rock that is saturated with hydrocarbons. Oil recovered from tar sands, commonly referred to as synthetic crude, is a potentially significant form of fossil fuel. Oil sands are distributed throughout the world but the largest deposit occurs in Alberta, Canada (the Atheabasca tar sands).

OIL SHALE: Any fine-grained sedimentary rock that has a relatively high content of a bituminous substance and 30-60% organic matter (kerogen) and yields significant quantities of oil when heated. Shale oil is a potentially valuable fossil fuel, but the present methods of mining and refining it are expensive, damage the land, pollute water and produce carcinogenic wastes.

OIL SPILL: A release of a liquid petroleum hydrocarbon into the environment due to human activity. It includes releases of crude oil from tankers, offshore platforms, drilling rigs and wells, refined petroleum products such as petrol, diesel and their by-products and heavier fuels used by large ships such as bunker fuel or the spill of any oily refuse or waste oil.

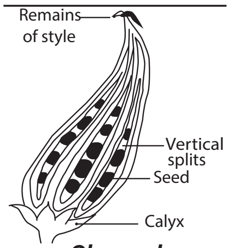
OIL TUBE: Narrow ducts in the walls of the fruits of many members of the Umbelliferae (Apiaceae), which contain volatile oils.

OIL-IMMERSION LENS: An optical microscopic object in which the front surface of the electron lens is immersed in a liquid such as cedar wood oil or sugar solution on the cover glass of the microscope specimen slide. An oil-immersion lens of a light microscope has a depth of 0.08 micron. The object, usually the cathode, lies deep within the electric field so that the index of refraction varies rapidly in its vicinity.

OKAZAKI FRAGMENT: A relatively short fragment of single-stranded DNA that is an intermediate in DNA synthesis. In eukaryotes, they are generally between 100 and 200 nucleotides long and in *E. Coli* are between 1,000 and 2,000 nucleotides long. They were first observed in 1968 by Reiji Okazaki and Tsuneko Okazaki, using pulse-labelling with radioactive thymidine.

OKLO REACTORS: Naturally occurring nuclear fission reactors of uranium deposit, where analysis of isotope ratios has shown that self-sustaining nuclear chain reactions have occurred. They are believed to have existed in uranium deposits at Oklo in Gabon, Central Africa, about 2000 million years ago. In 1972, French scientists noticed slight difference in the normal $^{235}U/^{238}U$ ratio in uranium ore from Oklo. A similar natural reactor has been found at Bangombe, some kilometres south of Oklo, but no other comparable reactors have been found anywhere in the world. It is thought that the geology of the mine was an important factor in the creation of the reactors, in particular, the seepage of water through overlying rock, which functioned as a moderator. The Oklo reactors are of considerable interest. More practically, Oklo can be regarded as a 2 billion-year experiment in the containment of nuclear waste. The reactors shut down naturally when the proportion of ^{235}U decreased and for the same reason natural reactors of this type could not occur today.

OKRO (OKRA; LADY'S FINGER; GUMBO): A tropical herbaceous plant belonging to the family Malvaceae and classified as *Abelmoschus esculentus*; or its many-seeded pod. Okra plants have small erect stems that can be bristly or hairless with heart-shaped leaves. The leaves are 10-20 cm long with 5-7 lobes. The plant produces flowers with five white to yellow petals which are 4-8 cm in diameter. The seed pod is a capsule up to 25 cm long, containing numerous seeds. Okro can grow between 1.2-1.8 m tall and as an annual plant, survives only one growing season. When fresh and green, okra releases a sticky substance with thickening properties, often used in soups and stews. The dry seed is a source of edible oil.

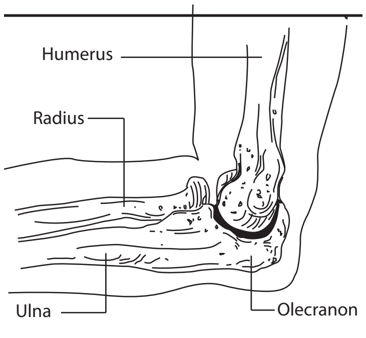


Okro pod

OLBERS' PARADOX: The paradox of why the sky is dark at night. If the universe is static, of infinite age, uniform and unchanging and the galaxies distributed isotropically, the sky at night would be bright, as in whatever direction one looked one would eventually see a star. The number of stars would increase in proportion to the square of the distance from the Earth. The intensity of light reaching the Earth from a given star is inversely proportional to the square of the distance. Consequently, the whole sky should be about as bright as the Sun. The paradox, that this is not the case, was stated by Heinrich Olbers (1758-1840) in 1826. It had been discussed earlier, in 1744, by J. P. L. Chesaux. The paradox is resolved by the fact that according to the big-bang theory, the universe is not infinite, not uniform and not unchanging. For instance, light from the most distant galaxies displays an extreme red shift and ceases to be visible.

OLEAGINOUS: Producing or containing oil or lipids. Oleaginous microorganisms such as some yeasts and fungi normally contain 20-25% of oil. This is of interest in biotechnology as alternative source of conventional oils because oils produced by oleaginous eukaryotic microorganisms are similar to plant oils. **Oleiferous** means oil-bearing.

OLECRANON PROCESS: A bony process on the ulna in the forelimb of vertebrates that projects beyond the joint between the humerus and the ulna. It forms the point of the elbow and attachment point for the triceps brachii muscle.



Olecranon Process

OLED (ORGANIC LIGHT-EMITTING DIODE): A type of light-emitting diode (LED) that uses an organic substance as the semiconductor. The organic material used is often films of polymer sandwiched between two conductors that emits light when an electric current is applied, and therefore does not require a backlight. OLED displays are either passive-matrix organic light-emitting diode (PMOLED) or active-matrix organic light-emitting diode (AMOLED). AMOLED has a storage capacitor that makes the line pixels light all the time allowing for higher resolution and larger display sizes. Passive-organic organic light-emitting diode (PMOLED) does not contain a storage capacitor and so the pixels in each line are actually off most of the time and needs more power to make them brighter. OLEDs are used for digital displays in devices such as television screens, computer monitors, mobile phones, and personal digital assistant (PDA). OLEDs enable emissive, bright, thin, flexible and efficient displays and are replacing liquid crystal displays (LCDs) in most display applications.

OLEFIN: Common name for alkene with the general formula C_nH_{2n} and any unsaturated hydrocarbon containing one or more pairs of carbon atoms linked by a double bond. They may be classified as **cyclic**, which means that the double bond is in a ring or as a **chain** (acyclic or aliphatic) or by the number of double bonds (monoolefin, diolefin, etc.). Olefins are obtained by the cracking of petroleum fractions at high temperatures and naturally rare. The simplest ones are ethylene, propylene, butylene, butadiene and isoprene and are the basis of the petrochemicals industry. They react by adding other chemical agents at the double bond to form derivatives or polymers.

OLEIC ACID (CIS-OCTADEC- 9- ENOIC ACID): An unsaturated fatty acid with one double bond and the molecular formula $C_{18}H_{34}O_2$, density 0.895 g/ml, melting point 13 °C (286 K) and boiling point 360 °C (633 K). It is found abundantly in animal and plant fats, occurring in butterfat, lard, tallow, groundnut oil, soya-bean oil, etc. It is used to make soap, as an excipient in pharmaceuticals and as an emulsifying agent in aerosol products.

OLERICULTURIST: A biological scientist who studies the cultivation of vegetables including the production, storage, processing and marketing. **Olericulture** is the applied life science of vegetable cultivation and production. It is usually associated with the cultivation of annual, herbaceous garden plants.

OLEUM (FUMING SULFURIC ACID): A solution of sulfur(VI) oxide in concentrated sulfuric acid or sulfuric acid with excess sulfur trioxide. It is a cloudy, grey, fuming, oily, corrosive liquid with a sharp, penetrating odour. The empirical formula is $H_2S_2O_7$. When it comes into contact with air following a spill, it releases sulfur trioxide. Oleum is primarily used in the chemical-technical industry, for example to produce organic interim products, in the chemical fibres industry, to upgrade sulfuric acids by means of the recycling procedure and to produce liquid sulfur trioxide.

OLFACTION: One of the chemical senses, specifically the sense of smell or the process of detecting smells. Olfaction registers chemical information in organisms ranging from insects to humans, including marine organisms. This is achieved by the receptors in olfactory organs such as the nose that are sensitive to air or water-borne chemicals. Stimulation of these receptors results in the transmission of information to the brain via the olfactory nerve.

OLFACTORY: It describes parts of the body and functions relating to the sense of smell. For example, **olfactory cell** is a receptor cell found high inside the nasal cavity, which is associated with the sense of smell. **Olfactory nerve** is the first cranial nerve, which conveys smell sensations as nerve impulses from the nose to the brain. Each of the two olfactory nerves detects smells by means of hair-like receptors in the mucous membrane lining the roof of the nasal cavity. **Olfactory lobe** is any of the pair of lobes in the forebrain at the anterior end of the cerebrum. They contain the end of the olfactory nerves and are concerned with the senses of smell. It is prominent in the dogfish and other animals that depend on this sense.

OLIGANDROUS: Having few stamens.

OLIGO: A prefix meaning 'a few' and used for compounds with a number of repeating units intermediate between those in monomers and those in high polymers. The limits are not precisely defined and in practice vary with the type of structure being considered, but are generally from 3 to 10, for example, oligopeptides and oligosaccharides.

OLIGOCARPIC (OLIGOCARPOUS): Bearing less than the normal amount of fruit.

OLIGOCENE EPOCH: The third epoch of the tertiary period of geological time, 35.5 – 3.25 million years ago. It corresponds to an interval of geological time from the close of the Eocene Epoch to the beginning of the Miocene Epoch. The epoch was characterised by the continued rise of mammals. The first to appear were pigs, rhinoceroses, etc.

OLIGOCHAETA: A class of hermaphrodite annelid worms that bear only a few bristles (chaetae). Examples include the earthworm (*Lumbricus*) and the freshwater bloodworm (*Tubifex*). The oligochaetes are primarily fresh-water and burrowing terrestrial animals. A few are marine and several species occur in the intertidal zone.

OLIGODENDROCYTE: A type of glial cell (type of cell in the central nervous system)that surrounds the neurons in the brain and spinal cord producing the insulating myelin sheath surrounding the nerve axon.

OLIGOLECITHAL: Of an egg, having a little yolk, typical of all higher mammals, some marsupials and many invertebrates.

OLIGOMER: A polymer formed from the combination of a few monomer molecules or by separation from the higher fractions of crude oil. Oil such as liquid paraffin is oligomeric. Another example of oligomeric oil is polybutene, which is used to make putty. Plasticisers, used to soften thermoplastics such as PVC are oligomeric esters.

OLIGOMER MOLECULE: A molecule of intermediate relative molecular mass, the structure of which essentially comprises a small plurality of units derived, actually or conceptually, from molecules of lower relative molecular mass. A molecule is regarded as having an intermediate relative molecular mass if it has properties which do vary significantly with the removal of one or a few of the units. If a part or the whole of the molecule has an intermediate relative molecular mass and essentially comprises a small plurality of units derived, actually or conceptually, from molecules of lower relative molecular mass, it may be described as oligomeric or by oligomer used adjectivally.

OLIGOMERIZATION: The process of converting a monomer or a mixture of monomers into an oligomer. An oligomerization by chain reaction, carried out in the presence of a large amount of chain-transfer agent, so that the end groups are essentially fragments of the chain transfer agent, is termed **telomerization**.

OLIGOMEROUS: Having less than the normal or usual number of parts.

OLIGONUCLEOTIDE: A deoxyribonucleic acid (DNA) or ribonucleic acid (RNA) sequence composed of two or more covalently linked nucleotides. Fragments containing up to 50 nucleotides are generally termed **oligonucleotides** and longer fragments are called **polynucleotides**. The linkages are the same as in polynucleotide. The only distinguishing feature is the small size or a short polymer of nucleotide.

OLIGOPHYLLOUS: Possessing few leaves.

OLIGOSACCHARIDE: Carbohydrates composed of between two and ten monosaccharides (simple sugars) units linked together through bonds called **glycosidic linkages**. Examples are **disaccharides** such as sucrose and maltose, which yield two monosaccharide molecules on hydrolysis; **Cellobiose**, the basic repeating unit of cellulose; **trehalose** (a sweet-tasting, crystalline disaccharide, $C_{12}H_{22}O_{11}$); and **gentiobiose** ($C_{12}H_{22}O_{11}$, composed of two units of glucose). Oligosaccharides may be intermediates in polysaccharide metabolism or may serve as storage compounds.

OLIGOTROPHIC: A term describing environments such as a lake, deep oceanic sediments, caves, glacial and polar ice, deep subsurface soil, aquifers, ocean waters and leached soils with a poor supply of nutrients and a low rate of formation of organic matter by photosynthesis. An **oligotroph** is an organism that lives in such an environment.

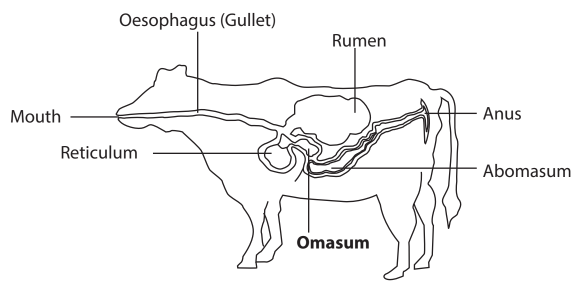
OLISPERMOUS: Having few seeds.

OLIVACEOUS: 1. Having a dusky, yellowish green colour. 2. Resembling an olive.

OLIVENITE: Basic or hydrated copper arsenate having brown, olive green or grey colour with the chemical formula $Cu_2(AsO_4)OH$, with Mohs hardness of 3. It occurs as a mineral in olive-green prisms and found in copper deposits.

OLIVINE (CHRYSOLITE): A greenish mineral, magnesium iron silicate with the chemical formula $(MgFe)_2SiO_4$. It is an important group of rock-forming silicate minerals crystallising in the orthorhombic system. It is obtained from igneous rocks and used as gems and refractories.

OMASUM (MANYPLIES; PSALTERIUM): The third and smallest compartment of the stomach of a cow or other ruminants, located between the abomasum and the reticulum. The inner surface has folds that break up food particles.



Omasum of a cow

OMEGA: The 24th letter of the Greek alphabet, denoted Ω for the upper case and ω for the lower case and used in many mathematical and scientific connotations. For example, in physics and electrical engineering Ω is the symbol for ohm, the International System of Units (SI) unit of electrical resistance; and ω is the symbol for angular velocity or angular frequency and in mathematics, it represents the smallest infinite ordinal; the ordinal of the natural order of the natural numbers.

O

OMEGA NAVIGATION SYSTEM: Long range radio base aid to navigation, which gives worldwide coverage, now replaced by the Global Positioning System (GPS).

OMEGA-MINUS (OMEGA HYPERON): A negatively charged elementary particle, the heaviest hyperon or heavy elementary particle with a rest mass of 3,272 times that of an electron.

OMNIBOX: An address bar, similar to the traditional browser address bar with additional features. For example, in the Google Chrome browser, the Omnibox (address bar) not only displays the address of the web page, but also can be used to search the Internet. Omnibox in Chrome can also perform mathematical calculations and even answer questions.

OMNIVORE: A type of animal such as a human being and pig that feeds on both plant and animal material as its primary food source.

ON SCREEN DISPLAY (OSD): A menu displayed on various devices that display options used to adjust the display or other options that may be available to that device.

ONCHOCERCIASIS (RIVER BLINDNESS; ROBLES' DISEASE): A parasitic disease transmitted by the blackfly of the genus *Simulium*. It is characterised by lumps under the skin, gradual wrinkling of skin, intense skin irritation and inflammation of the eyes leading to blindness. Onchocerciasis is the world's second-leading infectious cause of blindness after Trachoma, a contagious disease of the conjunctiva and cornea, caused by the gram-negative bacterium *Chlamydia trachomatis* is the leading infectious cause of blindness.

ONCOGENE: 1. A dominant mutant allele of a cellular gene that disrupts cell growth and division and is capable of transforming a normal cell into a cancerous cell. Proto-oncogenes typically encode proteins involved in the positive control of the cell division cycle such as growth factor receptors, signal transduction proteins and transcription factors. A change in their genetic sequence can result in uncontrolled cell growth, ultimately causing the formation of a cancerous tumour. In humans, proto-oncogenes can be transformed into oncogenes in three ways: **point mutation** which is the alteration of a single nucleotide base pair; **translocation** in which a segment of the chromosome breaks off and attaches to another chromosome; or **amplification** which is an increase in the number of copies of the proto-oncogene. 2. A gene carried by a virus that induces a cell to divide abnormally forming a tumour or a chemical organism or environmental factor that causes the development of cancer.

ONCOGENIC: A chemical organism or environmental factor that can cause the formation and development of cancer. Some viruses are oncogenic to vertebrates, notably the retroviruses. Many of these viruses contain genes that are responsible for the transformation of a normal host cell into a cancerous cell.

ONCOLOGY: The field of medical science concerned with the diagnosis and treatment of neoplasms (new, often uncontrolled growth of abnormal tissue). A specialist who practices oncology is called an oncologist.

ONCOTIC PRESSURE (COLLOID OSMOTIC PRESSURE): The part of the osmotic pressure exerted by proteins in blood plasma that usually tends to pull water into the circulatory system. The vessel walls are relatively impermeable to these molecules. Oncotic pressure largely offsets the hydrostatic pressure that tends to drive water from blood vessels into the extra cellular spaces and tissues. In conditions, where plasma proteins are reduced such as from being lost in the urine (proteinuria) or from malnutrition, there will be a reduction in oncotic pressure and an increase in filtration across the capillary, resulting in excess fluid build up in the tissues (oedema).

ONCOVIRUS: A class of retroviruses that cause a cell to become cancerous.

ONE: 1. The integer before two and after zero and the smallest natural number; the unit of counting or measurement; the first non-zero number and odd natural number. 2. The number of members of a set that has all of its members identical; unity. 3. The identity element under multiplication in a ring.

ONE GENE-ONE POLYPEPTIDE HYPOTHESIS: The theory that a single gene controls the production, specificity and activity of each enzyme that in turn affects a single step metabolic pathway. Thus, mutation of such a gene changes the ability of the cell to carry out a particular reaction and disrupts the entire pathway.

ONE TO ONE FUNCTION (ONE-TO-ONE FUNCTION): A function $y = f(x)$ is a one-to-one function if every value of x in the domain is associated with a unique value of y in the range, making it possible to find an inverse function.

ONE-DIMENSIONAL (1-D): Having only one dimension and therefore showing only linear information such as length or width or height. For example, 1-dimensional (1-D) figure is a figure such as a line segment, arc, or part of a curve that has length but no width or depth.

ONE-DIMENSIONAL (1-D) COORDINATE SYSTEM: A reference frame in which any point on a 1-dimensional figure can be located with one coordinate relative to the origin of a number line.

ONE-DIMENSIONAL RANDOM WALK: A Markov chain on the integers 0, 1, 2, 3 ... with one step transition probabilities
$P_{i\cdot j-1} = q_i$, $P_{ii} = r_i$, $P_{i,i+1} = P_i$ for I ≥ 0 with $q_0 = 0$.
For |i-j| ≥2, $P_{ij} = 0$ so that the parameters, q_i, r_i and P_i sum to one.

ONE-MANY: Of a function or mapping, capable of associating a single member of the domain with more than one member of the range of a set-valued function; holding between the same first argument and more than one second argument of a binary relation. For example, $\sin^{-1}x$ and $\sqrt{x}$ are one-many functions on the positive real numbers; and "... is the mother of ..." is a one-many relation, since a mother may have many children but every child has exactly one mother.

ONE-PHOTON PHOTOCHROMISM: Photochromic process involving a one-photon mechanism. In this case, the photoproduct B is formed from the singlet or from the triplet state of the thermodynamically-stable molecular entity A.

ONE-POINT COMPACTIFICATION (ALEXANDROFF COMPACTIFICATION): A compacification that adds a single point, designated ∞, to a Hausdorff space and adds complements of closed subsets of the original space as neighbourhoods of this point. The real numbers can be compactified in this manner.

ONE-POT SYNTHESIS: A way of improving the efficiency of a chemical reaction by allowing all the starting materials into the reaction chamber which react either in a single step or several steps without separating any intermediate product until the final product has been obtained. An example of a one-pot synthesis is the total synthesis of tropinone or the Gassman indole synthesis.

ONE-SIDED: 1. Of a surface, having the property that any two points can be joined without crossing an edge. 2. Of the limit of a function of a single real variable at a point, p, it is the limit as computed when the function is viewed as restricted to the half-line $]-\infty, p[$ or to the half-line $]p, -\infty[$. The function has a discontinuity at p if the one-sided limits do not exist or are not equal and is continuous at p if they are finite and equal, in which case that value is the two-sided limit at p. For example, the one-sided limit of $x/|x|$ at zero from above is +1 and the one-sided limit from below is -1.

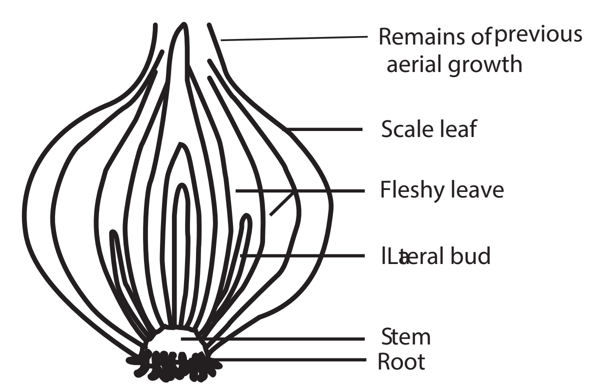
ONE-STEP WORD PROBLEMS: Word problems that can be solved using a single mathematical operation, for example, change problems, combine problems, and compare problems.

ONE-TAILED TEST (ONE TAIL TEST; ONE-SIDED TEST): A statistical test in which an observed value of a test statistic is either greater than or less than a certain value, but not both. For example, in testing whether weighing scales give short measure, a consumer does not regard overweight goods as a significant error.

ONE-TO-ONE CORRESPONDENCE (BIJECTION): A mapping between two sets, with each element of the first set made to correspond with exactly one element of the second set and vice versa. For example, the natural numbers n correspond one-to-one with the points $\langle n, n\rangle$ on the real plane.

ONE-TO-ONE MAPPING (INJECTION; INJECTIVE MAPPING): A mapping f: $S \to T$ is one-to-one if, whenever s_1 and s_2 are distinct elements of S, their images $f(s_1)$ and $f(s_2)$ are distinct elements of T. So f is one-to-one if $f(s_1) = f(s_2)$ implies that $s_1 = s_2$.

ONION: Herbaceous biennial plant, *Allium cepa*, of the family Alliaceae or its round bulb that has a strong smell and flavour. Onions are pungent and contain a sulfur-rich volatile oil which irritates the eyes. They are low in standard nutrients, but are valued for their flavour.



Longitudinal section of onion

ONION MORPHOLOGY: Multiphase morphology of roughly spherical shape that comprises alternating layers of different polymers arranged concentrically, all layers being of similar thickness.

ONIUM ION (ONIUM COMPOUNDS): An ion formed by addition of proton (H^+) to a molecule. An example is ammonium ion (NH_4^+) from ammonia and hydroxonium ion (H_3O^+) from water. Onium ions have a charge of +1 and double onium ions have a charge of +2.

ONLINE: A computer, which is connected to an internet or other devices. When a device is online it can send or receive information through that device. An **on-line system** is a large database, electronic mail and conferencing systems accessed via a dial-up modem.

ONLINE ANALYTICAL PROCESSING (OLAP): A process which allows users to analyse database information from multiple database systems at one time. OLAP data is multidimensional, in which case information can be compared in many different ways.

ONLY IF: Indicating a necessary condition, the converse of if, so that "P only if Q" is equivalent to "if P, Q".

ONSAGER RELATIONS: A set of conditions developed by the Swedish physicist, Lars Onsager in 1931, which states that the matrix, whose elements express various fluxes of a system such as diffusion and heat conduction as linear functions of the various conjugate affinities (such as mass and temperature gradients) for systems close to equilibrium, is symmetric when certain definitions are chosen for these fluxes and affinities. There may be a flow of energy from one part of the system to another and, at the same time, a flow of mass (diffusion). Flows of this type are coupled (each depends on the existence of other equations). Equations of this type can be generalised to any number of flows and are known as the **phenomenological relations**.

ONTO: Of, relating to or being a mapping such that every element of the target set referred to is the image of an element in the domain. For example, a function, f is said to map A onto B if for every b in B, there is some a in A: $f(a) = b$.

ONTOGENY: The process of development of a living organism, from the ovum to maturity. It starts with fertilisation and ends with the attainment of an adult state, usually expressed in terms of both maximal body size and sexual maturity.

ONTOLOGY: 1. A specification of the assumptions, terms or concepts underlying a particular field of knowledge. For example, the Gene Ontology (GO) project is an international collaboration between various databases in the field of genomics to standardise terminology used by researchers. 2. A branch of metaphysics, it is the philosophical study of the nature of being, existence or reality in general, as well as the basic categories of being and their relations.

ONYCHOPHORA: An enigmatic small elongated wormlike terrestrial invertebrates of damp dark habitats in warm regions; distinct from the phylum Annelida. These terrestrial animals are frequently referred to as *Peripatus*. They are in some respects intermediate between annelid worms and typical arthropods, that have an unsegmented vermiform body, numerous pairs of short unsegmented legs with terminal bifid claws and a head bearing a pair of segmented antennae, a pair of oral papillae on which slime glands open, a pair of simple eyes and a pair of jaws resembling blades, that possess a hemocoel as a body cavity and that breathe by means of tracheae and excrete by means of nephridia.

ONYX: A chalcedony that occurs in bands of different colours and is used as a gemstone, especially in cameos and intaglios.

OOCYST: A spherical structure, 50-60 microns in diameter that develops from the zygote in a sporozoan life cycle. The oocyst is surrounded by a tough cyst to prevent dehydration or other damage during its transit through the life cycle. It grows in size and its contents divide repeatedly to form sporozoites, which are released into the body cavity when the oocyst bursts.

OOCYTE: A cell in the ovary that develops into an ovum after two meiotic divisions. Primary oocytes develop from oogonia in the foetal ovary as they enter the early stages of meiosis. Only a fraction of the primary oocytes survive until puberty and even fewer will be ovulated.

OOGAMY: Sexual reproduction involving the formation and subsequent fusion of a large, usually stationary, female gamete and a small motile male gamete. The female gamete may contain nourishment for development of embryo, which is often retained and protected by the parent organism. The male gametes are characterised by highly motile spermatozoa, which compete for the fertilisation of the immotile egg. Oogamy occurs mostly in animals, but can also occur in many protists, certain orders of algae such as *Ochrophytes, Charophyceans* and some plants such as bryophytes, ferns and some gymnosperms like cycads and ginkgo.

OOGENESIS: The female form of gametogenesis in which the ova or egg cells are produced and developed in the animal ovary. Oogenesis consists of several sub-processes: oocytogenesis, ootidogenesis and finally maturation to form an ovum (oogenesis proper). Special cells called oogenia within the ovary divide repeatedly by mitosis to produce large numbers of prospective egg cells or oocytes. When matured, these undergo meiosis, which halves the number of chromosomes. During the first meotic division, a polar body and a secondary oocyte are produced. At the second meiotic division the secondary oocyte produces an ovum and a second polar body.

OOGONIUM (OOGONIA): 1. The female reproductive organ of algae and fungi. 2. Any of the immature sex cells in the animal ovary that gives rise to oocytes by mitotic division. They are formed in large numbers by mitosis early in foetal life from primordial germ cells, which are present in the foetus between 4 to 8 weeks.

OOLITE: Limestone or rock made up of small particles, usually of calcium carbonate, containing a nucleus and clearly defined concentric shells. Oolites are formed by chemical precipitation and accumulation of ancient sea floors.

OOMYCOTA (OOMYCETES): A phylogenetic lineage of fungus-like eukaryotic microorganisms that includes the water moulds and downy mildew formerly classified as a fungi. Oomycota are either saprotrophic or parasitic. They feed by extending hyphae into the food source or host's body. They reproduce both sexually and asexually. Asexual reproduction is by means of flaggelated zoospores, which are released from a sporangium. Sexual reproduction involves the fusion of an antheridium and an oogonium and results in the reproduction of a zygote, which can develop a protective wall made of cellulose and become a resistant oospore.

OORT CLOUD (ÖPIK-OORT CLOUD): A hypothetical spherical cloud of comets around the Sun a long way beyond the orbit of Pluto, extending to about half the distance to the nearest star. It consists of large numbers of objects and is the source of long-period comets. The outer extent of the Oort cloud defines the gravitational boundary of our Solar System. The Oort cloud can be regarded as 'rubble' left over from the formation of the solar system.

OOSPHERE: In botany, a large female gamete produced in the oogonia of algae and fungi; or a large nonmotile female gamete formed in an oogonium, such as the eggs formed in the oogonium of an oomycete.

OOSPORE: A thick-walled sexual spore that is produced as a result of oogamy in certain algae and fungi. It contains food reserves, develops a protective outer covering and enters a resting phase before germination.

OOTHECA: The egg case of some insects such as cockroaches and mantises. Some oothecae have only a thin covering but others are surrounded by a mass of foam that hardens soon after it has been produced.

OPACITY: 1. The measure of impenetrability to electromagnetic radiation, especially light. It is the reciprocal of the transmittance.

In radiative transfer, it describes the absorption and scattering of radiation in a medium, such as plasma, dielectric, shielding material and glass. 2. A character of paper that can be printed on one side without the image showing through on the other side. The thicker the paper stock, the greater the opacity. Opacity is important to the functioning of optical character recognition scanners, since they are highly sensitive to light and any stray marks that may show through the paper.

OPAL: 1. A natural hydrated form of silica with the chemical formula $SiO_2 \cdot nH_2O$, Mohs scale hardness 5 - 6, specific gravity 2.25 -1.99 and refractive index 1.455 - 1.435. It often occurs as stalactites and found in many types of rocks. The common opal is translucent, milk-white, yellow, red, blue or green and lustrous. For gemstone use, its natural colour is often enhanced by placing thin layers of opal on a darker underlying stone, like basalt. 2. **Open Physics Abstraction Layer** (OPAL) is an open source real-time physics engine used in games, robotics simulations and other 3D applications. It is the original name of software from Computer Associates that converts legacy output from mainframes and minicomputers into a graphical-based format. Now part of their Advantage Integration Server suite, the technology provides the development environment and supports 3270, 5250 and VT220 terminals and ODBC-compliant databases. Development can be done by drag and drop or by scripting in Opal Script or VBScript. It is currently supported only by ODE, but can be extended to run off on other engines.

OPALESCENCE: The iridescent milky appearance of certain dense transparent media when they are illuminated by visible polychromatic electromagnetic radiation in the visible range, such as sunlight. Opalescence occurs when particles are suspended in a solution. It also occurs in certain minerals, such as opal, in which there are thin surface films. It can occur in a system in which there are local fluctuations causing local fluctuations on the refractive index. The iridescence or rainbow-like display of interference of colours, arises because the intensity of scattered light is approximately proportional to the reciprocal fourth power of the wavelength of incident light (Rayleigh's law). Critical opalescence is opalescence that occurs at a critical point in various phase transitions, such as gas-liquid transitions due to density fluctuations.

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OPAQUE: A body that does not allow light to pass through it such that one cannot see through it or images cannot be seen through it. In other words, it is impenetrable by light, not transparent, impenetrable by a specific form of radiation other than visible light.

OPEN BALL: In mathematics (topology), it is an ε-neighbourhood or open set in a metric space.

OPEN CIRCULATORY SYSTEM: An arrangement of internal transport in which blood bathes the organs directly and there is no distinction between blood and interstitial fluid.

OPEN CLUSTER (GALACTIC CLUSTER): A diffuse group of between 20 and or few hundred stars that are concentrated along the central plane of the Milky Way. They are loosely bound to each other by mutual gravitational attraction and become disrupted by close encounters with other clusters and clouds of gas as they orbit the galactic centre.

OPEN COMMUNITY: A community in which the populations have different density peaks and range boundaries and are distributed more or less randomly.

OPEN COVER (OPEN COVERING): In mathematics (topology), it is a collection of open sets of a topological space whose union contains a given subset. For example, an open cover of the real line, with respect to the Euclidean topology, is the set of all open intervals $(-n, n)$, where n in $\mathbb{N}$.

OPEN DATABASE CONNECTIVITY (ODBC): A widely accepted application interface used for database access and is commonly used in various programs developed by Microsoft, such as Microsoft SQL, Microsoft Access and Microsoft Excel. It was originally developed by SQL Access Group and released in 1992.

OPEN DISC: In mathematics (topology), it is an open ball that is not compact, especially in the Cartesian plane $\mathbb{R}^2$.

OPEN FILM: A film in which mass transfer can occur between the film and the coexisting bulk phases, for all the components. The term **partly open film** applies to a film in which mass transfer can occur only for certain components. The term **closed film** applies to those films with fixed mass.

OPEN INTERVAL: A piece of a straight line that does not include its two endpoints. The open interval $\{x : a < x < b\}$ is denoted (a, b). A **half-open interval** contains one endpoint.

OPEN LABEL TRIAL (OPEN TRIAL): A type of clinical trial in which both the researchers and participants know which drug or treatment is being administered. This contrasts with double-blind experiment in which neither the researcher nor the participant knows what drug or treatment the participant is receiving.

OPEN LOOP CONTROL SYSTEM: A system that does not monitor or measure the condition of its output signal as there is no feedback and incapable of controlling automation processes. On the other hand, a closed loop system controls automation with a set of mechanical or electronic devices and automatically regulates an operation, process, or mechanism by feedback without human interaction.

OPEN MAPPING: In mathematics (topology), a function that sends open sets in the domain space to open sets in the range space.

OPEN MAPPING THEOREM: In mathematics (topology), a theorem giving conditions for a mapping to be an open mapping. The open mapping theorem for linear operators states that a continuous surjective linear mapping between Banach spaces is open. The open mapping theorem for analytic fundions states that an analytic function that is non-constant on a given domain is open theorem.

OPEN PROPORTION: A proportion with one or more variables. An open proportion is an open sentence and is neither true nor false. Examples are $\frac{2}{3} = \frac{a}{5}$ *and* $\frac{z}{15} = \frac{y}{3}$.

OPEN SENTENCE (PROPOSITIONAL FUNCTION; SENTENTIAL FUNCTION): In logic, a sentence containing one or more variables that can be either true or false depending on the value of the variable(s). Examples are $3x = 5$ and $3x + y = 4$.

OPEN SET: 1. In mathematics, in a metric space, a set of points around each of which there is an open ball lying within the set. In other words, a set is open if every point in the set has a neighbourhood lying in the set. 2. A member of a topology, one of the family of subsets of a topologic space that constitute the topology. 3. A set of which the complement is a closed set.

OPEN SEXTET: Species that have six electrons in the highest energy level of the central element.

OPEN SHORTEST PATH FIRST (OSPF): A method of finding the shortest path from one router to another in a local area network (LAN). As long as a network is IP-based, the OSPF algorithm will calculate the most efficient way for data to be transmitted.

OPEN SOURCE: A term used to describe software program that is freely available to the public and can be modified and distributed by users.

OPEN SYSTEM INTERCONNECTION (OSI): A network model developed by ISO in 1978 where peer-to-peer communications are divided into seven layers. Each layer performs a specific task or tasks and builds upon the preceding layer until the communications are complete.

OPEN THEORY: In logic, a theory that contains only open sentences.

OPEN-CAST MINING (OPEN-PIT MINING; STRIP MINING): Mining at the surface by removing the overlying material to recover the ore. It is used when deposits of commercially useful minerals or rock are found near the surface.

OPEN-HEARTH FURNACE: A smelting furnace, in which the charge is heated both by direct flame and by radiation from the roof and walls of the furnace, now largely superseded by the basic-oxygen process.

OPEN-SHELL SYSTEMS: Atomic or molecular systems in which the electrons are not completely assigned to orbitals in pairs.

OPEN-TUBULAR COLUMN: In chromatography, it is a column, usually having a small diameter, in which either the inner tube wall or a liquid or active solid held stationary on the tube wall acts as the stationary phase and there is an open, unrestricted path for the mobile phase.

OPERA GLASS (THEATRE BINOCULARS; GALILEAN BINOCULARS): Simple binoculars that consist of two Galilean telescopes which produce upright images without prisms. Magnification power below 5x is usually desired in these circumstances in order to minimise image shake and maintain a large enough field of view.

OPERAND: An operand is a number that is the subject of an operation. In the addition, 2 + 7 = 9, 2 and 7 are the operands.

OPERANT CONDITIONING (INSTRUMENTAL CONDITIONING; SKINNERIAN CONDITIONING): A type of associative learning that directly affects behaviour in a natural context, developed by B.F. Skinner. It is an association made between a behaviour and a consequence for that behaviour, in which the removal of a desirable outcome or the application of a negative outcome can be used to decrease or prevent undesirable behaviours. If the target response to be conditioned to occur spontaneously, the organism is given a reinforcer or reward immediately. After this procedure is repeated many times, the frequency of occurance of the targeted response will have significantly increased over its pre-experiment base rate.

OPERATING SYSTEM (OS): A programme that controls the basic operation of a computer, directs the input and output of data, keeps track of files and controls the processing of computer programs. Its roles include managing the functioning of the computer hardware, running the applications programmes, serving as an interface between the computer and the user and allocating computer resources to various functions.

OPERATION: 1. In mathematics, any process such as addition, multiplication, squaring, finding square root, set union and conjunction that generates a unique value according to the order of operations from one or more given numbers or values as arguments. 2. In mathematics, it is a function, especially one from a set to itself, such as differentiation of a differentiable function or rotation of a vector. 3. In medicine, it is the action performed by a surgeon on any part of the body, to treat or remove by cutting a diseased or an injured part. 4. A chromosomal segment of DNA that regulates the activity of the structural genes of an operon by interacting with a specific repressor.

OPERATION RESEARCH: A domain of applied mathematics that has scientific methods to provide a necessary basis for decision making. It is generally applied to the complex problems of people and equipment organisation in order to find the best solution to reach some goal. Its methods include simulation, linear programming, mathematical programming (nonlinear), game theory, etc.

OPERATION SYMBOL: In mathematics, it is a symbol used in expressions and number sentences to stand for a particular mathematical operation. Commonly used symbols include the following: addition: +, subtraction: -, multiplication: ×, ∗, •, division: ÷, /, powering: ^.

OPERATIONAL AMPLIFIER (OPAMP): An electronic device used with an external feedback to achieve high gain. It is the basic building block of many electronic circuits. It is a differential amplifier, because the output is directly proportional to the difference in voltage between the two inputs of the amplifier, which include the positive (+) input, referred to as "non-inverting"; and the negative (-) input, referred to as "inverting". A non-inverting amplifier has a positive gain and the output voltage is a positive multiple of the input voltage. An inverting amplifier has a negative gain; the output voltage is a negative multiple of the input voltage. It is used especially in circuits for performing mathematical operations such as addition, subtraction, multiplication and integration.

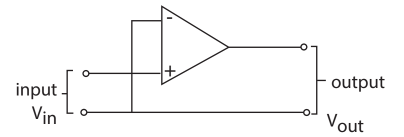

Fig I: ***Non-inverting amplifier***

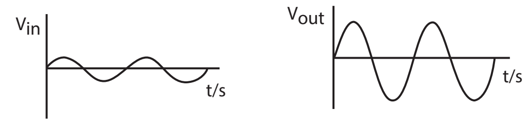

Fig ii: ***Output voltage (V_{out}) in phase with input voltage (V_{in}) for non-inverting amplifier***

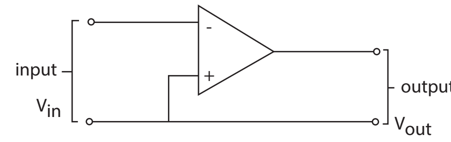

Fig iii: Inverting Amplifier

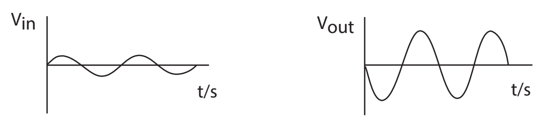

Fig iv: ***V_{out} in antiphase to V_{in} (inverting amplifier)***

OPERATIONAL pH CELL: An electrochemical cell which is the basis of practical pH measurement consisting of a hydrogen ion responsive electrode (hydrogen gas or glass) and a reference electrode immersed in the test solution.

OPERATIONAL TAXONOMIC UNITS (OTUS): The entities whose affinities are studied by numerical taxonomy. Depending on the level and the type of investigation an OTU may be of any taxonomic rank or an individual organism.

OPERATOR: 1. In mathematics, a symbol such as +, −, ×, ÷ and the integration operator, ∫ used to indicate an operation. 2. A mapping, such as a linear operator.

OPERATOR GENE: A site on the DNA adjacent to a structural gene that acts as part of a molecular 'switch' and determines whether or not transcription of the structural gene will occur. The switch may be turned 'off' by a repressor molecule binding to the operator, so preventing mRNA polymerase reaching the DNA. Conversely, absence of a repressor allows mRNA polymerase to bind to the DNA, i.e. the switch-is 'on'.

OPERATOR NORM: In mathematics, it refers to certain linear operators, for example, the operator norm of B(X, Y], between two normed spaces X and Y is given by the formula $\| T \| = \sup \{\| T(x) \| : \| x \| \leq 1 \}$.

OPERCULUM: 1. A lid or flap of skin covering an aperture such as the gill slit covering some fishes and the horny calcareous operculum secreted by gastropod molluscs, which closes the opening of the shell. 2. The cone shaped cap covering the spore of many mosses. It is forcibly pushed aside when the spores are matured to be released. 3. The bud cap covering the flowers of *Eucalyptus* and *Corymbia* trees which falls off as they open. 4. A lid covering the aperture of pollen grains. **Operculate** means having an operculum.

OPERON: A group of structural genes that are found next to each other on a chromosome and are turned on and off as an integrated unit and code for enzymes involved in a metabolic pathway, such as the biosynthesis of an amino acid. It is functionally a regulatory system of bacteria, in which genes coding for functionally related proteins are clustered along the DNA, enabling their expression to be coordinated in response to the cell's needs.

OPERON MODEL (JACOB-MONOD MODEL): A theory to explain how genetic regulation of protein synthesis in prokaryotes in which the structural genes that determine the organisation of the proteins are controlled by other regions of DNA upstream from the structural genes. It was proposed by F. Jacob and J. Monod in 1960. The theory is well substantiated for prokaryotes but there is little evidence that it is directly applicable to eukaryotic cells. The operon model shows how messenger RNA synthesis and thus the quantity of enzymes present, may be regulated.

OPHTHALMIC NERVE: The smallest of the three divisions of the trigeminal nerve. The ophthalmic nerve arises from the upper medial part of the trigeminal ganglion, enters the cavernous sinus and runs forward in the lateral wall of the sinus and divides into three branches, namely frontal, lacrimal and nasociliary, which pass into the orbit of the eye through the superior orbital fissure. The nasociliary usually arises first, then the lacrimal and the main part of the nerve is continued as the frontal. Near its origin, the ophthalmic nerve gives off a minute branch to the tentorium cerebelli; and, as it runs in the sinus, it receives filaments from the sympathetic plexus around the internal carotid artery and gives filaments to the oculomotor, trochlear and abducent nerves, the first two of which lie above it in the sinus.

OPHTHALMOLOGY: The branch of medicine and surgery concerned with diseases of vision and the eye.

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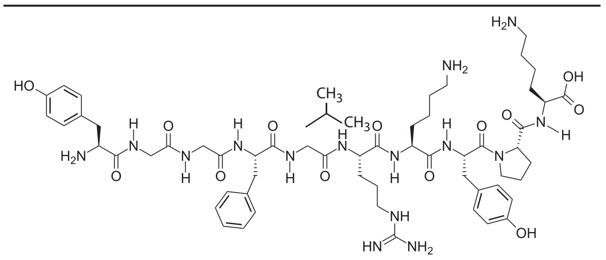
OPHTHALMOSCOPE: An instrument for examining the retina and structures of the inner eye. A powerful light and lens system, combined with the cornea and lens allows the retina and eye blood vessels to be seen at high magnification.

OPIATE: Any of a group of drugs derived from opium, which is an extract of the poppy plant, *Papaver somniferum*. Opiates include morphine, codeine and their synthetic derivatives, such as heroin (derived from morphine). They are used in medicine chiefly to relieve pain and induce sleep. But the use of morphine and heroin is strictly controlled since they can cause drug dependence and tolerance.

OPIATE ENDOGENOUS: A chemical produced naturally in the body by the pituitary gland and the hypothalamus in vertebrates during exercise, excitement, pain, consumption of spicy food, love and orgasm that have effects similar to morphine and other opiate drugs in their abilities to produce analgesia and a feeling of well-being. It is a type of neurotransmitter. Examples include endorphins and encephalins.



α-Neoendorphin

OPINION SURVEY: A procedure that aims to ascertain opinions possessed by members of some population with regard to particular topics.

OPIOID: Any one of a group of substances that work by binding to opioid receptors, which are found principally in the central nervous system and the gastrointestinal tract. They produce pharmacological and physiological effects similar to those of morphine. They are not necessarily structurally similar to morphine, although a sub-group of them, the opiates are morphine-derived compounds. Because it can produce a feeling of euphoria, some people use it recreationally.

OPIOID RECEPTOR: Any cell receptor molecule located in the brain and various organs that binds an opioid such as morphine or an endogenous opioid. There are three main types labelled μ (mu), κ (kappa) and delta (δ), based on the specific substances that they bind and the resulting physiologic effects. All are G-protein coupled receptors and their activation through binding of an appropriate ligand leads to analgesia (pain relief). It is important in medicine.

OPISTHOGLYPHOUS: It describes venomous snakes that have their grooved fangs behind the row of normal teeth in the mouth. Such snakes are described as back- or rear-fanged snakes.

OPISTHOKONTS: An assemblage of eukaryotic organisms that includes the animal and fungus kingdoms. Flagellated protists such as choanoflagellates also belong to this group.

OPISTHOSOMA: The posterior portion of the body of an arachnid and other arthropods of the phylum Chelicerata, behind the prosoma (cephalothorax), which consists of those body segments that do not bear legs.

OPIUM: The dried latex extracted from the unripe fruit of the opium poppy (*Papaver somniferum*). It is an addictive narcotic, containing several alkaloids including about 12% morphine, a powerful natural painkiller and addictive narcotic and codeine.

OPPENHEIMER-VOLKOFF LIMIT: The upper limit on the mass of a neutron star, above which it undergoes gravitational collapse to a black hole. It is more difficult to estimate this limit than the analogous Chandraskhar limit for white dwarf stars. It is thought that the Oppenheim - Volkoff limit is between two and three times the mass of the Sun.

OPPORTUNISTIC ORGANISMS: Organisms that can quickly exploit new resources as they arise by rapidly colonising a new environment and sustaining themselves from various different food sources. Such species characteristically exhibit selection. If conditions are not favourable, for example, they can postpone reproduction or stay dormant, until conditions make growth and reproduction possible.

OPPOSABLE: The capability of being placed against the remaining digits of a hand or foot. An example is the ability of the thumb to touch the tips of the fingers on that hand.

OPPOSING REACTIONS: Composite reactions, occurring in forward and reverse directions.

OPPOSITE: 1. Altogether different, as in nature, quality or significance. 2. A term used in botany to describe a form of leaf arrangement in which the leaves arise in pairs at each node. If each pair is at right angles to the pairs above and below it, as is usually the case, the arrangement is termed **decussate**. If the leaf pairs all arise in the same plane a **distichous arrangement** results. 3. In mathematics, the side of a right-angle triangle that is not an arm of a given angle.

OPPOSITE ANGLE: 1. An angle in a triangle immediately opposite a given side. 2. **Opposite angles** are the pair of angles formed between the opposite directions of two intersecting lines and sharing a vertex but having no arm in common.

OPPOSITE NUMBER: A number that is the same distance from 0 on a number line, but on the opposite side of zero. In symbols, the opposite of a number n is $-n$. If n is a negative number, $-n$ is a positive number. For example, the opposite of -5 is 5. The sum of a number n and its opposite is zero; $n + -n = 0$.

OPPOSITE RING: In mathematics, a reverse ring, which is constructed from a given non-commutative ring but with the multiplication performed in the reverse order.

OPPOSITE SIDE IN A TRIANGLE: The side opposite an angle of a triangle that is not a side of the angle. In a right angle triangle, the hypotenuse is always the opposite side. It is the longest side in a right angle triangle.

OPPOSITE SIDES IN A QUADRILATERAL: Two sides in a quadrilateral that do not share a vertex.

OPPOSITE-CHANGE RULE: An addition method that may be used to do mental calculations. It involves adding a number to one addend (the numbers added together) and subtracting from the other addend.

OPPOSITIFOLIOUS: Having opposite leaves that are situated on the same node but are separated by the diameter of the stem.

OPPOSITION: The moment at which a planet having its orbit outside that of the Earth is in line with the Earth and the Sun. In particular, two planets are in opposition to each other when their ecliptic longitudes differ by 180.

OPSIN: The light-sensitive pigment that occurs in the rod cells of the retina. In the vertebrate eye, opsin combines with retinal to form rhodopsin. Five classical groups of opsins are involved in vision, mediating the conversion of a photon of light into an electrochemical signal, the first step in the visual transduction cascade.

OPSONINS: Antibodies in the blood that coat invading microorganisms so that phagocytes can bind them more efficiently, rendering them more susceptible to phagocytes such as macrophages and neutrophils. **Opsonisation** (**opsonization**) is the process in which this occurs. Acute phase response proteins also act as opsonins, as do complement proteins to make bacteria or other cells more susceptible to the action of phagocytes.

OPTHALMICS: Agents or substances containing properties that clean the eyes and treat eye-related diseases. They include herbs such as carrot, chrysanthemum flower, lycium fruit, elderberry, turmeric root, wild asparagus root and blueberry.

OPTIC: 1. Relating or belonging to the eyes, vision or near or situated in the eye. 2. Of or relating to the science of optics or optical equipment.

OPTIC CHIASMA (OPTIC COMMISSURE): The x-shaped nerve tract underneath the brain where some of the fibres from both optic nerves (CN II) of each eye cross over to the opposite side. It is located at the bottom of the brain immediately below the hypothalamus. Sensory information from right side of both the right and left eye is transmitted to the right visual cortex in the brain, while information from the left side of each eye is carried to the left visual cortex.

OPTIC NERVE (CRANIAL NERVE II): Either of the second pair of cranial nerves of vertebrates transmitting visual stimuli received by the rods and cones in the retina to the brain.

OPTICAL ACTIVITY (OPTICAL ISOMERS): The ability of certain substances to rotate the plane of polarisation of a beam of light as it passes through a crystal, liquid or solution. It occurs when the molecules of the substance are asymmetric, so they can exist in two different structural forms, each being a mirror image of the other. The two forms are optical isomers or enantiomers and occur in pairs that are non-superimposable mirror images of one another. The existence of such forms is also known as **enantiomorphism**. One form will rotate the light in a clockwise direction as one faces the light source and it is labelled positive or **dextrorotatory**. The counterclockwise rotation is negative or **levorotatory**. Molecules that show optical activity have no plane of symmetry. Glucose is found in the dextrorotatory form. The other isomer, (-) glucose, can be synthesised in the laboratory, but cannot be synthesised by living organisms.

OPTICAL AXIS (OPTIC AXIS): 1. Also known as principal axis, it is the line passing through the optical centre and the centre of a curvature of a lens or spherical mirror. 2. The direction in a doubly refracting crystal in which light is transmitted without double refraction. 3. An imaginary straight line passing through the midpoint of the cornea (anterior pole) and the midpoint of the retina (posterior pole).

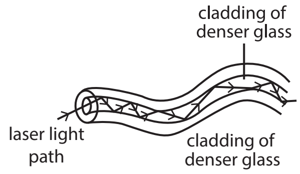
OPTICAL CENTRE: The point at the geometrical centre of a lens through which a ray of light entering the lens passes without deviation.

OPTICAL CHARACTER RECOGNITION, OCR (OPTICAL CHARACTER READER): A process for putting a text into a computer by means of a dominant reader in which a software attempts to identify characters by comparing shapes to those stored in the software library. It avoids the need to retype already printed material for data entry.

OPTICAL DENSITY (REFRACTIVE INDEX): A measure of a bending power of the medium as light passes through it. It is expressed as a ratio of the speed of light in vacuum to that in the considered medium. Light slows down as it passes from air or vacuum into any other medium. This causes the light to change its direction or be refracted. Refractive index (μ) = C/v, where C is the speed of light in vacuum and v is the speed of light in a medium. The statement that the refractive index of water is 1.33 means that light travels 1.33 times as fast in a vacuum than it does in water.

OPTICAL DISK: 1. The part of the retina where the axons of ganglion cells leave the eye to form the optic nerve. Because there are no light-sensitive rods or cones in the optic disk, this part of the eye is also known as the **blind spot**. 2. Also known as **optical media**; **optical storage**; or **optical disc drive** (**ODD**), it is any of a variety of information storage medium in which laser technology is used to record and read large volumes of digital data. They include CD-ROM, laser discs, digital versatile discs and DVDs and DVD-ROMs.

OPTICAL FIBRE: Flexible transparent and very fine optical pure glass fibre through which light can be reflected to transmit an image or information from one end to the other, with very little leakage through the side walls. Fibre optic systems use optical fibres to transmit information, in the form of coded pulses or fragmented images, from a source to a receiver. The optical fibre consists of a core of material with a refractive index higher than the surrounding cladding. They are used in medical instruments (endoscopes or fibrescopes) to examine internal body cavities, such as the stomach and the bladder, to transmit audio, video or data over long distances at high bit rates and also used, where a change in light transmission properties is used to sense or detect a change in some property, such as temperature, pressure or magnetic field.



Optical fibre

OPTICAL FILTER: A device which reduces the spectral range (bandpass, cut-off and interference filter) or radiant power of incident radiation (neutral density or attenuance filter) upon transmission of radiation.

OPTICAL FLAT: 1. A disk of high-grade quartz glass, approximately 2 centimetres thick that has very accurately polished surfaces so that the deviation from perfect flatness does not exceed 50 nanometres (0.05 micrometre). It is used in determinations of surface contour and in comparison of lineal measurement. 2. A plane surface, with deviations from a plane surface generally not exceeding one-tenth of a wavelength of light, used to redirect light in a telescope or other optical instrument.

OPTICAL GLASS: A type of clear homogeneous glass of known refractive index that is free from bubbles, strains and other imperfections, which can affect its transmission of light. It is used in the manufacture of lenses, prisms and other optical components.

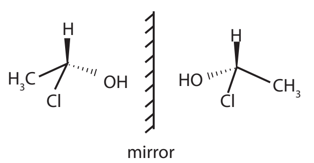
OPTICAL ILLUSION (VISUAL ILLUSION): Visually perceived images, scene or picture that differ from objective reality. There are three main types: (i) **literal optical illusions** that create images that are different from the objects that make them. (ii) **Physiological illusion** that are the effects on the eyes and brain of excessive stimulation of a specific type (brightness, tilt, colour, movement). (iii) **Cognitive illusions** where the eye and brain make unconscious inferences. An example of a natural optical illusion is that the Moon appears bigger when it is on the horizon than when it is high in the sky, owing to the refraction of light rays by the Earth's atmosphere.

OPTICAL INSTRUMENT: Single lenses or mirrors or combination of them used to produce images for some purpose, such as to form a real or virtual image, to form an optical spectrum or to produce light with a specified polarisation. Alternatively, it is the wavelength or an instrument that makes use of more lenses or mirrors or of a combination of these in order to change the path of light rays and produce an image. Examples are microscopes and telescopes.

OPTICAL INTERFEROMETER: An instrument that combines two separate light beams from the same object to form an interference pattern.

OPTICAL ISOMERISM: Existence of two forms of a molecule such that one is a mirror image of the other. The two molecules differ, in that, rotation of plain polarised light is equal but in opposite direction.

non-superimposable mirror images of 1-chloro-1-ethanol



Optical Isomerism

OPTICAL LATTICE: A periodic array produced by interference between laser beams propagating in opposite directions. The interference results in a periodic potential and is possible to trap atoms at minima of this potential. They are formed in two or three dimensions.

OPTICAL LEVER: An experimental device for measuring small angular displacements of a rotating body, for example, in a galvanometer or torsion balance. Typically, a small mirror is attached to the rotating object and a narrow fixed beam of light is directed onto a small mirror attached to the body and the reflected beam is directed onto a screen, producing a spot of light whose position is measured. The angle turned by the beam is twice the angle turned through by the mirror rotation of plane-polarised light by an optically active compound depending on the wavelength. A graph of rotation against wavelength has a characteristic shape showing peaks or troughs. This phenomenon is explained in the same way as optical activity.

OPTICAL MARK RECOGNITION (OPTICAL MARK READING; OMR): A process of scanning marks or characters, usually from hand-filled document forms such as surveys, reply cards, questionnaires, examination answer sheets and ballots in order for them to be processed.

OPTICAL MASER: A light amplifier usually used to produce monochromatic coherent radiation in the infrared, visible and ultraviolet regions of the electromagnetic spectrum or more simply, a laser that produces visible radiation. It is also sometimes simply called **maser**. Some common types of masers are the ammonia maser, rubidium maser and ruby maser. Masers uses include high precision frequency references, electronic amplifiers in radio telescopes, etc. Maser is the acronym of **m**icrowave **a**mplification by **s**timulated **e**mission of **r**adiation, which works on the same principle as the laser.

OPTICAL MICROSCOPE: An instrument used to produce an enlarged image of a small object by focusing light from a source. It consists of a light source, a condenser, an objective lens and an ocular or eyepiece, which can be replaced by a recording device.

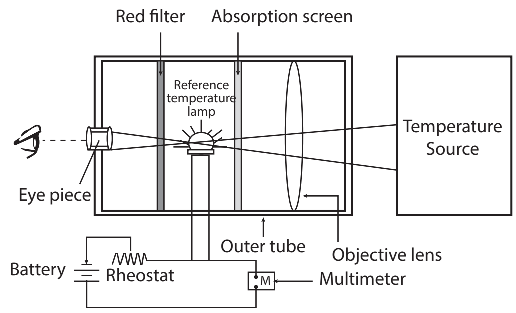
OPTICAL MOUSE: In computing, it is a device that controls the movement of the cursor or pointer on a monitor, using a small camera to take 1,500 pictures per second and emits light from a LED or laser. The light emitted is detected by a complementary metal-oxide semiconductor (CMOS) sensor as the mouse is moved. It was developed by Agilent Technologies and introduced in 1999.

OPTICAL PARAMETRIC AMPLIFICATION (OPA): Process of signal amplification by a parametric process and a pump wave.

OPTICAL PARAMETRIC OSCILLATOR (OPO): Light source similar to a laser, but based on 'non-linear optical gain' from parametric amplification rather than on stimulated emission.

OPTICAL PURITY: The ratio of the observed optical rotation of a sample consisting of a mixture of enantiomers to the optical rotation of one pure enantiomer.

OPTICAL PYROMETER: An instrument which determines the temperature of a very hot surface from its incandescent brightness; the image of the surface is focused in the plane of an electrically heated wire and current through the wire is adjusted until the wire blends into the image of the surface.



Optical pyrometer

OPTICAL SPECTROSCOPY: The study of systems by the electro-magnetic radiation with which they interact or that they produce. By convention, spectroscopic properties are expressed in terms of wavelength *in vacuo*, not the wavelength within the medium being studied.

OPTICAL TELESCOPE: An instrument that collects and focuses visible radiation from a distant object to produce an image of it or enables the radiation to be analysed. The three primary types of optical telescope are: **refractors** (dioptrics) which use lenses, **reflectors** (catoptrics) which use mirrors and **combined lens-mirror systems** (Catadioptrics) which use lenses and mirrors in combination. Examples of **combined lens-mirror system** are the Maksutov telescope and the Schmidt camera.

OPTICAL TEMPERATURE: The surface temperature of a celestial body as calculated by Stefan's law, assuming that the body behaves as a black body.

OPTICAL TRAP: A device for trapping small particles using electromagnetic radiation.

OPTICAL TWEEZERS: A device that is used to manipulate particles such as atoms or molecules. It works by a highly-focused laser beam to provide an attractive or repulsive force to physically hold and move microscopic dielectric objects and has been successful in studying a variety of biological systems.

OPTICAL YIELD: In a chemical reaction involving chiral reactants and products, the ratio of the optical purity of the product to that of the precursor, reactant or catalyst. The optical yield is not related to the chemical yield of the reaction.

OPTICALLY ACTIVE POLYMER: Polymer capable of rotating the polarisation plane of a transmitted beam of linear-polarised light.

OPTICALLY ACTIVE: Of some substances, capable of rotating the plane of polarisation of light that passes through them to the right (dextrorotatory) or left (levorotatory). Some substances are optically active in any state of aggregation. Examples of such substances are sugars, camphor, and tartaric acid. Some, such as quartz and cinnabar are active only in the crystal phase.

OPTICS: 1. A branch of physics that deals with production and propagation of light, the changes it undergoes and produces and closely related phenomena, for example, shadows and mirror images, lenses, microscopes, telescopes and cameras. 2. A term describing parts of the body related to the eye and its functions.

OPTIMAL: In mathematics, relating to, having or being a value that is an optimum (maxima or minima).

OPTIMAL CONTROL (CONTROL THEORY): The branch of mathematics and engineering that is relevant to the control of certain physical processes and systems and finding solutions to them, as well as cost effectiveness. In other words, it deals with the design, identification and analysis of systems with a view towards controlling them, in order to perform specific tasks or make them behave in a desired way.

OPTIMAL FORAGING THEORY: A theory in ecology stating that natural selection favours animals whose behavioural strategies maximise their net energy intake per unit time spent foraging. Such time includes the time spent both searching for prey and handling it.

OPTIMISATION (OPTIMIZATION): The determination of optimal values (maximum or minimum value) of a function, often subject to constraints. To **optimise** is to find the optimal values of a function.

OPTIMISATION METHODS: A procedure for finding the maxima or minima functions, generally, of several variables. It is most often encountered in statistics in the context of maximum likelihood estimation, where such methods are frequently needed to find the values of the parameters that maximise the likelihood.

OPTIMISATION THEORY: The branch of mathematics that is concerned with the analysis and solution of problems in mathematical programming. Some of its fields are game theory, linear programming, Markov chains, network analysis, control theory and calculus of variations.

OPTOACCOUSTIC SPECTROSOCOPY (THERMOACOUSTIC EFFECT): A spectroscopic technique in which a periodically interrupted beam of light generates sound in a gas through which it is passing. The energy in the light beam is firstly transformed into internal motions of the gas molecules, then into random translational motions of these molecules or heat and finally into periodic pressure fluctuations or sound. The principle underlying optoacoustic spectroscopy is that the absorbed electromagnetic radiation is converted into motion which is associated with the production of sound waves.

OPTOELECTRONICS: A branch of electronics concerned with the development of devices that modulate, transmit, respond and sense electromagnetic radiation in the gamma rays, x-rays, ultraviolet and infrared, in addition to visible light. Optoelectronic devices are electrical-to-optical or optical-to-electrical transducers or instruments that use such devices in their operation.

OPTOMETRY: The professional practice consisting of examination of the eyes to evaluate health and visual abilities, diagnosis of eye diseases and conditions of the eye and visual system and provision of necessary treatment by the use of eyeglasses, contact lenses and other functional, optical, surgical and pharmaceutical means.

OPUNTIA (PRICKLY PEAR): A genus in the cactus family, Cactaceae that grows in arid and semi-arid zones with stems that grow in distinctly jointed segments. The cacti have areoles that bear glochids, usually small to minute, barbed spines that are very sharp and brittle and very easily detached. These spines are modified leaves to reduce excessive water loss. *Opuntia ficus-indica*, commonly known as Indian fig opuntia, barbary fig, cactus pear, spineless cactus or prickly pear is the most commercially important species cultivated for its sweet fruit called tuna, which can be used to make jams and jellies; the young cladodes are used as a vegetable; as an ornamental; and for other uses such as fodder crop. It is a slow growing perennial shrub that grows up to 3-5 metres high. It is a xeromorphic plant with flat, round cladodes armed with spines. The flat green stem is succulent to conserve water and also takes over the photosynthetic functions of the leaves. Its fruit is succulent, reddish, ellipsoid and about 7 centimetres long. Another important species is *Opuntia aciculata* (*Opuntia engelmannii* var. aciculata) with the common names Chenille prickly pear, old man's whiskers and cowboy's red whiskers, a perennial dicot with stems which are jointed into circular fleshy greenish pads. It is grown for its edible fruits and as an ornamental tree and for its pads that are used as a vegetable. Examples of other species include *Opuntia monacantha*, commonly known as drooping prickly pear and *Opuntia stricta* (common prickly pear).

OR: The word "OR" is a connective word used in logic. The sentence "p OR q" is false only if both p and q are false: it is true if p or q or both are true. The symbol "$\vee$" is often used to represent OR. An OR sentence is also called a **disjunction**. The operation of OR is illustrated by the truth table shown in the diagram.

p	q	p v q
T	T	T
T	F	T
F	T	T
F	F	F

The operation of OR illustrated by the truth table

OR CIRCUIT: A logic circuit having two or more input wires and one output wire that gives a high-voltage output signal if one or more input signals are at a high voltage. It is used extensively as a basic circuit in computers.

OR RULE: The probability of a combination of mutually exclusive events, expressed as P(A or B) = P(A) + P(B).

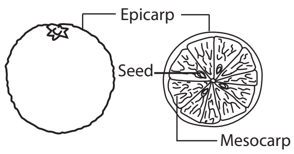
ORAL: A term describing parts of the body and functions related to the mouth.

ORAL CONTRACEPTIVE: Any hormonal preparation taken in the form of a pill to prevent conception. The most common form is the combined pill which contains an oestrogen and progestogen. Both act to suppress ovulation. The progestogen additionally causes changes in the viscosity of cervical mucus and alters the lining of the womb, both of which decrease the chances of fertilisation should ovulation occur. The 'minipill' contains only progestogen and has fewer side effects than the combined pill.

ORAL GROOVE: An indentation in the cytostome found in certain protozoa such as paramecium and aquatic invertebrates through which food is swept in by the cilia.

ORAL HYGIENE: The maintenance of healthy conditions in the human mouth by brushing and flossing to prevent tooth decay and gum disease.

ORANGE: Any of several species of small trees or shrubs in the genus *Citrus*, grown in tropical and subtropical regions. It is a thick-skinned juicy edible fruit with a reddish-yellow colour when ripe. Products from orange include essential oils, pectin, candied peel orange marmalade and stock feed.



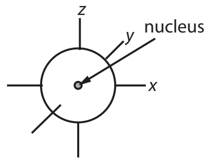
Orange, Citrus spp.

ORBICULAR: Almost circular and flattened, as are the leaves of the fringed water lily (*Nymphoides peltata*).

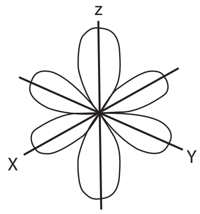
Orbicular Leaf

ORBIT: 1. In mathematics, the orbit of a mapping, f at a point, x is the sequence of points generated by repeated composition of f at x, expressed as $\{f^{[n]}(x) : n = 0, 1, 2, ...\}$, where $f^{[n]} = f \circ f \circ ... \circ f$ (n times) and $f \circ$ is the identity function. If this set is finite, the orbit is closed. The closure of the orbit will, under reasonable conditions, contain a fixed point of f. The term is especially used for permutations.

2. In group theory, the set of products, under an action of groups on a non-empty set, of all elements of the group with a given element of the set, expressed as $x^G = \mathrm{orb}_G(x) = (gx{:}\ g \varepsilon G)$. The distinct orbits of a set form a partition of the set. 3. The trajectory of an ordinary differential equation. 4. The path of an electron as it travels around the nucleus of an atom. 5. A path of one body in space revolving about another, such as the orbit of the Earth around the Sun or the Moon around the Earth. 6. Either of the two sockets in the skull of vertebrates that house the eyeballs.



S-orbital



P-orbital

ORBITAL: Atomic region around the nucleus of an atom or, in a molecule, around several nuclei in which there is a 95% probability of finding an electron. They are designated by a combination of numerals and letters such as 1*s*, 2*p*, 3*d*, 4*f*. The shapes of s and p orbitals are shown in the diagram.

ORBITAL CURVE: The trace of an orbit on a flattened map of Earth or another celestial body about which a spacecraft is orbiting. Each successive orbit is displaced by the amount of rotation of the body between each orbit.

ORBITAL ELECTRON: An electron which has a high probability of being in the vicinity (at distances on the order of 10^{-10} metre or less) of a particular nucleus, but has only a very small probability of being within the nucleus itself.

ORBITAL ENERGY: The sum of the potential energy and the kinetic energy of an object in orbit.

O

ORBITAL MUSCLES: Six muscles which are responsible for eye movement. Four of the orbital muscles move the eye up, down, left and right. The other two control the twisting motion of the eye when the head is tilted. Defects in these muscles and the nerves that control them may lead to conditions such as nystagmus and amblyopia.

ORBITAL PERIOD: The time taken for a body to go once around a closed orbit. The orbital period of a planet is described as a year.

ORBITAL PLANE: The plane defined by the motion of an object about a primary body. The position and velocity vectors lie within the orbital plane, whereas the angular momentum vector is at right angles to the orbital plane.

ORBITAL QUANTUM NUMBER (AZIMUTHAL QUANTUM NUMBER; SECOND QUANTUM NUMBER): The atomic orbital quantum number l, which determines the angular momentum of the electron. It is the second of a set of four quantum numbers which describe the unique quantum state of an electron. The possible values of l are (n – 1), (n – 2) ...1, 0. Thus in the first shell (n = 1) the electrons can only have angular momentum zero. In the second shell (n = 2), the value of l can be 1 or 0, giving rise to two sub shells of slightly different energy.

ORBITAL RESONANCE: A state in which the orbital period of one body is related to that of another by a simple integer fraction, such as $\frac{1}{2}, \frac{2}{3}, \frac{3}{5}$.

ORBIT STABILISER THEOREM: In mathematics, the theorem that for a group G that acts on a non-empty set X, the cardinality of the orbit of an element of X is the index of the stabiliser of that element in G.

ORBITAL STEERING: A concept expressing that the stereochemistry of approach of two reacting species is governed by the most favourable overlap of their appropriate orbitals.

ORBITAL SYMMETRY: The behaviour of an atomic or localised molecular orbital under molecular symmetry operations characterises its orbital symmetry. For example, under a reflection in an appropriate symmetry plane, the phase of the orbital may be unchanged (symmetric) or it may change sign (antisymmetric), for example, the positive and negative lobes are interchanged.

ORBITAL VELOCITY (ORBITAL SPEED): The speed of a planet, a satellite, a spacecraft or other bodies travelling in an orbit around the Earth or around some other celestial body.

ORBITER: The portion of a spacecraft that orbits a celestial body, such as a planet.

ORCHARD: A planted field of fruit trees such as citrus and cashew.

ORCHIDS: Perennial herbs which lack permanent woody structure, belonging to class Monocotyledoneae of the genus orchis, family Orchidaceae, a diverse or widespread family of flowering plants with 880 genera and about 26,049 species. There are epiphytic orchids and terrestrial orchids. They are easily distinguished from other plants by the fact that they show two growth patterns: monopodial and sympodial. They have highly modified petals (labellum), fused stamens and carpel and extremely small seeds. The leaves are simple with parallel venation, though some have reticulate venation. The leaves may be orbiculate, ovate or lanceolate. They are colourful and exude fragrance. They are mostly cultivated in the tropics and subtropical regions. The dried seedpods of some species of orchids are used as flavouring in baking, for perfumes manufacture and aromatherapy.

ORCHIOID: Resembling an orchid.

ORDER: 1. In biology, it is a taxonomic rank classification of a group of related families or a category used in the classification that consists of one or several similar or closely related families or unit. It is used in the classification of living organisms consisting of one or more families. Similar orders form a class. 2. In chemistry, it is the expression for the rate of a chemical reaction, the sum of the powers of the concentrations is the overall order of the reaction. For instance, in a reaction A + B → C, the rate equation may have the form R= k $[A][B]^2$. This reaction would be described as first order in A and second order in B. The overall order is three. The order of a reaction depends on the mechanism and it is possible for the rate to be independent of concentration (zero order) or for the order to be a fraction. 3. In mathematics, it is the number of times a variable is differentiated: dy/dx represents a first order derivative and dy^2/dx^2 is second order derivative. 4. In mathematics, the position of elements in a sequence. For example, in an ordered set, the order of elements makes a difference and $\langle a, b\rangle = \langle b, a\rangle$ if and only if $a = b$. 5. The number of rows or columns in a square matrix or determinant. 6. The number of elements of a finite group or set; its cardinality. 6. In mathematics, also called **period**, for an element a in a group, it is the smallest number of times the given element has to be multiplied by itself to yield the identity element of the group; the least positive integer n such that $a^n = e$. It is written $|a| = n$. If $\langle a\rangle$ is the set generated by a, then $|a| = |\langle a\rangle|$. If no such integer exists the element has infinite order. 7. The multiplicity of a zero or a pole. 8. The number of poles, counting multiplicity, in any fundamental parallelogram of a doubly periodic function, such as an elliptic function. 9. In physics, it is a category of phase transition. 10. In physics, it is a category of phase transition.

ORDER COMPLETE (COMPLETE; DEDEKIND COMPLETE): Of a partially ordered set, such that every subset has a supremum and infimum, for example, the reals are not complete but the interval [0,1] is. A partially ordered set is order complete for every countable non-empty set S of F that has an upper bound in F, a supremum exits in F and for countable non-empty set S of F, which has a lower bound in F, the supremum exits in F, an infimum exits in F.

ORDER IDEAL: In mathematics, denoted O(m), it is the ideal of a commutative ring with identity defined as the set of elements, r, of the ring such that $rm = 0$, where m is a given member of an R-module.

ORDER IRRELEVANCE: When counting a set of objects, it does not matter where one starts or in what order one counts, as long one counts every object once and only once.

ORDER NOTATION: A notation for indicating comparisons between the elements of one sequence and those of another or between two functions on a set. It is useful for replacing inequalities by equalities.

ORDER OF ACCURACY: The closeness of an approximate answer to the exact answer.

ORDER OF CONVERGENCE (RATE OF CONVERGENCE): The speed at which a convergent sequence approaches its limit. It is often measured by the number of terms or evaluations involved in obtaining a given accuracy. Convergence of a sequence subject to the condition, for $p > 1$ that $\frac{x_{n+1} - x}{x - x^p} = O(1)$ as n increases is called p^{th}-order convergence. An example is quadratic convergence when $p = 2$.

ORDER OF MAGNITUDE (ORDER): A class of scale or magnitude of any amount, where each class contains values of a fixed ratio to the class preceding it expressed in powers of ten. Orders of magnitude are generally used to make very approximate comparisons. If two numbers differ by one order of magnitude, one is about ten times larger than the other.

ORDER OF REACTION (REACTION ORDER): The classification of chemical reactions based on the power to which the concentration of components of the reaction is raised in the rate law. The overall order is all the exponents of the terms expressing concentrations of the molecules or atoms determining the rate of the reaction. For example, given a chemical reaction 2A + B → C which occurs at the rate determining step, the rate equation is given as r = k[A]2[B] and the reaction order with respect to A would be 2 and with respect to B would be 1, the total reaction order would be $2+1=3$. The order of reaction can be zero or even fractional. Reaction orders can be determined only by experiment.

ORDER OF ROTATIONAL SYMMETRY: The number of positions a shape can take, when rotated and still look the same. For example, a five point star can be rotated about its centre in five different ways and looks exactly like the same. The star is said to have a rotational symmetry of order 5.

ORDER OF OPERATIONS: In mathematics, rules that guide the order of simplifying expressions. It is the order in which we add, subtract, multiply or divide to solve a problem. They are as follows: (i) simplify all of the parentheses, including all forms of grouping symbols, such as brackets and braces; (ii) simplify all exponents; (iii) simplify all multiplication and division starting from left to right; and (iv) simplify all addition and subtraction, starting from left to right.

ORDER PARAMETER: A measure of the degree of the order of a phase of a system below its transition temperature. An order parameter has a non-zero value below the transition temperature and a zero value above the transition temperature.

ORDER STATISTIC: The order statistics of a random sample $X_1, \ldots, X_n$ are the sample values placed in ascending order and are usually independent and identically distributed. It is denoted by $X_{(1)}, \ldots, X_{(n)}$. In other words, they are random variables that satisfy

$X_{(1)} \leq X_{(2)} \leq \ldots \leq X_{(n)}$.

ORDER-DISORDER TRANSITION: A transition in which the degree of order of the system changes. Three principal types of disordering transitions may be distinguished: (i) positional disordering in a solid, (ii) orientational disordering which may be static or dynamic and (iii) disordering associated with electronic and nuclear spin states.

ORDER-PRESERVING MAP: In mathematics, it preserves the pointwise order among convex functions and is invertible, such that its inverse also preserves such order. An order-preserving map from a poset L to a poset M is a function f such that $\forall x, y \in L: [x \geq y \Longrightarrow f(x) \geq f(y)]$.

ORDERED PAIR: 1. A pair of numbers, in which one number is distinguished as the first number and the other as the second number of the pair. In the ordered pair (a, b), the object a is called the first entry and the object b the second entry of the pair. Alternatively, the objects are called the first and second coordinates or the left and right projections of the ordered pair. The order in which the objects appear in the pair is significant, for example, the ordered pair (a, b) is different from the ordered pair (b, a) unless $a = b$. 2. **Ordered coordinate pairs** are a set of number pairs (x, y) that indicates the position of a point on a graph, where x represents the number of units left or right of the origin and y represents the number of units up or down from the origin.

ORDERED SET: 1. In mathematics, a sequence of elements, usually enclosed in angle brackets, that is distinguished from the other sequences by the identity and order of the elements, so that $\langle a, b\rangle$ is not identical with $\langle b, a\rangle$ unless $a = b$. 2. Also known as **partially ordered set** or **poset**, it is a set endowed with an ordering.

ORDERED STRUCTURE: In mathematics, a structure such as a group, field or vector space having an ordering that preserves the underlying operations. Thus an ordered field is a field together with an order that is such that for $a < b$ and any c, one has $a + c < b +$ c and for $a < b$ and any $0 < c$ one has $ac < bc$.

ORDERED VECTOR SPACE: A vector space having a partial ordering that respects addition and positive multiplication operations. An example of ordered vector space is the real numbers with the usual order.

ORDERING: In logic, any of a number of categories of relations that permit at least some members of their domain to be placed in order. A **linear** or **simple ordering** is reflexive, antisymmetric, transitive and connected (complete) and thus enables every member to be ordered relative to every other; for example, "less than or equal to" on the integers. A **partial ordering** is reflexive, antisymmetric and transitive and thus generates chains of comparable elements; members of distinct chains may be incomparable, as with set inclusion. Either of these orderings is strong or strict if it is asymmetric instead of reflexive and antisymmetric, such as strictly less than or proper set inclusion. An ordering is a **well-ordering** if every non-empty subset has a least member under the ordering, i.e. a unique member that has the given relation to all other members of the subset. A **pre-ordering** or **quasi-ordering** is reflexive and transitive. The **reverse ordering** sets a less than (or equal to) b exactly when b is less than (or equal to) a.

ORDINAL: In mathematics, a set in which every member is also a subset (a transitive set) that contains only transitive elements. This can be used to generate the transfinite sequence, ø, {ø}, {ø, {ø}}, {ø, {ø}, {ø, {ø}}}, ...; the members of this sequence can be regarded as the canonical members of the sequence of ordinal numbers and therefore abbreviated 0, 1, 2, 3, ...

ORDINAL NUMBER: Any one of the series 1st, 2nd, 3rd, 4th, ... Ordinal numbers relate to order and the number of its elements, whereas cardinal numbers (1, 2, 3, 4 ...) relate to quantity or counting.

ORDINAL SCALE: In statistics, a scale on which data are shown simply in accordance with some order, in the absence of appropriate units of measurement. For example, a measure of consumers' satisfaction with a product may include "very dissatisfied," "somewhat dissatisfied," "somewhat satisfied," or "very satisfied." The items in this scale are ordered, ranging from least to most satisfied. Ordinal scales, therefore allow comparisons of the degree to which two subjects possess the dependent variable.

ORDINAL VARIABLE: A measurement that allows a sample of individuals to be ranked with respect to some characteristics but where differences at different points of the scale are not necessarily equivalent. For example, anxiety might be rated on a scale of none, mild, moderate and severe, with the values 0, 1, 2, 3, being used to label the categories. A patient with anxiety score of one could be ranked as less anxious than one with a given score of three, but patients with scores 0 and 2 do not necessarily have the same difference in anxiety as patients with scores 1 and 3.

ORDINALLY SIMILAR: In mathematics, of two relations, such that there is a one-to-one correspondence between their domains that preserves order under the given relations.

ORDINARY DIFFERENTIAL EQUATION (ODE): Equation containing total derivative but without partial derivative. For example, an ordinary differential equation in $x(t)$ might involve x, t, dx/dt, d^2x/dt^2 and perhaps other derivatives.

ORDINARY POINT: 1. In mathematics, a non-isolated point of a curve at which the curve has a smooth tangent and does not cross itself; a point that is not a singular point. 2. A point a of a second-order differential equation $y''+P(x)y'+Q(x)y-0$ for which both P and Q are analytic around a.

ORDINARY RAY (O RAY): The ray in a uniaxial crystal which vibrates in the basal plane and has a spherical ray velocity surface.

ORDINATE: The y-coordinate of a point, which is the vertical distance of the point from the horizontal or x-axis.

ORDINATION: A technique used in ecological work to relate the composition of different stands of vegetation to each other. It is used by those who consider that no two stands of natural vegetation are ever the same and thus they cannot be classified into distinct associations. Certain properties of each stand, namely data on the species present, are plotted against one or more axes, each axis representing some environmental gradient. The end result should be an arrangement in which the different vegetation stands are ordered in a manner that best reflects their similarities and differences.

O

ORDOVICIAN: The second period of geological time in the Palaeozoic era, about 500-440 million years ago. There are no known terrestrial organisms in this period.

ORE: A naturally occurring substance from which an element can be extracted or an aggregate such as a body of rock or a deposit of sediments mined for the economically valuable mineral it contains. It includes metallic minerals and nonmetallic substances such as sulfur, calcium fluoride (fluorite) and barium sulfate (barite).

OREGONATOR: A theoretical model for a type of chemical reaction mechanism that causes an oscillating reaction. It is the type of mechanism responsible for the B-Z reaction and involves five steps of the form:

$A + Y \rightarrow X$, $X + Y \rightarrow C$, $A + X \rightarrow 2X + Z$, $2X \rightarrow D$ and $Z \rightarrow Y$.

ORGAN: A collection of tissues joined in structural unit to perform one or a number of functions in a multicellular organism. Examples in mammals are the eyes, stomach, liver, heart and in plants, leaves, roots and flowers.

ORGAN CULTURE: The culture of complete living organs of animals or plants outside the body in a suitable culture medium in which the organ culture is able to accurately model functions of an organ in various states and conditions by the use of the actual in vitro organ itself. Animals' organs must be small enough to allow the nutrients in the culture medium to penetrate all the cells. Whole plant roots and even root systems can be kept alive in such conditions for a considerable period of time. The techniques for organ culture can be classified into (i) those employing a solid medium and (ii) those employing liquid medium.

ORGAN OF CORTI: The organ in the inner ear of mammals (cochlea) in which nerve cell endings which are sensitive to or respond to sound vibrations are located. It projects into the cochlear duct and consists of two membranes that run parallel to each other. Sensory hair cells, rooted in the basilar membrane, are in contact with the overlying tectorial membrane. It contains between 15,000-20,000 auditory nerve receptors. During the transmission of sound waves the basilar membrane vibrates, causing the sensory hairs to flex against the tectorial membrane. This results in the production of impulses, which are transmitted to the brain via the auditory nerve. The pinna and middle ear act as mechanical transformers, so that by the time sound waves reach the organ of corti, their pressure amplitude is 22 times that of the air impinging on the pinna.

ORGANELLE: One of several minute membrane-bound, discrete and specialised structures in a living cell that has a specific function and is usually separately enclosed within its own lipid bilayer. They are s found in eukaryotic cells and include mitochondria, chloroplast, lysosomes, ribosomes and the nucleus.

ORGANIC: 1. A substance containing chains of carbon linked to atoms of hydrogen produced by or found in plants or animals. 2. Plants grown without the use of synthetic pesticides, fungicides or inorganic fertilisers and prepared without the use of preservatives.

ORGANIC ACIDS: Organic compounds with acidic properties, the most common ones being carboxylic acids, whose acidity are associated with their carboxyl group –COOH. They are weak acids with pH values ranging from as low as 3 for carboxylic to as high as 9 for phenolic. The relatively stronger acids are the sulfonic acids containing the group SO2OH. Organic acids are derived from biological origin and they contain carbon atoms while inorganic acids originate from inorganic compounds and minerals. Citric and malic acids are the primary organic acids found in fruit juices.

ORGANIC AEROSOL: Aerosol particles consisting predominantly of organic compounds, mainly carbon (C), hydrogen (H), oxygen (O), and lesser amounts of other elements. **Secondary Organic Aerosols** (SOAS) are a major component of atmospheric particulates, emitted from natural and man-made sources; produced through a complex interaction of sunlight, volatile organic compounds from trees, plants, cars or industrial emissions, and other airborne chemicals. SOAs have been found to cause lung and heart problems and other health effects.

ORGANIC BASE: Organic compound which acts as a base and can donate a pair of electrons to form a bond. Examples are an amine, pyridine, imidazole, benzimidazole, histidine and phosphazene bases.

ORGANIC CHEMISTRY: A branch of chemistry that deals with the scientific study of the structure, preparation, properties and reactions of carbon-based compounds, except oxides of carbon, carbonate and carbide. Organic compounds form the chemical basis of life and are more abundant than inorganic compounds. Substances such as drugs, vitamins, plastics, carbohydrates, protein and fat consist of organic molecules and these are studied to determine their structures for benefits to mankind.

ORGANIC COMPOUNDS: Compounds containing the carbon that are typically found in all living organisms or one of the classes of compounds whose behaviour is influenced by the chemistry of carbon. The major organic compounds are carbohydrates, fats, proteins, nucleic acids and vitamins.

ORGANIC DYE LASER: A liquid laser in which the active medium is an organic dye in a solvent.

ORGANIC EVOLUTION: The process by which modification in the genetic makeup of population of organisms occur in response to environmental changes. Organic evolution includes two major processes: **anagenesis**, which is the alteration of the genetic properties of a single lineage over time; and **cladogenesis** or **branching** in which a single lineage splits into two or more distinct lineages that continue to change anagenetically.

ORGANIC FARMING: The form of agriculture without the use of synthetic fertilisers, food additives or pesticides, plant growth regulators such as hormones or other agrochemicals. In place of synthetic fertiliser, compost, manure, crop rotation, seaweed, biological pest control or other substances derived from living things are used.

ORGANIC FOOD: Crops or livestock that have been grown without the application of synthetic fertilisers or pesticides and without using genetically modified organisms.

ORGANIC LIGHT-EMITTING DIODE (OLED): A type of light-emitting diode (LED) that uses an organic substance as the semiconductor. The organic material used is often films of polymer sandwiched between two conductors that emits light when an electric current is applied, and therefore does not require a backlight. OLED displays are either passive-matrix organic light-emitting diode (PMOLED) or active-matrix organic light-emitting diode (AMOLED). AMOLED has a storage capacitor that makes the line pixels light all the time allowing for higher resolution and larger display sizes. Passive-organic organic light-emitting diode (PMOLED) does not contain a storage capacitor and so the pixels in each line are actually off most of the time and needs more power to make them brighter. OLEDs are used for digital displays in devices such as television screens, computer monitors, mobile phones, and personal digital assistant (PDA). OLEDs enable emissive, bright, thin, flexible and efficient displays and are replacing liquid crystal displays (LCDs) in most display applications.

ORGANIC MATTER (ORGANIC MATERIAL): Dead plant or animal tissue that originates from living sources such as plants, animals, insects and microbes capable of decay or the product of decay.

ORGANIC PHOSPHORUS: Phosphorus present as a constituent of an organic compound. It occurs in a variety of organic compounds including nucleic acids and lipids.

ORGANISER (ORGANIZER): In biology, it is any part of the embryo which causes changes to occur in another part through induction, thus directing development and differentiation. The primary organiser, blastopore lip or archenteron roof directs the overall development of the gastrula. A secondary organiser directs the development of a subsidiary part or tissue. An example is the induction of the medullary plate by the dorsal lip of the blastopore.

ORGANISM: Any contiguous living system that exhibits life processes such as respiration, excretion, reproduction and response to stimulus. Examples are animal, plant, fungus or micro-organisms such as bacteria and viruses. It may be single-celled or unicellular such as bacteria or composed of many cells or multicellular such as humans.

ORGANOGENESIS: The formation of internal organs during embryonic development. In animals, this begins following the rearrangement of the cell at gastrulation, when the germ layers are fully formed in their correct position. Internal organs initiate development in humans within the 3rd to 8th weeks in utero. Dividing cells of the gastrula begin to differentiate and the rudimentary organs and organ systems begin to form.

ORGANOHETERYL GROUPS: Univalent groups containing carbon, which are thus organic, but which have their free valence at an atom other than carbon.

ORGANOMERCURIAL FUNGICIDE: An organic fungicide containing mercury. Most of this group are phenylmercury derivatives, for example, phenylmercuric acetate and are used as seed dressings. Mercury-containing fungicides are highly toxic to mammals and treated grain must be clearly labelled or even dyed to distinguish it from edible grain.

ORGANOMETALLIC COMPOUND: A chemical compound in which a metal is directly bound to carbon in an organic group. Most organometallic compounds are solids, although some are liquids and others are gases. Examples of organometallic compounds are compounds in which the carbon atoms of organic group are bonded directly to a metal atom through either a σ- or π- bond. For example, $C_6H_5Ti\ (OC_5H_7)_3$ is an organometallic compound because a metal (Ti) is bonded directly with carbon. Ferrocene, $Fe(C_5H_5)_2$, is a stable compound in which an iron atom is sandwiched between two hydrocarbon rings. Organometallic compounds containing tin are used as pharmaceuticals, pesticides and fire retardants.

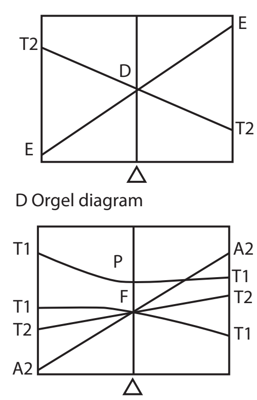
ORGANOPHOSPHATE INSECTICIDE: A group of organic compounds containing phosphorus, which is very toxic to humans as they cause the irreversible inhibition of the cholinesterase enzymes that break down acetylcholine. An example is malathion, used to kill aphids and other insects such as fruit flies.

ORGANOPHOSPHOROUS COMPOUND: A general way of referring to all organic compounds in which at least one hydrogen atom has been replaced by a phosphorous containing compound. Usually the phosphorus should be directly bonded to a carbon on the organic residue. The term also applies to organic compounds in which the phosphorus is not directly bonded to the organic residue as in phosphate esters which occur in adenosine triphosphate (ATP). Organophosphorous compounds are used in pesticides such as malathion and parathion, often used to control aphids, mealybugs, mites, etc.

ORGANOSILICON COMPOUND: A chemical compound in which silicon is directly bound to carbon in an organic group. The first organosilicon compound, tetraethylsilane was discovered by Charles Friedel and James Crafts in 1863 by reaction of tetrachlorosilane with diethylzinc. They are used as water repellents, lubricants and sealants.

ORGASM: The climax of sexual excitement in human, which in males coincides with ejaculation. A sense of physiological and emotional release is accompanied by a feeling of extreme pleasure. Orgasm is controlled by the involuntary or autonomic, limbic system and is accompanied by quick cycles of muscle contraction in the lower pelvic muscles, which surround the primary sexual organs and the anus.

ORGEL DIAGRAMS: Diagrams showing how the energy levels of a transition-metal atom split when it is placed in a ligand field. The vertical axis shows the energy and the horizontal axis shows the strength of the ligand field, with zero ligand field strength at the centre of the horizontal axis. They are named after Leslie Orgel, who developed them in 1955. Orgel diagrams are restricted to only show weak field (i.e. high spin) cases and offer no information about strong field (low spin) cases. Because Orgel diagrams are qualitative, no energy calculations can be performed from these diagrams. Also, they only show the symmetry states of the highest spin multiplicity instead of all possible terms, unlike a Tanabe-Sugano diagram. Orgel diagrams, however, show the number of spin allowed transitions, along with their respective symmetry designations. In an Orgel diagram, the parent term (P, D or F) in the presence of no ligand field is located in the centre of the diagram, with the terms due to that electronic configuration in a ligand field at each side. For the D Orgel diagram, the left side contains d^1 and d^6 tetrahedral and d^4 and d^9 octahedral complexes. The right side contains d^4 and d^9 tetrahedral and d^1 and d^6 octahedral complexes. For the F Orgel diagram, the left side contains d^2 and d^7 tetrahedral and d^3 and d^8 octahedral complexes. The right side contains d^3 and d^8 tetrahedral and d^2 and high spin d^7 octahedral complexes.



D Orgel diagram

F and P Orgel diagram

Orgel diagrams

ORIENTABLE SURFACE: A surface is orientable if a consistent concept of clockwise (or anticlockwise) rotation can be defined on the surface in a continuous manner. An orientable surface contains no subset that is homeomorphic to the Möbius band. Examples of orientable surfaces include the sphere, the torus and the tori of higher genus. Every orientable surface has an even Euler characteristic and can be embedded in three-space. When mapped into a Euclidean space, an orientable surface has two distinct sides.

ORIENTATION: 1. In biology, it is the change in position of the whole or part of an organism in response to a stimulus, such as light. 2. In chemistry, it is the relative dispositions of atoms, ions or groups in molecules or crystals. 3. In mathematics, the position of a geometric figure relative to a coordinate system; especially, the sense of a directed line. Thus a simple closed curve in the plane may be given either a clockwise or an anticlockwise orientation. In three dimensions, a set of coordinate axes may form a left- or right-handed trihedron and orientation is reversed in a mirror.

ORIFICE: An opening, hole or vent.

ORIGAMI: The Japanese art of paper-folding, without cutting and joining, to form definite objects.

ORIGIN: A fixed point of reference in a coordinate system, in which the values of the Cartesian coordinates are zero.

ORIGIN OF ELEMENTS: The nuclear process that gives rise to chemical elements. There is not one single process that can account for all the elements. The abundance of the chemical elements is determined not just by the stability of the nuclei of the atoms but also how readily the nuclear processes leading to the existence of these atoms occur. Most of the helium in the universe was produced by fusion in the early universe when the temperature and the pressure were very high. Most of the elements between helium and iron were made in nuclear fusion reactions inside stars. Since iron is at the bottom of the energy valley of stability, energy needs to be put into a nucleus heavier than iron for a fusion reaction to occur. Inside stars, some heavy elements are built up by the s-process, where s stands for slow in which high-energy neutrons are absorbed by a nucleus, with the resulting nucleus undergoing beta decay to produce a nucleus with a higher atomic number. Some heavy elements are produced by the r-process, where r stands for rapid, which occurs in supernova explosion when a great deal of gravitational energy is released when a large star that has exhausted its nuclear fuel collapses to a neutron star, with the release of a very large number of neutrons.

ORIGIN OF LIFE: The process by which living organisms developed from inanimate matter which is generally thought to have occurred on Earth between 3,500 and 4,000 million years ago.

ORIGIN OF REPLICATION (ORI): A sequence of DNA at which replication is initiated on a chromosome, plasmid or virus.

ORIMULSION: Fuel made by mixing bitumen and water, which can be burnt in the same way as heavy oil.

ORION: A very prominent constellation in the equatorial region of the sky, identified with the hunter of the Greek mythology.

ORMOLU: A gold coloured alloy made up of copper, zinc and sometimes tin, used for furniture decoration, jewellery and mouldings.

ORNITHINE (ORN): An amino acid with the chemical formula $H_2N(CH_2)_3CH(NH_2)COOH$ that plays a role in the urea cycle. It is not a constituent of proteins but is important in living organisms as a metabolic intermediate in the synthesis of urea.

ORNAMNTAL PLANTS: Plants that are grown for decorative purposes in gardens, landscape design projects, as houseplants, for cut-flowers and specimen display. The cultivation of these, called **floriculture** forms a major branch of horticulture.

ORNITHOLOGY: The branch of zoology concerned with the scientific study of birds relating to their structure, classification and their habits, songs, flight and value to agriculture as destroyers of insect pests.

ORNITHOPHILY: The pollination of flowers by birds. It is an important form of pollination for many tropical species, the most common pollinators being hummingbirds (in South America) and sunbirds (in Africa). Bird-pollinated flowers are often red in colour. Pollination by insects (entomophily) is believed to be derived from ornithophily.

OROGENY (OROGENESIS): The formation of mountains, brought about by the movements of the rigid plates making up the Earth's crust. Orogenies may result from continental collisions, the under thrusting of continents by oceanic plates, the overriding of oceanic ridges by continents and other causes.

OROPHILOUS: Thriving in mountainous areas. This is usually used to describe plants.

ORPIMENT: A natural yellow mineral form of arsenic (III)sulfide with the chemical formula AS_2,S_3. Crystals are small, tabular and rarely distinct. The mineral occurs more commonly in foliated or columnar masses. The name is also used for the synthetic compound, which is used as a pigment.

ORRERY: A mechanical device that illustrates the relative positions and the motions of the heavenly bodies.

ORTHANT: Any of the eight regions into which three-space is divided by the coordinate planes; especially the positive orthant consisting of all vectors with non-negative coordinates. Similarly in n-dimensional space, the positive orthant consists of all vectors with non-negative coordinates.

ORTHIC TRIANGLE: The triangle whose vertices are the feet of the altitudes of a given triangle.

ORTHO: 1. Prefix indicating that one of three possible isomers of a benzene ring has two substituted groups in the 1.2 positions (i.e. on adjacent carbon atoms). The abbreviation o- used, for example, as o-dichlorobenzene is 1,2 dichlorobenezene. 2. Prefix formerly used to indicate the most hydrated form of an acid, H_3PO_4, called **orthophosphoric acid** to distinguish it from the lower **metaphosphoric acid**, HPO_3 (which is actually $(HPO_3)n$). 3. Prefix denoting the form of diatomic molecules in which nuclei have the same spin direction, for example orthohydrogen.

ORTHO ACIDS: Hypothetical compounds having the structure $RC(OH)_3$, for example, hydrated forms of carboxylic acids. Orthocarbonic acid, $C(OH)_4$, is generically included.

ORTHO AMIDES: Hypothetical compounds having the structure $RC(NH_2)_3$ and *N*-substituted derivatives thereof. $(R_2N)_4C$ are generically included, but such use is obsolescent.

ORTHO ESTERS: Compounds having the structure $RC(OR')_3$ ($R' \neq H$) or the structure $C(OR')_4$ ($R' \neq H$), for example, $HC(OCH_3)_3$ trimethyl orthoformate, $C(OCH_3)_4$ tetramethyl orthocarbonate.

ORTHO- AND PERI-FUSED: Polycyclic compounds in which one ring contains two and only two, atoms in common with each of two or more rings of a contiguous series of rings. Such compounds have *n* common faces and less than 2*n* common atoms.

ORTHO-FUSED: Polycyclic compounds in which two rings have two and only two, atoms in common. Such compounds have *n* common faces and 2*n* common atoms.

ORTHOCENTRE: The point of intersection of the three altitudes of a triangle. All three altitudes always intersect at the same point. The orthocentre is not always inside the triangle. If the triangle is obtuse, it will be outside. To make this happen the altitude lines have to be extended so that they cross.

ORTHOCHROMATIC: 1. Any spectrum of light that is devoid of red light. 2. Photographic emulsion that is sensitive to only blue and green light and can only be processed with a red safe light. Using it, blue objects appear lighter and red ones darker because of increased blue sensitivity.

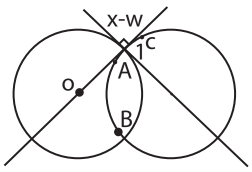
ORTHOCLADOUS: Having straight branches.

ORTHOCLASE: A common mineral in igneous rocks, consisting of potassium aluminium silicate ($KAlSi_2O_8$). It occurs in the form of vitreous crystals of various colours. Orthoclase is one of the end-members (pure compounds) of the feldspar group.

ORTHODONTICS: The branch of dentistry that is concerned with the irregularities of the teeth for aesthetic and clinical reasons, for example, by using braces.

ORTHOGENESIS (ORTHOGENETIC EVOLUTION; PROGRESSIVE EVOLUTION; AUTOGENESIS): The hypothesis of the nature of evolutionary change, which proposed that organisms evolved along particular paths predetermined by some factor in their genetic make-up rather than the external forces of natural selection.

ORTHOGONAL (ORTHOGRAPHIC): 1. In mathematics, of two lines, planes or curves, perpendicular at their point of intersection; they are at right angles to each other. 2. Of two vectors in an inner product space, having the property that the inner product is zero. 3. Of a pair of Latin squares, such that each pair of corresponding symbols occurs exactly once.



Orthogonal circles

ORTHOGONAL CIRCLES: Two circles that intersect at right angles to each other.

ORTHOGONAL COMPLEMENT: In mathematics, of a vector or set of vectors, the subspace of vectors orthogonal to the given vector or set. For example, in a Euclidean vector space, the orthogonal complement of x = (1, 0, 0) has as a basis y = (0, 1, 0) and z = (0, 0, 1). Equivalently, in three-dimensional Euclidean geometry, the y-z plane is the orthogonal complement of the x-axis.

ORTHOGONAL CURVES: Curves or lines that meet at right angles or 90°.

ORTHOGONAL FUNCTIONS: In mathematics, a set of functions $f_1, \ldots, f_n$ that may be infinite, of which every pair of distinct functions satisfies the identity $\int_a^b f_i(x)\, f_j(x)\,dx = 0$ on some given range of integration $]a, b[$ or for a more general set and measure. The Legendre polynomials are orthogonal on]-1, 1[.

ORTHOGONAL GROUP: In mathematics, the group of n x n orthogonal matrices over the reals, denoted O(n). It is a group and a manifold and therefore a Lie group. The matrices form a group because they are closed under multiplication and taking inverses.

ORTHOGONAL LINES: Lines that meet at 90°.

ORTHOGONAL MATRIX: A matrix whose transpose is the same as its inverse. A square matrix A is orthogonal, if A^TA = I, where A^T is the transpose of the matrix A and I is the identity matrix. It has the following properties: (i) if A is orthogonal $A^{-1} = A^t$, then AA^t = I; (ii) the determinant of an orthogonal matrix, i.e. det(A) = ±1; (iii) if two matrices A and B are orthogonal matrices of the same order, then their product is also orthogonal, i.e. AB is orthogonal.

ORTHOGONAL PROJECTION: A geometrical drawing that produces an image on a line or plane by perpendicular lines crossing the plane. If A is the point and A' its image, this means that AA' will be orthogonal or at right angles to the line or plane.

ORTHOGONAL SEQUENCE: In mathematics, a sequence of polynomials such as the Legendre polynomials with the property that any two different polynomials in the sequence are orthogonal to each other under some inner product.

ORTHOGONAL VECTORS: Vectors that are perpendicular, for example, if x and y are non-zero vectors and y = 0, then x and y are orthogonal.

ORTHOLOGOUS: Homologous genes that are derived from descendants of a single gene in a common ancestor. Thus, when a lineage splits to form two new species any gene gives rise to two orthologues, which may subsequently diverge in their DNA sequence and function.

ORTHOMORPHIC MAP (CONFORMAL MAP; ANGLE-PRESERVING TRANSFORMATION; BIHOLOMORPHIC MAP; CONFORMAL TRANSFORMATION): A transformation that preserves the angles between curves. An analytic function is conformal at any point where it has a nonzero derivative. An example is the Mercator map, a two-dimensional representation of the surface of the Earth that preserves compass directions. Conformal mapping is extremely important in complex analysis, as well as in many areas of physics and engineering.

ORTHONORMAL: In mathematics, of two vectors in an inner product space, having the properties that the inner product is zero and the product of any element with itself is unity.

ORTHONORMAL SEQUENCE: An orthogonal sequence of vectors, often functions, all of unit norm. The standard basis in Euclidean space is orthonormal.

ORTHOPAEDICS: The branch of surgery concerned with disorders of the spine and joints and the repair of deformities of these parts. Methods range from the use of splints, physiotherapy and manipulation, to surgical correction of deformity, fixing of fractures and refashioning or replacement of joints; also suture or transposition of tendons, muscles or nerves is performed.

ORTHOPTERA (ORTHOPTEROIDEA): A large order of generalised insect orders comprising the grasshoppers, locusts, crickets, stick insects, praying mantids and in some classification systems, the cockroaches. They are characterised by gradual metamorphosis, enlarged hind legs, modified for jumping and biting mouthparts and produce sounds by stridulation. The crickets and long-horned grasshoppers have long threadlike antennae and stridulate by rubbing together modified veins on their forewings.

ORTHOPTEROUS: Belonging to the order Orthopthera, a large order of insects. Insects in this order possess leathery forewing and membranous hind wings. Hind legs are adapted for leaping. Examples are crickets, locusts and grasshoppers.

ORTHOPYROXENE: A group of pyroxene minerals that have crystallised in the orthorhombic system. Examples include enstatite and hypersthene.

ORTHOSTICHOUS: Having parts that are arranged in straight rows or ranks.

ORTHOTROPIC: Exhibiting an essentially vertical growth habit.

ORTHOTROPISM: A tropism, which is orientated directly in line with the stimulus concerned. An example is the vertical growth of main stems and roots in response to gravity (orthogeotropism). **Tropism** is the directional growth movement of a plant or plant part that occurs in response to an external stimulus from a specific direction.

ORTHOTROPOUS (ATROPOUS): A term used to describe a form of ovule orientation in the ovary in which the ovule develops in an upright position from the placenta. This arrangement is rare but may be found in some members of the Polygonaceae.

OSBORNE'S RULE: A rule that is used to transform relations between trigonometric identities and hyperbolic functions. It states that trigonometric identities can be transformed into the corresponding identities for hyperbolic functions by multiplying out fully, replacing all trigonometric functions with their hyperbolic analogues and then changing the signs of any terms involving the product of two hyperbolic sines. For example, given the identity cos(x - y) = cos$x$cos$y$ + sin$x$sin$y$ the rule permits gives the corresponding identity cosh(x - y) = cosh$x$cosh$y$ - sinh$x$sinh$y$.

OSCILLATE: 1. In mathematics, of a function, sequence or series, not to tend either to a finite limit or to infinity or else to do so in such a way that every neighbourhood of the limit contains both values greater than and values less than it. For example, the sequence ⟨0, 1, 0, 1, 0, ... ⟩ and the series $1-\frac{1}{2}+\frac{1}{4}-\frac{1}{8}+\ldots$ oscillate; the function $x\sin x^2$ oscillates, but is not periodic. 2. In physics, to vary in magnitude or position in a regular manner about a central point.

OSCILLATING REACTION (CLOCK REACTION): A complex mixture of reacting chemical in which the concentrations of the products and reactants change periodically, either with time or with position in the reacting medium. Thus, the concentration of a component may increase with time to maximum, decrease to a minimum, then increase again and so on, continuing the oscillation over a period of time. They are a class of reactions that serve as an example of non-equilibrium thermodynamics, resulting in the establishment of a nonlinear oscillator.

OSCILLATING UNIVERSE: A closed-universe model in which the gravitational attraction of the mass within the universe will eventually slow down causing the expansion of the galaxies to reverse, eventually resulting in a 'Big Crunch' where all the matter in the universe would be contracted into small volume of high density. The theory suggests that the universe would repeatedly expand and collapse through alternate Bangs and Big Crunches.

OSCILLATING WATER COLUMN (OWC) OCEAN-WAVE ENERGY CONVERTER: A type of turbine built on land that uses the kinetic energy of waves to force air in and out of a turbine to generate electrical energy.

OSCILLATION: 1. In mathematics, a measure of the spread of a bounded set, the supremal difference between pairs of elements, defined as the difference between the supremum and infimum of the set. The oscillation of a function on an interval is also known as irs **saltus**. 2. In physics, the repetitive motion, typically in time, past a position of equilibrium or a central neutral position. Examples include a swinging pendulum and AC power. If the output voltage or current has the form of a sine wave with respect to time, the device is called a **sinusoidal** or **harmonic oscillator**. If the output voltage changes abruptly from one level to another such as a square wave or saw tooth wave, it is called **relaxation oscillator**. Oscillations occur not only in physical systems but also in biological systems and in human society.

OSCILLATOR: An electronic device that generates a periodic output, often a sinusoid or a square wave or any device producing a desired oscillation (vibration). Examples of their uses in electronic circuits are to produce and decode radio signals, to produce the scanning signals for television tubes and to convert signals from transducers into a readily transmitted form.

OSCILLOGRAPH: An instrument for displaying or recording the values of rapid variations between two states, electrical or mechanical.

OSCILLOSCOPE (CATHODE RAY OSCILLOSCOPE, CRO): An electronic instrument, used to measure voltages that vary over time and to display the waveforms of electrical oscillations or signals, by means of the deflection of a beam of electrons. Oscillocopes are used to display a changing electric signal as a picture on the screen of a cathode-ray tube. Because they can convert almost any physical phenomenon into a corresponding electric voltage, they are used widely in scientific and engineering applications such as acoustic research, television-production engineering and electronics design.

OSCINES: Members of the suborder Oscines, of the order Passeriformes, comprising songbirds that have highly developed vocal organs.

OSCULATE: To share the same tangent and curvature and at a given point, as in the case of two osculating curves. For example, the curve $y = x^2$ and the x-axis osculate at the origin.

OSCULATING CIRCLE (CIRCLE OF CURVATURE): The circle that touches a curve on the concave side and has the same tangent and curvature as a given curve at a given point with its radius equal to the radius of curvature at the point. For a function $y = f(x)$, the formula for the curvature and radius of curvature (its radius), at a point $[x, f(x)]$ on the curve $y = f(x)$, the circle of curvature is tangent to the curve and has a radius $r(x)$. The radius of curvature is normal to the concave side of the curve at the point and is equal to the reciprocal of its curvature.

OSCULATING PLANE: The plane in a Euclidean space or affine space spanned by the unit tangent and unit principal normal vectors (intersection of the osculating plane with the normal plane) to a space curve at a given point; the curvature of the curve is evaluated in this plane. The unit binormal is the normal vector to the osculating plane.

OSCULATION (TACNODE): In mathematics, a point at which two osculating circles to the curve at that point have a common tangent, when both branches extend on both sides of the point.

OSCULUM (OSCULA): 1. The mouth-like excretory aperture in the body wall of a sponge through which the current of water leaves the body cavity. 2. Any of the suckers on the head of a tape worm, by which it attaches itself to the gut wall of its host.

OSMIRIDIUM (IRIDOSMINE): A hard, white, naturally occurring mineral, which is an alloy consisting principally of osmium (17 to 48%) and iridium (49%), with small inclusions of platinum, rhodium and other metals. It is used in producing pen nibs, needles and wearing points.

OSMIUM: A hard, blue-white metallic transition element with the chemical symbol Os, atomic number 76 and atomic weight 190.2. It is found associated with platinum. It is used in certain alloys with platinum and iridium. Osmium forms a number of complexes in a range of oxidation states. In density of the elements, it is exceeded only by iridium and its uses include pen points, light bulb filaments and as a catalyst.

OSMIUM DIOXIDE (OSMIUM(IV) OXIDE): A poisonous, volatile, black or yellow brown crystalline solid with a characteristic pungent smell with the chemical formula OsO_2, molar mass 222.229 g/mol, density 11.4 g/cm^3 and decomposes at its melting point of 500 °C (773.15 K). It is made by heating osmium in air or by the reduction of osmium tetroxide. Its aqueous solutions are used as catalyst in organic reactions. It is used as a stain to fix biological material, especially lipids.

OSMIUM TETROXIDE (OSMIC ACID; (OSMIUM(VIII) OXIDE): A crystalline oxide of osmium with a pungent smell and a poisonous irritating vapour with the chemical formula OsO_4 and forms monoclinic crystals. It has a density of 4.9 g/cm^3, melting point 40.25 °C (313.40 K) and a boiling point of 129.7 °C (402.8 K). It is formed slowly when osmium powder reacts with oxygen at ambient temperature. It is used as a catalyst, oxidising agent and biological fixative and stain.

OSMOCONFORMER: A marine invertebrate whose body fluids are in osmotic balance with its environment. The osmolarity and ionic concentrations of their body fluids are similar to those of the seawater in which they live. Osmoconformers maintain their internal salinity balance with the surrounding seawater by actively regulating their internal concentrations of amino acids, ions and proteins to match the osmolarity of the environment. Such animals avoid the need to expend much energy on osmoregulation.

OSMOLALITY: Quotient of the negative natural logarithm of the rational activity of water and the molar mass of water. Its symbol is m.

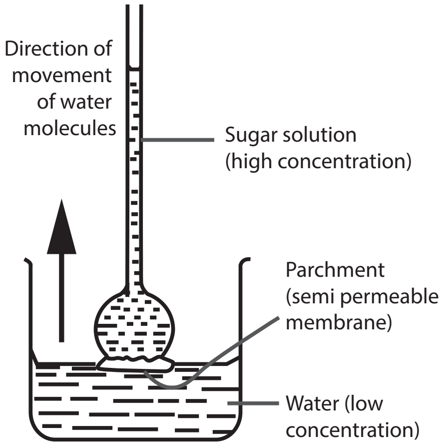
OSMOLYTE: Any organic compound, soluble in the solution within a cell or in the surrounding fluid such as plasma osmolytes. It protects cells from desiccation by maintaining a high intracellular osmolality. Osmolytes are active in the process of osmoregulation and are of a particular use in the kidney, where cells are exposed to highly concentrated fluids. For example, when a cell swells due to external osmotic pressure, membrane channels open and allow efflux of osmolytes which carry water with them, restoring normal cell volume. Compounds that are known as osmolytes include polyols, amines, certain amino acids and urea.

OSMOMETER: A device used to measure osmotic pressure of a solution, colloid or compound. It is useful in determining the concentration of dissolved salts or sugars in blood or urine samples and for determining the molecular weight of unknown compounds and polymers.

OSMORECEPTOR: A sensory receptor situated in the hypothalamus of most homeothermic organisms that detects changes in osmotic pressure, contributing to fluid balance in the body. When the osmotic pressure of blood changes, water diffusion into and out of the osmoreceptor cells changes. It expands when the blood plasma is more dilute and contracts when the concentration is higher. This results in the release of antidiurectic hormone and the subsequent conservation of water, thereby maintaining the homeostasis of the body fluids.

OSMOREGULATOR: An active regulation of the osmotic pressure of an organism's fluids to maintain the homeostasis of the organism's water content. It keeps the organism's fluids from becoming too diluted or too concentrated. Vertebrates and some aquatic invertebrates, especially freshwater invertebrates, expend energy on osmoregulation, which involves getting rid of metabolic wastes and other substances such as hormones that would be toxic if allowed to accumulate in the blood via organs such as the skin and the kidneys. It keeps the amount of water and dissolved solutes in balance to maintain cell volumes and achieve optimum conditions for metabolism.

OSMOSIS: The transfer of water molecules from a region of higher water concentration to a region of lower water concentration through a semi-permeable membrane OR the transfer of water molecules from a region of lower solute concentration to a region of higher solute concentration through a semi-permeable membrane OR a process by which molecules of a solvent, usually water, diffuse from a solution of low concentration to a solution of a higher concentration through a semi-permeable membrane. The distribution of water in living organism is dependent to a large extent on osmosis. If a solution is separated from a pure solvent by a membrane that is permeable to the solvent but not to the solute, the solution will turn to become more dilute by absorbing the solvent through the solute.



An experimental set up to show the process of osmosis

OSMOTIC COEFFICIENT: Quantity characterising the deviation of the solvent from ideal behaviour referenced to Raoult's law. Its symbol is Φ.

OSMOTIC CONCENTRATION (OSMOLARITY): Product of the osmolality and the mass density of water. Its symbol is *c*.

OSMOTIC POTENTIAL: A difference in osmotic pressure that draws water from an area of less osmotic pressure to an area of greater osmotic pressure. Osmotic potential in plants is a factor which affects water movement, hence the amount of dissolved salts in soil. Soils high in soluble salt tend to reduce uptake of water by plant roots and seeds. The osmotic potential and pressure potential combined make up the water potential of a plant cell. If there are two cells next to each other of different water potentials, water will move from the cell with the higher water potential to the cell with the lower water potential. Water enters plant cells from the environment via osmosis. Water moves because the overall water potential in the soil is higher than the water potential in the roots and plant parts. If the soil is desiccated then there will be no net movement into the plant cells and the plant will die.

OSMOTIC PRESSURE: The pressure required to prevent the flow of a solvent into a solution through a semi-permeable membrane or pressure that must be applied to stop osmosis from happening. It can be demonstrated in an osmometer. This process is of vital importance in biology as the cell's membrane is selective towards many of the solutes found in living organisms.

OSMOTROPH: Any heterotrophic organism that obtains nourishment by absorbing dissolved organic from its surroundings.

OSSICLES (AUDITORY OSSICLES): The three small bones, namely the malleus, incus and stapes in the mammalian middle ear, which form a link between the eardrum (tympanum) and the oval window (fenestra ovalis). The ossicles transmit and amplify vibrations of the eardrum (tympanum) across the middle ear to the oval window (fenestra ovalis), which transfers them to the inner ear through the fluid-filled labyrinth or cochlea. The absence of the ossicles leads to a moderate-to-severe hearing loss.

OSSICULUS: The stone or pit of a drupe; a pyrene.

OSSIFICATION (OSTEOGENESIS): The process in which new bone material is formed in vertebrate animal by special cells called **osteoblasts** that secrete layers of extra cellular matrix on the surface of the existing cartilage. There are two processes resulting in the formation of normal, healthy bone tissue: intra-membranous ossification and endochondral ossification. In **intra-membranoeus** ossification, bones are formed directly in connective tissue. In **endochondral ossification**, the bones are formed by the replacement of cartilage.

OSTEICHTHYES: Class of bony fishes of the subphylum Vertebra (Craniata) comprising the bony fishes, marine and fresh water fishes with a bony skeleton. Osteichthyes is divided into the ray-finned fish, **Actinopterygii** and the lobe-finned fish, **Sarcopterygii**. All have gills covered with bony operculum and a layer of thin overlapping bony scales covers the entire body surface. Bony fish have a swim bladder, which acts as a hydrostatic organ enabling the animals to remain suspended in the water at any depth. In some fish, this bladder acts as a lung.

OSTEITIS DEFORMANS (PAGET'S DISEASE): A chronic disease caused by excessive breakdown and formation of bone tissue that can cause bones to weaken. It commonly affects the pelvis, lumbar spine (lower back), skull and long bones. It normally occurs in people of over 40 years. Depending on which part of the body is affected, signs and symptoms include bone pain, headaches and hearing loss, damage to cartilage of joints and hip pain. Paget's disease is named after English surgeon James Paget (1814-1899).

OSTEOARTHRITIS: A degenerative condition in which the protective cap of cartilage at the end of bones wears away. It may progress into the bone itself, causing pain and stiffness, restricting movement.

OSTEOBLAST: Any of the cells found in bone that secretes osteoid, which is composed mainly of Type I collagen and other substances responsible for bone formation and also for mineralisation of the osteoid matrix. They are derived from osteoprogenitor cells in the bone marrow. They eventually become osteocytes.

OSTEOCLAST: Any of the cells found on the surface of bone that are involved in the removal of mineralised bone matrix and breaking up the organic bone to enhance the further development and restructuring of bones during growth and repair. Osteoclasts are formed by the fusion of cells of the monocyte-macrophage cell line.

OSTEOCYTE: Any of the star-shaped cells found in the bone that performs cellular activities such as respiration and exchange of materials with the blood that are required for maintenance of bone tissues. Osteocytes are formed, when osteoblasts become trapped in the matrix they secrete.

OSTEOID: A soft material consisting mainly of collagen that is secreted by osteoblast and constitutes the uncalcified organic portion of the bone matrix that forms prior to the maturation of bone tissue.

Osteoid develops into hard bone matrix when it combines with calcium phosphate deposited from the blood.

OSTEOLOGY: A branch of anatomy concerned with the structure, morphology, disease, pathology, function and development of bones.

OSTEOMALACIA: A condition in which the bones become soft due to lack of calcium or phosphate. It is caused by inadequate calcium in the diet, phosphorus or vitamin D deficiency and renal disease. The bone becomes weak and brittle resulting in disorders such as pain and muscle cramps, bone deformity and a tendency to spontaneous fracture.

OSTEOMYELITIS: A bacterial infection of bone, usually caused by staphylococcus, streptococcus or salmonella carried to the bone by blood or gaining access through open fractures. Osteomyelitis commonly affects children, causing fever and local pain. If untreated or partially treated, it may become chronic with bone destruction and a discharging sinus.

OSTEONECTIN: A glycoprotein in bone secreted by osteoblasts during bone formation. It binds to collagen and is involved in the formation of growth sites for calcium phosphate crystals required for forming the hard bone matrix.

OSTEOPATHY: A system of treatment based on theory that disease arises from the mechanical and structural disorder of the body skeleton. Prevention and treatment are practiced by manipulation, often of the spinal column. Osteopathy can play an important role in the treatment of both chronic and cute musculo-skeletal pain.

OSTEOPOROSIS: A bone disorder characterised by loss of bone density, causing skeletal weakness and brittleness when the mineral portion of the bone is lost. It occurs most commonly in post-menopausal women.

OSTEOSARCOMA: A malignant tumour of bone that rapidly spreads to the lungs and, less commonly to other areas. Osteosarcoma occurs mainly in adolescents, when the body is developing rapidly and in the elderly.

OSTIOLE: A small pore found in the reproductive bodies of certain algae and fungi, through which spores are discharged.

OSTIUM (OSTIA): 1. Any of the small openings or pores in the body wall of a sponge through which water enters the body cavity. 2. Any one of the apertures in the arthropod heart through which blood enters from the haemocoel.

OSTRACODERM: An extinct agnathan; a fishlike creature encased in an armour of bony plates.

OSTRACUM: The calcified portion of an invertebrate's shell. While the organism is living, the ostracum is covered by layers of protein forming a periostracum (the thin outer organic covering of a mollusc shell).

OSTRICH: A large flightless bird, *Struthio camelus* that lives in the dry grasslands of Africa. The ostrich makes up the family Struthionidae in the order Struthioniformes. The largest of all birds, it is 2-3 metres tall, has a long thin neck with a small flat head and weighs more than 150 kg. Its egg weighs about 1.4 kg. They are fast runners and can attain a speed of about 65 km/h. Ostriches are reared for their meat and plumes, which are made into hats and dresses; but demand for the plumes is now negligible. Ostrich hide is used as luxury leather.

OSTWALD PROCESS: A process for the industrial production of nitrogen oxide and nitric acid by the oxidation of ammonia.

OSTWALD RIPENING: A process used in crystal growth in which, over time, small crystals or sol particles dissolve and redeposit onto larger crystals. The large crystals grow and the small crystals disappear. The reason behind this is that the smaller crystals are associated with higher energy. When they dissolve the heat associated with this, higher energy is released, enabling recrystallisation to occur on the large crystals. Ostwald ripening is used in applications such as photography, requiring crystals with specific properties. This phenomenon was first observed by Russian-German physical chemist, Wilhelm Ostwald in 1896.

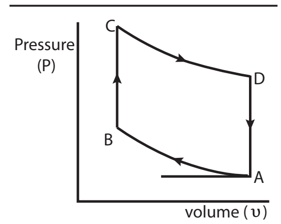
OSTWALD, FRIEDRICH WIHELM (1853-1932): A Russian-German physical chemist, who was highly influential in the development of physical chemistry as a subject. In 1887 he went to Leipzig where he worked on a wide range of topics including hydrolysis. In 1888 he discovered Ostwald's law of dilution of an electrolyte. He gave the first modern definition of a catalyst in 1894. He was awarded the 1909 Noble Prize in Chemistry for his work on catalysis. His process for the conversion of ammonia to nitric acid was of great industrial importance.

OTG (USB OTG; USB ON-THE-GO): An extension of the Universal Serial Bus (USB) that enables peripheral devices to be connected directly to each other without the use of a computer. For example, a mobile phone can act as a host, allowing other external peripheral devices such as flash drives, mice or keyboards to be connected to it.

OTOLITH (STATOCONIUM; OTOCONIUM): A gelatinous structure containing a high concentration of particles of calcium carbonate, which form part of the macula of the inner ear. The otolith organs are made up of the saccule and utricle. The utricle is sensitive to a change in horizontal movement and the saccule gives information about vertical acceleration such as when in an elevator.

OTOSCOPE: An instrument for examining the outer ear and eardrum. The otoscope has a light, a lens and a conical earpiece. Examining the ear can reveal the cause of symptoms such as an earache, the ear feeling full or hearing loss.

OTTO ENGINE (FOUR-STROKE ENGINE): A large stationary single-cylinder internal combustion four-stroke engine designed by Nikolaus Otto (1832-1891), German inventor. The phases of suction, compression, combustion, expansion and exhaust occur sequentially in a four-stroke-cycle or two-stroke-cycle reciprocating mechanism. It was a low-RPM machine and only fired every other stroke due to the Otto cycle, also designed by Otto. In the diagram, it is compressed adiabatically, AB and fired at B after which the pressure and temperature are increased rapidly over a constant volume BC. The piston then moves out, causing adiabatic expansion, CD after which the exhaust valve opens reducing the pressure to the atmosphere, DA. The next inward motion of the piston sweeps over the exhaust gas to complete the cycle.



Otto cycle

OUABAIN (G-STROHANTHIN): A white toxic glycoside with molecular formula $C_{29}H_{44}O_{12}$ derived from the seeds of certain African plants, such as *Strophantus gratus* and *Acokanthera ouabaio*. It is used as a drug to stimulate contractions of heart muscle in treating heart failure and other conditions and as an arrow poison by certain African people. It blocks the action of sodium/potassium adenosine triphosphate ATPase, a crucial membrane-bound transport protein that maintains differential ion concentrations between a cell's interior and external environment. Used widely in biomedical research, it is now known to occur naturally in the body at very low concentrations.

OUNCE: Unit of weight in the avoirdupois system, which is one sixteenth of a pound (avoirdupois). The avoirdupois ounce is equal to 28.35g and the troy once is equal to 31.1g. In volume measurements, one fluid once in British Imperial system is equal to 28.413 ml and in the U.S. system it is equal to 29.57 ml.

OUT OF AFRICA HYPOTHESIS: The theory that modern human populations (*Homo sapiens*) are all derived from a single speciation event that took place in a restricted region in Africa. Genetic and fossil evidence, have shown that archaic *Homo sapiens* evolved anatomically to modern humans between 200,000 and 100,000 years ago in Africa. One branch was supposed to have left Africa by 60,000 years ago and over time, replacing earlier human populations such as Neanderthals and *Homo erectus*. The concept was first noticed and published by Charles Darwin in 1871. It was only speculative until the 1980s, when it was corroborated by a study of present-day mitochondrial DNA, combined with evidence based on physical anthropology of archaic specimens.

OUTBOX: A term used to describe the location or folder, where e-mail is temporarily stored before being sent.

OUTBREEDING (CROSSING): The mating between unrelated or distantly related individuals of a species. Outbreeding populations have a greater potential for adapting to environmental changes and often producing a hybrid of superior quality. Outbreeding increases the number of heterozygous individuals; therefore recessive characteristics tend to be masked by dominant alleles.

OUTCOME: A possible result of an event that depends on probability. For example, "heads" or "tails" are the two possible outcomes of flipping a coin.

OUTCROSSING: The process of introducing new and unrelated genetic variation into a flock usually by introducing a male. This process tends to reduce genetic abnormalities and the probability of all individuals being subject to inherited diseases because of increase diversity. Another advantage is that in line-breeding, outcrossing brings in vigour and fertility to the stock. An **outcross** is a cross between individuals of different strains. For example, in plants, the transfer of pollen from the anthers of one plant to the stigma of another.

OUTER CORE: The liquid layer of the Earth's core that lies beneath the mantle and surrounds the inner core. It is fluid layer about 2,200 km thick composed of mostly iron and nickel. Its outer boundary lies 2,890 km beneath Earth's surface.

OUTER EAR (EXTERNAL EAR): The part of the ear external to the eardrum consisting of the pinna, concha and auditory meatus. It is present in mammals, birds, some reptiles and consists of a tube (the external auditory meatus). It gathers sound energy and focuses it on the eardrum or tympanium. For frequencies around 3 kHz, the outer ear selectively boosts the sound pressure 30 to 100 times, making humans most sensitive to frequencies in this range. Humans are prone to ear damages and loss of hearing at this frequency.

OUTER ORBITAL COMPLEX: A metal coordination compound in which the metal ion *d* orbital used in forming the coordinate bond is at the same energy level as the *s* and *p* orbitals.

OUTER MEASURE: 1. In mathematics, also called **Caratheodory outer measure**, it is a set funtion constructed preliminary to and sharing many of the properties of, a measure. Consider an outer measure μ defined on the class $P(X)$ of subsets of a metric space (X,d), μ is a Caratheodory outer measure, if $\mu(A\cup B)=\mu(A)+\mu(B)$ for every pair of sets A,$B\subset X$ which has positive distance $\inf[d(x,y):x\in B)>0]$. A theorem due to Caratheodory shows then that the Borel sets are μ-measurable. 2. **Lebesgue outer measure** is the particular outer measure of a set in Euclidean n-space computed by taking the infimum of the sum of the volumes (content) of any Lebesgue covering of the set by countable families of open finite order intervals (box): $\mu^*(E)=\inf\{\sum\left|I_n\right|: E \text{ lies in} \cup I_n\}$. The restriction of this function to the Lebesgue measurable subsets defines Lebesgue measure in n-space. **Jordan outer measure** is an analogous measure defined using only finite covers.

OUTER PLANET: Any planet whose orbit lies outside the asteroid belt. They are Jupiter, Saturn, Uranus, or Neptune.

OUTER SPACE: The void that exists beyond any celestial body including the Earth. It is not completely empty or a perfect vacuum, but contains a low density of particles, predominantly hydrogen plasma, as well as electromagnetic radiation, magnetic fields and neutrinos. Theoretically, it also contains dark matter and dark energy.

OUTER-SPHERE ELECTRON TRANSFER: A reaction in which the electron transfer takes place with no or very weak (4 – 16 kJ mol^{-1}) electronic interaction between the reactants in the transition state. It involves electron transfer from reductant to oxidant with the coordination shells or spheres of each staying intact. In other words, it is a reaction in which a small change takes place in the coordination spheres of the redox centres. If the donor and the acceptor exhibit a strong electronic coupling, the reaction is described as **inner-sphere electron transfer**. The two terms derive from studies concerning metal complexes and it has been suggested that for organic reactions, the terms 'nonbonded' and 'bonded' electron transfer should be used.

OUTGASSING: 1. Entrapped gasses within a solid material that escape to the atmosphere. 2. A form of pre-treatment in which a catalyst is heated in vacuo to remove adsorbed or dissolved gas.

OUTGROUP: A comparison of species or higher taxon used in systematics to a group of closely related species or taxa in order to assess whether a particular character shared by the group members are derived or ancestral.

OUTLIER: In statistics, an observation significantly different from other observations. It is worth investigating to see why there is such an observation. It might have occurred because of a measurement error or there might be some interesting story to be learned about that observation.

OUTPUT: 1. The end product of a machine, such as a generator (power in watts), a worker or by other means. 2. Information produced from a computer such as printouts, labels, reports, etc. 3. A number paired to an input by an imaginary function machine applying a rule. 4. The values for y in a function consisting of ordered pairs (x,y).

OUTPUT DEVICE: Any computer hardware equipment used for displaying information in a form intelligible to the user or the results of processing carried out by a computer. Examples are a screen, speaker, headphones and printer.

OUTPUT RATE: The number of results that is produced by an instrument divided by time of operation.

OUTSIDE: In mathematics, for a contour in the Euclidean plane, it is the set denoted $\text{out}\Gamma$, defined as the set of points not lying on the curve for which the winding number is zero.

OUTWASH: Sands and gravels deposited by streams of molten water from the melting ice of a glacier and laid down in stratified deposits. At the edge of a glacier, it may be about 100 metres thick and can cover very wide distances in kilometres.

OVAL WINDOW (FENESTRA OVALIS; VESTIBULAR WINDOW): A membrane-covered opening between the middle ear and the vestibule of the inner ear to which the base of the stapes is connected and through which the ossicles of the ear transmit sound vibrations to the cochlea. Vibrations coming into contact with the tympanum are transmitted through the middle ear by the three ear ossicles and transmitted to the inner ear by the oval window which is connected to the third ear ossicle. Vibrations from the tympanum reaching the oval window are amplified over twenty times.

OVALS OF CASSINI (CASSINI OVALS): In mathematics, the locus of the vertex of a triangle when the product of the sides adjacent to the vertex is held constant and the opposite side is held fixed. When the constant is one quarter of the length of the fixed side, this produces a lemniscate; for a smaller ratio, two distinct ovals are obtained.

OVARIAN CYCLE: The normal monthly sex cycle accompanied by a series of changes in the ovary that includes development of an ovarian follicle, rupture of the follicle, discharge of the ovum, and the development of the corpus luteum.

OVARIAN FOLLICLE (GRAAFIAN FOLLICLE; SECONDARY FOLLICLE; VESICULAR OVARIAN FOLLICLE): The female reproductive unit composed of an oocyte, granulosa cells and theca layers found inside the ovary. The oocyte inside the follicle would mature into an ovum or egg cell, which is then released from the ovary into the fallopian tube during ovulation.

OVARY: 1. The female reproductive organ that produces the female gamete (ovum or egg cells). It also produces the hormones oestrogen and progesterone in human females. Human females have two ovaries, containing hollow balls of cells (follicles) situated close to the opening of a fallopian tube in which the eggs develop and from which they are released in a regular cycle. At birth, between 150,000 to 500,000 follicles usually are present, but these numbers decrease with age. At menopause the few remaining follicles decay and the ovaries shrink and produce far less oestrogen. Biologically, during a female's development stages, only 300 - 400 follicles mature and release an egg, which develops into an embryo if fertilised. If not fertilised, the eggs are expelled from the body by menstruation. 2. The enlarged base of a flower's female organ, the pistil, consisting of the carpel or a number of fused carpels containing one or more ovules. After fertilisation, the ovary wall develops into the fruit enclosing the seeds.


Ovate leaf

OVATE: A term used to describe an organ, such as a leaf, that is egg shaped, with the broadest part nearest the point of attachment.

OVEN-DRY SOIL: One of the energy status of soil water in which soil has been dried at 105 °C until it reaches constant weight and does not contain any water with all the pore spaces filled with gases. The three others are field capacity, wilting point and air-dry (hygroscopic coefficient). A saturated soil has all of the pore space filled with water.

OVER: In mathematics, with respect to, especially where one algebraic structure is defined in terms of another. For example, if F is a field with respect to which the vector space V is defined, then V is said to be a vector space over F.

OVER DOMINANCE: The condition in which heterozygous (two or more different genes) organisms have phenotypes such as greater fitness better adapted than the corresponding homozygote (two identical genes) organisms. This is explained as the heterozygous organism having enhanced gene expression at a particular locus or set of loci relative to either homozygous alternative. This has been suggested as an explanation of hybrid vigour.

OVER EXPLOITATION: Unsustainable use of renewable natural resources such as timber and some wild life, especially by the acts of humans, leading to the point of diminishing returns and the extinction of the resource.

OVER FERTILISATION: Application of too much fertiliser to land. This has negative effects, for example, the plant looks wilted and can even turn brown; it can reverse the fertility of the soil and reduce its productivity; it can also detoriate the water table below the soil and absorption of such water by crops can lead to its being practically inedible since the chances of it developing a low pH value via acidic characteristic is very high. Also, rivers can convey surplus agricultural nutrients to the oceans. This causes a massive proliferation of algae.

OVER FISHING: Fishing at rates in any body of water, from a pond to the oceans that exceeds the sustained yield cropping of fish species, resulting in a net population decline. Overfishing leads to the depletion of fish resources, low biological growth rates and critical low biomass levels. This can cause environmental problems, such as the emergence of some pest fishes.

OVER GRAZING: Damage to plants or soil due to animals' intensive grazing for extended periods per unit time or reducing the plant's ability to hold the soil for sufficient recovery periods. The causes are the activities of livestock in poorly managed agricultural applications or by overpopulations of wild animals. Overgrazing can reduce the fertility of the soil, the invasion of foreign species, such as unwanted plants and other organisms, desertification and erosion, etc.

OVER KILL: The shooting, trapping or poisoning of certain populations, usually for sport or economic reasons.

OVER POTENTIAL (OVERVOLTAGE): 1. The difference between the electrode potential required to cause electrolysis and the thermodynamic value of the electrode potential in the absence of electrolysis. 2. A voltage that exceeds the normal voltage at which a device or circuit is designed to operate; or the extra voltage required to cause an ion to be discharged.

OVER-RING: A ring in which a given ring can be embedded. An example is quotient ring.

OVERDETERMINED: In mathematics, of a system of equations, usually linear, involving more equations than variables or unknowns. An overdetermined system is almost always inconsistent (it has no solution) when constructed with random coefficients. However, an overdetermined system will have solutions in some cases, for example, if some equation occurs several times in the system or if some equations are linear combinations of the others.

OVERCLOCK (OVERCLOCKING): A method of setting a computer to perform faster than its manufacturer's specified speed. It is done by changing the clock ratio of the front side bus (FSB) in the BIOS setup, by setting or changing jumpers, dip switches, CMOS settings, firmware updates or using software utilities. Overclocking allows users to get an overall performance boost and is most often performed on the computer's CPU and Video card.

OVERCONSOLIDATED SOIL DEPOSIT: A soil deposit that has been subjected to an effective pressure greater than the present overburden pressure.

OVERCROWDING: A state of being filled with more people, animals or things than is desirable.

OVERDRIVE: 1. A user-installable microprocessor from Intel Pentium which fits into a socket on 486-based PCs designed to accommodate an OverDrive chip to upgrade to a faster microprocessor. 2. In automotive engineering, it is the highest gear in a transmission, which allows the engine to operate at a lower RPM (revolutions per minute) for a given road speed. This allows the vehicle to achieve better fuel efficiency, and reduced noise level.

OVERLAP: 1. The interaction of orbitals on different atoms in the same region of space. 2. The amount by which one photograph includes the same area covered by another, customarily expressed as a percentage. 3. The portion of a map or chart that overlaps the area covered by another of the same series. 4. In mathematics, of sets, having some elements in common. 5. Of geometrical figures, having two parts that intersect or overlay one another. 6. Of two angles that are not adjacent, sharing a side and a vertex because they overlap.

OVERPOPULATION: The situation that arises when the number of individuals of a given species exceeds the number that its environment can sustain, in terms of resources such as space and food. Some of the causes are birth rate exceeding death rate and immigration exceeding emigration. Overpopulation can lead to damage to the environment, reduced life quality and a population crash (reduced numbers due to high mortality and failure to produce viable offspring).

OVERRIDING PLATE: A crustal plate which collides with a more dense plate and moves above the plate. A crustal plate is a rigid layer of the Earth's crust that is believed to drift slowly.

OVERSEEDING: Sowing seeds where other crops are already growing. This applies usually to vegetable gardens.

OVERTONE: A faint higher tone that has a frequency or pitch higher than the fundamental frequency (the natural frequency of the main body producing the sound). Overtones contribute greatly to the musical tone quality of a sound source.

OVERTYPE MODE (OVR): A text mode in Microsoft Word that overwrites text to the right of the cursor as the user types. This mode is entered and exited by pressing the insert key.

OVERVOLTAGE (OVERPOTENTIAL): 1. The difference between the electrode potential and the actual voltage required to cause electrolysis. 2. A voltage that exceeds the normal voltage at which a device or circuit is designed to operate. 3. The amount by which the applied voltage is in excess of the Geiger threshold in a radiation counter tube.

OVERWRITE: 1. In general, it is a term used to describe when new information or data replaces old information or data. 2. **Overwrite mode** is one of two typing settings on a keyboard, which allows new input to replace existing characters. The Insert keyboard key is commonly used to toggle this mode, enabling or disabling it. 3. An **overwrite virus** is a computer virus that overwrites a file with its own code, helping spread the virus to other files and computers.

OVIDUCT (FALLOPIAN TUBE): Any of the narrow tubes in female mammals which carries eggs from the ovaries into the uterus. Some mammals have two and others have only one, as the second tube fails to develop.

OVIPARITY (OVIPAROUS): A type of development in which the young hatch from fertilised eggs laid outside the mother's body as occurs in birds and some reptiles; or the eggs are fertilised externally, as occurs in amphibians and many lower forms.

OVIPORE: A pore-like sexual organ found in females of various animal species. During copulation the male transfers a spermatophore to the female's ovipore; which is usually stored in a spermatheca.

OVIPOSITOR: An organ at the hind end of the abdomen of female insects through which eggs are laid. It consists of a maximum of three pairs of appendages. In some insects such as wasps, sawflies and grasshoppers, the appendages are modified to perform tasks such as digging and piercing, so that the eggs can be laid in other inaccessible places for protection. The sting of bees and wasps is a modified ovipositor.

OVO-LACTO VEGETARIAN (LACTO-OVO-VEGETARIAN; LACTOOVOVEGETARIAN): A type of vegetarian, whose diet excludes animal flesh, but eats foods such as eggs, honey and dairy products that can be produced from a living animal without causing death or undue suffering to the animal or on health grounds.

OVOFLAVIN (LACTOCHROME; LACTOFLAVIN; RIBOFLAVIN; VITAMIN B2): The flavin in milk. It is a yellow crystalline compound with the molecular formula $C_{17}H_{20}N_4O_6$ that is a growth-promoting member of the vitamin B complex and occurs both free such as in milk and combined as in liver.

OVOID: Egg-shaped, a term applied to three-dimensional surfaces.

OVOVIVIPARITY (OVOVIVIPAROUS): A type of development in which the young is retained in the mother's uterus after fertilisation for physical protection until it is born or hatched. The embryo derives food from the yolk in the egg. It occurs in some invertebrates, fishes and reptiles.

OVULATION: The release of mature egg cells (oocytes, female gametes, ova) from the ovary. In mammals this is stimulated by luteinizing hormone. The developing egg cell within its follicle migrates to the ovary surface. During a female's menstrual cycle the mature ovarian follicle ruptures, releasing the mature egg cells from the follicle into the body cavity, from where it passes into the oviduct. In female mammals, ovulation occurs in the human female's equivalent of menstrual cycle known as oestrous cycle. **Ovulate** means to produce ovules.

OVULE: Unfertilised seed or structure found in seed plants and flowering plants which develops into a seed after fertilisation. It consists of three parts: the **integuments**, which forms the outer layer; the **nucellus** also known as **megasporangium** and the **megagametophyte** also known as the **embryo sac** in flowering plants. The ovules of gymnosperms are situated on the ovuliferous scales of the female cones while those of angiosperms are enclosed in the carpel. After fertilisation, the ovule develops into a seed.

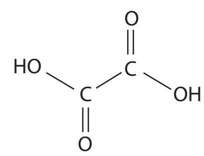
OVULE CULTURE: The culture of excised ovules on suitable media in vitro. Ovules have been cultured with pollen grains of the same species in order to observe the processes of fertilisation and embryo development. Culture with pollen of different species may enable certain crosses to be made that are normally impossible due to incompatibility factors in the stigma.

OVULIFEROUS SCALE: Any one of a group of large woody specialised leaves that form the female cone of a coniferous plant such as a pine that bears ovules and hence seeds.

OVUM (EGG CELL): A mature female reproductive cell in animals and plants or more specifically, a fertilised ovum in egg-laying animals such as birds and insects, especially a female sex cell when it is ready for fertilisation. The mature ova (plural of ovum) are generally spheroidal and large and the number of ova produced at one time varies from millions in many marine animals that spawn into the surrounding sea water to about a dozen or less in mammals in which adaptations for internal nourishment of the developing embryo and care of the young are highly developed. In lower plants and algae, the ovum is also called oosphere.

OX-EYE: A small cloud which occasionally appears off the eastern coast of Africa and spreads quickly to cover the whole of the sky, a sign of a severe storm accompanied by a violent wind.

OXALIC ACID (ETHANEDIOIC ACID): A colourless poisonous crystalline solid with the chemical formula HO_2CCO_2H, density 1.90 g cm^{-3} (anhydrous) and 1.653g cm^{-3} (dihydrate), melts at 189°C (462 K) and sublimes. It is soluble in water, alcohol and ether. Oxalic acid has two carboxyl groups joined directly, making it one of the strongest organic acids. It is prepared by heating sodium formate with sodium hydroxide to form sodium oxalate, which is converted to calcium oxalate and treated with sulfuric acid to obtain free oxalic acid. It occurs in certain plants such as sorrel and the leaf blades of rhubarb usually as its calcium or potassium salts. It is used in the leather and textile industries, in dyeing and bleaching, photography, ink manufacture, metal polishes and for removing rust and ink stains.

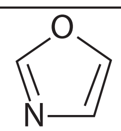


Chemical Structure of oxalic acid (ethanedioic acid)

OXALOACETATE: A four-carbon molecule found in mitochondria that condenses with acetyl conenzyme A to form citrate in the first reaction of the citric acid cycle (Krebs cycle). Oxaloacetate must be constantly regenerated in order for the citric acid cycle and the electron transport chain to continue.

OXALOACETIC ACID (OXOBUTANEDIOIC ACID; OXOSUCCINIC ACID): A colourless crystalline dicarboxylic acid with the chemical formula $C_4H_4O_5$ that is formed by oxidation of malic acid in the Kreb's cycle and by transamination from aspartic. It plays an important role as an intermediate in the metabolism of carbohydrates and as a precursor in the synthesis of amino acids.

OXAZOLE (1,3-OXAZOLE): A colourless heterocyclic compound with the chemical formula C_3H_3ON and boiling point 70 °C (343 K). It has a nitrogen atom and an oxygen atom in a five-membered ring. It is miscible with organic solvents and water and is used to prepare other organic compounds.



Chemical structure of Oxazole

OXBOW LAKE: A small curved lake developed in the meander loop of a river or stream channel that has abandoned its course. It generally forms as a river and shortens its course by cutting through a meander neck. Oxbow lakes silt up to form marshes and finally meander scars.

OXIDANT (OXIDISING AGENT; OXIDIZING AGENT): A substance in a chemical reaction which absorbs agents or electrons from the substance undergoing oxidation and gets reduced. Examples

O

of oxidising agents are ozone (O_3), permanganate ion (MnO_4^-), nitric acid (HNO_3), as well as oxygen.

OXIDASE: Any enzyme that catalyses oxidation-reduction reactions that involves the transfer of electrons to molecular oxygen. In these reactions, oxygen is reduced to water (H_2O) or hydrogen peroxide (H_2O_2). Examples of oxidases are glucose oxidase, monoamine oxidase, cytochrome P450 oxidase, NADPH oxidase and xanthine oxidase. One important oxidase is the cytochrome c oxidase, which is the main enzyme which makes the body use oxygen to generate energy and the final component of the electron transfer chain.

OXIDATION: 1. A chemical reaction in which a substance such as an atom, ion or molecule gives up electrons, gains oxygen or loses hydrogen where oxygen joins with a substance. An example is copper reacting with oxygen to form copper oxide. Some other examples are **burning**, which is a rapid oxidation; and **rusting**, which is a slow oxidation. 2. A form of chemical weathering, caused by the chemical reaction that takes place between certain iron-rich minerals in rock and the oxygen in water.

OXIDATION OF ORGANIC MATTER: Breakdown of organic matter by microbial activity into simpler organic molecules that require further decomposition or into mineralised nutrients. Soil organisms involved that perform the chemical and physical changes include microorganisms, earthworms, microarthropods, ants beetles etc).

OXIDATION HALF-EQUATION: The separate equation showing the substances that become oxidised by losing one or more electrons; or an equation that shows explicitly the total number of electrons lost. For example, in a reaction: $CuSO_{4(aq)} + Zn_{(s)} \rightarrow ZnSO_{4(aq)} + Cu_{(s)}$, the oxidation half-reaction is given by $Zn_{(s)} \rightarrow Zn^{2+}_{(aq)} + 2e$; the reduction half equation is given by $Cu^{2+}_{(aq)} + 2e \rightarrow Cu$.

OXIDATION NUMBER (OXIDATION STATE): A number assigned to a particular atom in a molecular or ionic entity that represents, actually or notionally, the charge on that atom. For example, the oxidation number of a central atom in a coordination compound is the charge it would have if all the ligands were removed along with the electron pairs that were shared with the central atom. It may be represented by a Roman numeral such as III and IV, placed as a superscript in front of the element symbol or in parentheses after the name of the element indicating the valency of the element immediately before the number, for example Fe^{III} or iron(III). In the latter case, there is no space between the element name and the oxidation number. For example, in water, the central atom is oxygen and the ligands are the two hydrogens which come to the oxygen to share its electrons. If the central atom is stripped, the hydrogens go away along with the two electrons, leaving the oxygen minus 2 electrons. Thus, oxygen has an oxidation number of -2. Its sign indicates whether the control has increased (negative) or decreased (positive) placed before the number.

OXIDATION-REDUCTION (REDUCTION-OXIDATION; REDOX): A chemical reaction in which atoms have their oxidation state changed. An example is the oxidation of carbon to yield carbon dioxide. When an atom of carbon combines with oxygen, the carbon loses electrons to oxygen and the carbon atom's oxidation state or number increases. Similarly, when it loses oxygen, it tends to gain electrons and hydrogen and its oxidation state or number decreases. Oxidation is defined as a loss of electrons or an increase in oxidation number, while reduction is defined as a gain of electrons or a decrease in oxidation number, whether or not oxygen itself is actually involved in the reaction. Oxidation-reduction reactions are balanced in the form of chemical equations by arranging the quantities of the substances involved so that the number of electrons lost by one substance is equalled by the number gained by another substance. The substance that loses electrons or undergoes oxidation is described as an **electron donor** or **reductant**. The substance that gains electrons or undergoes reduction is described as an **electron acceptor** or **oxidant**. Examples of reductants are the active metals, hydrogen, carbon and sulfurous acid. Oxidants are halogens, especially fluorine and chlorine, oxygen, ozone, potassium permanganate, potassium dichromate, nitric acid and concentrated sulfuric acid. When magnesium oxide is formed from magnesium and oxygen, the magnesium atoms have lost two electrons or the oxidation number has increased from zero to +2. This also applies when magnesium reacts with chlorine to form magnesium chloride. In solution, ferrous iron with oxidation number +2, may be oxidised to ferric iron with oxidation number +3, by the loss of an electron. In the reduction of cupric oxide the oxidation number of copper has changed from +2 to zero by the gain of two electrons. Oxidation-reduction reactions are important in combustion of fuels, electrolysis and extraction of metals ores. Metals corrode because of redox reaction.

OXIDATIVE ADDITION: The insertion of a metal complex into a covalent bond involving formally an overall two-electron losing one metal or a one-electron loss on each of two metals.

OXIDATIVE BURST (RESPIRATORY BURST): The rapid release of various highly reactive, toxic oxygen derivatives by certain cells, especially macrophages and neutrophils of the vertebrate immune system, as they come into contact with different bacteria or fungi. They are also released from the ovum of higher animals after the ovum has been fertilised and also from plant cells. These toxic compounds are used to kill bacteria engulfed by the macrophage or secreted to attack larger parasites outside the macrophage. To combat infections, immune cells use the NADPH oxidase, which is an enzyme family in the vasculature to reduce dioxygen (O_2) to oxygen free radical and then hydrogen peroxide (H_2O_2). Neutrophils and monocytes utilise myeloperoxidase to further combine hydrogen peroxide (H_2O_2) with Cl^- to produce hypochlorite, which plays a role in destroying bacteria. Absence of NADPH oxidase prevents the formation of reactive oxygen species and will result in chronic granulomatous disease.

OXIDATIVE COUPLING: The coupling of two molecular entities through an oxidative process, usually catalysed by a transition metal compound and involving dioxygen as the oxidant.

OXIDATIVE DEAMINATION: A reaction involved in the catabolism of amino acid that assists their excretion from the body. An example is the conversion of glutamate to a ketaglutarate, a reaction catalysed by the enzyme glutamine dehydrogenase. This process results in the production of two toxics: hydrogen peroxide and ammonia.

OXIDATIVE DECARBOXYLATION: The catabolic process in the Kreb's cycle in which both hydrogen and carbon dioxide are released during the conversion of citric acid to oxaloacetic acid. The released hydrogen enters the respiratory system whilst the carbon dioxide is expelled from the lungs.

OXIDATIVE DETERIORATION: Decay resulting from exposure to air.

OXIDATIVE PHOSPHORYLATION: A reaction occurring during the final stages of aerobic respiration, in which adenosine triphosphate (Adenosine triphosphate (ATP)), the molecule that supplies energy to metabolism is produced from adenosine diphosphate (ADP) by addition of a third phosphate group coupled to electron transport in the electron transport chain. It is an enzymatic process that occurs in both prokaryotes and eukaryotes. In eukaryotes, the process occurs as part of cellular respiration within the mitochondrion. In prokaryotes, it occurs in the cell membrane itself. It is the cell's principal method of storing the energy released by the oxidation of nutrients. Oxidative phosphorylation is one of the two ways in which a phosphate group is introduced into a molecule; the other one is substrate-level phosphorylation.

OXIDE: A binary compound formed between an element and oxygen. They are classified as according to their acid-base reaction as basic acid (Na_2O), acidic oxide (CO_2), neutral oxide (NO) and amphoteric oxide (Al_2O_3). Many different types of oxides can be prepared from a given element, except the noble gases. Some oxides exist naturally in the atmosphere and in the earth. Examples of these oxides are silicon dioxide (SiO_2) in quartz, aluminium oxide (Al_2O_3) in corundum, iron oxide (Fe_2O_3) in hematite, carbon dioxide (CO_2) gas and water (H_2O).

OXIDE FILM: The film of oxide formed on the surface of some metals as soon as they are exposed to air. Example is aluminium oxide.

OXIDE NETWORK: Network comprising only metal–oxygen linkages.

OXIDISED SPECIES: A term used to characterise the degree of oxidation (or reduction) in atoms, molecules and ions. It can be applied to an atom in a molecule or an ion which has a high oxidation state. An element or atom in a compound can be oxidised by reaction with oxygen, while it can be reduced by reaction with hydrogen. An oxidised species may be formed also through the loss of electrons (either to the positive electrode in a cell or through transfer to another atom or group of atoms). For example, the sulfur in H_2S is reduced sulfur relative to elementary sulfur, while SO_2 and SO_3 are oxidised. Metallic iron (Fe) is a reduced state of iron, while the Fe^{2+} ion (ferrous ion) and Fe^{3+} ion (ferric ion) are oxidised states of iron. Fe^{3+} is in a higher oxidation state than Fe^{2+}, which is in a higher oxidation state than Fe.

OXIDISING ACID: Brønsted acid that can act as a strong oxidising agent as well as an acid. Because the acidic proton can be reduced to hydrogen gas, all Brønsted acids are moderately strong oxidising agents. Acids such as nitric acid, perchloric acid, chloric acid, chromic acid and concentrated sulfuric acid are stronger oxidising agent because they contain oxygen in their anionic structure. Elements such as copper that are below hydrogen in the electromotive series cannot be oxidised and dissolved in a non-oxidising acid. However, an oxidising acid such as nitric acid can oxidise the copper and allow it to dissolve: $2HNO_3 + Cu \rightarrow CuO + H_2O + 2NO_2$, followed by the reaction between the acid and the oxide:

$2HNO_3 + CuO \rightarrow Cu(NO_3)_2 + H_2O$.

OXIDISING AGENT (OXIDISER; OXIDANT): A substance in a chemical reaction that causes oxidation to occur to an atom or group of atoms by removing electrons from it. This causes the component atom of the substance to undergo a decrease in oxidation number and the substance undergoes reduction. When this happens the oxidising agent is reduced. Examples of oxidising agents are ozone (O_3), permanganate ion (MnO_4^-), nitric acid (HNO_3) and oxygen (O_2). Fluorine (F_2) is the strongest oxidising agent because it has the highest reduction potential.

OXIDOREDUCTASE: Any of a class of enzymes that catalyse oxidation-reaction (the transfer of hydrogen or electrons between molecules). The electron donor is the **reductant** and the electron acceptor is the **oxidant**. For example, an enzyme that catalysed this reaction would be an oxidoreductase: $A^- + B \rightarrow A + B^-$, where A is the reductant (electron donor) and B is the oxidant (electron acceptor).

OXIME: Any of a group of compounds containing the group >C=NOH, produced by the condensation of ketones or aldehydes with hydroxylamine. Oximes are useful as intermediates in organic syntheses.

OXO CARBOXYLIC ACIDS (OXO ACIDS): Compounds having a carboxy group as well as an aldehydic or ketonic group in the same molecule, for example, $HC(=O)CH_2CH_2CH_2C(=O)OH$ 5-oxopentanoic acid. In an organic context the term is generally shortened to oxo acids.

OXO COMPOUNDS: Compounds containing an oxygen atom, =O, doubly bonded to carbon or another element. The term embraces aldehydes, carboxylic acids, ketones, sulfonic acids, amides and esters. Oxo used as an adjective (and thus separated by a space) modifying another class of compound, as in oxo carboxylic acids, indicates the presence of an oxo substituent at any position. To indicate double-bonded oxygen that is part of a ketonic structure, the term keto is sometimes used as a prefix, but such use has been abandoned by IUPAC for naming specific compounds. A traditional use of keto is for indicating oxidation of CHOH to C=O in a parent compound that contains OH groups such as carbohydrates, for example, 3-ketoglucose.

OXO PROCESS (OXO REACTION; HYDROFORMYLATION): A reaction that involves the addition of hydrogen and the formyl group -CHO, derived from hydrogen and carbon monoxide, to an alkene in the presence of a catalyst ($HCo(CO)_4$). The reaction is used in the conversion of the alkenes into aldehydes, alcohols and other oxygenated organic compounds.

OXO-: Prefix indicating the presence of oxygen in a chemical compound.

OXO-ION: An ion containing oxygen and other atoms. An example is sulfate (SO_4^-).

OXOANION (OXYANION): An anion in which an element, usually a non-metal, is bonded to one or more oxygen atoms. Examples are sulfates (SO_4^{2-}) such as magnesium sulfate and copper sulfate; and sulfites (SO_3^{2-}), for example, potassium bisulfite and sodium bisulfite.

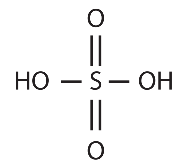


Fig. I:
Chemical Structure of sulphuric acid (an example of sulphate, an oxoanion)

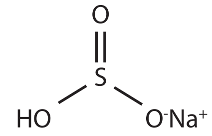


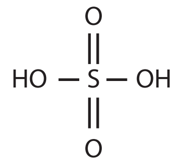
Fig II:
Chemical Structure of sodium bisulphite (an example of sulphite, an oxoanion)

OXOCHLORATE(I) ION: The International Union of Pure and Applied Chemistry (IUPAC) name for **hypochlorite ion** with the chemical formula ClO^-. It is any salt or ester of hydrochlorous acid. Examples include sodium hypochlorite (chlorine bleach or bleaching agent) and calcium hypochlorite (bleaching powder or swimming pool chlorination compound). Hypochlorites decompose in sunlight, giving chlorides and oxygen.

OXONIUM ION: Any oxygencation with three bonds, formed by the loss of an electron from a reaction with hydrogen atom. An example is hydronium ion (H_3O^+). Another type of oxonium ion is obtained by protonation or alkylation of a carbonyl group. An example is R - C = O^+ - R′ which forms a resonance structure with the fully fledged carbocation R - C^+ - O - R′. It is very stable.

OXONIUM YLIDES: 1. Compounds having the structure $R_2O^+–C^-R_2$. 2. A class of 1,3-dipolar compounds of general structure $R_2C=O^+–Y^-$, comprising carbonyl imides, carbonyl oxides and carbonyl ylides.

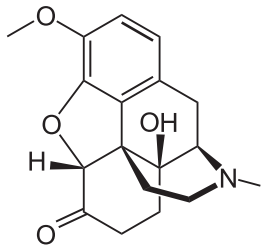
OXYACETYLENE (OXYFUEL): A type of fuel that burns acetylene in pure oxygen, producing temperatures of about 3000 °C (3273 K). It is widely used in welding to fuse metals and for cutting steel. To cut steel, the point is preheated with the oxyacetylene flame and a powerful jet of oxygen is then directed onto the point. The oxygen reacts with the hot steel to form iron oxide and the heat of this reaction melts more iron, which is blown away by the force of the jet.



Chemical Structure of sulphuric acid (an example of oxoacid)

OXYACID (OXOACID): An acid in which the acidic hydrogen atom is part of a hydroxyl group and formed from the reaction of a nonmetal oxide with water. Other oxyacids are nitric acid, nitrous acid and phosphorous acid.

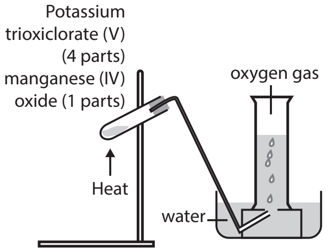
OXYCODONE: An opioid medication with the chemical formula $C_{18}H_{21}N_2$. It is synthesised from opium-derived thebaine. It is similar in structure to codeine but with the -OH group in codeine replaced by a carbonyl group. It is an analgesic medication used for the treatment of moderate to severe pain. It is also a controlled substance in most countries.



Chemical structure of Oxycodone

OXYFUEL (OXYACETYLENE): A type of fuel that burns acetylene in pure oxygen, producing temperatures of about 3000 °C (3273 K). It is widely used in welding to fuse metals and for cutting steel. To cut steel, the point is preheated with the oxyacetylene flame and a powerful jet of oxygen is then directed onto the point. The oxygen reacts with the hot steel to form iron oxide and the heat of this reaction melts more iron, which is blown away by the force of the jet.

OXYGEN: A colourless, odourless, tasteless, non-metallic gaseous chemical element with the atomic chemical symbol O, atomic number 8, atomic weight 15.9994, solidifies at -218 °C (55 K), melting point -218.4 °C (54.75 K), boiling point -183 °C (90 K), density at 0 °C 1.429 grams per litre and a valence of 2. It occurs as the diatomic molecule O_2. Ozone, O_3, is an allotrope of oxygen O_2. It is the most abundant element in the earth's crust and constitutes about 21% by volume and more than 46% by weight. It combines with hydrogen to form water and with many other elements to form oxides. Atmospheric dioxygen is of vital importance for all organisms that carry out aerobic respiration. It is important in the photosynthesis process of green plants, which assimilate carbon dioxide in the presence of sunlight and give off oxygen. A small amount of oxygen dissolves in water which is essential for the respiration of fish and other aquatic life. It forms oxides and is part of many other molecules and functional groups, including nitrate, sulfate, phosphate and carbonate, alcohols, aldehydes, carboxylic acids and ketones and peroxides. For industrial use, it is obtained by distillation of liquefied air. Industrially, it is used in steelmaking and other metallurgical processes. In the laboratory, oxygen gas can be prepared using either hydrogen peroxide (H_2O_2) reacting with manganese(IV) oxide or potassium trioxoclorate(V) reacting with manganese(IV) oxide.



The production of Oxygen gas

OXYGEN CYCLE: The movement of oxygen between the biotic and abiotic components of the environment. The oxygen cycle is closely linked to the carbon cycle and the water cycle. In the process of respiration, oxygen is taken in by living organisms and released into the atmosphere as carbon dioxide. The carbon dioxide is then absorbed by plants through photosynthesis to form carbohydrates and oxygen. During photosynthesis, oxygen is evolved by the chemical splitting of water and returned to the atmosphere. In the upper atmosphere, ozone is formed from oxygen and dissociates to release oxygen.

OXYGEN DEBT: The volume of extra oxygen consumed in aerobes when intensive work is done with inadequate oxygen supply or a physiological state produced by vigorous exercise in which the lungs cannot supply all the oxygen that the muscles need or the amount of oxygen required to remove the products of anaerobic respiration. This is observed after intensive physical activities when a person breaths heavily to restore his normal rhythm of breathing, which is a sort of paying back the oxygen debt. To meet the body's increased demand for energy, pyruvate is converted anaerobically to lactic acid, which is in turn converted into glucose and glycogen (animal starch) or lactic acid is converted into carbon dioxide and water, which accumulates in the tissues and requires oxygen for its breakdown. When oxygen is available again, lactic acid is oxidised in the liver, thus repaying the debt.

OXYGEN DISSOCIATION CURVE (OXYGEN–HAEMOGLOBIN DISSOCIATION CURVE): The S-shaped curve produced when the percentage of haemoglobin in its saturated form with oxygen (SO_2) on the vertical axis is plotted against the partial pressure of oxygen (PO_2) on the horizontal axis. It is a measure of the oxygen concentration in the surrounding medium and helps in understanding how our blood carries and releases oxygen. It is determined by how readily haemoglobin acquires and releases oxygen molecules into the fluid that surrounds it (haemoglobin's affinity for oxygen). In the lungs, where the partial pressure of oxygen is high, the blood is rapidly saturated with oxygen. On the contrary, a small drop in the partial pressure of oxygen results in a large drop in percentage saturation of haemoglobin.

OXYGEN EVOLVING COMPLEX, OEC (WATER-SPLITTING COMPLEX): A water oxidising enzyme consisting of a small complex of proteins and manganese ions associated with the reaction centres of photosystem II (shorter wavelengths) that splits water molecules to yield oxygen during the light reactions of photosynthesis. The complex exists in five states: S_0 to S_4. Photons trapped by photosystem II move the system from state S_0 to S_4. Because S_4 is unstable, it reacts with water to produce free oxygen. The oxidation provides the low-energy electrons from water that are excited by photosystem II (PSII) using the light energy trapped by chlorophyll pigments. The resultant high energy electrons can then flow through the electron transport chain. The OEC is located on the lumen side of the thylakoid membrane in the chloroplast and requires chloride ions to function.

OXYGEN LANCE: A pipe that directs oxygen under pressure into a molten pig iron during the manufacture of steel. The tip of the oxygen lance is water-cooled and directs thousands of cubic metres of pressurised oxygen into the furnace, where it joins with carbon and other substances to create a high temperature to burn and remove impurities and convert the pig iron to steel.

OXYGEN SAG CURVE: The resultant curve when the concentration of dissolved oxygen in a river into which sewage or some other pollutants has been discharged is plotted against the distance downstream from the sewage outlet. Samples of water are taken upstream and downstream from the sewage outlet. Because of the action of saprotrophic organisms that decompose the organic matter in the sewage and in the process use up the available oxygen, sewage presence reduces the oxygen content and increases the biological oxygen demand.

OXYGEN-FLASK COMBUSTION: A method for the determination of halogens and sulfur in organic compounds consisting of a combustion procedure followed by appropriate titrimetric determination. Combustion of the organic material in oxygen yields water-soluble inorganic products, which are determined as directed for the individual element. The combustion is carried out in a suitable conical flask into the stopper of which is fused one end of a piece of platinum wire. A flask of 500 ml is used, unless otherwise specified in the monograph. Towards the other end of the wire, a piece of platinum gauze is attached to provide a means of holding the substance clear of the absorbing liquid during combustion.

OXYGENATES: Organic compounds containing oxygen, which are either alcohols such as methanol, ethanol, isopropyl alcohol or ethers such as diisopropyl ether and methyl tert-butyl ether. They are used as additives in motor fuels to reduce carbon monoxide.

OXYGENATION: The process of adding a molecule of oxygen to another molecule, especially where oxygen is added to haemoglobin to become oxyhaemoglobin.

OXYHAEMOGLOBIN: The oxygenated form of haemoglobin, the pigment found in the red blood cells. It is a bright red substance and transports most of the oxygen in the bloodstream from the lungs to the tissues. Each molecule of haemoglobin carries four molecules of oxygen.

OXYLOPHYTE: A plant such as purple saxifrage (*Saxifraga oppositifolia*) that thrives in acidic soils.

OXYNTIC CELL (PARIETAL CELL): Any of the cells in the wall of the stomach that secrete hydrochloric acid and intrinsic factor, in response to histamine (via H_2receptors), acetylcholine (M_3 receptors) and gastrin (CCK2 receptors). Intrinsic factor is involved in the absorption of vitamin B_{12} in the small intestine. Hydrochloric acid forms part of the gastric juice.

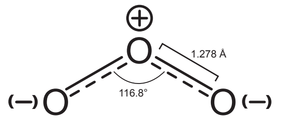
OXYTOCICS: Agents or substances containing properties that assist labour and hasten or promote easy childbirth. They include herbs such as cotton root bark, blue/black cohosh, mistletoe, parsley, sage and turmeric.

OXYTOCIN: A pituitary hormone, secreted by the posterior pituitary gland in mammals that stimulates the flow of milk in lactating females and the contraction of the smooth muscles in the uterus, intestines and the blood arterioles during parturition. Like antidiuretic hormone, oxytocin is produced in the neurosecretory cells of the hypothalamus but is stored and secreted by the posterior pituitary gland. It was first synthesised in 1953 and the first artificially produced pituitary hormone.

OYSTER MUSHROOM (*Pleurotus ostreatus*): An edible mushroom with a broad, fan or oyster-shaped cap that grows in many temperate and subtropical forests on dying hardwood trees. The flesh is white, firm and varies in thickness due to stipe arrangement. It possesses gills that are white or cream. Its mycelia can kill and digest nematodes, which is a way in which the mushroom obtains nitrogen, making it one of a few known carnivorous mushrooms.

OZONATION: 1. The formation of ozone in the Earth's atmosphere. In the stratosphere, about 20-50 km above the surface of the Earth, oxygen molecules dissociate into their constituent atoms under the influence of ultraviolet radiation of short wavelength. These atoms combine with oxygen molecules to form ozone. Ozone is also formed in the lower atmosphere from nitrogen oxides and other pollutants by photochemical reactions. Through its oxidative damage to leaves and their photosynthetic apparatus, tropospheric ozone is increasingly a cause of impaired plant growth and reduced crop yields. 2. A process of water treatment that destroys bacteria and other microorganisms through an infusion of ozone, a gas produced by subjecting oxygen molecules to high electrical voltages. It also reduces the formation of trihalomethanes (THMs), which result from the interaction of chlorine and naturally-occurring organic material in the source water.

OZONE: A highly reactive, extremely poisonous, pale blue atmospheric gas with the chemical formula O_3 that has a penetrating odour. It is an allotrope of oxygen in which the molecule contains three atoms instead of two as in the common form and accounts for the penetrating odour of the air after a thunderstorm or around electrical equipment. It is formed naturally in the ozone layer from atmospheric oxygen by electric discharge or exposure to ultraviolet radiation. It is also produced in the lower atmosphere (troposphere) naturally, and by photochemical reactions involving gases resulting from human activities. Tropospheric ozone acts as a greenhouse gas. In high concentrations, tropospheric ozone can be harmful to a wide-range of living organisms. Ozone also exits in limited quantities in the stratosphere, at about 10-50 km above the Earth's surface, created by the interaction between solar ultraviolet radiation and molecular oxygen (O_2). Ozone usually is manufactured by passing an electric discharge through a current of oxygen or dry air. The resulting mixtures of ozone and original gases are suitable for most industrial purposes, although purer ozone may be obtained from them by various methods. For example, upon liquefaction, an oxygen-ozone mixture separates into two layers, of which the denser one contains about 75% ozone. Although it resembles oxygen in many respects, ozone is much more reactive; hence, it is an extremely powerful oxidising agent, particularly useful in converting olefins into aldehydes, ketones or carboxylic acids. Because it can decolorise many substances, it is used commercially as a bleaching agent for organic compounds; as a strong germicide it is used to sterilise drinking water as well as to remove objectionable odours and flavours.



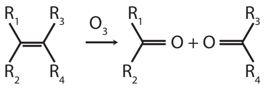
Chemical structure of Ozone

OZONE DEPLETER: Any chemical such as chlorofluorocarbons that destroys the ozone in the stratosphere.

OZONE LAYER: A region of the upper atmosphere (the stratosphere), about 10-50 km high which contains significant concentrations of ozone (O_3), extending from about 12 to 40 km, with ozone concentration reaching a maximum of between 20 and 25 km. This layer is where the concentration of ozone is greatest. The ozone layer is also present in trace quantities elsewhere in the Earth's atmosphere. This layer absorbs 93-99% of the solar high frequency, UV rays, which is potentially damaging to life on Earth. In this layer, most of the solar UV radiation is absorbed by the ozone molecules, causing a rise in the temperature of the stratosphere and preventing vertical mixing so that the stratosphere forms a stable layer. Ozone holes (an area where the ozone is up to 50% thinner than normal) develops periodically in the ozone layer above the Antarctica. They have been caused by a series of complex photochemical reaction involving nitrogen oxides produced from aircraft and more seriously, chlorofluorocarbons (CFCs) and halons. This may have led to an increase in skin cancer caused by UV exposure. CFCs rise to the stratosphere where they react with ultraviolet light to release chlorine atoms, which are highly reactive to catalyse the destruction of ozone.

OZONIDES: Any of a class of chemical compounds formed by reactions of ozone with other compounds. Organic ozonides, often made from olefins, are unstable, most of them decomposing rapidly into oxygen compounds, such as aldehydes, ketones and peroxides or reacting rapidly with oxidising or reducing agents. A few inorganic ozonides are known, containing the negatively charged ion O^-_3. An example is potassium ozonide (KO_3), an unstable orange-red solid formed from potassium hydroxide and ozone that, upon heating, decomposes into oxygen and potassium superoxide (KO_2).

OZONOLYSIS: A process that uses ozone to cleave unsaturated organic bonds to form ozonides to identify double bonds or synthesising chemicals. Some products may undergo rearrangement thereafter, depending on the medium to give a final product.


Generalised reaction scheme of Ozonolysis

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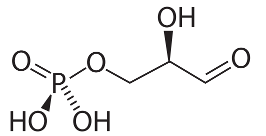
P

1,3,4,5,6-PENTAHYDROXY-2-HEXANONE (FRUCTOSE; LAEVULOSE; L-FRUCTOSE; FRUIT SUGAR): A simple sugar with the molecular formula $C_6H_{12}O_6$ that is found in honey, most sweet fruits and starches that gives fruits their sweet taste. It is the sweetest of all naturally occurring carbohydrates and has low glycaemic value. Fructose, together with glucose and galactose are absorbed directly into the bloodstream during digestion. Crystalline fructose obtained from processing corn or sugar is used in food and beverages as a nutritive component.

3-PHOSPHOGLYCERALDEHYDE (GLYCERALDEHYDE 3-PHOSPHATE; TRIOSE PHOSPHATE; PGAL): A compound with the chemical formula $C_3H_7O_6P$, formed during the dark reaction of photosynthesis and occurs as an intermediate in several central metabolic pathways of all organisms. It is a phosphate ester of the 3-carbon sugar glyceraldehyde. Some PGAL molecules exit the cycle and can be converted to glucose, while other PGAL molecules are used to reform ribulose biphosphate (RuBP) to continue the dark reaction.



Chemical Structure of 3-Phosphoglyceraldehyde

p: 1. The symbol for an unspecified prime number. It is also used as a prefix, for example, p-element. 2. Abbreviation for pico-, a prefix used for units of measurement to denote a factor of 10^{-12} in the S.I. system.

P: 1. Abbreviation for peta-, a prefix used in units of measurement to denote a factor of 10^{15} in the S.I. system. 2. Also written in lower case p, it is the symbol for pressure. 3. In logic, also written p, the symbol for an unspecified sentence or proposition. 4. The class of decision problems for which there are polynomial time algorithms.

P EQUALS NP: The conjecture, generally disbelieved, that every NP-decision problem possesses a polynomial time algorithm.

***p*-BLOCK ELEMENTS:** The elements consisting of the last six groups except helium. In the elemental form of the p-block elements, the highest energy electron occupies a p-orbital. This block comprises all the non-metals, semi-metals and some metals. It includes the Boron group in group 13 (IIIB, IIIA); the Carbon group in group 14 (IVB, IVA); the Nitrogen group or pnictogens in group 15 (VB, VA); the Oxygen group or the chalcogens in group 16 (VIB, VIA); the Halogens occupy group 17 (VIIB, VIIA). The last one is group 18 (Group 0), the Noble gases excluding helium.

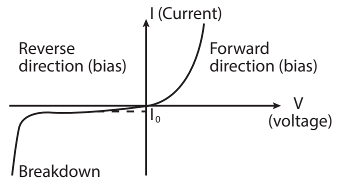
P-CODE (PSEUDOCODE): A computer programming language that resembles plain English that cannot be compiled or executed, but explains a resolution to a problem.

P-ELEMENT: In mathematics, the p-element of a group is an element of order p^{α} for p a prime and α a positive integer.

***P*-GROUP:** A group in which the order of every element is a power of p, where p is prime. A finite p-group has order p^{α} for some natural number α. The orders of different elements may be different powers of p. For each element g of a p-group, there exists a non-negative integer n such that the product of p^n copies of g is equal to the identity element.

P-MACHINE (PSEUDOMACHINE): A computer or computer processor created through software and not hardware.

P-N JUNCTION DIODE: A two-terminal semiconductor device made of materials such as silicon, to which impurities have been added. This allows electric current to flow in only one direction. Materials of *p*-type contain "holes" that behave like positively charged particles, whereas *n*-type materials are negatively biased and contain free electrons. If the positive pole of a battery is connected to the *p*-side of the junction and the negative pole to the *n*-side, charge flows across the junction. If the battery is connected in the opposite direction, very little charge can flow. The *p-n* junction forms the basis for computer chips, solar cells and other electronic devices.



Voltage-current characteristic of p-n junction

P-PRIMARY MODULE: In mathematics, a module over an integral domain in which for each element of the module there is a positive α such that $p^{\alpha x} = 0$, where p is a prime of the integral domain. the sequence in which formulas are evaluated.

P-PROTEIN (PHLOEM PROTEIN): A protein found in large amounts in the sieve elements tube of phloem tissue in plants. It takes various forms in the mature sieve tube element, depending on plant species, ranging from a network of filaments to discrete crystalline bodies. One general property is its ability to form a gel and functions as a puncture repair substance, forming a plug at any site of damage in the sieve tube element, thus preventing loss of food materials being translocated by the phloem. In an intact functioning sieve tube element, the P-protein is mainly found lining the interior wall.

***p*-SUBGROUP:** In mathematics, a subgroup that is a p-group.

P-TYPE CONDUCTIVITY: The conductivity associated with holes in a semiconductor, which are equivalent to positive charges and having impurity atoms with a valency of three such as boron, aluminium, indium and gallium. One hole per atom is created by an unsatisfied bond. The majority of carriers are, therefore, holes.

***p*-VALUE:** The probability, estimated under the null hypothesis, having outcome as extreme as the observed value in the sample.

P-WAVE (COMPRESSIONAL WAVE; PRIMARY WAVE): A compressional wave in which the displacement of the medium's particles is in line with or parallel to the direction of travel of the wave motion. It shakes the ground back and forth in the same direction and the opposite direction as the direction the wave is moving. It is the first wave to be recorded by a seismograph. It is the fastest of all seismic waves, travelling at a velocity of 5-7 kilometres per second in the crust and 8-9 kilometres per second in the upper mantle.

P/O RATIO (PHOSPHORUS OXYGEN RATIO): A measure of oxidative phosphorylation showing the number of inorganic phosphate molecules used for adenosine triphosphate (ATP) synthesis for every oxygen consumed. It may also indicate the number of ATPs produced for every pair of electrons travelling through the electron transport system.

P1 (PARENTAL GENERATION): The generation of individuals of different genotypes that is mated, usually for scientific purposes, to produce hybrids from which subsequent generations, known as the F_1, F_2, etc, will arise. Crosses between P1 yield the F1 Generation (First Filial Generation).Only pure-breeding (homozygous) individuals are selected for the P generation.